

Treatment effect estimation with missing outcomes and many covariate categories

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Abstract

We characterize the precision of population average treatment-effect estimation in balanced randomized trials with selectively observed binary outcomes, a binary surrogate, and a finite baseline alphabet. Outcomes are missing at random conditional on treatment, baseline category, and surrogate, with unknown cell-specific arrival probabilities bounded below by a supplied positive floor q . This floor indexes and tunes the procedure. For n , the sample size, and d , the number of baseline categories, we give an explicit mathematical attainment construction: a deterministic mixed-count estimator with squared-error risk bounded by a universal constant times the comparison rate

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{nq} + \left[\frac{d}{nq \log(e + nq)} \right]^2 \right\}.$$

This guarantee holds for every positive arrival floor. Under $q \log^2(e + nq) \leq 1/64$, matching lower bounds establish minimax squared-error order $r_{n,d,q}$ and optimal maximal expected length of order $\sqrt{r_{n,d,q}}$ for uniformly honest connected intervals at each fixed coverage level strictly between zero and one. At each fixed $0 < q < 1/2$, the point-risk comparison also matches, with constants depending on q . Near complete ascertainment, an observed-outcome contrast attains risk at most $4/n + (1 - q)^2$. For $d \geq 1024n^2$, the matching minimax order is $n^{-1} + (1 - q)^2$, making $1 - q = O(n^{-1/2})$ necessary and sufficient along arrival-floor sequences for parametric risk uniformly over all alphabet sizes. A target-preserving synthetic assignment transfers the construction to discrete observational regression contrasts and closes the stated logarithmic minimax gap at each fixed strict-interior treatment-positivity margin.

1 Introduction

The precision of a randomized trial at a prespecified analysis horizon depends on both the primary outcomes that have arrived and the information recorded for the full sample. Baseline categories, treatment assignments, and short-term measurements can remain available while primary outcomes are being collected. When outcome arrival varies across these observed characteristics, estimating a population treatment effect requires combining population cell weights with outcome information within cells. This paper characterizes the resulting statistical precision in a balanced randomized experiment with binary primary outcomes, a binary surrogate, and a finite baseline alphabet.

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The model imposes independent and identically distributed sampling, treatment assignment independent of baseline characteristics and potential outcomes and surrogates, assignment probability one half, and consistency. Outcome arrival is missing at random conditional on treatment, baseline category, and realized surrogate. Each occupied cell has an unknown arrival probability at least a supplied value q , where $0 < q \leq 1$. This lower bound indexes the model and is used to tune the estimator and interval; individual cell probabilities remain unknown. Baseline probabilities may vary arbitrarily, including zero probabilities. The surrogate supplies observed conditioning information for ascertainment, and the target is the population average treatment effect on the primary outcome. Under these conditions, [Lemma 3](#) expresses the effect as a signed sum of cell membership probabilities multiplied by outcome means among arrivals. Membership information from every sampled individual enters this representation.

Our first result is a constructive squared-error guarantee across the entire positive-arrival domain. Write n for the sample size, d for the number of baseline categories, and $N = nq$ for the effective sample-size scale. The mixed-count estimator in [Algorithm 1](#) satisfies a uniform risk bound of order

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{N} + \left[\frac{d}{N \log(e + N)} \right]^2 \right\},$$

where $r_{n,d,q}$ is the comparison rate, with a universal comparison constant, as established in [Theorem 6](#). The construction separates membership, pilot, and outcome estimation into three auxiliary streams. Membership counts estimate cell weights; the pilot selects between empirical arrived-outcome means and a Chebyshev polynomial statistic built from marked factorial counts. Exact conditional averaging over the auxiliary randomization produces a deterministic estimator.

Under the rare-arrival restriction

$$q \log^2(e + nq) \leq \frac{1}{64},$$

[Theorem 4](#) establishes matching minimax squared-error order $r_{n,d,q}$, with universal constants. The decision class includes all bounded parameter-independent randomized estimators. Its minimax value agrees with that for deterministic estimators by [Lemma 1](#). The lower bound retains the full observed experiment, including memberships whose outcomes have yet to arrive. A one-category family supplies the reciprocal effective-sample-size contribution, while an activated moment-matching construction supplies the category contribution. Along sequences satisfying the rare-arrival restriction and $N \rightarrow \infty$, [Proposition 5](#) consequently gives consistency exactly when $d = o\{N \log(e + N)\}$, and squared-error order N^{-1} exactly when $d = O\{\sqrt{N} \log(e + N)\}$.

The same rare-arrival experiment admits a matching uncertainty-quantification comparison. For each fixed miscoverage level $\alpha \in (0, 1)$, [Theorem 7](#) establishes optimal maximal expected length of order $\sqrt{r_{n,d,q}}$ among connected intervals with uniform coverage at least $1 - \alpha$. The comparison constants may depend on α . Coverage and expected length integrate jointly over the sample and any endpoint randomization. A deterministic squared-error envelope supplies an attaining interval, and the converse covers the full randomized endpoint decision class.

Near complete ascertainment, the remaining deficit supplies a complementary precision scale. Throughout the arrival-floor model, [Theorem 1](#) gives

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+ \leq \min \left\{ 1, \bar{C} r_{n,d,q}, \frac{4}{n} + (1 - q)^2 \right\},$$

where $\mathfrak{R}_{n,d,q}^+$ is the minimax squared-error risk and $\bar{C}$ is a universal constant. The observed-outcome arm contrast attains the last upper bound. For $d \geq 1024n^2$, [Theorem 2](#) establishes

matching order $n^{-1} + (1 - q)^2$ uniformly in q . Hence, along arrival-floor sequences, an eventual n^{-1} risk guarantee holding simultaneously over every alphabet size is equivalent to $1 - q_n = O(n^{-1/2})$, by [Proposition 1](#). Complete arrival gives n^{-1} order uniformly over arbitrary alphabet growth.

The construction also connects randomized outcome arrival to observational adjustment. An independent fair assignment retains an observational outcome when the synthetic assignment agrees with the observed treatment. Under a fixed strict-interior treatment-positivity margin, this transformation preserves the baseline-weighted regression contrast and supplies a randomized missing-at-random completion. Averaging the mixed-count estimator over synthetic assignments yields the deterministic observational procedure in [Theorem 8](#). At each fixed margin strictly between zero and one half, its minimax squared-error order is

$$\min \left\{ 1, \frac{1}{n} + \left[\frac{d}{n \log(e + n)} \right]^2 \right\},$$

with margin-dependent constants. Consistency and conditional exchangeability give the observational contrast its population causal interpretation. The same experiment comparison supplies the matched randomized point-risk order at every fixed $0 < q < 1/2$ in [Theorem 5](#), with constants depending on that floor.

The statistical mechanism is a weighted-cell-mean problem with many scarce arrived counts. Ordinary cell ratios accumulate a category-dependent bias; on the difficult large-alphabet branch, a Chebyshev correction reduces the aggregate bias to the scale $d/\{N \log(e + N)\}$. The converse uses moment-matched reciprocal priors whose activated observation laws remain close even though all cell memberships are retained. Each augmented record carries an independent activation flag with probability qH , where H is the upper endpoint of the latent prior support and has squared-logarithmic order. This explains the rare-arrival condition ensuring a valid same-experiment construction. [Section 5](#) and [Theorem 12](#) present these two sides.

These results connect three strands of existing work. Surrogate-assisted analysis explains how early measurements inform treatment-effect estimation and efficiency under appropriate observation and nuisance conditions [[Kallus and Mao, 2025](#), Theorems 2.1–2.2 and 4.1–4.2]. Decaying-labelling analysis establishes asymptotic inference under effective-sample-size and nuisance-rate conditions [[Zhang et al., 2025](#), Theorems 2.2 and 4.1, equation (2.3)]. Our comparisons characterize uniform squared error and honest expected interval length as category frequencies, alphabet size, and ascertainment vary over the specified finite models. For discrete observational adjustment, [Zeng et al. \[2026\]](#), Assumption 2, Theorems 1–2, Section 3.2, and Appendix C.5] provide the closest rate benchmark. The synthetic-assignment construction supplies an attaining polynomial estimator, while their lower-bound conclusion supports the fixed-positivity comparisons. The polynomial and moment-matching methods build on discrete functional estimation [[Jiao et al., 2015](#), [Wu and Yang, 2016, 2019](#)]; the expected-length criterion connects to honest confidence-interval theory [[Donoho, 1994](#), [Cai and Low, 2004](#), [Armstrong and Kolesár, 2020](#)].

The exposition proceeds through related work, the experiment and identification conditions, and the point-risk comparisons across ascertainment regimes. It then develops the constructive estimator, honest intervals under rare arrival, and the observational extension through synthetic assignment. Discussion and open questions follow. The appendices collect identification and decision-theory preliminaries, risk certificates, full-experiment lower bounds, interval and observational arguments, the verification note, and proofs of the main results.

2 Related work

The availability of short-term measurements before primary outcomes motivates both surrogate analysis and methods for incomplete trial data. [Prentice \[1989\]](#) formulates criteria for surrogate endpoints, while [Fleming and DeMets \[1996\]](#) emphasizes the substantive evidence required to connect surrogate responses to clinical effects. Short-term outcomes can also inform treatment-effect estimation while primary outcomes are still being collected, as in [Van Lancker et al. \[2020\]](#). Here, the surrogate enters the conditioning information for outcome arrival, and the target remains the population average treatment effect. Identification rests on randomized assignment, consistency, and outcome missingness at random conditional on baseline category, treatment, and surrogate, as specified in [Assumptions 2 and 4 to 6](#).

The closest comparison for estimation rates is [Zeng et al. \[2026, Assumption 2, Theorems 1–2, Section 3.2, and Appendix C.5\]](#). Their observational experiment comprises independent observations of discrete covariates, treatment, and outcome, with treatment propensities bounded away from zero and one by a fixed positivity margin. Writing n for the sample size and d for the number of covariate categories, their plug-in analysis gives a squared-error upper bound of order $n^{-1} + d^2/n^2$. Their minimax lower bound is of order $n^{-1} + d^2/\{n^2 \log^2 n\}$ in the stated regime $d \lesssim n \log n$, with constants depending on positivity. These conclusions give a logarithmic comparison between the plug-in guarantee and the minimax benchmark. Appendix C.5 develops the moment-matching argument underlying that benchmark. For the corresponding observational regression contrast under fixed strict-interior positivity on occupied categories, [Theorem 8](#) supplies a constructive polynomial-estimation guarantee through synthetic assignment. Its positivity margin, ϵ , is held fixed in this comparison.

An earlier conference announcement by [Kennedy \[2019, abstract 300411\]](#) considers an average causal effect with arbitrarily high-dimensional discrete confounders. It reports plug-in risk bounds, a minimax lower bound, and a variation with propensity-score positivity deteriorating with sample size, thereby establishing earlier consideration of the shrinking-positivity observational problem. The theorem-level comparison permitted by the available abstract is recorded in the limitations section.

The randomized analysis addresses a different observation mechanism. In the randomized fixed-horizon trial with a finite baseline alphabet, treatment is assigned with known probability one half, and primary outcomes arrive with unknown probabilities conditional on the observed cells. The arrival floor q is imposed on occupied cells in [Assumption 7](#). Under the rare-arrival restriction in [Assumption 8](#), [Theorem 4](#) matches upper and lower squared-error bounds at the rate defined in [Definition 17](#),

$$\min \left\{ 1, \frac{1}{nq} + \left[\frac{d}{nq \log(e + nq)} \right]^2 \right\}.$$

This comparison separates fixed treatment positivity in the observational experiment from outcome ascertainment in a balanced randomized experiment. The latter analysis quantifies the joint contribution of the effective sample size and the number of categories, while the synthetic-assignment construction connects the two experiments.

Surrogate-assisted efficiency provides a complementary perspective. [Kallus and Mao \[2025, Theorems 2.1–2.2 and 4.1–4.2\]](#) characterize efficiency bounds with different patterns of surrogate and outcome observation, under treatment unconfoundedness, outcome missingness at random, and the overlap conditions appropriate to each observation pattern. Their labelled-sample asymptotics allow the labelled fraction to vanish while the expected number of labelled

observations diverges, with fixed conditional distributions within the labelled and unlabelled populations. Attainment of the efficiency bound requires their nuisance convergence conditions: relevant products of regression and propensity or density-ratio errors must be negligible relative to the inverse square root of the expected labelled sample size. Treatment propensity estimates satisfy the stated boundedness condition, and estimated labelling propensities remain positive. Their moment requirements include uniformly bounded moments of order greater than two for the outcome residual, density ratio, and specified regression terms [Kallus and Mao, 2025, Assumption 7 and Appendix D.1, Assumption 9]. These results explain how predictive surrogates improve asymptotic precision under those conditions. The finite-alphabet minimax analysis here characterizes uniform risk as category frequencies and the alphabet size vary over the stated model class.

Decaying outcome observation is also central to Zhang et al. [2025, Theorems 2.2 and 4.1, equation (2.3)]. Their triangular-array observational framework combines treatment overlap with positive, potentially vanishing labelling probabilities conditional on treatment and covariates. Theorem 2.2 establishes asymptotic normality when both nuisance models are correctly specified, the inverse-propensity effective sample size diverges, and the product-rate condition in equation (2.3) holds. Its conditions also include finite inverse-propensity expectations, bounded outcome variance, conditional residual variance bounded above and away from zero, and the stated weighted nuisance and tail requirements. Their bias-reduced estimator in Theorem 4.1 accommodates nuisance approximation error under its product restriction and alternative sparsity conditions, together with sub-Gaussian design and conditional noise assumptions, covariance lower bounds, propensity moment control, and bounded offset signals [Zhang et al., 2025, Assumptions 1–3 and 5–8]. These conditions support asymptotic inference with estimated nuisance functions. Our randomized rare-arrival comparison instead measures worst-case squared error over a finite categorical model, and the interval analysis imposes uniform coverage over that same experiment.

The approximation methods have established antecedents in the estimation of discrete-distribution functionals. Polynomial estimators and moment-matched lower bounds are developed by Jiao et al. [2015], Wu and Yang [2016], and Wu and Yang [2019]. Their pairing reflects the dual relationship between polynomial approximation and separation of distributions with matching moments. Classical approximation theory supplies the reciprocal-function approximation used in the converse argument [Meinardus, 1967]; explicit reciprocal approximation formulas are also presented in Plonka and Tasche [2026, Section 3]. The application here combines these tools with observed cell memberships and selectively observed outcomes. The roles of reciprocal approximation and moment-matched priors are stated in Lemmas 6 and 7, and the resulting comparison concerns the full observed experiment in Theorem 12.

Bias correction for regression functionals supplies another comparison. Higher-order influence functions use higher-order U-statistics to obtain rate-optimal point and interval procedures for nonlinear functionals under smoothness and nuisance-estimation conditions [Robins et al., 2008, 2009]. Augmented minimax linear estimation corrects a regression plug-in estimator by a minimax linear estimate of its error and obtains semiparametric efficiency under its function-class conditions [Hirshberg and Wager, 2021]. Weak-overlap analysis instead thresholds doubly robust scores and derives asymptotic normality and Wald inference under propensity-tail, outcome-smoothness, and nuisance-rate restrictions [Dorn, 2025]. The present model replaces smooth nuisance classes by arbitrary masses and heterogeneous arrival probabilities on a finite, growing alphabet. Its guarantees are nonasymptotic worst-case squared-error and honest-length compar-

isons, and its logarithmic improvement comes from polynomial correction of scarce arrived-cell counts rather than smooth regression approximation.

The interval criterion connects the analysis to uncertainty quantification through maximal expected length subject to uniform coverage. [Donoho \[1994\]](#) relates statistical estimation to optimal recovery, [Cai and Low \[2004\]](#) studies expected-length comparisons and adaptation for confidence intervals, and [Armstrong and Kolesár \[2020\]](#) develops honest intervals that account for worst-case bias in nonparametric regression. For the present missing-outcome experiment, [Definition 29](#) specifies honesty for connected intervals, and [Theorem 7](#) gives the corresponding length comparison under rare outcome arrival. This places estimation error and honest interval length within a common finite-alphabet decision problem.

At complete outcome arrival, [Sudijono et al. \[2026\]](#) provides a useful comparison from randomized-experiment theory. They jointly optimize the assignment design and estimator for the sample average treatment effect in a finite population with binary potential outcomes, obtaining a sharp minimax analysis that includes a second-order risk expansion. Our complete-arrival result in [Theorem 3](#) concerns a population average treatment effect under independent and identically distributed sampling and a fixed balanced assignment design. The common first-order sampling scale therefore arises in two distinct decision problems: joint design and estimation for a finite-population sample effect, and estimation of a population effect under a specified randomized sampling experiment.

3 Experiment, identification, and risk

Throughout, $n \geq 1$ is the sample size, $d \geq 1$ is the number of baseline categories, and $q \in (0, 1]$ is a supplied lower bound on occupied-cell outcome arrival. Generic positive constants c and C are independent of these parameters unless dependence is stated explicitly. We write $u \lesssim v$ for $u \leq Cv$ and $u \asymp v$ for two such inequalities, uniformly over the stated parameter domain. Scalar errors are measured by absolute value. The arm index a and surrogate index s range over $\{0, 1\}$, while the baseline index x ranges over $\{0, \dots, d-1\}$. We use N , the effective sample-size scale, for $N = nq$; it combines the sample size with the guaranteed within-cell arrival probability.

3.1 Records and cells

The experiment has a finite baseline category X , a binary treatment assignment A , binary potential surrogates $S(a)$, binary potential outcomes $Y(a)$, and an outcome-arrival indicator R . The following definitions specify the record space and the coordinates selected by treatment assignment.

Assumption 1. `[ass:iid-sampling]` (Independent and identically distributed sampling) *The observed records $O_1, \dots, O_n$ are independent and identically distributed according to the observed-data law induced by P , the full-data probability law.*

Treatment assignment selects the realized surrogate S from the two potential surrogate coordinates.

Definition 1. `[synth_2]` (Full-data record coordinates) *A full-data record, for alphabet size $d \in \mathbb{N}$, consists of a baseline category X , a binary arm indicator A , control and treated surrogate coordinates $S(0)$ and $S(1)$, control and treated outcome coordinates $Y(0)$ and $Y(1)$, and an*

arrival flag R , with domains

$$X \in \{x \in \mathbb{Z} : 0 \leq x < d\}, \quad (A, S(0), S(1), Y(0), Y(1), R) \in \{0, 1\}^6.$$

The assigned-arm outcome Y is selected in the same way, and the finite record space carries its discrete measurable structure.

Definition 2. [synth_3] (Potential and assigned surrogates) *The potential surrogate $S(a)$, for $a \in \{0, 1\}$, denotes the record's binary control surrogate when $a = 0$ and binary treated surrogate when $a = 1$. The assigned-arm surrogate S , for any full-data record with binary assignment $A \in \{0, 1\}$, is defined by*

$$S = \begin{cases} S(1), & A = 1, \\ S(0), & A = 0. \end{cases}$$

A full-data probability law P , defined in [Definition 4](#), specifies the joint distribution of these coordinates.

Definition 3. [synth_4] (Potential and assigned-arm outcomes) *The binary potential outcomes $Y(0)$ and $Y(1)$ are the control and treated outcome coordinates of a full-data record. For a nonnegative integer d , such a record consists of a baseline category X in an alphabet of size d and binary coordinates $A, S(0), S(1), Y(0), Y(1), R$. The assigned-arm outcome Y is defined by*

$$Y = \begin{cases} Y(1), & A = 1, \\ Y(0), & A = 0. \end{cases}$$

The space of full-data records is equipped with the measurable structure in which every subset is measurable.

Outcome ascertainment is described within treatment–baseline–surrogate cells. The cell-index set $\mathcal{J}_d$ contains a cell j for each possible triple of these observed coordinates.

Assumption 2. [ass:randomized-independence] (Independence of treatment assignment) *Under P , the full-data probability law, treatment assignment A satisfies*

$$A \perp\!\!\!\perp \{X, S(0), S(1), Y(0), Y(1)\}.$$

Here X is the baseline covariate, $S(0), S(1)$ are the potential surrogates, and $Y(0), Y(1)$ are the potential outcomes.

Two probabilities distinguish membership from outcome arrival: $p_{axs}(P)$ is the full-membership cell probability, and $\rho_{axs}(P)$ is the conditional arrival probability in an occupied cell.

Assumption 3. [ass:balanced-randomization] (Balanced treatment assignment) *Under P , the full-data probability law, treatment assignment A satisfies*

$$P(A = 1) = \frac{1}{2}.$$

Thus an occupied cell has positive membership probability, and its arrived-cell probability is the product of its membership and conditional arrival probabilities. For the treatment contrast, the treatment sign $g(a)$ assigns opposite signs to the two arms.

Definition 4. [synth_1] (Full-data probability law) *The full-data law P , for a nonnegative integer d , is a probability measure on the finite record space*

$$(X, A, S(0), S(1), Y(0), Y(1), R) \in \{0, \dots, d-1\} \times \{0, 1\}^6.$$

Here X is the baseline category, A is the treatment arm, $S(0)$ and $S(1)$ are the potential surrogates, $Y(0)$ and $Y(1)$ are the potential outcomes, and R is the arrival flag. The baseline alphabet is empty when $d = 0$.

The observed record O is (X, A, S, R, RY) . Its sampled copies O_i retain cell membership and arrival status for every individual; when $R = 1$, the final coordinate records the outcome. The canonical observed sample law $\mathbf{O}_n$, defined in Definition 11, is the corresponding product distribution.

Assumption 4. [ass:surrogate-consistency] (Consistency of the surrogate) *The realized surrogate S equals the potential surrogate corresponding to treatment assignment A :*

$$S = S(A) \quad \text{almost surely under } P,$$

where P is the full-data probability law.

3.2 Sampling, randomization, and outcome arrival

Outcome consistency completes the link between observed and potential outcomes.

Assumption 5. [ass:outcome-consistency] (Consistency of the outcome) *The realized outcome Y equals the potential outcome corresponding to treatment assignment A :*

$$Y = Y(A) \quad \text{almost surely under } P,$$

where P is the full-data probability law.

Independent and identically distributed sampling is imposed within each triangular-array row, allowing the underlying law and alphabet size to vary across experiments [Zeng et al., 2026]. Outcome arrival is governed by the following conditional independence condition.

Assumption 6. [ass:arrival-mar] (Conditionally independent outcome arrival) *Under P , the full-data probability law, the outcome-arrival indicator R and realized outcome Y are conditionally independent given baseline X , treatment assignment A , and realized surrogate S :*

$$R \perp\!\!\!\perp Y \mid (X, A, S).$$

For later aggregation, we index treatment–baseline–surrogate cells explicitly.

Definition 5. [synth_6] (Cell alphabet and membership) *The cell alphabet $\mathcal{J}_d$, for a nonnegative integer d , is*

$$\mathcal{J}_d = \{0, 1\} \times \{x \in \mathbb{Z} : 0 \leq x < d\} \times \{0, 1\}.$$

For a full-data record with baseline category X , binary assignment A , and realized surrogate S , membership in a cell $j = (a, x, s) \in \mathcal{J}_d$ is defined by

$$A = a \quad \wedge \quad X = x \quad \wedge \quad S = s.$$

The next definition records each cell's population mass and its conditional arrival probability.

Definition 6. [synth_16] (Cell probabilities $p_{axs}(P)$ and $\rho_{axs}(P)$) *The full-membership probability of a cell is*

$$p_{axs}(P) := P(A = a, X = x, S = s), \quad (a, x, s) \in \mathcal{J}_d = \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\},$$

where P is a full-data law. On every occupied cell, meaning $p_{axs}(P) > 0$, define its conditional outcome-arrival probability by

$$\rho_{axs}(P) := P(R = 1 \mid A = a, X = x, S = s) = \frac{P(A = a, X = x, S = s, R = 1)}{p_{axs}(P)}.$$

The supplied arrival floor is imposed only on occupied cells.

Assumption 7. [ass:occupied-cell-arrival] (Occupied-cell arrival floor) *For each cell*

$$(a, x, s) \in \mathcal{J}_d = \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\},$$

let $p_{axs}(P)$, the full-membership cell probability, and $\rho_{axs}(P)$, the conditional outcome-arrival probability on an occupied cell, be defined as in [Definition 6](#). The occupied-cell arrival condition is

$$p_{axs}(P) > 0 \implies \rho_{axs}(P) \geq q.$$

The preceding randomization, consistency, and arrival restrictions define the full positive-arrival model.

Definition 7. [def:unrestricted-arrival-model-class] (Arrival-floor model $\mathcal{M}_{n,d,q}^+$) *The class $\mathcal{M}_{n,d,q}^+$ consists of full-data laws P , for integers $n, d \geq 1$ and an arrival floor $0 < q \leq 1$, satisfying the following conditions:*

- (Randomized independence.) [Assumption 2](#).
- (Balanced randomization.) [Assumption 3](#).
- (Surrogate consistency.) [Assumption 4](#).
- (Outcome consistency.) [Assumption 5](#).
- (Arrival MAR.) [Assumption 6](#).
- (Occupied-cell arrival.) [Assumption 7](#).

The arrival floor ranges over the entire interval $0 < q \leq 1$. Independent, identically distributed sampling is imposed separately on the experiment through [Assumption 1](#).

Outcome arrival missing at random given treatment, baseline, and surrogate equates the outcome distribution among arrivals with the full-cell outcome distribution whenever arrivals have positive probability [[Rubin, 1976](#), [Robins et al., 1994](#), [Kallus and Mao, 2025](#)]. The surrogate therefore serves as an observed conditioning variable for ascertainment: its realized value can capture variation in arrival associated with the assigned treatment and the outcome. Positive ascertainment within occupied cells is imposed next.

Definition 8. [def:ate-functional] (Average treatment effect $\tau(P)$) *The average treatment effect is*

$$\tau(P) := E_P[Y(1) - Y(0)],$$

where $Y(1)$ and $Y(0)$ are the potential outcomes under treatment and control.

The occupied-cell arrival floor is specific to this analysis. It guarantees an outcome-arrival probability of at least the supplied value q in every cell represented in the population, while permitting both cell membership probabilities and arrival probabilities to vary across cells. This condition supplies positive arrived-cell mass wherever the population cell contribution can be nonzero. A prospective analysis must justify this lower bound; the stated tuning and guarantees apply when it is valid.

Together, the randomization, consistency, and arrival restrictions define the arrival-floor model $\mathcal{M}_{n,d,q}^+$ in Definition 7.

Definition 9. [synth_19] (Treatment sign function) *Define the treatment sign function $g : \{0, 1\} \rightarrow \{-1, 1\}$ by*

$$g(a) := 2a - 1, \quad a \in \{0, 1\}.$$

Thus $g(0) = -1$ and $g(1) = 1$.

This family covers the full range of positive arrival floors, including complete outcome arrival at $q = 1$. Sampling determines the product experiment induced by each member of the family.

3.3 Identification and squared-error risk

The causal target is the population average treatment effect $\tau(P)$ in Definition 8.

Definition 10. [def:cell-functional] (Signed cell functional $\Phi(P)$) *The total signed observed-cell functional is*

$$\Phi(P) := 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a) c_{axs}(P),$$

where $\mathcal{J}_d$ is the treatment-baseline-surrogate cell-index set and $g(a)$ is the treatment sign function. On this index set, with $x \in \{0, \dots, d-1\}$, the full-membership cell probability $p_{axs}(P)$ is defined in Definition 6.

For a cell with positive arrived-cell probability,

$$P(A = a, X = x, S = s, R = 1) > 0,$$

define its arrived-cell outcome mean and weighted contribution by

$$\mu_{axs}(P) := E_P[Y \mid A = a, X = x, S = s, R = 1], \quad c_{axs}(P) := p_{axs}(P) \mu_{axs}(P).$$

When the arrived-cell probability is zero, including whenever $p_{axs}(P) = 0$, set

$$\mu_{axs}(P) := 0, \quad c_{axs}(P) := 0.$$

Thus Φ is defined on every full-data law.

Binary potential outcomes place this target in $[-1, 1]$. Its observed-data representation combines full-membership probabilities with outcome means among arrivals. The total signed cell functional $\Phi(P)$, defined in [Definition 10](#), uses an arrived-cell outcome mean $\mu_{axs}(P)$ and a membership-weighted contribution $c_{axs}(P)$, with explicit conventions for cells having zero arrived mass.

Definition 11. [\[synth_5\]](#) (Observed sample and its canonical law) *The random observed sample is $\mathbf{O}_n = (O_1, \dots, O_n)$, where $O_i = (X_i, A_i, S_i, R_i, R_i Y_i)$. For nonnegative integers n, d and a full-data probability law P with d baseline categories, its canonical law is*

$$\mathcal{L}_P(\mathbf{O}_n) := (\mathcal{L}_P(X, A, S, R, RY))^{\otimes n}.$$

Here $\mathcal{L}_P(X, A, S, R, RY)$ is the distribution under P of the observed record (X, A, S, R, RY) , whose last four coordinates are binary and whose final coordinate is the product of arrival and outcome. Thus $\mathcal{L}_P(\mathbf{O}_n)$ is the law of n independent observed records with this common distribution.

To distinguish the two roles in subsequent formulas, we write $\mathbf{O}_n = (O_1, \dots, O_n)$ for the random observed-record tuple and $\mathcal{L}_P(\mathbf{O}_n)$ for its product law. The canonical-law display above specifies the latter distribution.

For every law in [Definition 7](#), [Lemma 3](#) establishes $\tau(P) = \Phi(P)$; its proof appears in the appendix. The representation assigns zero contribution to null cells and uses observable arrived-cell means on occupied cells. Full cell membership remains observed even when the outcome is missing, so the representation separates population cell weights from within-cell outcome ascertainment. Balanced assignment supplies the factor of two in the signed contrast.

We evaluate estimation over the entire arrival-floor model through the minimax squared-error risk $\mathfrak{R}_{n,d,q}^+$ in [Definition 12](#). Its decision class $\mathcal{T}_n$ contains all parameter-independent bounded statistical decision rules, including a possibly randomized estimator $\mathbb{T}$.

Definition 12. [\[def:unrestricted-minimax-risk\]](#) (Unrestricted minimax squared-error risk) *The minimax squared-error risk $\mathfrak{R}_{n,d,q}^+$, for $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$, is defined by*

$$\mathfrak{R}_{n,d,q}^+ := \inf_{\mathbb{T} \in \mathcal{T}_n} \sup_{P \in \mathcal{M}_{n,d,q}^+} \int \int_{[-1,1]} (h - \tau(P))^2 \mathbb{T}(o, dh) \mathcal{L}_P(\mathbf{O}_n)(do).$$

Here $\mathcal{M}_{n,d,q}^+$ is the full-data model class in [Definition 7](#), and $\tau(P)$ is the population average treatment effect in [Definition 8](#). The class $\mathcal{T}_n$ consists of all parameter-independent Markov kernels from the observed sample space to $[-1, 1]$, including deterministic estimators as Dirac kernels.

The random observed-record tuple $\mathbf{O}_n$ and its product law under the sampling in [Assumption 1](#) are

$$\mathbf{O}_n = (O_1, \dots, O_n), \quad \mathcal{L}_P(\mathbf{O}_n) = (\mathcal{L}_P(O_1))^{\otimes n},$$

where each O_i is an observed record induced by P . The risk integrates over both the observed sample and the kernel draw.

By [Lemma 1](#), the defining infimum equals the infimum over deterministic estimators taking values in $[-1, 1]$. At $q = 1$, the same decision class $\mathcal{T}_n$ defines the minimax risk over $\mathcal{M}_{n,d,1}^+$, whose laws satisfy $R = 1$ almost surely.

The inner integral averages over the procedure's conditional output distribution given the sample, and the outer integral averages over sampling. Parameter independence means that the procedure is common to all laws in the specified model. Under squared loss, replacing a randomized output by its conditional mean given the sample weakly reduces risk for each law, as stated in [Lemma 1](#). This comparison makes the all-procedure criterion equivalent to its deterministic counterpart.

3.4 The rare-arrival experiment and comparison rate

A second model family restricts the arrival floor relative to the effective sample-size scale. Its defining condition uses the logarithmic scale ℓ , which equals $\log(e + N)$ on the statistical parameter domain fixed above.

Definition 13. [\[synth_7\]](#) (Total real logarithmic scale) *The logarithmic scale ℓ , for $n \in \mathbb{N}$ and $q \in \mathbb{R}$, is defined by*

$$\ell = \begin{cases} \log|\exp(1) + nq|, & \exp(1) + nq \neq 0, \\ 0, & \exp(1) + nq = 0, \end{cases}$$

where $\log$ on positive arguments is the natural logarithm.

The rare-arrival regime is selected by the following restriction on q and this logarithmic scale.

Assumption 8. [\[ass:rare-arrival-slice\]](#) (Rare-arrival restriction) *The arrival floor q and logarithmic scale ℓ satisfy*

$$q\ell^2 \leq \frac{1}{64}.$$

The rare-arrival restriction is specific to this analysis and selects parameter triples for which the arrival floor is small relative to the squared logarithmic scale. It restricts the guaranteed floor; individual occupied cells may have larger arrival probabilities. The rare-arrival model $\mathcal{M}_{n,d,q}$ in [Definition 14](#) combines this parameter restriction with the preceding statistical conditions.

Definition 14. [\[def:model-class\]](#) (Rare-arrival model $\mathcal{M}_{n,d,q}$) *$\mathcal{M}_{n,d,q}$ is the class of full-data probability laws P satisfying the following conditions:*

- *(Parameters.)* $n, d \geq 1$ and $0 < q \leq 1$.
- *(Sampling.)* [Assumption 1](#) holds.
- *(Randomization.)* [Assumptions 2](#) and [3](#) hold.
- *(Consistency.)* [Assumptions 4](#) and [5](#) hold.
- *(Arrival.)* [Assumptions 6](#) and [7](#) hold.
- *(Rare-arrival restriction.)* [Assumption 8](#) holds.

At parameter triples satisfying [Assumption 8](#), this model uses the same full-data restrictions as [Definition 7](#), together with the stated sampling condition. The observed experiment $\mathcal{E}_{n,d,q}$ in [Definition 15](#) is explicitly indexed by this rare-arrival model.

Definition 15. [def:observed-experiment] (Observed sample experiment) *The observed experiment $\mathcal{E}_{n,d,q}$, for $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$, is the family*

$$\mathcal{E}_{n,d,q} := \{\mathcal{L}_P(\mathbf{O}_n) : P \in \mathcal{M}_{n,d,q}\},$$

where $\mathcal{M}_{n,d,q}$ is the model class defined in [Definition 14](#). Here $\mathbf{O}_n$, the random observed-record tuple, and its law are specified by

$$\mathbf{O}_n = (O_i)_{i=1}^n, \quad O_i = (X_i, A_i, S_i, R_i, R_i Y_i), \quad \mathcal{L}_P(\mathbf{O}_n) = (\mathcal{L}_P(X, A, S, R, RY))^{\otimes n}.$$

Thus the records are independent and identically distributed with the observed marginal law induced by P . Each record retains the baseline category $X \in \{0, \dots, d-1\}$, assignment A , realized surrogate S , arrival indicator R , and arrival-multiplied outcome RY ; the last four coordinates are binary. The record space carries the discrete sigma-algebra.

The decision problem retains every sampled cell membership, including those whose outcomes have yet to arrive. Its minimax squared-error risk $\mathfrak{R}_{n,d,q}$, defined in [Definition 16](#), uses the same decision class and joint sample-and-procedure expectation as the unrestricted criterion.

Definition 16. [def:minimax-risk] (Minimax squared-error risk) *The minimax squared-error risk $\mathfrak{R}_{n,d,q}$, for $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$, is*

$$\mathfrak{R}_{n,d,q} := \inf_{\mathbf{T} \in \mathcal{T}_n} \sup_{P \in \mathcal{M}_{n,d,q}} \int \int_{[-1,1]} (h - \tau(P))^2 \mathbf{T}(o, dh) \mathcal{L}_P(\mathbf{O}_n)(do).$$

Here $\mathcal{M}_{n,d,q}$ is the full-data model class of [Definition 14](#), and $\tau(P)$ is the population average treatment effect of [Definition 8](#). The observed sample $\mathbf{O}_n = (O_i)_{i=1}^n$ denotes the random tuple of observed records. Its law, under [Assumption 1](#), is

$$\mathcal{L}_P(\mathbf{O}_n) = \bigotimes_{i=1}^n \mathcal{L}_P(O_i).$$

The decision class $\mathcal{T}_n$ consists of all parameter-independent Markov kernels from the observed-sample space to $[-1, 1]$, including deterministic estimators represented by Dirac kernels.

The notation $E_P[(\mathbf{T} - \tau(P))^2]$ denotes the displayed joint sample-and-kernel risk. By [Lemma 1](#), the minimax value also equals the infimum of the maximal squared-error risk over deterministic estimators taking values in $[-1, 1]$.

The two risk definitions distinguish the parameter domains over which the subsequent comparisons are stated. For the rare-arrival analysis, the comparison-rate expression $r_{n,d,q}$ in [Definition 17](#) combines the effective sample-size scale with the number of baseline categories.

Definition 17. [def:frontier-rate] (Frontier rate $r_{n,d,q}$) *The frontier rate is*

$$r_{n,d,q} := \min \left\{ 1, \frac{1}{nq} + \left[\frac{d}{nq \log(e + nq)} \right]^2 \right\}.$$

In terms of the setup notation, the expression is $\min\{1, N^{-1} + [d/(N\ell)]^2\}$. Its first term records the reciprocal effective sample-size scale, while its second term expresses the joint dependence on alphabet size and outcome ascertainment. The truncation at one matches the uniform squared-error bound of the estimator that always reports zero for a target in $[-1, 1]$. The estimation results that follow compare the defined risks with these scales under their stated ascertainment regimes.

4 Point estimation across ascertainment regimes

The squared-error comparisons depend jointly on outcome ascertainment and the number of baseline categories. For every arrival floor $0 < q \leq 1$, the unrestricted minimax risk $\mathfrak{R}_{n,d,q}^+$ of [Definition 12](#) admits an upper envelope combining the mixed-count rate in [Definition 17](#) with a dimension-free observed-outcome bound. The latter separates a sampling contribution of order n^{-1} from the squared ascertainment deficit $(1 - q)^2$. We first specify the estimators supporting this envelope, then give matched comparisons for large alphabets, rare arrival, and a fixed arrival floor $0 < q < 1/2$.

4.1 An all-arrival envelope and its estimators

The clipped observed-outcome arm contrast $\hat{\tau}_n^{\text{HT}}$ uses balanced assignment to form a signed average of arrived outcomes. Its projection $\Pi_{[-1,1]}$ keeps the estimate in the range of the population treatment effect. The following definition also fixes the clipping convention used by the mixed-count estimator.

Definition 18. `[def:complete-arrival-ht-estimator]` (Clipped observed-outcome arm contrast) *The clipped observed-outcome arm contrast is the map*

$$\hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) := \Pi_{[-1,1]} \left\{ \frac{2}{n} \sum_{i=1}^n (2A_i - 1) \mathbf{1}\{R_i = 1, R_i Y_i = 1\} \right\}, \quad n \geq 1.$$

Here n, d are nonnegative integers, and $\mathbf{O}_n$, the observed sample, consists of records $O_i = (X_i, A_i, S_i, R_i, R_i Y_i)$, with $X_i \in \{0, \dots, d - 1\}$ and all four remaining coordinates binary; the baseline support is empty when $d = 0$. The notation $R_i Y_i$ denotes the stored arrived-outcome coordinate; the indicator equals one exactly when both this coordinate and the arrival coordinate R_i equal one. Projection and clipping coincide:

$$\Pi_{[-1,1]}(u) = \text{clip}_{[-1,1]}(u) = \max\{-1, \min\{1, u\}\}, \quad u \in \mathbb{R}.$$

For $n = 0$, the map is defined to equal zero.

The mixed-count construction combines cell membership information with cell-specific arrived-outcome statistics. It uses the effective sample size $N = nq$ and the logarithmic scale $\ell = \log(e + N)$ in [Definition 13](#). Its polynomial degree k grows with this logarithmic scale, according to the following total tuning rule.

Definition 19. `[synth_8]` (Polynomial degree tuning) *The degree k , for $n \in \mathbb{N}$ and $q \in \mathbb{R}$, is defined by*

$$k = \begin{cases} \max \left\{ 0, \left\lfloor \frac{\log |e + nq|}{64} \right\rfloor \right\}, & e + nq \neq 0, \\ 0, & e + nq = 0. \end{cases}$$

On the polynomial branch, the first-kind Chebyshev polynomial T_k determines the needle polynomial $Q_k(t)$. Here t denotes an arrived-count intensity, and the threshold B sets the scale of the construction. The next two definitions specify the polynomial and its coefficient sequence a_v , where v indexes the degree.

Definition 20. [synth_18] (Chebyshev needle polynomial) Put $N = nq$, $\ell = \log(e + N)$, $k = \lfloor \ell/64 \rfloor$, and $B = 256\ell$. Let T_k be the first-kind Chebyshev polynomial, characterized by $T_k(\cos u) = \cos(ku)$. On the branch $\ell \geq 128$ and $d \geq \sqrt{N}$, where $k \geq 2$, define

$$Q_k(t) := \begin{cases} \frac{1 - T_k(1 - 2t/B)}{2k^2t/B}, & t > 0, \\ 1, & t = 0. \end{cases}$$

Here $t \geq 0$ is an arrived-count intensity, and the value at zero is the continuous polynomial extension of the displayed quotient. On every other branch, set $Q_k(t) := 1$ for every $t \geq 0$.

Definition 21. [synth_13] (Polynomial coefficient rule) The coefficients a_v , for $n, d, v \in \mathbb{N}$ and $q \in \mathbb{R}$, are defined by

$$a_v = \begin{cases} [t^v](1 - Q_k(t)), & 128 \leq \ell \text{ and } \sqrt{N} \leq d, \\ 0, & \text{otherwise.} \end{cases}$$

Here $[t^v]$ denotes the coefficient of t^v , $N = nq$ is the effective sample size, ℓ is the logarithmic scale, and Q_k is the Chebyshev needle polynomial. On the indicated branch, this polynomial is the needle radius divided by $2k^2$, multiplied by the polynomial quotient of

$$1 - T_k\left(1 - \frac{2t}{\text{needle radius}}\right)$$

upon division by t , with the remainder discarded. The degree k and needle radius are the construction's values at (n, q) , and T_k is the first-kind Chebyshev polynomial.

The estimator assigns observations to membership, pilot, and estimation streams. The pilot arrived count C'_j selects the cell statistic, while the estimation arrived count C_j and arrived-one count U_j supply its outcome information. Their definitions retain the observed cell memberships and distinguish the two uses of arrived outcomes.

Definition 22. [synth_17] (Arrived-outcome counts C'_j and C_j) The pilot and estimation arrived-outcome counts are

$$C'_j := \sum_{i \in \mathcal{I}_{\text{pil}}} \mathbf{1}\{A_i = a, X_i = x, S_i = s, R_i = 1\}, \quad C_j := \sum_{i \in \mathcal{I}_{\text{est}}} \mathbf{1}\{A_i = a, X_i = x, S_i = s, R_i = 1\}, \quad j = (a, x, s)$$

Here $\mathbf{O}_n$, the random observed sample, consists of the records

$$O_i = (X_i, A_i, S_i, R_i, R_i Y_i), \quad \mathbf{O}_n = (O_i)_{i=1}^n, \quad \mathcal{L}_P(\mathbf{O}_n) = (\mathcal{L}_P(O_1))^{\otimes n}.$$

Given a prefix length $K_0 \leq n$ and an assignment of its indices to membership, pilot, and estimation streams, $\mathcal{I}_{\text{pil}}$ and $\mathcal{I}_{\text{est}}$ denote the sets of prefix indices assigned to the pilot and estimation streams, respectively. The cell alphabet and observed-sample law are defined in [Definitions 5](#) and [11](#); the prefix and stream assignments are those used in [Algorithm 1](#).

Definition 23. [synth_10] (Estimation arrived-one count) The estimation arrived-one count U_j is defined for any $k, d \in \mathbb{N}$, collection of k observed records, assignment of each record to a stream in $\{0, 1, 2\}$, and cell $j = (a, x, s) \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}$ by

$$U_j := \#\{0 \leq i < k : i \text{ is assigned to stream 2, } A_i = a, X_i = x, S_i = s, R_i = 1, (RY)_i = 1\}.$$

Here $O_i = (X_i, A_i, S_i, R_i, (RY)_i)$ denotes the i th observed record, with baseline coordinate $X_i \in \{0, \dots, d-1\}$ and binary coordinates $A_i, S_i, R_i, (RY)_i$.

The selected cell statistic G_j uses a polynomial expression when the pilot count is small and an empirical arrived-outcome mean on the remaining cells. The associated factorial statistic H_j records the polynomial expression separately, and the elementary ratio D_j specifies the empirical-mean branch, including its value when the arrived count is zero.

Definition 24. [synth_15] (Selected polynomial cell statistic) *The selected cell statistic G_j , for natural-number parameters n, d , a real parameter q , a finite collection of observed records with baseline coordinate in $\{0, \dots, d-1\}$ and binary assignment, surrogate, arrival, and arrived-outcome coordinates, an assignment of each record to one of three streams, and a cell $j \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}$, is defined by*

$$G_j = \begin{cases} \sum_{v=1}^{\max\{k-1, 0\}} a_v U_j \prod_{t=0}^{v-2} \max\{C_j - 1 - t, 0\}, & 128 \leq \ell, \quad \sqrt{N} \leq d, \quad C'_j \leq B/4, \\ U_j / C_j, & C_j > 0 \text{ and the preceding branch condition fails,} \\ 0, & C_j = 0 \text{ and the preceding branch condition fails.} \end{cases}$$

Here $N = nq$ is the effective sample size, ℓ is the construction's logarithmic scale, $B = 256\ell$ is its threshold, k is its polynomial degree, and a_v are its polynomial coefficients. The quantities C'_j , C_j , and U_j are, respectively, the pilot-stream arrived count, the estimation-stream arrived count, and the estimation-stream arrived count with outcome equal to one in cell j . An empty sum equals zero and an empty product equals one.

Definition 25. [synth_14] (Factorial cell statistic) *The cell statistic H_j , for a finite collection of observed records, assignments to streams 0, 1, 2, a cell $j \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}$, $n, d \in \mathbb{N}$, and $q \in \mathbb{R}$, is*

$$H_j = \begin{cases} \sum_{v=1}^{\max\{k-1, 0\}} a_v U_j \prod_{r=1}^{v-1} \max\{C_j - r, 0\}, & \ell \geq 128 \text{ and } \sqrt{N} \leq d, \\ 0, & \text{otherwise.} \end{cases}$$

Here $N = nq$ is the effective sample size, ℓ is the logarithmic scale, and the degree parameter is

$$k = \max\{\lfloor \ell/64 \rfloor, 0\}.$$

The estimation-stream arrival count C_j and arrived-one count U_j are

$$C_j = \sum_i \mathbf{1}\{\text{record } i \text{ is assigned to stream 2, } (A_i, X_i, S_i) = j, R_i = 1\}, \quad U_j = \sum_i \mathbf{1}\{\text{record } i \text{ is assigned to stream 2, } (A_i, X_i, S_i) = j, R_i = 1, Y_i = 1\}$$

The coefficients associated with the Chebyshev polynomial Q_k are

$$a_v = \begin{cases} [t^v](1 - Q_k(t)), & \ell \geq 128 \text{ and } \sqrt{N} \leq d, \\ 0, & \text{otherwise,} \end{cases}$$

where $[t^v]$ denotes the coefficient of t^v . Empty sums equal zero and empty products equal one.

Definition 26. [synth_12] (Elementary cell ratio statistic) *The elementary cell statistic D_j , for any finite collection of observed records, any assignment of those records to three streams indexed by 0, 1, 2, and any cell $j \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}$, is*

$$D_j = \begin{cases} U_j/C_j, & C_j > 0, \\ 0, & C_j = 0. \end{cases}$$

Here the records have baseline coordinate $X \in \{0, \dots, d-1\}$ and binary coordinates A, S, R, RY , and the counts are

$$C_j = \#\{\text{records assigned to stream 2} : (A, X, S) = j, R = 1\},$$

$$U_j = \#\{\text{records assigned to stream 2} : (A, X, S) = j, R = 1, RY = 1\}.$$

Cell weights use the membership count M_j , which includes observations with either arrival status. The weight W_j divides this count by the stream intensity m . The following definitions therefore separate estimation of cell mass from estimation of its outcome mean.

Definition 27. [synth_9] (Membership stream cell count) *The membership count M_j , for any nonnegative integers k, d , a collection of k observed records O_i with baseline coordinates in $\{0, \dots, d-1\}$ and binary assignment, surrogate, arrival, and recorded-outcome coordinates, an assignment of each record to one of the streams 0, 1, 2, and a cell $j = (a, x, s) \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}$, is*

$$M_j := \#\{i : 0 \leq i < k, i \text{ is assigned to stream } 0, A_i = a, X_i = x, S_i = s\}.$$

Definition 28. [synth_11] (Cell membership weight) *The cell-weight statistic W_j is defined for nonnegative integers k, d, n , a collection of k observed records O_i with baseline coordinate $X_i \in \{0, \dots, d-1\}$ and binary coordinates A_i, S_i, R_i, RY_i , an assignment of each record to a stream labelled 0, 1, 2, and a cell*

$$j \in \{0, 1\} \times \{0, \dots, d-1\} \times \{0, 1\}.$$

Using the membership count and stream intensity

$$M_j = \#\{i \in \{0, \dots, k-1\} : O_i \text{ is assigned to stream } 0, (A_i, X_i, S_i) = j\}, \quad m = \frac{n}{6},$$

define

$$W_j = \frac{M_j}{m}.$$

The count M_j is interpreted as a real number in this quotient, and division by zero is assigned the value zero, so $W_j = 0$ when $n = 0$. The index sets $\{0, \dots, k-1\}$ and $\{0, \dots, d-1\}$ are empty when their respective sizes are zero.

These ingredients define the mixed-count estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$. A Poisson prefix length K_0 and independent stream assignments produce an auxiliary estimate; averaging over this randomization yields a deterministic function of the observed sample.

Algorithm 1 (Mixed-count estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$)

Given the observed sample $\mathbf{O}_n$, parameters n, d, q , effective sample size $N = nq$, logarithmic scale ℓ , and the declared cell-statistic and polynomial formulas, define the estimator by the following steps.

1. Draw an auxiliary prefix length independently of the data:

$$K_0 \sim \text{Poisson}(n/2).$$

When $K_0 \leq n$, independently assign each of the first K_0 observations to the membership, pilot, or estimation stream, each with probability $1/3$. Denote the prefix and stream-assignment randomization by ω .

2. On the event $K_0 > n$, set

$$\tilde{\tau}(\mathbf{O}_n; \omega) = 0.$$

When $K_0 \leq n$, form the declared statistics $M_j, C'_j, C_j, U_j, W_j, D_j$. If $N \leq 1$, set

$$\tilde{\tau}(\mathbf{O}_n; \omega) = 0.$$

3. When $K_0 \leq n$, $N > 1$, and either $\ell < 128$ or $d < \sqrt{N}$, set

$$\tilde{\tau}(\mathbf{O}_n; \omega) = \text{clip}_{[-1,1]} \left\{ 2 \sum_{j=(a,x,s)} g(a) W_j D_j \right\}.$$

On this branch, the Chebyshev quotient and its coefficients are left unevaluated.

4. When $K_0 \leq n$, $N > 1$, $\ell \geq 128$, and $d \geq \sqrt{N}$, form the declared total quantities Q_k, a_v, H_j, G_j and set

$$\tilde{\tau}(\mathbf{O}_n; \omega) = \text{clip}_{[-1,1]} \left\{ 2 \sum_{j=(a,x,s)} g(a) W_j G_j \right\}.$$

5. Output the conditional average

$$\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) := E_\omega[\tilde{\tau}(\mathbf{O}_n; \omega) \mid \mathbf{O}_n].$$

This average is a finite sum over prefix lengths $K_0 \leq n$ and their stream assignments, together with the zero contribution from the Poisson tail. It defines a total, deterministic, sample-measurable estimator whose squared-error risk is no larger than that of the auxiliary randomized output.

The resulting deterministic risk envelope $\mathfrak{U}_{n,d,q}$ is specified branch by branch in the next result. Its certificate uses the effective stream size N_0 , equal to mq . Together with the observed-outcome contrast, it provides two upper bounds applicable throughout the unrestricted arrival domain.

Theorem 1. [thm:unrestricted-near-complete-arrival-envelope] (Near-complete arrival risk envelope) *There exists a universal constant $\bar{C} > 0$ with the following properties. For every $n, d \geq 1$ and $0 < q \leq 1$, let $r_{n,d,q}$ be the rate in [Definition 17](#), let ℓ be the logarithmic scale in [Definition 13](#), let k be the polynomial degree specified in [Definition 19](#), and let B be the threshold*

specified in [Definition 20](#). Define the effective sample size N , stream intensity m , effective stream size N_0 , and prefix-tail constant γ_0 by

$$N = nq, \quad m = \frac{n}{6}, \quad N_0 = mq, \quad \gamma_0 = \log 2 - \frac{1}{2}.$$

Define the deterministic squared-error envelope $\mathfrak{U}_{n,d,q}$ as follows. When $N \leq 1$, set

$$\mathfrak{U}_{n,d,q} = 1.$$

When $N > 1$ and the estimator in [Algorithm 1](#) uses its elementary branch, set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, \left(\frac{8d}{eN_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\}.$$

When $N > 1$ and that estimator uses its polynomial branch, define the certificate quantities

$$b_{n,d,q} = 2 \left(\frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right),$$

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d \left(e^{\ell/8} + 1 \right) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right),$$

and set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \right\}.$$

For every full-data law $P \in \mathcal{M}_{n,d,q}^+$, the unrestricted-arrival model class in [Definition 7](#), suppose that $\mathbf{O}_n$, the observed sample, satisfies [Assumption 1](#) under the canonical product law of n observed records induced by P . The mixed-count estimator of [Algorithm 1](#) satisfies

$$E_P \left[\left\{ \hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q},$$

where $\tau(P)$ is the average treatment effect in [Definition 8](#). It also satisfies

$$E_P \left[\left\{ \hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right] \leq E_P \left[E_\omega \left[\left\{ \tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P) \right\}^2 \right] \right],$$

where $\tilde{\tau}(\mathbf{O}_n; \omega)$ is the auxiliary randomized output in [Algorithm 1](#), and the inner expectation uses its specified auxiliary randomization, including the zero output when the Poisson prefix length exceeds n .

For every $n, d \geq 1$ and $0 < q \leq 1$, the minimax squared-error risk $\mathfrak{R}_{n,d,q}^+$ over the parameter-independent bounded estimator kernels in [Definition 12](#) obeys the combined envelope

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+ \leq \min \left\{ 1, \overline{C} r_{n,d,q}, \frac{4}{n} + (1-q)^2 \right\}.$$

The clipped observed-outcome contrast $\hat{\tau}_n^{\text{HT}}$ of [Definition 18](#) satisfies

$$\sup_{P \in \mathcal{M}_{n,d,q}^+} E_P \left[\left\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right] \leq \frac{4}{n} + (1-q)^2,$$

where each expectation is taken under the canonical product law of n observed records induced by P .

The universal comparison constant $\overline{C}$ is independent of sample size, alphabet size, and arrival floor. The two upper bounds express different ways to control squared error: the mixed-count bound accounts for category complexity through $r_{n,d,q}$, whereas the observed-outcome bound accounts for missing outcomes through the ascertainment deficit. As q approaches one, the latter supplies a dimension-free guarantee. Conditional averaging in [Algorithm 1](#) preserves the mixed-count guarantee while eliminating the auxiliary randomization from the reported estimate.

4.2 Large alphabets and the uniform near-complete criterion

For alphabets at least quadratic in sample size, the sampling and ascertainment-deficit terms also give a matching lower bound. The next result establishes their joint order with universal constants throughout the stated large-alphabet region.

Theorem 2. [\[thm:unrestricted-large-alphabet-deficit-lower\]](#) (Unrestricted risk with large alphabets) *Let n, d be nonnegative integers and q a real number satisfying:*

- *(Sample size.)* $n \geq 1$.
- *(Alphabet size.)* $d \geq 1024n^2$.
- *(Arrival floor.)* $0 < q \leq 1$.

Then $\mathfrak{R}_{n,d,q}^+$, the unrestricted minimax squared-error risk defined in [Definition 12](#), satisfies

$$\max \left\{ \frac{1}{256n}, \frac{(1-q)^2}{1024} \right\} \leq \mathfrak{R}_{n,d,q}^+ \leq \frac{4}{n} + (1-q)^2.$$

These bounds also give the envelope $\frac{1}{2048} \left\{ \frac{1}{n} + (1-q)^2 \right\} \leq \mathfrak{R}_{n,d,q}^+ \leq 4 \left\{ \frac{1}{n} + (1-q)^2 \right\}$.

The comparison identifies two sources of difficulty on the same scale: ordinary sampling variation and the remaining ascertainment deficit. When $(1-q)^2$ exceeds n^{-1} , it determines the risk order in this region; when it is at most of order n^{-1} , sampling variation determines that order. Thus the dimension-free observed-outcome guarantee in [Theorem 1](#) is sharp up to universal constants for $d \geq 1024n^2$.

A guarantee uniform over every alphabet size must cover this large-alphabet region. Combining its lower bound with the all-arrival upper envelope yields an exact sequence-level criterion for parametric squared-error risk.

Proposition 1. [\[prop:unrestricted-uniform-near-complete-elbow\]](#) (Uniform near-complete arrival elbow) *Let $(q_n)_{n \geq 0}$ satisfy $0 < q_n \leq 1$ for every n , and let $\mathfrak{R}_{n,d,q}^+$ be the unrestricted-arrival minimax squared-error risk defined in [Definition 12](#). For every integer $n \geq 1$,*

$$n\mathfrak{R}_{n,d,q_n}^+ \leq 4 + n(1 - q_n)^2 \quad \text{for every integer } d \geq 1.$$

Moreover, for every real number

$$b < \frac{1}{2048} \{1 + n(1 - q_n)^2\},$$

there exists an integer $d \geq 1$ such that

$$b < n\mathfrak{R}_{n,d,q_n}^+.$$

The following two conditions are equivalent:

- **(Arrival rate.)** There exists a real constant $M \geq 0$ such that, for all sufficiently large integers n ,

$$1 - q_n \leq \frac{M}{\sqrt{n}}.$$

- **(Uniform risk bound.)** There exists a real constant D such that, for all sufficiently large integers n and every integer $d \geq 1$,

$$\mathfrak{R}_{n,d,q_n}^+ \leq \frac{D}{n}.$$

For every pair of integers $n, d \geq 1$,

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q_n}^+.$$

If the sequence $(\sqrt{n}(1 - q_n))_{n \geq 0}$ is unbounded above, then for every real constant C , there are infinitely many integers n such that

$$C < n\mathfrak{R}_{n,1024n^2,q_n}^+.$$

The quantifier over all d is central to this criterion. A deficit of order $n^{-1/2}$ is necessary and sufficient for an eventual n^{-1} upper bound holding simultaneously over every alphabet size. The explicit sequence $d_n = 1024n^2$ witnesses the converse when the rescaled deficit is unbounded. For any fixed deficit multiplier, the sufficient window also gives the following numerical comparison along arbitrary alphabet-size sequences.

Proposition 2. [prop:unrestricted-near-complete-phase-boundary] (Near-complete arrival phase boundary) *Let $\mathfrak{R}_{n,d,q}^+$ be the unrestricted minimax squared-error risk defined in Definition 12. Fix a real constant M and sequences $(d_n)_{n \geq 0}$ and $(q_n)_{n \geq 0}$ satisfying:*

- **(Constant.)** $M \geq 0$.
- **(Alphabet size.)** d_n is a natural number with $d_n \geq 1$ for every n .
- **(Arrival floor.)** $q_n \in (0, 1]$ for every n .
- **(Arrival deficit.)** For all sufficiently large n ,

$$1 - q_n \leq \frac{M}{\sqrt{n}}.$$

Then, for all sufficiently large n ,

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d_n,q_n}^+ \leq \frac{4 + M^2}{n}.$$

At complete ascertainment, the deficit is zero. The following endpoint specialization records the observed-outcome contrast as a direct witness to the dimension-free parametric benchmark.

Theorem 3. [thm:unrestricted-complete-arrival-frontier] (Complete-arrival minimax risk frontier) *For every sample size $n \geq 1$ and alphabet size $d \geq 1$, let $\mathcal{M}_{n,d,1}^+$ be the complete-arrival model class of Definition 7, and let $\mathbf{O}_n$ consist of n independent, identically distributed observed records induced by the full-data law P . Let $\mathfrak{R}_{n,d,1}^+$ be the minimax squared-error risk of Definition 12, and let $\hat{\tau}_n^{\text{HT}}$ be the clipped observed-outcome arm contrast of Definition 18. Writing $\tau(P)$ for the population average treatment effect, we have*

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,1}^+ \leq \sup_{P \in \mathcal{M}_{n,d,1}^+} E_P \left[\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \}^2 \right] \leq \frac{4}{n}.$$

Consequently, at the complete-arrival endpoint $q = 1$, the minimax squared-error risk has order n^{-1} with universal constants, uniformly over $d \geq 1$.

Complete arrival makes the observed-outcome arm contrast sufficient to attain the minimax order for every alphabet size. The sampling lower bound remains of the same order, so the endpoint comparison is uniform even when the alphabet grows arbitrarily quickly. Its sequence formulation follows directly.

Proposition 3. [prop:unrestricted-complete-arrival-phase-boundary] (Complete-arrival point-risk boundary) *There exist real constants $0 < c \leq C$ such that, for every sequence of alphabet sizes $d_n \geq 1$, the unrestricted minimax squared-error risk $\mathfrak{R}_{n,d_n,1}^+$ defined in Definition 12 at complete arrival $q = 1$ satisfies, for all sufficiently large n ,*

$$\frac{c}{n} \leq \mathfrak{R}_{n,d_n,1}^+ \leq \frac{C}{n}.$$

The comparison constants are independent of the chosen alphabet-size sequence.

4.3 Rare arrival and effective sample size

Under Assumption 8, the rate in Definition 17 gives a matched comparison for the rare-arrival risk of Definition 16. This comparison retains the full observed experiment, including memberships whose outcomes have yet to arrive, and applies to the entire decision class in that risk definition.

Theorem 4. [thm:matched-minimax-frontier] (Matched minimax frontier) *There exist universal constants $0 < \underline{c} \leq \overline{C} < \infty$ such that, for every $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$ satisfying*

- (Sample size and dimension.) $n \geq 1$ and $d \geq 1$.
- (Arrival parameter.) $0 < q \leq 1$.
- (Rare-arrival restriction.) Assumption 8 holds.

the minimax squared-error risk $\mathfrak{R}_{n,d,q}$ defined in Definition 16 and the frontier rate $r_{n,d,q}$ defined in Definition 17 satisfy

$$\underline{c} r_{n,d,q} \leq \mathfrak{R}_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

The universal lower constant $\underline{c}$ makes the two components of the rate jointly informative about minimax difficulty. The reciprocal effective sample size captures the sampling scale, while

the squared category term captures the cost of estimating the treatment contrast across many cells:

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{N} + \left(\frac{d}{N\ell} \right)^2 \right\}.$$

At fixed alphabet size, the category term becomes asymptotically negligible as N grows. The next result also establishes this effective-sample-size order within the submodel having a constant surrogate in both treatment arms.

Proposition 4. [prop:fixed-d-effective-sample-reduction] (Fixed-alphabet effective sample reduction) *Fix an integer $d \geq 1$. Let $n(t)$ and $q(t)$ be sequences indexed by $t \in \mathbb{N}$ satisfying:*

- *(Sample size and arrival.)* For every t , $n(t) \in \mathbb{N}$, $n(t) \geq 1$, and $0 < q(t) \leq 1$.
- *(Rare-arrival slice.)* For every t , the pair $(n(t), q(t))$ satisfies [Assumption 8](#).
- *(Effective sample size.)* The effective sample size

$$N(t) = n(t)q(t)$$

satisfies $N(t) \rightarrow \infty$ as $t \rightarrow \infty$.

Then the frontier rate and minimax squared-error risk of [Definitions 16](#) and [17](#) satisfy

$$r_{n(t),d,q(t)} \asymp N(t)^{-1}, \quad \mathfrak{R}_{n(t),d,q(t)} \asymp N(t)^{-1} \quad (t \rightarrow \infty).$$

The minimax squared-error risk with the same estimator class and loss as in [Definition 16](#), taken over the restricted full-data laws

$$\{P \in \mathcal{M}_{n(t),d,q(t)} : P(S(0) = 0, S(1) = 0) = 1\},$$

is also of order $N(t)^{-1}$ as $t \rightarrow \infty$. Each order comparison means eventual upper and lower bounds by positive constant multiples of $N(t)^{-1}$.

Allowing the alphabet to grow separates consistency from effective-sample-size precision. The category term vanishes when d is small relative to $N \log(e + N)$, and it remains at most of order N^{-1} when d is at most of order $\sqrt{N} \log(e + N)$. The matched comparison turns these rate calculations into exact equivalences for the minimax risk.

Proposition 5. [prop:phase-boundaries] (Rare-arrival minimax phase boundaries) *Let $(n_t)_{t \in \mathbb{N}}$ and $(d_t)_{t \in \mathbb{N}}$ be sequences of natural numbers, and let $(q_t)_{t \in \mathbb{N}}$ be a sequence of real numbers. Define the effective sample size by*

$$N_t = n_t q_t.$$

Suppose that:

- *(Positive sizes.)* $n_t \geq 1$ and $d_t \geq 1$ for every $t \in \mathbb{N}$.
- *(Arrival probabilities.)* $0 < q_t \leq 1$ for every $t \in \mathbb{N}$.
- *(Rare-arrival slice.)* (n_t, q_t) satisfies [Assumption 8](#) for every $t \in \mathbb{N}$.

- (*Effective sample growth.*) $N_t \rightarrow \infty$ as $t \rightarrow \infty$.

Then the minimax squared-error risk $\mathfrak{R}_{n_t, d_t, q_t}$ of [Definition 16](#) satisfies both equivalences:

$$\begin{aligned} \mathfrak{R}_{n_t, d_t, q_t} \longrightarrow 0 &\iff d_t = o(N_t \log(e + N_t)), \\ \mathfrak{R}_{n_t, d_t, q_t} = \Theta(N_t^{-1}) &\iff d_t = O\left(\sqrt{N_t} \log(e + N_t)\right), \end{aligned}$$

where all limits and asymptotic comparisons are taken as $t \rightarrow \infty$.

These thresholds quantify how many baseline categories can be accommodated at a given outcome-maturity floor. They apply along sequences satisfying the rare-arrival restriction and growing effective sample size. In particular, they distinguish the broader category-growth range permitting consistency from the smaller range preserving squared-error risk of order N_t^{-1} .

4.4 A fixed strict-interior arrival floor

A separate matched comparison applies to the unrestricted model when the arrival floor is fixed at $0 < q < 1/2$. Its lower comparison uses the discrete observational lower bound of [Zeng et al. \[2026, Theorem 2, Section 3.2, p. 9, and its proof in Section C.5, pp. 39–44\]](#), including the restriction setting the control regression to zero. The resulting comparison holds for every $n, d \geq 1$, with constants depending on the fixed arrival floor.

Theorem 5. [\[thm:unrestricted-fixed-q-minimax-frontier\]](#) (Unrestricted fixed-arrival-floor minimax frontier)¹ *Fix an arrival floor q with $0 < q < 1/2$. There exist real constants $0 < \underline{c}(q) \leq \overline{C}(q) < \infty$, depending only on q , such that for every pair of integers $n, d \geq 1$, the unrestricted minimax squared-error risk $\mathfrak{R}_{n, d, q}^+$ defined in [Definition 12](#) satisfies*

$$\underline{c}(q) \min \left\{ 1, \frac{1}{n} + \left(\frac{d}{n \log(e + n)} \right)^2 \right\} \leq \mathfrak{R}_{n, d, q}^+ \leq \overline{C}(q) \min \left\{ 1, \frac{1}{n} + \left(\frac{d}{n \log(e + n)} \right)^2 \right\}.$$

With the same constants, the frontier rate $r_{n, d, q}$ defined in [Definition 17](#) satisfies

$$\underline{c}(q) r_{n, d, q} \leq \mathfrak{R}_{n, d, q}^+ \leq \overline{C}(q) r_{n, d, q}.$$

For a fixed arrival floor, effective sample size is proportional to sample size, so the comparison can be written using n and $\log(e + n)$. The lower constant $\underline{c}(q)$ and upper constant $\overline{C}(q)$ may depend on that floor. This is a second logarithmic matched regime: [Theorem 4](#) gives universal constants under the rare-arrival restriction, while the present result gives a comparison over the unrestricted model at each fixed $q \in (0, 1/2)$.

The fixed-floor comparison yields corresponding sequence-level thresholds. The next statement records both the exact consistency condition and the necessary and sufficient category-growth order for parametric squared-error risk.

¹**Formalization scope.** This result uses the cited conclusion [[Zeng et al., 2026](#)] (Theorem 2, Section 3.2, p. 9, together with its proof in Section C.5, pp. 39–44, which sets the control regression to zero). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Proposition 6. [prop:unrestricted-fixed-q-phase-boundaries] (Unrestricted fixed-arrival-floor phase boundaries)² Fix $0 < q < 1/2$, where q is the arrival floor, and let $\mathfrak{R}_{n,d,q}^+$ be the unrestricted-arrival minimax squared-error risk defined in Definition 12. For every sequence of integers $d_n \geq 1$,

$$\mathfrak{R}_{n,d_n,q}^+ \rightarrow 0 \iff d_n = o(n \log(e + n)), \quad n \rightarrow \infty.$$

There exists a real constant $c > 0$ such that, for every real $M > 0$, there exists a real constant $C \geq c$ with the following property: for every sequence of integers $d_n \geq 1$, if

$$d_n \leq M\sqrt{n} \log(e + n) \quad \text{for all sufficiently large } n,$$

then

$$\frac{c}{n} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \frac{C}{n} \quad \text{for all sufficiently large } n.$$

Here c may depend on the fixed q , and C may depend on q and M ; the thresholds for sufficiently large n may depend on the sequence.

Conversely, for every sequence of integers $d_n \geq 1$, if there exist real constants $0 < c_* \leq C_*$ such that

$$\frac{c_*}{n} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \frac{C_*}{n} \quad \text{for all sufficiently large } n,$$

then

$$d_n = O(\sqrt{n} \log(e + n)), \quad n \rightarrow \infty.$$

The fixed-floor thresholds express consistency on the $n \log(e + n)$ category scale and parametric precision on the $\sqrt{n} \log(e + n)$ scale. For a prescribed category-growth multiplier, the eventual upper constant may depend on that multiplier and on q . This dependence differs from the universal constants in the rare-arrival comparison and from the explicit deficit-dependent constant in the near-complete window.

The following table collects the comparisons while keeping the universal upper envelope, the matched large-alphabet region, and the two logarithmic matched regimes separate. Here $N = nq$, $\ell = \log(e + N)$, and $r_{n,d,q}$ is the rate in Definition 17.

5 A constructive estimator

The risk comparisons admit an explicit statistical construction. The mixed-count estimator in Algorithm 1 combines information about cell membership with information about arrived outcomes, assigning these two tasks to separate parts of each cell contribution. Its elementary branch uses empirical cell means; its polynomial branch modifies the contribution of cells with few arrivals. We collect the ingredients by their statistical role before describing the resulting estimate and its risk guarantee.

²**Formalization scope.** This result uses the cited conclusion [Zeng et al., 2026] (Theorem 2, Section 3.2, p. 9, together with its proof in Section C.5, pp. 39–44, which sets the control regression to zero). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Table 1: Squared-error comparisons across ascertainment regimes

Domain	Risk comparison	Constants	Supporting results
$n, d \geq 1, 0 < q \leq 1$	$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+ \leq \min\{1, \bar{C}r_{n,d,q}, 4/n + (1-q)^2\}$	Universal	Theorem 1
$d \geq 1024n^2, 0 < q \leq 1$	$\mathfrak{R}_{n,d,q}^+ \asymp n^{-1} + (1-q)^2$	Universal	Theorem 2
$1 - q_n \leq M/\sqrt{n}$ eventually, all $d \geq 1$	$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q_n}^+ \leq \frac{4+M^2}{n}$; uniform n^{-1} order iff $1 - q_n = O(n^{-1/2})$	Explicit dependence on M	Propositions 1 and 2
$q = 1, d \geq 1$	$\mathfrak{R}_{n,d,1}^+ \asymp n^{-1}$, uniformly in d	Universal	Theorem 3 and Proposition 3
Rare-arrival restriction	$\mathfrak{R}_{n,d,q} \asymp \min\{1, N^{-1} + [d/(N\ell)]^2\}$	Universal	Theorem 4 and Proposition 5
Fixed $0 < q < 1/2$	$\mathfrak{R}_{n,d,q}^+ \asymp \min\{1, n^{-1} + [d/\{n \log(e+n)\}]^2\}$	Depend on fixed q	Theorem 5 and Proposition 6

5.1 Mechanism of the mixed-count correction

The canonical formulas are collected once in [Algorithm 1](#) and its immediate ingredients [Definitions 13 and 19 to 28](#). Their statistical roles are simple. Membership-labelled records estimate the population cell weights, while pilot and estimation labels separate branch selection from estimation of each arrived-cell mean. An ordinary ratio assigns zero to a cell with no arrived outcome. Across many rare cells, those zero-count events create the category-dependent bias that appears in the elementary certificate.

On the large-alphabet branch, the polynomial extension of $\{1 - Q_k(t)\}/t$ supplies the reciprocal correction implicit in a cell mean. Marked factorial counts estimate its monomials without discarding the association between an outcome and its cell: for cell intensity t_j and arrived-cell mean μ_j , the factorial statistic satisfies $E(H_j) = \mu_j\{1 - Q_k(t_j)\}$. Thus Q_k controls the residual bias, and the bound on $t_j|Q_k(t_j)|$ controls its membership-weighted aggregate. With degree k proportional to $\log(e + N)$, this changes the aggregate bias scale from the elementary scarce-cell scale to $d/\{N \log(e + N)\}$, while a pilot count sends well-observed cells back to the stable empirical ratio. Independent membership, pilot, and estimation streams justify the comparison calculation; the capped-prefix transfer and conditional averaging turn that comparison into the deterministic fixed-sample estimator.

The construction is explicit on every parameter triple. Its polynomial branch activates at $\log(e + nq) \geq 128$, equivalently $nq \geq e^{128} - e$. Exact evaluation averages 3^{k_0} assignments at each retained prefix length k_0 ; the limitations section discusses this computational scale separately from the attainment guarantee.

5.2 Risk control and the deterministic envelope

The construction supplies a squared-error guarantee throughout the unrestricted arrival class in [Definition 7](#). Its deterministic envelope $\mathfrak{U}_{n,d,q}$ depends on the sample size, alphabet size, and arrival floor. The following result gives the envelope explicitly, compares it with the rate in [Definition 17](#), and records the risk improvement from conditional averaging.

Theorem 6. [thm:unrestricted-mixed-count-upper] (Unrestricted mixed-count risk bound)
There exists a universal constant $0 < \overline{C} < \infty$ such that, for every sample size n , alphabet size d , arrival floor q , and full-data law P satisfying

- **(Domain.)** $n, d \geq 1$ and $0 < q \leq 1$.
- **(Model.)** $P \in \mathcal{M}_{n,d,q}^+$, the model class defined in [Definition 7](#).
- **(Sampling.)** The observed sample $\mathbf{O}_n$ has the canonical product law induced by P , with its coordinate observations satisfying [Assumption 1](#).

the deterministic estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$ of [Algorithm 1](#) satisfies

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q},$$

where $\tau(P)$ is the average treatment effect defined in [Definition 8](#), $r_{n,d,q}$ is the rate defined in [Definition 17](#), and $\mathfrak{U}_{n,d,q}$ is the deterministic risk envelope specified below.

Set

$$N = nq, \quad m = \frac{n}{6}, \quad N_0 = mq = \frac{N}{6}, \quad \gamma_0 = \log 2 - \frac{1}{2}.$$

Use the logarithmic scale ℓ defined in [Definition 13](#), the polynomial degree k defined in [Definition 19](#), and the threshold B defined in [Definition 20](#). Define the certificate quantities

$$b_{n,d,q} = 2 \left(\frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right),$$

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d \left(e^{\ell/8} + 1 \right) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right).$$

For $N \leq 1$, set $\mathfrak{U}_{n,d,q} = 1$. For $N > 1$ on the elementary branch of [Algorithm 1](#), set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, \left(\frac{8d}{eN_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\}.$$

For $N > 1$ on its polynomial branch, set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \right\}.$$

Moreover, the estimator's conditional average over the auxiliary randomization satisfies

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq E_P \left[E_\omega \left[\left(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P) \right)^2 \mid \mathbf{O}_n \right] \right],$$

where ω is the auxiliary prefix and stream-assignment randomization from [Algorithm 1](#), and $\tilde{\tau}(\mathbf{O}_n; \omega)$ is its output, including the zero output when the Poisson prefix length exceeds n .

The bound separates effective-sample-size error from category complexity through

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{N} + \left(\frac{d}{N\ell} \right)^2 \right\},$$

as defined in [Definition 17](#). The polynomial certificate reflects the mechanism of the construction: its leading squared contribution involves $dB/(N_0 k^2)$, which scales as $d/(N\ell)$ on the polynomial

branch. Membership weights and arrived-outcome statistics thus enter a common aggregate guarantee even when many cells have few arrived observations. The final inequality in [Theorem 6](#) also shows why the conditional average is the reported estimator: averaging the bounded auxiliary outputs preserves their risk guarantee and can reduce squared-error risk.

The detailed cellwise moment bounds and their aggregation are collected in [Theorem 9](#). For the subsequent interval construction, the essential feature is that $\mathfrak{U}_{n,d,q}$ is a known, deterministic upper bound on squared-error risk, uniform over the stated model class. It supplies the radius $\sqrt{\mathfrak{U}_{n,d,q}/\alpha}$, where $\alpha \in (0, 1)$ is the miscoverage level, used by the clipped interval in [Definition 30](#).

6 Honest intervals under rare outcome arrival

The deterministic squared-error envelope in [Theorem 6](#) also supports uncertainty quantification for the average treatment effect. The relevant criterion combines uniform coverage over the rare-arrival model with maximal expected interval length. Fix a miscoverage level $\alpha \in (0, 1)$, so that the required coverage is $1 - \alpha$. To include procedures with auxiliary randomization, both coverage and expected length are evaluated under the joint distribution of the observed sample and the reported endpoints. The following definition specifies this decision problem.

Definition 29. `[def:interval-length-risk]` (Minimax expected interval length) *The minimax maximal expected interval length, $\mathfrak{L}_{n,d,q,\alpha}$, is defined for $n, d \in \mathbb{N}$ and $q, \alpha \in \mathbb{R}$ as follows. Let $\mathcal{M}_{n,d,q}$ be the model class in [Definition 14](#), let $\mathbf{O}_n$ be the observed sample with law $\mathcal{L}_P(\mathbf{O}_n)$ from [Definition 15](#), and let $\tau(P)$ be the average treatment effect in [Definition 8](#).*

The class $\mathcal{C}_{n,d,q,\alpha}$ consists of parameter-independent Markov kernels l from the observed-sample space to endpoint pairs (l, u) satisfying $-1 \leq l \leq u \leq 1$, with uniform joint sample-and-randomization coverage: for every $P \in \mathcal{M}_{n,d,q}$,

$$\int \int \mathbf{1}\{l \leq \tau(P) \leq u\} \mathsf{l}(o, d(l, u)) \mathcal{L}_P(\mathbf{O}_n)(do) \geq 1 - \alpha.$$

The realized connected interval is $[l, u]$, with length $u - l$. Define

$$\mathfrak{L}_{n,d,q,\alpha} := \inf_{\mathsf{l} \in \mathcal{C}_{n,d,q,\alpha}} \sup_{P \in \mathcal{M}_{n,d,q}} \int \int (u - l) \mathsf{l}(o, d(l, u)) \mathcal{L}_P(\mathbf{O}_n)(do),$$

where the infimum and supremum are real-valued, with the convention that an infimum or supremum over an empty set equals zero. In particular, $\mathfrak{L}_{n,d,q,\alpha} = 0$ when $\mathcal{C}_{n,d,q,\alpha}$ is empty. Both coverage and expected length integrate over the sample and the kernel draw. Deterministic connected-interval procedures are included through Dirac kernels.

Uniform honesty requires the coverage inequality separately for every law in [Definition 14](#). For a randomized endpoint procedure l , this requirement averages over its auxiliary draw as well as sampling variation. The same averaging governs length. Thus the admissible class $\mathcal{C}_{n,d,q,\alpha}$ treats deterministic and randomized procedures within one criterion, and the minimax length $\mathfrak{L}_{n,d,q,\alpha}$ measures the smallest attainable worst-case expected width at the specified coverage.

A concrete honest interval uses the mixed-count estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$ from [Algorithm 1](#) and its deterministic risk envelope $\mathfrak{U}_{n,d,q}$ in [Theorem 6](#). Both are calibrated to the supplied arrival floor q . The envelope determines a radius before observing the sample; intersecting the resulting interval with the target range gives the following construction. Its coverage guarantee therefore requires the supplied floor to be no larger than every occupied-cell arrival probability.

Definition 30. [def:risk-envelope-interval] (Risk-envelope interval I_α^{mix}) *The risk-envelope interval is*

$$I_\alpha^{\text{mix}} := \left[\hat{\tau}_{n,d,q}^{\text{mix}} - \sqrt{\frac{\mathfrak{L}_{n,d,q}}{\alpha}}, \hat{\tau}_{n,d,q}^{\text{mix}} + \sqrt{\frac{\mathfrak{L}_{n,d,q}}{\alpha}} \right] \cap [-1, 1],$$

where $\hat{\tau}_{n,d,q}^{\text{mix}}$ is the mixed-count estimator and $\mathfrak{L}_{n,d,q}$ is its explicit deterministic squared-error envelope.

The interval I_α^{mix} is deterministic conditional on the observed sample and therefore enters [Definition 29](#) through a Dirac endpoint rule. Its radius converts uniform squared-error control into coverage at least $1 - \alpha$. The next result compares its maximal expected length with the best uniformly honest connected interval, using the comparison rate $r_{n,d,q}$ defined in [Definition 17](#).

Theorem 7. [thm:connected-interval-frontier] (Connected interval length frontier) *For every miscoverage level $\alpha \in (0, 1)$, there exist constants $0 < \underline{c}_\alpha \leq \overline{C}_\alpha < \infty$ such that the following holds for every $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$ satisfying:*

- *(Sample size and dimension.)* $n \geq 1$ and $d \geq 1$.
- *(Arrival floor.)* $0 < q \leq 1$.
- *(Rare-arrival slice.)* The restriction in [Assumption 8](#) holds.

Let $r_{n,d,q}$ be the frontier rate in [Definition 17](#), let $\mathfrak{L}_{n,d,q,\alpha}$ be the minimax expected interval length in [Definition 29](#), and let I_α^{mix} be the interval in [Definition 30](#). Over the model class $\mathcal{M}_{n,d,q}$ in [Definition 14](#),

$$\underline{c}_\alpha \sqrt{r_{n,d,q}} \leq \mathfrak{L}_{n,d,q,\alpha} \leq \sup_{P \in \mathcal{M}_{n,d,q}} E_P[\text{length}(I_\alpha^{\text{mix}})] \leq \overline{C}_\alpha \sqrt{r_{n,d,q}}.$$

Here expectations and probabilities are taken under the law of the independent observed sample induced by P , and interval length is defined by

$$\text{length}([l, u]) = u - l.$$

Moreover, for every $P \in \mathcal{M}_{n,d,q}$, the average treatment effect $\tau(P)$ from [Definition 8](#) satisfies

$$\Pr_P\{\tau(P) \in I_\alpha^{\text{mix}}\} \geq 1 - \alpha.$$

The upper bound reflects the square-root conversion from squared-error risk to an interval radius: the deterministic envelope controls coverage, while clipping bounds the realized length by the width of the untruncated interval. The converse connects statistical indistinguishability with the geometry of a connected interval. Coverage transfers between competing laws in the joint sample-and-endpoint distribution, and a realized interval that meets several separated target neighborhoods must span their separation. Working with this joint distribution retains the auxiliary endpoint draw throughout both the coverage comparison and the length calculation. The parameter-independent decision-rule comparison in [Lemma 2](#) supplies the corresponding control when an endpoint procedure is appended to the observed experiment.

For each fixed coverage level $1 - \alpha \in (0, 1)$, [Theorem 7](#) establishes expected-length order $\sqrt{r_{n,d,q}}$, with constants that may depend on α , uniformly over the sample sizes, dimensions,

and arrival floors satisfying its conditions. Alongside the squared-error order in [Theorem 4](#), this gives an uncertainty-quantification interpretation of the same comparison rate: its square root is the attainable and necessary order of maximal expected width for uniformly honest connected intervals. In particular, along sequences satisfying [Assumption 8](#), the optimal expected length tends to zero exactly when the rate in [Definition 17](#) tends to zero.

7 An observational extension through synthetic assignment

The discrete observational experiment connects adjustment for many covariate categories to the randomized experiment with missing outcomes. Its target is a baseline-weighted regression contrast under a fixed strict positivity margin. An independent synthetic assignment converts observational records into randomized records whose outcomes arrive when the synthetic and observational treatments agree. This construction preserves the contrast and transfers the mixed-count estimator to observational data.

We begin by specifying the observational law and its occupied-category positivity condition.

Definition 31. `[def:zeng-discrete-ate-class]` (Observational law class $\mathcal{D}_{d,\epsilon}$) *The class $\mathcal{D}_{d,\epsilon}$, for $0 < \epsilon < 1/2$, consists of one-record laws P_Z of the baseline category, observational treatment, and outcome*

$$(X, B^{\text{obs}}, Y) \in \{1, \dots, d\} \times \{0, 1\}^2$$

with baseline probabilities

$$p_x = P_Z(X = x)$$

and, on each occupied category $p_x > 0$, conditional treatment probabilities and outcome means

$$\pi_x = P_Z(B^{\text{obs}} = 1 \mid X = x), \quad \mu_{bx} = E_{P_Z}[Y \mid X = x, B^{\text{obs}} = b],$$

satisfying

$$\epsilon \leq \pi_x \leq 1 - \epsilon, \quad 0 \leq \mu_{bx} \leq 1 \quad (b \in \{0, 1\}).$$

The conditional quantities π_x and μ_{bx} are specified on occupied categories.

For a law P_Z in the class $\mathcal{D}_{d,\epsilon}$, both values of the observational treatment B^{obs} have positive conditional probability in every occupied category. The margin bounds these probabilities away from zero while allowing arbitrary baseline probabilities, including null categories. Consequently, both conditional outcome means are defined wherever the baseline distribution assigns positive mass.

The estimand aggregates the within-category regression contrasts using that baseline distribution.

Definition 32. `[def:zeng-ate-functional]` (Observed-law contrast θ_Z) *The total observed-law functional θ_Z is defined, for every $P_Z \in \mathcal{D}_{d,\epsilon}$, by*

$$\theta_Z(P_Z) := \sum_{x:p_x>0} p_x(\mu_{1x} - \mu_{0x}).$$

Null baseline categories contribute zero.

The contrast $\theta_Z(P_Z)$ is a functional of the observational law. Under outcome consistency, $Y = Y(B^{\text{obs}})$, and conditional exchangeability,

$$\{Y(0), Y(1)\} \perp\!\!\!\perp B^{\text{obs}} \mid X,$$

positivity identifies each occupied-category regression mean with the corresponding conditional potential-outcome mean. The contrast then equals the population average treatment effect $E[Y(1) - Y(0)]$, giving the usual causal interpretation of covariate adjustment [Rosenbaum and Rubin, 1983].

Let $\mathbf{Z}_n$ denote the sample of n independent observational records. The decision problem allows estimation procedures to use independent auxiliary randomness as well as the sample.

Definition 33. [def:zeng-minimax-risk] (Observational minimax risk $\mathfrak{R}_{n,d,\epsilon}^Z$) *The minimax squared-error risk for the observational contrast is*

$$\mathfrak{R}_{n,d,\epsilon}^Z := \inf_{\hat{\theta}} \sup_{P_Z \in \mathcal{D}_{d,\epsilon}} E_{P_Z} \left[(\hat{\theta}(\mathbf{Z}_n) - \theta_Z(P_Z))^2 \right],$$

where $\mathbf{Z}_n$ denotes the independent, identically distributed sample of n observational records with law P_Z . The infimum ranges over all possibly randomized estimators measurable with respect to $\mathbf{Z}_n$ and auxiliary randomness independent of P_Z ; the expectation integrates both sources of randomness.

The observational minimax risk $\mathfrak{R}_{n,d,\epsilon}^Z$ therefore measures the smallest worst-case squared error over the full decision class in Definition 33. To connect this problem to the randomized construction, we present together the transformation in Lemma 5 and the estimation procedure in Theorem 8.

Synthetic assignment and observational estimation. For an observational record (x, b, y) , draw a fair coin U , independently of the record, and define

$$o(x, b, y; u) := (x - 1, u, 0, 1\{b = u\}, 1\{b = u\}y), \quad u \in \{0, 1\}.$$

The five coordinates are the randomized baseline, assignment, surrogate, arrival indicator, and arrived outcome. For a sample realization $\mathbf{Z}_n = ((x_i, b_i, y_i))_{i=1}^n$, use independent fair coins $U_1, \dots, U_n$ and set

$$o_i(u_i) := o(x_i, b_i, y_i; u_i).$$

The transformation uses the observed record and a known coin distribution, so its rule is independent of the unknown observational law.

The induced randomized full-data law P_Z^* in Lemma 5 can be specified through a latent representation. Draw the baseline with probabilities p_x . Conditional on an occupied category x , draw B^{obs} with success probability π_x , and draw independent binary potential outcomes with means μ_{0x} and μ_{1x} , independently of B^{obs} . Draw an independent fair assignment A , retain $X - 1$ as the randomized baseline, and set

$$S(0) = S(1) = 0, \quad Y = Y(A), \quad R = 1\{B^{\text{obs}} = A\}.$$

This representation reproduces the synthetic observed-record distribution. Its arm-specific arrival probabilities are

$$P_Z^*(R = 1 \mid X - 1 = x - 1, A = 1, S = 0) = \pi_x, \quad P_Z^*(R = 1 \mid X - 1 = x - 1, A = 0, S = 0) = 1 - \pi_x,$$

and its target satisfies

$$\tau(P_Z^*) = \theta_Z(P_Z).$$

Both arrival probabilities are at least ϵ . Independence of the latent observational treatment and potential outcomes conditional on the baseline supplies the missing-at-random condition in this constructed law. The causal interpretation of the original observational contrast continues to rest on consistency and conditional exchangeability.

Apply the mixed-count estimator $\hat{\tau}_{n,d,\epsilon}^{\text{mix}}$ from [Algorithm 1](#) to the synthetic sample. Averaging over all fair-coin assignments defines the deterministic observational estimator $\hat{\theta}_{n,d,\epsilon}^{\text{poly}}$:

$$\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{Z}_n) := \frac{1}{2^n} \sum_{(u_1, \dots, u_n) \in \{0,1\}^n} \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)).$$

This finite average defines the procedure for every observational sample. Conditional averaging weakly reduces squared-error risk relative to retaining the auxiliary coin draws.

The resulting risk bound is uniform over sample size and alphabet size. At each fixed strict positivity margin, it also yields a matching all-estimator comparison.

Theorem 8. [\[thm:zeng-fixed-positivity-polynomial-frontier\]](#) (Fixed positivity polynomial frontier)³ *There exists a universal constant $\bar{C} > 0$ such that the following holds for every pair of integers $n, d \geq 1$ and every strict positivity margin $0 < \epsilon < 1/2$. Let $\mathcal{D}_{d,\epsilon}$ be the observational class in [Definition 31](#), and let $\mathbf{Z}_n = ((X_i, B_i^{\text{obs}}, Y_i))_{i=1}^n$ be an independent sample from $P_Z \in \mathcal{D}_{d,\epsilon}$. Write the observational baseline categories as $1, \dots, d$. For a realization $\mathbf{Z}_n = ((x_i, b_i, y_i))_{i=1}^n$, define the synthetic records, with baseline coordinates indexed from zero, by*

$$o_i(u_i) = (x_i - 1, u_i, 0, 1\{b_i = u_i\}, 1\{b_i = u_i\}y_i), \quad u_i \in \{0, 1\},$$

and define the deterministic estimator

$$\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{Z}_n) = \frac{1}{2^n} \sum_{(u_1, \dots, u_n) \in \{0,1\}^n} \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)),$$

where $\hat{\tau}_{n,d,\epsilon}^{\text{mix}}$ is the mixed-count estimator in [Algorithm 1](#). Then

$$\sup_{P_Z \in \mathcal{D}_{d,\epsilon}} E_{P_Z} \left[\left(\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{Z}_n) - \theta_Z(P_Z) \right)^2 \right] \leq \bar{C} r_{n,d,\epsilon},$$

where $\theta_Z(P_Z)$ is the observational contrast in [Definition 32](#) and $r_{n,d,\epsilon}$ is the rate in [Definition 17](#).

Moreover, for each fixed $0 < \epsilon < 1/2$, there exist constants $0 < c_\epsilon \leq C_\epsilon < \infty$ such that, for every pair of integers $n, d \geq 1$, the minimax risk in [Definition 33](#) satisfies

$$c_\epsilon \min \left\{ 1, \frac{1}{n} + \left[\frac{d}{n \log(e+n)} \right]^2 \right\} \leq \mathfrak{R}_{n,d,\epsilon}^Z \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \left[\frac{d}{n \log(e+n)} \right]^2 \right\}.$$

³**Formalization scope.** This result uses the cited conclusion [[Zeng et al., 2026](#)] (Theorem 2, Section 3.2, p. 9, together with its proof in Section C.5, pp. 39–44, which sets the control regression to zero). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

The statistical mechanism is the conversion of observational treatment imbalance into selective outcome arrival under independent assignment. Positivity supplies a common arrival floor, and the target-preserving transformation makes the randomized MAR risk certificate applicable. Averaging over the synthetic assignments then removes their auxiliary variation. For fixed ϵ , the two terms in the comparison describe sampling error and the cost of adjusting for many categories; the truncation captures the bounded scale of the contrast.

Under consistency and conditional exchangeability, [Theorem 8](#) therefore characterizes minimax squared-error risk for the observational average treatment effect. As n tends to infinity at a fixed strict positivity margin, uniform mean-square consistency is attainable exactly when

$$d = o\{n \log(e + n)\}.$$

The constants in the two-sided comparison may depend on that margin, while the constructive bound retains the dependence on ϵ through the rate in [Definition 17](#).

The observational lower-bound input comes from [Zeng et al. \[2026, Theorem 2, Section 3.2, p. 9, together with its proof in Section C.5, pp. 39–44\]](#). Their argument sets the control regression to zero on occupied categories and supplies the lower order

$$\frac{1}{n} + \left(\frac{d}{n \log n} \right)^2$$

for fixed strict positivity, sufficiently large n , and $1 \leq d \leq c_0 n \log n$, where $c_0 > 0$ is a constant from that result. The comparison in [Theorem 8](#) extends the risk characterization to every pair $n, d \geq 1$, with a total estimator and the bounded-risk truncation.

Finally, the same construction fixes the direction of the experiment comparison supporting the fixed-arrival analysis. At the fixed arrival floor $q = \epsilon \in (0, 1/2)$, the induced laws belong to the randomized class in [Definition 7](#). Any estimator for their randomized observed samples can be applied to observational records by appending independent synthetic assignments, with the same target and risk. The observational sample is therefore at least as informative as its synthetic image. Taking worst-case risk over the containing randomized class gives a risk at least as large as the observational minimax risk, so the published observational lower bound transfers in this direction to the fixed-arrival lower comparison in [Theorem 5](#). The formal target-preserving map is given in [Lemma 5](#).

8 Discussion and extensions

The precision comparisons distinguish three features of the observed experiment: the availability of primary outcomes, the number of covariate categories across which ascertainment must be accommodated, and the remaining ascertainment deficit. Their relative importance changes with the arrival regime. The resulting interpretation is a comparison of attainable precision over the stated model classes, with each regime retaining its own conditions and quantifiers.

8.1 Informative outcomes and adjustment complexity

The effective sample size $N = nq$ measures outcome availability at the guaranteed occupied-cell arrival floor. Because individual cells may have arrival probabilities exceeding that floor, it is a

conservative scale for outcome information. Under [Assumption 8](#), the matched comparison in [Theorem 4](#) separates the sampling contribution N^{-1} from the category contribution

$$\left\{ \frac{d}{N \log(e + N)} \right\}^2,$$

with their sum truncated at a constant. Thus increasing outcome availability improves both components, whereas increasing the alphabet size raises the cost of estimating the aggregate contrast across sparsely observed cells. The comparison incorporates the memberships observed for individuals whose primary outcomes have yet to arrive.

Along sequences satisfying the rare-arrival restriction and $N \rightarrow \infty$, [Proposition 5](#) gives two distinct measures of manageable adjustment complexity. Consistency corresponds to $d = o\{N \log(e + N)\}$, while squared-error risk of order N^{-1} corresponds to $d = O\{\sqrt{N} \log(e + N)\}$. These thresholds distinguish category growth compatible with learning the treatment effect from category growth compatible with preserving the effective-sample-size precision benchmark. At each fixed coverage level, [Theorem 7](#) gives the corresponding interpretation for uniformly honest connected intervals: optimal maximal expected length has the order of the square root of the matched squared-error rate.

Near complete ascertainment, the deficit $1 - q$ provides a complementary measure of difficulty. The observed-outcome contrast attains the upper bound $4/n + (1 - q)^2$ throughout the unrestricted model in [Theorem 1](#). For $d \geq 1024n^2$, [Theorem 2](#) establishes the matching order $n^{-1} + (1 - q)^2$. At complete arrival, [Theorem 3](#) gives squared-error order n^{-1} uniformly over every alphabet size. Together, these results explain how sufficiently complete ascertainment supports precision governed by sampling variation even under arbitrary category growth.

The quantifiers distinguish a uniform guarantee from a guarantee along a specified dimension sequence. In [Proposition 1](#), a single risk constant and an eventual sample-size threshold apply simultaneously to every $d \geq 1$. Such an n^{-1} guarantee is equivalent to $1 - q_n = O(n^{-1/2})$. A specified sequence can satisfy a different sufficient condition through its category growth: at any fixed $q \in (0, 1/2)$, [Proposition 6](#) characterizes n^{-1} risk by $d_n = O\{\sqrt{n} \log(e + n)\}$. The comparison constants may depend on the fixed floor and the category-growth multiplier. The near-complete criterion therefore describes precision uniformly over arbitrary alphabets, while the fixed-floor criterion describes precision along sequences with controlled adjustment complexity.

8.2 Surrogates and prospective trial interpretations

The surrogate serves as an observed variable for conditioning outcome arrival. Under [Assumption 6](#), arrivals represent the outcome distribution within treatment–baseline–surrogate cells; [Assumption 7](#) supplies positive ascertainment in every occupied cell. Its realized value may depend on assignment through [Assumption 4](#). The target remains the population average treatment effect on the primary outcome in [Definition 8](#), with the cell aggregation specified in [Definition 10](#).

The constant-surrogate specialization provides a useful benchmark for this interpretation. When both potential surrogates equal zero, the arrival condition reduces to conditioning on treatment and baseline category. For fixed alphabet size, sequences in the rare-arrival slice with diverging effective sample size retain minimax squared-error order N^{-1} within this specialization, as established by [Proposition 4](#). That benchmark isolates the precision scale associated with outcome availability in a model where the surrogate supplies a single category.

Clinical-trial applications can be interpreted prospectively through these conditions. At a prespecified analysis horizon, a binary primary endpoint would supply the outcome, an available short-term measurement would supply the binary surrogate, and recorded baseline categories would determine the adjustment alphabet. Applying the comparisons would require independent sampling, balanced randomized assignment, consistency, and the stated conditional arrival and occupied-cell positivity conditions. The delayed binary endpoints considered by [Marschner and Becker \[2001, Abstract\]](#) provide motivation for this interpretation. The use of short-term outcomes and baseline covariates for interim decisions in [Van Lancker et al. \[2020, Abstract\]](#) supplies related motivation. Here, these examples connect the statistical observation scheme to prospective trial settings, while the precision conclusions retain the conditions of the cited internal results.

Further questions concerning scope and research priorities are collected in [Section 9](#).

9 Limitations and open questions

The risk comparisons leave several questions about joint changes in outcome arrival and alphabet size. The unrestricted upper bound in [Theorem 6](#) provides an envelope across arrival regimes, while [Theorems 4 and 5](#) give matching comparisons under their respective conditions. Matching the unrestricted envelope throughout intermediate moving-arrival and alphabet regimes remains open. This requires lower bounds that track simultaneous changes in the arrival floor and the number of baseline categories.

A related question concerns the distinction between a uniform near-complete requirement and a requirement for a particular alphabet sequence. [Proposition 1](#) characterizes the uniform comparison, and [Theorem 2](#) supplies the large-alphabet obstruction. These results leave open the precise near-complete requirement along individual smaller-alphabet sequences. A sequence-specific characterization would determine how the necessary degree of ascertainment depends on the growth of the alphabet.

For uncertainty quantification, [Theorem 7](#) compares uniformly honest connected intervals under the rare-arrival restriction in [Assumption 8](#). Extending matching expected-length comparisons beyond that restriction is another open problem. Such an extension would require interval lower bounds appropriate to the additional arrival regimes, together with procedures attaining the corresponding lengths under uniform coverage.

The observational comparison in [Theorem 8](#) concerns a fixed positivity margin strictly between zero and one half, as specified in [Definition 31](#). The endpoint at which every occupied category has treatment propensity one half calls for a separate risk characterization. Determining how this endpoint relates to the strict-interior comparisons would clarify the role of treatment-propensity variation in the observational problem.

The conditional arrival assumption in [Assumption 6](#) also defines an important boundary of the model. Extending the analysis beyond missing at random requires specifying how arrival may depend on the outcome after conditioning on the recorded covariates, assignment, and surrogate. Alternative identifying restrictions or sensitivity parameters would then determine the target and the precision comparison. The present risk results do not supply those extensions.

Statistical attainment and practical evaluation are separate questions. The procedure in [Algorithm 1](#) averages 3^{k_0} stream assignments at each retained prefix length k_0 , and the observational procedure in [Theorem 8](#) additionally averages over 2^n synthetic assignments. Their risk guarantees concern those exact procedures. Moreover, the polynomial branch begins only when

$\log(e + nq) \geq 128$, or $nq \geq e^{128} - e$. These scales make the displayed formulas mathematical attainment constructions rather than evidence of numerical usefulness at ordinary sample sizes. Computational evaluation should seek more efficient representations and quantify additional error from approximations. Prospective clinical evaluation must also justify the supplied arrival floor used for tuning and coverage, examine the choice of baseline categories and the conditional arrival assumption, and compare with established trial analyses.

Kennedy [2019, abstract 300411] earlier announced average-causal-effect estimation with arbitrarily high-dimensional discrete confounders, including a variation with sample-size-dependent weakening of propensity-score positivity. The available abstract does not state its precise model class, rate, loss, or attaining construction. It therefore supports priority for studying that shrinking-positivity observational setting but cannot settle theorem-level overlap with the fixed-margin observational result here. A comparison with a complete theorem statement or manuscript remains open.

Appendices

A Identification and decision-theory preliminaries

This appendix collects the decision-theoretic comparisons, cell-identification identities, and synthetic-assignment map used in the risk analysis. The generic experiment and testing notation introduced below is local to this appendix.

A.1 Finite decision experiments

For the decision experiment in Lemma 1, let Λ denote a parameter set, $\mathcal{Z}$ a finite observation space with measurable singletons, Q_λ the observation law at parameter λ , and $\vartheta(\lambda) \in [-1, 1]$ the scalar target. A randomized procedure is a parameter-independent Markov kernel κ : conditional on observation o , it draws an action from $\kappa(o, \cdot)$. Its squared risk averages the squared estimation error over both the observation and the action. A deterministic procedure is represented by a kernel concentrated at one action for each observation. For actions in $[-1, 1]$, the barycenter is the conditional mean of the action. The following comparison establishes the risk effect of replacing a randomized action by this mean, together with the corresponding clipping and uniform-seed representations.

Lemma 1. [lem:randomized-kernel-barycenter] (Barycenters and uniform randomization)
Let Λ be an arbitrary parameter set, and suppose:

- *(Sample space.) $\mathcal{Z}$ is a finite measurable space with measurable singletons.*
- *(Experiment.) For each $\lambda \in \Lambda$, Q_λ is a probability law on $\mathcal{Z}$.*
- *(Target.) For each $\lambda \in \Lambda$, the target satisfies $\vartheta(\lambda) \in [-1, 1]$.*

For a parameter-independent Markov kernel κ from $\mathcal{Z}$ to $\mathbb{R}$, supported on $[-1, 1]$ at every observation, define its barycenter by

$$b_\kappa(o) = \int_{\mathbb{R}} h \kappa(o, dh).$$

Then $b_\kappa(o) \in [-1, 1]$ for every $o \in \mathcal{Z}$, and, for every $\lambda \in \Lambda$,

$$\int_{\mathcal{Z}} (b_\kappa(o) - \vartheta(\lambda))^2 Q_\lambda(do) \leq \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do).$$

The minimax squared risk over these bounded kernels equals that over deterministic maps into $[-1, 1]$:

$$\inf_{\kappa} \sup_{\lambda \in \Lambda} \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do) = \inf_{b: \mathcal{Z} \rightarrow [-1, 1]} \sup_{\lambda \in \Lambda} \int_{\mathcal{Z}} (b(o) - \vartheta(\lambda))^2 Q_\lambda(do),$$

where the first infimum ranges over the bounded kernels just specified.

Define clipping by

$$\text{clip}_{[-1, 1]}(h) = \max\{-1, \min\{1, h\}\}.$$

For every parameter-independent Markov kernel κ from $\mathcal{Z}$ to $\mathbb{R}$, and every $\lambda \in \Lambda$, its image under clipping satisfies

$$\int_{\mathcal{Z}} \int_{\mathbb{R}} (\text{clip}_{[-1, 1]}(h) - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do) \leq \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do),$$

with both risks interpreted as nonnegative extended integrals. The extended minimax squared risk over all such real-valued Markov kernels, converted to a real value, equals the bounded-kernel minimax squared risk above.

The uniform law on $[0, 1]$, defined by restricting Lebesgue measure to this interval, is a probability law. Every bounded kernel κ admits a measurable map $f : \mathcal{Z} \times [0, 1] \rightarrow [-1, 1]$ such that, for every $o \in \mathcal{Z}$, $\kappa(o, \cdot)$ is the distribution of $f(o, U)$, where U has this uniform law. Conversely, every such measurable map induces a bounded Markov kernel with precisely these conditional distributions. The same uniform seed law serves every parameter.

Proof of Lemma 1. 1. Fix a bounded kernel κ and an observation $o \in \mathcal{Z}$. Its support condition gives

$$\kappa(o, [-1, 1]) = 1, \quad -1 \leq h \leq 1 \quad \text{for } \kappa(o, dh)\text{-almost every } h.$$

The coordinate function is therefore integrable. Integrating these bounds against the probability measure $\kappa(o, \cdot)$ yields

$$-1 = \int_{\mathbb{R}} (-1) \kappa(o, dh) \leq b_\kappa(o) \leq \int_{\mathbb{R}} 1 \kappa(o, dh) = 1.$$

Every map on the finite space $\mathcal{Z}$ is measurable because its singletons are measurable. Thus b_κ is a measurable map into $[-1, 1]$, determined by κ .

2. Fix $\lambda \in \Lambda$. Bounded support gives square integrability of the coordinate and its translate by $\vartheta(\lambda)$, and

$$\int_{\mathbb{R}} (h - \vartheta(\lambda)) \kappa(o, dh) = b_\kappa(o) - \vartheta(\lambda).$$

The second-moment formula for the variance of the translated coordinate, together with invariance of variance under subtraction of a constant, gives

$$\begin{aligned} \int_{\mathbb{R}} (h - b_\kappa(o))^2 \kappa(o, dh) &= \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) \\ &\quad - \left(\int_{\mathbb{R}} (h - \vartheta(\lambda)) \kappa(o, dh) \right)^2. \end{aligned}$$

Substituting the preceding mean identity and rearranging yields

$$\int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) = (b_\kappa(o) - \vartheta(\lambda))^2 + \int_{\mathbb{R}} (h - b_\kappa(o))^2 \kappa(o, dh).$$

The last integral is nonnegative, so

$$(b_\kappa(o) - \vartheta(\lambda))^2 \leq \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh).$$

Both sides are integrable as functions of o , since $\mathcal{Z}$ is finite and Q_λ is a probability measure. Integrating gives

$$\int_{\mathcal{Z}} (b_\kappa(o) - \vartheta(\lambda))^2 Q_\lambda(do) \leq \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do).$$

3. Define the estimator classes and their real-valued risks by

$$\begin{aligned} \mathcal{K}_{\text{bd}} &:= \{\kappa : \mathcal{Z} \rightsquigarrow \mathbb{R} : \kappa \text{ is a parameter-independent Markov kernel and } \kappa(o, [-1, 1]) = 1 \text{ for every } o\}, \\ \mathcal{B}_{\text{det}} &:= \{b : \mathcal{Z} \rightarrow [-1, 1]\}, \end{aligned}$$

$$\mathcal{R}_{\text{bd}}(\kappa, \lambda) := \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do), \quad (\kappa, \lambda) \in \mathcal{K}_{\text{bd}} \times \Lambda,$$

$$\mathcal{R}_{\text{det}}(b, \lambda) := \int_{\mathcal{Z}} (b(o) - \vartheta(\lambda))^2 Q_\lambda(do), \quad (b, \lambda) \in \mathcal{B}_{\text{det}} \times \Lambda.$$

All real suprema over an empty parameter set are taken to be zero.

For every bounded kernel, its support condition and the target bound imply

$$-2 \leq h - \vartheta(\lambda) \leq 2, \quad 0 \leq (h - \vartheta(\lambda))^2 \leq 4 \quad \text{for } \kappa(o, dh)\text{-almost every } h.$$

Integrating first against $\kappa(o, \cdot)$ and then against Q_λ gives

$$0 \leq \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) \leq 4, \quad 0 \leq \mathcal{R}_{\text{bd}}(\kappa, \lambda) \leq 4.$$

For every $b \in \mathcal{B}_{\text{det}}$, define its deterministic kernel by

$$\kappa_b(o, \cdot) := \delta_{b(o)}, \quad o \in \mathcal{Z}.$$

Measurability of b makes this a Markov kernel, and its interval-valued range gives

$$\kappa_b(o, [-1, 1]) = 1, \quad \mathcal{R}_{\text{bd}}(\kappa_b, \lambda) = \mathcal{R}_{\text{det}}(b, \lambda).$$

In particular, both estimator classes are nonempty: take

$$b_{\text{zero}} : \mathcal{Z} \rightarrow [-1, 1], \quad b_{\text{zero}}(o) := 0,$$

and its kernel $\kappa_{b_{\text{zero}}}$. The risk identity also gives $0 \leq \mathcal{R}_{\text{det}}(b, \lambda) \leq 4$. Consequently, all worst-case risks lie in $[0, 4]$, including when Λ is empty, and their infima are well defined.

The deterministic-kernel construction gives the first comparison:

$$\inf_{\kappa \in \mathcal{K}_{\text{bd}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda) \leq \inf_{b \in \mathcal{B}_{\text{det}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{det}}(b, \lambda).$$

Conversely, each $\kappa \in \mathcal{K}_{\text{bd}}$ has the barycenter map $b_\kappa \in \mathcal{B}_{\text{det}}$, whose pointwise risk inequality gives

$$\sup_{\lambda \in \Lambda} \mathcal{R}_{\text{det}}(b_\kappa, \lambda) \leq \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda).$$

For a nonempty parameter set this follows from monotonicity of bounded suprema; for an empty parameter set both sides equal zero. Taking infima gives

$$\inf_{b \in \mathcal{B}_{\text{det}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{det}}(b, \lambda) \leq \inf_{\kappa \in \mathcal{K}_{\text{bd}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda).$$

The two comparisons establish the asserted equality.

4. Fix an arbitrary real-valued kernel κ and $\lambda \in \Lambda$. First suppose $h \leq 1$. If $-1 \leq h$, clipping fixes h , so the squared losses agree. If $h \leq -1$, clipping gives -1 , and

$$\begin{aligned} (h - \vartheta(\lambda))^2 - (-1 - \vartheta(\lambda))^2 &= (-1 - h)(1 - h + 2\vartheta(\lambda)) \\ &\geq 0, \end{aligned}$$

because both factors are nonnegative. In the remaining case $1 \leq h$, clipping gives 1 , and

$$\begin{aligned} (h - \vartheta(\lambda))^2 - (1 - \vartheta(\lambda))^2 &= (h - 1)(h + 1 - 2\vartheta(\lambda)) \\ &\geq 0. \end{aligned}$$

Thus, for every $h \in \mathbb{R}$,

$$(\text{clip}_{[-1,1]}(h) - \vartheta(\lambda))^2 \leq (h - \vartheta(\lambda))^2.$$

Define the clipped kernel by the pushforward

$$\kappa^{\text{cl}}(o, \cdot) := \kappa(o, \cdot) \circ \text{clip}_{[-1,1]}^{-1}, \quad o \in \mathcal{Z},$$

and define the extended risk for any real-valued Markov kernel by

$$\mathcal{R}_{\text{ext}}(\kappa, \lambda) := \int_{\mathcal{Z}} \int_{\mathbb{R}} (h - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do) \in [0, \infty],$$

where both integrals are nonnegative extended integrals. Clipping is measurable. The pushforward integration formula and monotonicity of these integrals give

$$\begin{aligned} \mathcal{R}_{\text{ext}}(\kappa^{\text{cl}}, \lambda) &= \int_{\mathcal{Z}} \int_{\mathbb{R}} (\text{clip}_{[-1,1]}(h) - \vartheta(\lambda))^2 \kappa(o, dh) Q_\lambda(do) \\ &\leq \mathcal{R}_{\text{ext}}(\kappa, \lambda). \end{aligned}$$

This comparison includes infinite original risks.

5. Define the full kernel class by

$$\mathcal{K}_{\text{real}} := \{\kappa : \mathcal{Z} \rightsquigarrow \mathbb{R} : \kappa \text{ is a parameter-independent Markov kernel}\}.$$

For a bounded kernel, squared loss is nonnegative and integrable at every observation. Its inner integral is integrable over the finite observation space. Applying the identification of a nonnegative integrable function's real and extended integrals at both levels gives

$$\mathcal{R}_{\text{ext}}(\kappa, \lambda) = \mathcal{R}_{\text{bd}}(\kappa, \lambda) \quad \text{in } [0, \infty], \quad \kappa \in \mathcal{K}_{\text{bd}}.$$

Inclusion of bounded kernels into the full class therefore gives

$$\inf_{\kappa \in \mathcal{K}_{\text{real}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{ext}}(\kappa, \lambda) \leq \inf_{\kappa \in \mathcal{K}_{\text{bd}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda),$$

with the right side here evaluated in $[0, \infty]$.

For the reverse comparison, clipping takes values in $[-1, 1]$, since

$$-1 \leq \max\{-1, \min\{1, h\}\} \leq 1 \quad (h \in \mathbb{R}).$$

A measurable pushforward preserves probability, and

$$\text{clip}_{[-1,1]}^{-1}([-1, 1]) = \mathbb{R}, \quad \kappa^{\text{cl}}(o, [-1, 1]) = \kappa(o, \mathbb{R}) = 1.$$

Thus $\kappa^{\text{cl}} \in \mathcal{K}_{\text{bd}}$ for every $\kappa \in \mathcal{K}_{\text{real}}$. The preceding risk comparison yields

$$\inf_{\kappa \in \mathcal{K}_{\text{bd}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda) \leq \inf_{\kappa \in \mathcal{K}_{\text{real}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{ext}}(\kappa, \lambda).$$

Hence the two extended minimax values agree.

For the conversion to a real value, use

$$\text{rv} : [0, \infty] \rightarrow \mathbb{R}, \quad \text{rv}|_{[0, \infty)} = \text{id}, \quad \text{rv}(\infty) = 0.$$

The bounds $0 \leq \mathcal{R}_{\text{bd}} \leq 4$ place every bounded-kernel supremum, and their infimum, in $[0, 4]$. On this interval the real and extended orders agree, so conversion commutes with these suprema and the infimum. For an empty parameter set both kinds of supremum equal zero. It follows that

$$\text{rv} \left(\inf_{\kappa \in \mathcal{K}_{\text{real}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{ext}}(\kappa, \lambda) \right) = \inf_{\kappa \in \mathcal{K}_{\text{bd}}} \sup_{\lambda \in \Lambda} \mathcal{R}_{\text{bd}}(\kappa, \lambda),$$

where the right side is now evaluated in $\mathbb{R}$.

6. Define the seed law, viewed as a measure on $[0, 1]$, by

$$\nu_{\text{seed}} := \text{Leb}_{\mathbb{R}}|_{[0,1]}.$$

Its total mass is

$$\nu_{\text{seed}}([0, 1]) = \text{Leb}_{\mathbb{R}}([0, 1]) = 1 - 0 = 1,$$

so it is a probability law.

7. Fix $\kappa \in \mathcal{K}_{\text{bd}}$. The measurable randomization theorem for Markov kernels with a standard Borel output space, applied to the real output space, supplies a jointly measurable map satisfying

$$\tilde{f} : \mathcal{Z} \times [0, 1] \rightarrow \mathbb{R}, \quad \kappa(o, \cdot) = \nu_{\text{seed}} \circ \tilde{f}(o, \cdot)^{-1} \quad (o \in \mathcal{Z}).$$

Define the interval-valued realization by

$$f : \mathcal{Z} \times [0, 1] \rightarrow [-1, 1], \quad f(o, v) := \text{clip}_{[-1,1]}(\tilde{f}(o, v)).$$

It is jointly measurable by measurability of clipping. For $h \in [-1, 1]$, clipping fixes h ; the bounded support condition consequently gives

$$\text{clip}_{[-1,1]}(h) = h \quad \text{for } \kappa(o, dh)\text{-almost every } h, \quad \kappa(o, \cdot) \circ \text{clip}_{[-1,1]}^{-1} = \kappa(o, \cdot).$$

Composition of measurable pushforwards now yields

$$\begin{aligned} \nu_{\text{seed}} \circ f(o, \cdot)^{-1} &= (\nu_{\text{seed}} \circ \tilde{f}(o, \cdot)^{-1}) \circ \text{clip}_{[-1,1]}^{-1} \\ &= \kappa(o, \cdot) \circ \text{clip}_{[-1,1]}^{-1} = \kappa(o, \cdot). \end{aligned}$$

Thus $f(o, U)$, for $U \sim \nu_{\text{seed}}$, has the required distribution at every observation.

8. Conversely, let $f : \mathcal{Z} \times [0, 1] \rightarrow [-1, 1]$ be jointly measurable, and define

$$\kappa_f(o, \cdot) := \nu_{\text{seed}} \circ f(o, \cdot)^{-1}, \quad o \in \mathcal{Z}.$$

Every section $f(o, \cdot)$ is measurable, so its pushforward is a probability measure. Finiteness of $\mathcal{Z}$ and measurability of its singletons give the kernel measurability condition. Moreover,

$$f(o, \cdot)^{-1}([-1, 1]) = [0, 1], \quad \kappa_f(o, [-1, 1]) = \nu_{\text{seed}}([0, 1]) = 1.$$

Hence κ_f is a bounded Markov kernel with exactly the displayed conditional distributions. Both constructions use the same law ν_{seed} , independently of the parameter. For each λ , the auxiliary seed can be sampled independently of the observation under the product law

$$Q_\lambda \otimes \nu_{\text{seed}}.$$

□

The comparison holds at every parameter, so conditional averaging also preserves a uniform squared-risk bound. It supplies the decision-theoretic justification for the auxiliary average in [Algorithm 1](#) and the synthetic-assignment average in [Theorem 8](#). Clipping weakly reduces squared loss for targets in the stated interval; on the real line, this clipping map coincides with Euclidean projection onto that interval. The uniform-seed representation places bounded randomized procedures and procedures implemented with independent auxiliary randomization in the same decision class.

For the testing comparison in [Lemma 2](#), let Q_0, Q_1 denote two laws on $\mathcal{Z}$, and let $\mathcal{H}$ be a measurable action space. The notation $Q_i \times \kappa$ denotes the joint observation–action law, whereas $Q_i \kappa$ denotes its action marginal. Total variation, written TV, is the supremum of absolute probability differences over measurable events. For binary actions, a randomized test has conditional probability $\psi(o) = \kappa(o, \{1\})$ of selecting alternative 1. The next comparison applies both to events that retain the observation and to events determined by the action alone.

Lemma 2. [[lem:randomized-kernel-tv-comparison](#)] (Total variation under randomization) *Let Q_0, Q_1 be probability laws on a finite measurable space $\mathcal{Z}$ with measurable singletons, and let κ be a common Markov kernel from $\mathcal{Z}$ to a measurable action space $\mathcal{H}$. For $i \in \{0, 1\}$, write $Q_i \times \kappa$ for the joint law obtained by drawing an input from Q_i and then an action from κ conditional on that input, and write $Q_i \kappa$ for its action marginal. For two probability laws on the*

same measurable space, define their total variation distance as the supremum of their absolute probability differences over measurable events. Then

$$\mathrm{TV}(Q_0\kappa, Q_1\kappa) \leq \mathrm{TV}(Q_0 \times \kappa, Q_1 \times \kappa) = \mathrm{TV}(Q_0, Q_1).$$

For every measurable event $T \subseteq \mathcal{Z} \times \mathcal{H}$,

$$(Q_0 \times \kappa)(T) + (Q_1 \times \kappa)(T^c) \geq 1 - \mathrm{TV}(Q_0, Q_1),$$

and for every measurable event $T \subseteq \mathcal{H}$,

$$(Q_0\kappa)(T) + (Q_1\kappa)(T^c) \geq 1 - \mathrm{TV}(Q_0, Q_1).$$

Proof of Lemma 2. 1. We first establish the bounded-function estimate used for the kernel probabilities. For any measurable function

$$\varphi : \mathcal{Z} \longrightarrow [0, 1], \quad L_{\varphi, t} := \{z \in \mathcal{Z} : t \leq \varphi(z)\} \quad (t \in \mathbb{R}),$$

the level sets are measurable, and the definition of total variation gives

$$|Q_0(L_{\varphi, t}) - Q_1(L_{\varphi, t})| \leq \mathrm{TV}(Q_0, Q_1).$$

The layer-cake identity gives, for $i \in \{0, 1\}$,

$$\int_{\mathcal{Z}} \varphi(z) Q_i(dz) = \int_{(0,1]} Q_i(L_{\varphi, t}) dt.$$

All integrands are measurable and bounded. Subtracting these identities and integrating the preceding event bound yields

$$\begin{aligned} \left| \int_{\mathcal{Z}} \varphi dQ_0 - \int_{\mathcal{Z}} \varphi dQ_1 \right| &= \left| \int_{(0,1]} [Q_0(L_{\varphi, t}) - Q_1(L_{\varphi, t})] dt \right| \\ &\leq \int_{(0,1]} |Q_0(L_{\varphi, t}) - Q_1(L_{\varphi, t})| dt \\ &\leq \mathrm{TV}(Q_0, Q_1). \end{aligned}$$

For any measurable event $T_{\mathrm{act}} \subseteq \mathcal{H}$, define

$$\varphi_{\mathrm{act}}(z) := \kappa(z, T_{\mathrm{act}}) \quad (z \in \mathcal{Z}).$$

This function is measurable and takes values in $[0, 1]$, and

$$(Q_i\kappa)(T_{\mathrm{act}}) = \int_{\mathcal{Z}} \varphi_{\mathrm{act}} dQ_i \quad (i \in \{0, 1\}).$$

Applying the bounded-function estimate and taking the supremum over measurable action events proves

$$\mathrm{TV}(Q_0\kappa, Q_1\kappa) \leq \mathrm{TV}(Q_0, Q_1).$$

2. For any measurable joint event $T_{\text{joint}} \subseteq \mathcal{Z} \times \mathcal{H}$, define its sections and their kernel probabilities by

$$T_{\text{joint},z} := \{h \in \mathcal{H} : (z, h) \in T_{\text{joint}}\}, \quad \varphi_{\text{joint}}(z) := \kappa(z, T_{\text{joint},z}) \quad (z \in \mathcal{Z}).$$

The sections are measurable, and φ_{joint} is measurable with values in $[0, 1]$. The joint-law identity is

$$(Q_i \times \kappa)(T_{\text{joint}}) = \int_{\mathcal{Z}} \varphi_{\text{joint}} dQ_i \quad (i \in \{0, 1\}).$$

Thus the bounded-function estimate, followed by the supremum over joint events, gives

$$\text{TV}(Q_0 \times \kappa, Q_1 \times \kappa) \leq \text{TV}(Q_0, Q_1).$$

For the reverse inequality, every measurable input event $T_{\text{in}} \subseteq \mathcal{Z}$ satisfies

$$(Q_i \times \kappa)(T_{\text{in}} \times \mathcal{H}) = Q_i(T_{\text{in}}) \quad (i \in \{0, 1\}),$$

because $\kappa(z, \mathcal{H}) = 1$. Consequently,

$$\begin{aligned} |Q_0(T_{\text{in}}) - Q_1(T_{\text{in}})| &= |(Q_0 \times \kappa)(T_{\text{in}} \times \mathcal{H}) - (Q_1 \times \kappa)(T_{\text{in}} \times \mathcal{H})| \\ &\leq \text{TV}(Q_0 \times \kappa, Q_1 \times \kappa). \end{aligned}$$

Taking the supremum over input events proves

$$\text{TV}(Q_0 \times \kappa, Q_1 \times \kappa) = \text{TV}(Q_0, Q_1).$$

Combining this equality with the action-marginal bound gives

$$\text{TV}(Q_0 \kappa, Q_1 \kappa) \leq \text{TV}(Q_0 \times \kappa, Q_1 \times \kappa) = \text{TV}(Q_0, Q_1).$$

3. The joint laws are probability laws. Hence, for any measurable joint event T_{joint} ,

$$(Q_1 \times \kappa)(T_{\text{joint}}^c) = 1 - (Q_1 \times \kappa)(T_{\text{joint}}),$$

and the event bound defining their total variation gives

$$\begin{aligned} &(Q_0 \times \kappa)(T_{\text{joint}}) + (Q_1 \times \kappa)(T_{\text{joint}}^c) \\ &= 1 + (Q_0 \times \kappa)(T_{\text{joint}}) - (Q_1 \times \kappa)(T_{\text{joint}}) \\ &\geq 1 - \text{TV}(Q_0 \times \kappa, Q_1 \times \kappa) = 1 - \text{TV}(Q_0, Q_1). \end{aligned}$$

4. Likewise, the action marginals are probability laws, so every measurable action event T_{act} satisfies

$$(Q_1 \kappa)(T_{\text{act}}^c) = 1 - (Q_1 \kappa)(T_{\text{act}}).$$

Using their total-variation event bound and then the action-marginal contraction established above,

$$\begin{aligned} (Q_0 \kappa)(T_{\text{act}}) + (Q_1 \kappa)(T_{\text{act}}^c) &= 1 + (Q_0 \kappa)(T_{\text{act}}) - (Q_1 \kappa)(T_{\text{act}}) \\ &\geq 1 - \text{TV}(Q_0 \kappa, Q_1 \kappa) \\ &\geq 1 - \text{TV}(Q_0, Q_1). \end{aligned}$$

□

Retaining the observation alongside the randomized action preserves total variation, while taking the action marginal weakly reduces it. When the event represents selection of alternative 1, the displayed probability sum is the sum of the two testing errors. These comparisons allow the lower-bound analysis in [Theorem 12](#) and the interval analysis in [Theorem 7](#) to cover parameter-independent randomized procedures using the same observation-law comparison.

A.2 Identification through observed cells

The cell functional in [Definition 10](#) expresses the target through membership probabilities and arrived-outcome means, with zero contributions assigned to null cells. Its connection to the average treatment effect uses randomized independence, balanced assignment, consistency, conditional missing at random, and positive arrival in occupied cells, as specified in [Assumptions 2 to 7](#). The first identity applies to the unrestricted-arrival class.

Lemma 3. [[lem:unrestricted-cell-identification](#)] (Identification by cell contributions) *Let $n, d \in \mathbb{N}$, $q \in \mathbb{R}$, and let $P \in \mathcal{M}_{n,d,q}^+$ satisfy [Definition 7](#). The average treatment effect $\tau(P)$ and the total observed-cell functional $\Phi(P)$, defined in [Definitions 8 and 10](#), satisfy*

$$\tau(P) = \Phi(P) = 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a) c_{axs}(P).$$

Proof of [Lemma 3](#). Fix $P \in \mathcal{M}_{n,d,q}^+$. We use the conditions in [Definition 7](#).

1. By [Definition 3](#), the potential outcomes are binary. Linearity of integration for these bounded variables therefore gives

$$\tau(P) = E_P[Y(1)] - E_P[Y(0)] = P(Y(1) = 1) - P(Y(0) = 1),$$

using [Definition 8](#).

2. We establish, for every $(a, x, s) \in \mathcal{J}_d$, the identity

$$c_{axs}(P) = P(A = a, X = x, S = s, Y = 1).$$

First suppose $p_{axs}(P) > 0$. By [Definition 6](#) and [Assumption 7](#) and $q > 0$,

$$P(A = a, X = x, S = s, R = 1) = p_{axs}(P) \rho_{axs}(P) \geq q p_{axs}(P) > 0.$$

Conditional independence in [Assumption 6](#), applied within this occupied cell, yields the cross-multiplied identity

$$\begin{aligned} & P(A = a, X = x, S = s, R = 1, Y = 1) p_{axs}(P) \\ &= P(A = a, X = x, S = s, R = 1) P(A = a, X = x, S = s, Y = 1). \end{aligned}$$

Since Y is binary and the arrived-cell probability is positive, [Definition 10](#) gives

$$\begin{aligned} c_{axs}(P) &= p_{axs}(P) \frac{P(A = a, X = x, S = s, R = 1, Y = 1)}{P(A = a, X = x, S = s, R = 1)} \\ &= \frac{P(A = a, X = x, S = s, R = 1, Y = 1) p_{axs}(P)}{P(A = a, X = x, S = s, R = 1)} \\ &= P(A = a, X = x, S = s, Y = 1). \end{aligned}$$

In the remaining case, nonnegativity implies $p_{axs}(P) = 0$. Monotonicity of probability gives

$$\begin{aligned} 0 &\leq P(A = a, X = x, S = s, Y = 1) \leq p_{axs}(P) = 0, \\ 0 &\leq P(A = a, X = x, S = s, R = 1) \leq p_{axs}(P) = 0. \end{aligned}$$

Thus both probabilities vanish, and the zero-arrival convention in [Definition 10](#) gives $c_{axs}(P) = 0$, proving the identity also in this case.

3. For each $a \in \{0, 1\}$, define the finite measure obtained by restricting P to the arm- a , outcome-one event:

$$\nu_a^{\text{out}}(\cdot) := P(\cdot \cap \{A = a, Y = 1\}).$$

The events $\{X = x, S = s\}$, with $x \in \{0, \dots, d-1\}$ and $s \in \{0, 1\}$, are disjoint and cover the full-data record space. Finite additivity of this restricted measure therefore yields

$$\begin{aligned} \sum_{x=0}^{d-1} \sum_{s \in \{0,1\}} P(A = a, X = x, S = s, Y = 1) \\ = \sum_{x=0}^{d-1} \sum_{s \in \{0,1\}} \nu_a^{\text{out}}(X = x, S = s) = P(A = a, Y = 1). \end{aligned}$$

4. Expanding [Definition 10](#) over the two treatment arms, using $g(1) = 1$ and $g(0) = -1$ from [Definition 9](#), and substituting the preceding identities gives

$$\begin{aligned} \Phi(P) &= 2 \sum_{x=0}^{d-1} \sum_{s \in \{0,1\}} (c_{1xs}(P) - c_{0xs}(P)) \\ &= 2(P(A = 1, Y = 1) - P(A = 0, Y = 1)). \end{aligned}$$

5. Fix $a \in \{0, 1\}$. Projecting the independence asserted in [Assumption 2](#) onto the selected potential outcome gives

$$A \perp\!\!\!\perp Y(a), \quad P(A = a, Y(a) = 1) = P(A = a) P(Y(a) = 1).$$

By [Assumption 3](#),

$$P(A = 1) = \frac{1}{2}, \quad P(A = 0) = 1 - P(A = 1) = \frac{1}{2}.$$

The assigned-arm outcome in [Definition 3](#) satisfies $Y = Y(a)$ on $\{A = a\}$. Consequently,

$$\begin{aligned} 2P(A = a, Y = 1) &= 2P(A = a, Y(a) = 1) \\ &= 2P(A = a) P(Y(a) = 1) = P(Y(a) = 1). \end{aligned}$$

Applying this identity first to $a = 1$ and then to $a = 0$ gives

$$2P(A = 1, Y = 1) = P(Y(1) = 1), \quad 2P(A = 0, Y = 1) = P(Y(0) = 1).$$

6. Subtracting the arm-zero identity from the arm-one identity and substituting into the expressions for $\tau(P)$ and $\Phi(P)$ proves

$$\tau(P) = \Phi(P) = 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a) c_{axs}(P),$$

where the final equality is [Definition 10](#).

□

The identity makes the observed-cell functional an exact population target throughout the unrestricted-arrival class. The surrogate enters through the cells on which arrival is conditioned, and the signed sum aggregates their contributions into the treatment contrast. This is the identification relation used by the unrestricted risk bound in [Theorem 6](#).

For the rare-arrival class, the same relation is recorded under the model conditions governing its minimax analysis.

Lemma 4. [lem:cell-identification-background] (Identification by cell contributions) *Let $n, d \in \mathbb{N}$, $q \in \mathbb{R}$, and let $P \in \mathcal{M}_{n,d,q}$ satisfy the model conditions in [Definition 14](#). The average treatment effect $\tau(P)$ of [Definition 8](#) equals the total signed observed-cell functional $\Phi(P)$ of [Definition 10](#):*

$$\tau(P) = \Phi(P) = 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a)c_{axs}(P).$$

Here the cell contributions and their null-cell convention are as defined in [Definition 10](#).

Proof of Lemma 4. 1. By [Lemma 3](#), it suffices to establish

$$P \in \mathcal{M}_{n,d,q}^+$$

where $\mathcal{M}_{n,d,q}^+$ is the arrival-floor model of [Definition 7](#). For such a law, the cited result supplies

$$\tau(P) = \Phi(P) = 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a)c_{axs}(P).$$

2. Since $P \in \mathcal{M}_{n,d,q}$, [Definition 14](#) gives

$$n \geq 1, \quad d \geq 1, \quad 0 < q \leq 1,$$

together with [Assumptions 2](#) and [3](#), [Assumptions 4](#) and [5](#), and [Assumptions 6](#) and [7](#). These are precisely the conditions defining the arrival-floor model. Thus

$$P \in \mathcal{M}_{n,d,q}^+$$

and the identity in the preceding step follows. □

The rare-arrival restriction in [Assumption 8](#) specifies the regime for the risk comparison, while this identity fixes its estimand. In particular, the null-cell convention gives a total cell functional across laws with different occupied categories. The mixed-count analysis underlying [Theorem 4](#) therefore targets the same average treatment effect throughout the rare-arrival model.

A.3 Synthetic assignment and experiment comparison

The observational class in [Definition 31](#) supplies an experiment that can be mapped into the unrestricted randomized model. The map assigns an independent fair binary arm to each observational record, records outcome arrival when that arm matches the observational treatment, and sets the surrogate to zero. The following statement establishes a parameter-independent observation map together with a full-data completion preserving the target.

Lemma 5. [lem:synthetic-randomization-kernel] (Synthetic randomization kernel) *Let the inputs satisfy:*

- **(Sample size.)** $n, d \in \mathbb{N}$ and $n \geq 1$.
- **(Arrival floor.)** $0 < q < 1/2$.
- **(Observational law.)** $P_Z \in \mathcal{D}_{d,q}$, the discrete positivity class of [Definition 31](#).

Index observational baseline categories by $x \in \{1, \dots, d\}$ and full-data baseline categories by their corresponding zero-based indices $x - 1 \in \{0, \dots, d - 1\}$. For an observational record (x, b, y) , define

$$o(x, b, y; u) = (x - 1, u, 0, \mathbf{1}\{b = u\}, \mathbf{1}\{b = u\}y), \quad u \in \{0, 1\}.$$

Applying this map independently to each input record with independent fair binary draws defines a Markov kernel independent of P_Z .

There exists a full-data law $P_Z^* \in \mathcal{M}_{n,d,q}^+$, as defined in [Definition 7](#), such that applying this kernel to $\mathbf{Z}_n$, an i.i.d. sample of n observational records from P_Z , yields exactly the n -fold product of the observed-record law induced by P_Z^* . The resulting observed sample $\mathbf{O}_n$ satisfies [Assumption 1](#) under P_Z^* , and

$$\tau(P_Z^*) = \theta_Z(P_Z),$$

with the functionals defined in [Definitions 8](#) and [32](#).

Proof of Lemma 5. 1. Construct the full-data completion. For observational categories $x \in \{1, \dots, d\}$ and $b \in \{0, 1\}$, use the following total versions of the quantities in [Definition 31](#):

$$p_x := P_Z(X = x), \quad \pi_x := \begin{cases} P_Z(X = x, B^{\text{obs}} = 1)/p_x, & p_x > 0, \\ 0, & p_x = 0, \end{cases}$$

and

$$\mu_{bx} := \begin{cases} \frac{P_Z(X = x, B^{\text{obs}} = b, Y = 1)}{P_Z(X = x, B^{\text{obs}} = b)}, & P_Z(X = x, B^{\text{obs}} = b) > 0, \\ 0, & P_Z(X = x, B^{\text{obs}} = b) = 0. \end{cases}$$

The two observational arm probabilities sum to p_x . Positivity therefore gives, when $p_x > 0$,

$$P_Z(X = x, B^{\text{obs}} = 1) \geq qp_x > 0, \quad P_Z(X = x, B^{\text{obs}} = 0) \geq qp_x > 0.$$

Thus these means agree with those in [Definition 31](#) on occupied categories. Each ratio lies in $[0, 1]$, since its numerator is the probability of a subset of the event in its denominator.

Define the binary mass functions

$$\beta_{bx}^{\text{mark}} := \begin{cases} 1 - \pi_x, & b = 0, \\ \pi_x, & b = 1, \end{cases} \quad \beta_{bxy}^{\text{out}} := \begin{cases} 1 - \mu_{bx}, & y = 0, \\ \mu_{bx}, & y = 1, \end{cases} \quad (x \in \{1, \dots, d\}, b, y \in \{0, 1\}).$$

Their masses are nonnegative and satisfy

$$\sum_{b=0}^1 \beta_{bx}^{\text{mark}} = 1, \quad \sum_{y=0}^1 \beta_{bxy}^{\text{out}} = 1.$$

Let $\mathbb{P}_{\text{lat}}$ be the probability law on $\{0, \dots, d-1\} \times \{0, 1\}^4$, with coordinates $(X, B^{\text{lat}}, Y(0), Y(1), A)$, specified by

$$\mathbb{P}_{\text{lat}}(X = x-1, B^{\text{lat}} = b, Y(0) = y_0, Y(1) = y_1, A = a) = \frac{p_x}{2} \beta_{bx}^{\text{mark}} \beta_{0xy_0}^{\text{out}} \beta_{1xy_1}^{\text{out}},$$

for $x \in \{1, \dots, d\}$ and $a, b, y_0, y_1 \in \{0, 1\}$. The preceding normalization identities and $\sum_x p_x = 1$ show that this is a probability law.

On this probability space set

$$S(0) := S(1) := 0, \quad S := S(A), \quad Y := Y(A), \quad R := \mathbf{1}\{B^{\text{lat}} = A\},$$

and define the full-data law

$$P_Z^* := \mathcal{L}_{\mathbb{P}_{\text{lat}}}(X, A, S(0), S(1), Y(0), Y(1), S, Y, R).$$

Here $\mathcal{L}$ denotes the distribution of the displayed coordinates.

2. Verify membership in the arrival-floor model. The product factor $1/2$ in the construction gives

$$A \perp\!\!\!\perp (X, S(0), S(1), Y(0), Y(1)), \quad P_Z^*(A = 1) = \frac{1}{2}.$$

Together with $S = S(A)$ and $Y = Y(A)$, these establish [Assumptions 2](#) to [5](#).

For $a, r_{\text{arr}} \in \{0, 1\}$ and $x \in \{1, \dots, d\}$, define the arrival mass

$$\beta_{axr_{\text{arr}}}^{\text{arr}} := \begin{cases} \beta_{ax}^{\text{mark}}, & r_{\text{arr}} = 1, \\ \beta_{1-a,x}^{\text{mark}}, & r_{\text{arr}} = 0. \end{cases}$$

Fix also $y \in \{0, 1\}$. The event $R = r_{\text{arr}}, A = a$ fixes the latent mark to a when $r_{\text{arr}} = 1$, and to $1-a$ when $r_{\text{arr}} = 0$. Summing over the unselected potential outcome gives

$$P_Z^*(A = a, X = x-1, S = 0, R = r_{\text{arr}}, Y = y) = \frac{p_x}{2} \beta_{axr_{\text{arr}}}^{\text{arr}} \beta_{axy}^{\text{out}}. \quad (1)$$

Indeed, the mass of the unselected potential outcome sums to one. Summing this identity over the binary outcome and arrival coordinates yields

$$\begin{aligned} p_{a,x-1,0}(P_Z^*) &= \frac{p_x}{2}, \\ P_Z^*(A = a, X = x-1, S = 0, R = r_{\text{arr}}) &= \frac{p_x}{2} \beta_{axr_{\text{arr}}}^{\text{arr}}, \\ P_Z^*(A = a, X = x-1, S = 0, Y = y) &= \frac{p_x}{2} \beta_{axy}^{\text{out}}. \end{aligned}$$

Consequently,

$$\begin{aligned} P_Z^*(A = a, X = x-1, S = 0, R = r_{\text{arr}}, Y = y) p_{a,x-1,0}(P_Z^*) \\ = P_Z^*(A = a, X = x-1, S = 0, R = r_{\text{arr}}) P_Z^*(A = a, X = x-1, S = 0, Y = y). \end{aligned}$$

For $S = 1$, every probability in this identity is zero, since $S = 0$ throughout the construction. This proves [Assumption 6](#), including cells of zero mass.

The same calculation gives

$$P_Z^*(A = a, X = x - 1, S = 0, R = 1) = \frac{p_x}{2} \beta_{ax}^{\text{mark}}, \quad p_{a,x-1,1}(P_Z^*) = 0.$$

Thus an occupied cell has $S = 0$ and $p_x > 0$. By the observational positivity bounds,

$$\begin{aligned} \rho_{0,x-1,0}(P_Z^*) &= 1 - \pi_x \geq q, \\ \rho_{1,x-1,0}(P_Z^*) &= \pi_x \geq q. \end{aligned}$$

This establishes [Assumption 7](#). Finally, $n \geq 1$, membership of P_Z in the observational class supplies $d \geq 1$, and $0 < q < 1/2$ implies $q \leq 1$. Hence

$$P_Z^* \in \mathcal{M}_{n,d,q}^+$$

by [Definition 7](#).

3. Record the sampling property of the canonical observed product law. Define the observed-record alphabet and its one-record law by

$$\mathcal{O}_d := \{0, \dots, d-1\} \times \{0, 1\}^4, \quad \nu_{\text{obs}} := \mathcal{L}_{P_Z^*}(X, A, S, R, RY).$$

For $i \in \{1, \dots, n\}$, let

$$\text{pr}_i : \mathcal{O}_d^n \longrightarrow \mathcal{O}_d, \quad \text{pr}_i(\mathbf{o}) := \mathbf{o}_i.$$

For arbitrary subsets $E_i \subseteq \mathcal{O}_d$, the defining property of the product measure gives

$$\nu_{\text{obs}}^{\otimes n} \left(\bigcap_{i=1}^n \text{pr}_i^{-1}(E_i) \right) = \prod_{i=1}^n \nu_{\text{obs}}(E_i).$$

The coordinate projections are measurable on these finite spaces. Taking all but one E_i to be the entire alphabet also gives the common marginal ν_{obs} . Thus this product experiment satisfies [Assumption 1](#).

4. Identify the kernel output law. First, the selected latent record has exactly the observational law:

$$\mathcal{L}_{\mathbb{P}_{\text{lat}}}(X + 1, B^{\text{lat}}, Y(B^{\text{lat}})) = P_Z.$$

To verify this identity, sum over A and the unselected potential outcome. For each $x \in \{1, \dots, d\}$ and $b, y \in \{0, 1\}$, the resulting singleton mass is

$$\mathbb{P}_{\text{lat}}(X = x - 1, B^{\text{lat}} = b, Y(b) = y) = p_x \beta_{bx}^{\text{mark}} \beta_{bxy}^{\text{out}}.$$

When $p_x > 0$,

$$p_x \beta_{bx}^{\text{mark}} = P_Z(X = x, B^{\text{obs}} = b).$$

For $y = 1$, multiplying by μ_{bx} gives the corresponding observational singleton probability by its defining ratio. For $y = 0$, subtracting that singleton probability from the arm probability gives the same conclusion:

$$p_x \beta_{bx}^{\text{mark}} \beta_{bxy}^{\text{out}} = P_Z(X = x, B^{\text{obs}} = b, Y = y).$$

When $p_x = 0$, both sides vanish because the singleton event is contained in the null baseline category. Equality of all singleton masses proves the selected-record identity.

Define the fair binary law and one-record kernel by

$$\nu_{\text{coin}} := \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1, \quad \kappa_{\text{syn}}((x, b, y), \cdot) := \frac{1}{2}\delta_{o(x, b, y; 0)} + \frac{1}{2}\delta_{o(x, b, y; 1)}.$$

This kernel is a measurable probability kernel on finite alphabets and depends solely on the input record. Under the completion, the observed tuple satisfies the pointwise identity

$$(X, A, S, R, RY) = o(X + 1, B^{\text{lat}}, Y(B^{\text{lat}}); A).$$

For a matching assignment, the selected outcome equals $Y(A)$; for a mismatching assignment, both arrived-outcome coordinates are zero. Moreover, A is independent of the selected latent record. The preceding distributional identity therefore gives

$$\nu_{\text{obs}} = P_Z \kappa_{\text{syn}} := \int \kappa_{\text{syn}}((x, b, y), \cdot) P_Z(d(x, b, y)).$$

For a deterministic input sample

$$\mathbf{z} = ((x_i, b_i, y_i))_{i=1}^n \in (\{1, \dots, d\} \times \{0, 1\}^2)^n$$

and $\mathbf{u} = (u_i)_{i=1}^n \in \{0, 1\}^n$, define

$$\mathbf{o}_n(\mathbf{z}, \mathbf{u}) := (o(x_i, b_i, y_i; u_i))_{i=1}^n, \quad \kappa_{\text{syn}, n}(\mathbf{z}, \cdot) := \bigotimes_{i=1}^n \kappa_{\text{syn}}((x_i, b_i, y_i), \cdot).$$

Independent fair draws give

$$\kappa_{\text{syn}, n}(\mathbf{z}, \cdot) = (\mathbf{o}_n(\mathbf{z}, \cdot))_{\#} \nu_{\text{coin}}^{\otimes n},$$

where the subscript $\#$ denotes pushforward. Integrating over the input sample, regrouping the product coordinates into record-coin pairs, and then applying the record map coordinatewise gives

$$\begin{aligned} \int \kappa_{\text{syn}, n}(\mathbf{z}, \cdot) P_Z^{\otimes n}(d\mathbf{z}) &= (\mathbf{o}_n)_{\#} (P_Z^{\otimes n} \otimes \nu_{\text{coin}}^{\otimes n}) \\ &= \bigotimes_{i=1}^n \left[\int \kappa_{\text{syn}}((x, b, y), \cdot) P_Z(d(x, b, y)) \right] \\ &= \nu_{\text{obs}}^{\otimes n}. \end{aligned}$$

The regrouping is justified by the product identity

$$\left(\prod_{i=1}^n P_Z(dz_i) \right) \left(\prod_{i=1}^n \nu_{\text{coin}}(du_i) \right) = \prod_{i=1}^n (P_Z(dz_i) \nu_{\text{coin}}(du_i)),$$

and coordinatewise pushforward preserves this product form. Thus the kernel produces exactly the observed product law identified in the preceding step, and its output satisfies [Assumption 1](#).

5. Compute the target through cell contributions. Since P_Z^* belongs to the arrival-floor model, [Lemma 3](#) supplies

$$\tau(P_Z^*) = \Phi(P_Z^*) = 2 \sum_{(a,x,s) \in \mathcal{J}_d} g(a) c_{axs}(P_Z^*).$$

Here the contributions and their zero convention are given by [Definition 10](#), and [Definition 9](#) gives $g(0) = -1$, $g(1) = 1$.

For $p_x > 0$, the arrival mass in either arm is strictly positive:

$$P_Z^*(A = a, X = x - 1, S = 0, R = 1) = \frac{p_x}{2} \beta_{ax}^{\text{mark}} \geq \frac{qp_x}{2} > 0.$$

The joint-mass identity in [Equation \(1\)](#) therefore yields

$$\mu_{a,x-1,0}(P_Z^*) = \frac{\frac{p_x}{2} \beta_{ax}^{\text{mark}} \mu_{ax}}{\frac{p_x}{2} \beta_{ax}^{\text{mark}}} = \mu_{ax}, \quad c_{a,x-1,0}(P_Z^*) = \frac{p_x}{2} \mu_{ax}.$$

When $p_x = 0$, the arrived-cell probability is zero, so the defining convention gives the same contribution identity. Every cell with $S = 1$ also has zero arrived-cell probability. Hence, for every category and arm,

$$c_{a,x-1,0}(P_Z^*) = \frac{p_x}{2} \mu_{ax}, \quad c_{a,x-1,1}(P_Z^*) = 0.$$

Substituting these identities into the signed cell sum gives

$$\begin{aligned} \tau(P_Z^*) &= 2 \sum_{x=1}^d \left(\frac{p_x}{2} \mu_{1x} - \frac{p_x}{2} \mu_{0x} \right) \\ &= \sum_{x:p_x > 0} p_x (\mu_{1x} - \mu_{0x}) = \theta_Z(P_Z), \end{aligned}$$

where the last equality uses [Definition 32](#); null categories contribute zero. □

The map preserves both the product observation law and the target at the corresponding full-data completion. Its parameter independence permits composition with any randomized estimator in [Definition 12](#). This fixes the direction of the experiment comparison used in [Theorem 5](#): observational minimax lower bounds transfer to the unrestricted randomized model at the same arrival floor.

For construction, the same map supplies the synthetic records used by the observational procedure in [Theorem 8](#). The target equality connects its squared loss to that of the randomized estimator, and [Lemma 1](#) supplies the risk comparison for averaging over the synthetic assignments. Thus the observation map supports both the fixed-arrival lower-bound comparison and the constructive observational extension.

B Risk certificates for the constructive estimator

The risk analysis separates estimation of cell membership probabilities from estimation of arrived-cell outcome means. Independence holds in the uncapped Poisson comparison experiment used for this analysis. The reported finite-sample estimator instead averages assignments

of a capped prefix, and the prefix-transfer argument below connects its risk to that independent-stream comparison. The polynomial branch controls the contribution of sparsely observed cells. The resulting certificate bounds squared error for the observed-cell functional in [Definition 10](#); identification then connects that certificate to the causal upper bound in [Theorem 6](#).

For the deterministic mixed-count estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$ of [Algorithm 1](#), write $N = nq$ for the effective sample size, $m = n/6$ for the intensity of each auxiliary stream, and $N_0 = mq = N/6$ for the effective stream size. The logarithmic scale ℓ , degree k , and threshold B are specified in [Definitions 13](#), [19](#) and [20](#). Define the prefix-tail constant by

$$\gamma_0 = \log 2 - \frac{1}{2}.$$

On the polynomial branch, where $N > 1$, $\ell \geq 128$, and $d \geq \sqrt{N}$, introduce the certificate quantities

$$b_{n,d,q} = 2 \left\{ \frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right\}$$

and

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d(e^{\ell/8} + 1) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right).$$

Here $b_{n,d,q}$ is the bias allowance and $v_{n,d,q}$ is the additive second-moment allowance. These quantities summarize the polynomial approximation, marked-count fluctuations, and estimation of membership weights.

The deterministic squared-error envelope $\mathfrak{U}_{n,d,q}$ combines these allowances with the elementary branch and the prefix-overflow contribution. The following statement gives its full formula and compares it with the rate $r_{n,d,q}$ of [Definition 17](#) throughout the unrestricted-arrival class $\mathcal{M}_{n,d,q}^+$ of [Definition 7](#).

Theorem 9. [thm:unrestricted-uniform-mixed-count-certificate] (Uniform mixed-count risk certificate) *There exists a universal constant $0 < \bar{C} < \infty$ such that the following holds.*

- **(Domain.)** Let $n, d \geq 1$ be integers and let $0 < q \leq 1$.
- **(Model.)** Let $P \in \mathcal{M}_{n,d,q}^+$, the unrestricted-arrival model class of [Definition 7](#).
- **(Sampling.)** Let $\mathbf{O}_n$, the sample of observed records, satisfy [Assumption 1](#) under P .

Use the mixed-count estimator of [Algorithm 1](#). Let ℓ , the logarithmic scale, be defined by [Definition 13](#); let k , the polynomial degree, be defined by [Definition 19](#); and let B , the threshold, be defined by [Definition 20](#). Put

$$N = nq, \quad m = \frac{n}{6}, \quad N_0 = mq = \frac{N}{6}, \quad \gamma_0 = \log 2 - \frac{1}{2}.$$

Define the deterministic envelope $\mathfrak{U}_{n,d,q}$ as follows. If $N \leq 1$, set

$$\mathfrak{U}_{n,d,q} = 1.$$

If $N > 1$ and either $\ell < 128$ or $d < \sqrt{N}$, set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, \left(\frac{8d}{eN_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\}.$$

If $N > 1$, $\ell \geq 128$, and $d \geq \sqrt{N}$, put

$$b_{n,d,q} = 2 \left\{ \frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right\},$$

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d(e^{\ell/8} + 1) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right),$$

and set

$$\mathfrak{U}_{n,d,q} = \min \{ 4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \}.$$

Then, with $\Phi(P)$ the cell functional of [Definition 10](#) and $r_{n,d,q}$ the rate of [Definition 17](#), expectation under the canonical i.i.d. observed-sample law satisfies

$$E_P \left[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P))^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \bar{C} r_{n,d,q}.$$

The certificate distinguishes information from arrived outcomes, measured by N_0 , from information about cell membership, measured by m . On the polynomial branch, the squared bias allowance captures the aggregate effect of estimating means in sparsely observed cells, while the additive allowance accounts for fluctuations in both components of the cell contribution. The elementary branch supplies the corresponding bound in its stated parameter range. Together, the branches give a deterministic precision benchmark depending on n, d, q , with a universal comparison constant.

B.1 Polynomial bias and marked-count second moments

The polynomial statistic H_j and the pilot-selected statistic G_j are specified in [Definitions 24](#) and [25](#). Their mathematical role is to replace the elementary cell statistic where the pilot count indicates sparse outcome arrival. The degree and threshold jointly determine the approximation contribution $4dB/(N_0 k^2)$ in the bias allowance. The exponentially small term in that allowance accounts for branch control. The certificate combines these contributions before squaring, preserving the distinction between bias and second-moment terms.

Marked counts retain the association between an arrived outcome and its cell. Consequently, the second-moment analysis concerns the marked polynomial statistic, together with the pilot-selected branch, rather than an unmarked count alone. The expression $v_{n,d,q}$ records the resulting allowance: its terms involve the effective arrived-outcome intensity, the membership-stream intensity, the alphabet size, and the polynomial tuning parameters. This decomposition explains why the degree enters the bias allowance while the coefficient growth associated with the polynomial construction enters the additive allowance.

B.2 Membership weights and branch control

The membership weight W_j in [Definition 28](#) is formed from the membership-labelled records. In the uncapped Poisson comparison experiment, that stream is independent of the pilot and estimation streams. The finite-sample estimator uses the capped prefix and conditional averaging in [Algorithm 1](#); the overflow allowance and prefix-transfer comparison connect it to the independent-stream analysis. The weight supplies the population mass attached to each arrived-cell mean while retaining information from all observed memberships.

The two sample scales appear explicitly in the certificate through $1/N_0$, $1/m$, and the mixed term $B/(mN_0)$. These terms retain the cost of estimating membership weights alongside the

cost of estimating arrived-cell means. The pilot-selected statistic in [Definition 24](#) organizes the latter task within the polynomial branch, whereas the elementary ratio in [Definition 26](#) supplies the estimator on the elementary branch. The branch conditions in [Theorem 9](#) specify where each deterministic expression applies.

B.3 Transfer to the fixed observed sample and causal target

The auxiliary construction in [Algorithm 1](#) uses an independent Poisson prefix and uniform stream assignments. It assigns zero output when the prefix exceeds the available n records and clips the output to $[-1, 1]$. The term $4e^{-n\gamma_0}$ in [Theorem 9](#) is the prefix-overflow allowance. Its dependence on the full record count n reflects the supply of records for the auxiliary construction, while $N = nq$ governs the effective outcome information. Clipping supplies the bounded-loss allowance represented by the minimum with 4; the zero estimator on the small-effective-sample branch supplies the bound 1.

Conditional averaging over the auxiliary randomization produces the deterministic estimator in [Algorithm 1](#). The squared-loss comparison for such averaging is supplied by [Lemma 1](#). Thus the certificate concerns the estimator evaluated on the fixed observed sample, with the auxiliary construction serving its risk analysis.

For the causal interpretation, [Lemma 3](#) identifies the observed-cell functional with the average treatment effect under the unrestricted model conditions. Substituting that identity into [Theorem 9](#) gives the causal risk control used in [Theorem 6](#). The same identification step applies throughout the rare-arrival class $\mathcal{M}_{n,d,q}$ of [Definition 14](#). The following specialization records the resulting envelope for that class.

Theorem 10. [thm:uniform-mixed-count-certificate] (Uniform mixed-count risk certificate) *There exists a universal constant $0 < \bar{C} < \infty$ such that the following holds for every $n, d \in \mathbb{N}$, $q \in \mathbb{R}$, and full-data law $P \in \mathcal{M}_{n,d,q}$ satisfying [Definition 14](#), including $n \geq 1$, $d \geq 1$, and $0 < q \leq 1$.*

Let $\hat{\tau}_{n,d,q}^{\text{mix}}$ be the estimator in [Algorithm 1](#), and let $r_{n,d,q}$ be the rate in [Definition 17](#). Let ℓ be the logarithmic scale defined in [Definition 13](#), k the polynomial degree defined in [Definition 19](#), and B the threshold defined in [Definition 20](#). Put

$$N = nq, \quad m = \frac{n}{6}, \quad N_0 = mq = \frac{N}{6}, \quad \gamma_0 = \log 2 - \frac{1}{2},$$

where N is the effective sample size, m the stream intensity, N_0 the effective stream size, and γ_0 the prefix-tail constant. Define the deterministic envelope $\mathfrak{U}_{n,d,q}$ by the following branches:

- **(Small effective sample.)** *If $N \leq 1$, set*

$$\mathfrak{U}_{n,d,q} = 1.$$

- **(Elementary branch.)** *If $N > 1$ and either $\ell < 128$ or $d < \sqrt{N}$, set*

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, \left(\frac{8d}{eN_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\}.$$

- **(Polynomial branch.)** *If $N > 1$, $\ell \geq 128$, and $d \geq \sqrt{N}$, define the certificate quantities*

$$b_{n,d,q} = 2 \left(\frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right),$$

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d \left(e^{\ell/8} + 1 \right) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right),$$

and set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \right\}.$$

Then, writing $\mathbf{O}_n$ for the observed sample induced by P and $\tau(P)$ for the average treatment effect in [Definition 8](#),

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \bar{C} r_{n,d,q}.$$

The rare-arrival restriction places the model within the domain of the unrestricted certificate, and [Lemma 4](#) fixes the causal target there. Accordingly, the specialization retains the same numerical envelope. Its polynomial bias, second-moment, and prefix-transfer components are those of [Theorem 9](#); model inclusion and identification supply the specialization step.

The next statement records this causal upper bound together with the squared-loss comparison between the deterministic estimator and its auxiliary randomized output. It makes explicit the role of conditional averaging in the final estimator.

Theorem 11. [\[thm:mixed-count-upper\]](#) (Mixed-count risk upper bound) *There exists a universal constant $0 < \bar{C} < \infty$ such that the following holds for every choice of parameters and law satisfying:*

- **(Sample size and dimension.)** $n, d \in \mathbb{N}$ satisfy $n, d \geq 1$.
- **(Arrival floor.)** $q \in \mathbb{R}$ satisfies $0 < q \leq 1$.
- **(Model.)** The full-data law P belongs to $\mathcal{M}_{n,d,q}$ in [Definition 14](#).

Let $\hat{\tau}_{n,d,q}^{\text{mix}}$ and $\tilde{\tau}(\mathbf{O}_n; \omega)$ be the deterministic estimator and auxiliary randomized output of [Algorithm 1](#), where $\mathbf{O}_n$ is the observed sample and ω is the auxiliary randomization. Let ℓ be the logarithmic scale defined in [Definition 13](#), k the polynomial degree defined in [Definition 19](#), and $B = 256\ell$ the threshold defined in [Definition 20](#). Use the elementary and polynomial branch rules of [Algorithm 1](#). Define the effective sample size N , stream intensity m , effective stream size N_0 , and prefix-tail constant γ_0 by

$$N = nq, \quad m = \frac{n}{6}, \quad N_0 = mq = \frac{N}{6}, \quad \gamma_0 = \log 2 - \frac{1}{2}.$$

Define the certificate quantities

$$b_{n,d,q} = 2 \left(\frac{4dB}{N_0 k^2} + 3e^{-16\ell} \right)$$

and

$$v_{n,d,q} = 8 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 64d \left(e^{\ell/8} + 1 \right) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) + 32e^{-32\ell} \left(1 + \frac{1}{m} \right).$$

The deterministic risk envelope $\mathfrak{U}_{n,d,q}$ is defined as follows. For $N \leq 1$, set

$$\mathfrak{U}_{n,d,q} = 1.$$

For $N > 1$ on the elementary branch, set

$$\mathfrak{U}_{n,d,q} = \min \left\{ 4, \left(\frac{8d}{eN_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\}.$$

For $N > 1$ on the polynomial branch, set

$$\mathfrak{U}_{n,d,q} = \min \{ 4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \}.$$

Then, with $\tau(P)$, the average treatment effect, defined in [Definition 8](#), and $r_{n,d,q}$, the frontier rate, defined in [Definition 17](#),

$$E_P \left[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

The deterministic estimator also satisfies

$$E_P \left[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2 \right] \leq E_{P,\omega} \left[(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P))^2 \right],$$

where the auxiliary expectation uses the independent Poisson prefix and uniform stream assignments, including the zero output when the prefix exceeds n , specified in [Algorithm 1](#).

Averaging over the prefix and stream assignments removes auxiliary variation under squared loss. The first inequality in [Theorem 11](#) is the certificate of [Theorem 10](#), and the final comparison is the conditional-averaging relation of [Lemma 1](#) applied to the specified auxiliary output. These references preserve the same construction and envelope across the two statements. For a prospective choice of n, d, q , the envelope therefore supplies a deterministic causal risk benchmark that incorporates cell complexity, outcome arrival, and the fixed-sample implementation.

C Lower bounds with full observed memberships

The lower comparisons concern the observed experiment in [Definition 15](#), which retains baseline category, assignment, surrogate, arrival status, and arrived outcome for every sampled individual. Two constructions account for the two components of the rate in [Definition 17](#): a one-category family isolates the effective sample-size contribution, while an activated many-category experiment supplies the category-complexity comparison. We first introduce the activated moment-matching construction, then state its approximation inputs and the one-category bounds. A separate comparison addresses the ascertainment deficit in the unrestricted large-alphabet regime.

C.1 The activated comparison experiment

The construction in [Algorithm 2](#) augments each observed record with an activation flag whose distribution is common to the two hypotheses. Conditional on a latent vector drawn once for the sample, it generates exactly n independent records with balanced treatment assignment. The following algorithm specifies both augmented laws and the conditions on their finite priors.

Algorithm 2 (Activated fuzzy-hypothesis experiment)

The activated handle $\mathfrak{H}_{\text{act}}$ is the ordered pair of augmented n -record laws constructed below, with the following inputs:

- **(Parameters.)** A constant $\eta > 0$, fixed before integers $n, d \geq 1$ and an arrival floor $0 < q \leq 1$, satisfying [Assumption 8](#). Define the effective sample size, logarithmic scale, moment-matching degree, support endpoint, nominal rare-cell mass, and rare-cell count by

$$N = nq, \quad \ell = \log(e+N), \quad K = \lceil \ell \rceil, \quad H = K^2, \quad b_0 = \frac{\eta}{N\ell}, \quad J = \min \left\{ d-1, \left\lfloor \frac{1}{2b_0} \right\rfloor \right\}.$$

- **(Finite priors.)** An ordered pair (Π_0, Π_1) of probability laws, each assigning probability one to a finite subset of $[1, H]$.
- **(Moment matching and separation.)** The priors satisfy

$$\int z^v d\Pi_0(z) = \int z^v d\Pi_1(z) \quad \text{for every integer } 0 \leq v \leq K,$$

and

$$\left| \int z^{-1} d\Pi_1(z) - \int z^{-1} d\Pi_0(z) \right| \geq \frac{1}{12}.$$

For each $i \in \{0, 1\}$, independently draw a latent coordinate $Z_x \sim \Pi_i$ for every baseline index $x \in \{0, \dots, d-1\}$. Conditional on this single latent vector, generate n independent records using the following common rule. Draw the baseline index with probabilities

$$P(X = x) = \begin{cases} b_0, & 0 \leq x < J, \\ 1 - Jb_0, & x = J, \\ 0, & J < x < d. \end{cases}$$

The first J indices are the rare cells, and index J is the reservoir. Draw treatment independently with

$$A \sim \text{Bernoulli}(1/2), \quad S = 0.$$

Independently of the latent vector, baseline index, and treatment, draw and reveal the activation flag

$$\mathbf{B}_{\text{act}} \sim \text{Bernoulli}(qH).$$

For a rare cell x , the same coordinate Z_x governs both treatment arms and all records in that cell.

If $\mathbf{B}_{\text{act}} = 0$, set $(R, RY) = (0, 0)$. Conditional on $\mathbf{B}_{\text{act}} = 1$, draw (R, RY) according to

	$(R, RY) = (1, 1)$	$(R, RY) = (1, 0)$	$(R, RY) = (0, 0)$
$A = 1, X = x < J$	$1/H$	$(Z_x - 1)/H$	$1 - Z_x/H$
$A = 0, X = x < J$	0	Z_x/H	$1 - Z_x/H$
$X \geq J$, either arm	0	1	0

Each augmented record is

$$(\mathbf{B}_{\text{act}}, (X, A, S, R, RY)).$$

The i -th component of $\mathfrak{H}_{\text{act}}$ is the law of these n augmented records after integrating over the product prior $\Pi_i^{\otimes d}$. Thus the latent vector is drawn once for the entire sample, and the records

are independent conditional on that vector. The handle is defined for every prior pair satisfying the stated conditions.

The augmented pair $\mathfrak{H}_{\text{act}}$ in [Algorithm 2](#) preserves all original observations. Its revealed activation flag $\mathbf{B}_{\text{act}}$ has a distribution depending on the experiment's parameters and independent of the latent coordinates and prior index. Projection onto the original record coordinates is therefore a common, parameter-independent observation map. The total-variation comparison in [Lemma 2](#) supplies the decision-theoretic connection between an observed-law comparison and any common randomized procedure.

The tuning constant η , moment-matching degree K , support endpoint H , nominal rare-cell mass b_0 , and rare-cell count J are specified in [Algorithm 2](#). The mass cap reserves at least half of the baseline probability for the reservoir. Each rare cell has a single latent coordinate Z_x , shared by its two treatment arms and all its sampled records. In the table, this coordinate governs arrival, while its reciprocal governs the treated arrived-outcome mean; the control outcome is zero. This arrangement places reciprocal separation and polynomial moment matching in the same experiment.

C.2 Reciprocal approximation and finite priors

The approximation inputs concern the interval and degree used by [Algorithm 2](#). Best uniform approximation quantifies the discrepancy between the reciprocal function and polynomials of the matching degree. The first input gives that discrepancy exactly.

Lemma 6. [`lem:reciprocal-best-approximation`] (Reciprocal best uniform approximation)
For every integer $K \geq 2$, set $H = K^2$. Then

$$\inf_{\substack{p \text{ real polynomial} \\ \deg(p) \leq K}} \sup_{z \in [1, H]} |z^{-1} - p(z)| = \frac{H-1}{2H} \left(\frac{K-1}{K+1} \right)^K.$$

Proof of [Lemma 6](#). 1. Fix an integer $K \geq 2$ and set

$$H := K^2 > 1.$$

We first establish the approximation formula on a general positive interval. Let

$$0 < a_{\text{app}} < b_{\text{app}}, \quad m_{\text{app}} \in \mathbb{N}_0,$$

and define

$$\begin{aligned} \Delta_{\text{app}} &:= b_{\text{app}} - a_{\text{app}}, & v_{\text{app}} &:= \frac{a_{\text{app}} + b_{\text{app}}}{\Delta_{\text{app}}}, & s_{\text{app}} &:= \sqrt{v_{\text{app}}^2 - 1}, \\ \rho_{\text{app}} &:= v_{\text{app}} - s_{\text{app}}, & E_{\text{app}} &:= \frac{2\rho_{\text{app}}^{m_{\text{app}}}}{\Delta_{\text{app}}(v_{\text{app}}^2 - 1)}. \end{aligned}$$

Since $\Delta_{\text{app}} > 0$ and $a_{\text{app}} + b_{\text{app}} > \Delta_{\text{app}}$, we have $v_{\text{app}} > 1$. Consequently,

$$0 \leq s_{\text{app}} < v_{\text{app}}, \quad s_{\text{app}}^2 = v_{\text{app}}^2 - 1, \quad (v_{\text{app}} - s_{\text{app}})(v_{\text{app}} + s_{\text{app}}) = 1.$$

Thus $\rho_{\text{app}} > 0$, and $v_{\text{app}} + s_{\text{app}} > 1$ gives

$$0 < \rho_{\text{app}} < 1, \quad \rho_{\text{app}}^{-1} = v_{\text{app}} + s_{\text{app}}, \quad \rho_{\text{app}} + \rho_{\text{app}}^{-1} = 2v_{\text{app}}.$$

In particular,

$$\rho_{\text{app}}^2 - 2v_{\text{app}}\rho_{\text{app}} + 1 = 0, \quad E_{\text{app}} > 0.$$

2. Define the Chebyshev polynomials, with integer indices, by

$$T_0(t_{\text{app}}) = 1, \quad T_1(t_{\text{app}}) = t_{\text{app}}, \quad T_{\nu+2}(t_{\text{app}}) = 2t_{\text{app}}T_{\nu+1}(t_{\text{app}}) - T_{\nu}(t_{\text{app}}) \quad (\nu \in \mathbb{N}_0),$$

$$T_{-\nu} = T_{\nu} \quad (\nu \in \mathbb{N}_0).$$

Here $t_{\text{app}} \in \mathbb{R}$. These polynomials satisfy

$$\deg T_{\nu} = |\nu|, \quad T_{\nu}(\cos \theta_{\text{app}}) = \cos(\nu \theta_{\text{app}}) \quad (\nu \in \mathbb{Z}, \theta_{\text{app}} \in \mathbb{R}).$$

For every $\nu \in \mathbb{N}_0$, define the residual polynomial

$$\mathcal{R}_{\text{app},\nu}(t_{\text{app}}) := \rho_{\text{app}}^{-1}T_{\nu+1}(t_{\text{app}}) - 2T_{\nu}(t_{\text{app}}) + \rho_{\text{app}}T_{\nu-1}(t_{\text{app}}).$$

The degree bounds for its three terms give

$$\deg \mathcal{R}_{\text{app},\nu} \leq \nu + 1.$$

The Chebyshev recurrence also gives

$$\mathcal{R}_{\text{app},\nu+2}(t_{\text{app}}) = 2t_{\text{app}}\mathcal{R}_{\text{app},\nu+1}(t_{\text{app}}) - \mathcal{R}_{\text{app},\nu}(t_{\text{app}}).$$

We claim that

$$\mathcal{R}_{\text{app},\nu}(v_{\text{app}}) = \frac{2(v_{\text{app}}^2 - 1)}{\rho_{\text{app}}^{\nu}} \quad (\nu \in \mathbb{N}_0).$$

For the two initial indices, the identities from the preceding step yield

$$\mathcal{R}_{\text{app},0}(v_{\text{app}}) = (\rho_{\text{app}}^{-1} + \rho_{\text{app}})v_{\text{app}} - 2 = 2(v_{\text{app}}^2 - 1),$$

$$\mathcal{R}_{\text{app},1}(v_{\text{app}}) = \rho_{\text{app}}^{-1}(2v_{\text{app}}^2 - 1) - 2v_{\text{app}} + \rho_{\text{app}} = \frac{2(v_{\text{app}}^2 - 1)}{\rho_{\text{app}}}.$$

If the claim holds at ν and $\nu + 1$, the recurrence gives

$$\begin{aligned} \mathcal{R}_{\text{app},\nu+2}(v_{\text{app}}) &= 2(v_{\text{app}}^2 - 1) \left(\frac{2v_{\text{app}}}{\rho_{\text{app}}^{\nu+1}} - \frac{1}{\rho_{\text{app}}^{\nu}} \right) \\ &= \frac{2(v_{\text{app}}^2 - 1)}{\rho_{\text{app}}^{\nu+2}}, \end{aligned}$$

because $2v_{\text{app}}\rho_{\text{app}} - \rho_{\text{app}}^2 = 1$. This proves the claim by induction, including the degree-zero case.

3. Define the affine map

$$t_{\text{app}}(z) := v_{\text{app}} - \frac{2z}{\Delta_{\text{app}}} \quad (z \in \mathbb{R}).$$

Since $\mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}}) > 0$, the polynomial

$$1 - \frac{\mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z))}{\mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}})}$$

is well defined and vanishes at $z = 0$. It therefore determines a unique real polynomial p_{app} through

$$z p_{\text{app}}(z) = 1 - \frac{\mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z))}{\mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}})} \quad (z \in \mathbb{R}).$$

The numerator has degree at most $m_{\text{app}} + 1$, so division by z gives

$$\deg p_{\text{app}} \leq m_{\text{app}}.$$

For every $z \neq 0$, rearranging the defining identity yields

$$z^{-1} - p_{\text{app}}(z) = \frac{\mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z))}{z \mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}})}.$$

4. For $\theta_{\text{app}} \in \mathbb{R}$, define

$$\alpha_{\text{app},+}(\theta_{\text{app}}) := \frac{(m_{\text{app}} + 1)\theta_{\text{app}}}{2}, \quad \alpha_{\text{app},-}(\theta_{\text{app}}) := \frac{(m_{\text{app}} - 1)\theta_{\text{app}}}{2},$$

$$F_{\text{app}}^{\sin}(\theta_{\text{app}}) := \sin \alpha_{\text{app},+}(\theta_{\text{app}}) - \rho_{\text{app}} \sin \alpha_{\text{app},-}(\theta_{\text{app}}),$$

$$F_{\text{app}}^{\cos}(\theta_{\text{app}}) := \cos \alpha_{\text{app},+}(\theta_{\text{app}}) - \rho_{\text{app}} \cos \alpha_{\text{app},-}(\theta_{\text{app}}).$$

The Chebyshev cosine identity gives

$$\begin{aligned} \mathcal{R}_{\text{app}, m_{\text{app}}}(\cos \theta_{\text{app}}) &= \rho_{\text{app}}^{-1} \cos((m_{\text{app}} + 1)\theta_{\text{app}}) - 2 \cos(m_{\text{app}}\theta_{\text{app}}) \\ &\quad + \rho_{\text{app}} \cos((m_{\text{app}} - 1)\theta_{\text{app}}). \end{aligned}$$

To factor this expression, use

$$\cos((m_{\text{app}} \pm 1)\theta_{\text{app}}) = 1 - 2 \sin^2 \alpha_{\text{app},\pm}(\theta_{\text{app}}) = 2 \cos^2 \alpha_{\text{app},\pm}(\theta_{\text{app}}) - 1,$$

$$\cos(m_{\text{app}}\theta_{\text{app}}) - \cos \theta_{\text{app}} = -2 \sin \alpha_{\text{app},+}(\theta_{\text{app}}) \sin \alpha_{\text{app},-}(\theta_{\text{app}}),$$

$$\cos(m_{\text{app}}\theta_{\text{app}}) + \cos \theta_{\text{app}} = 2 \cos \alpha_{\text{app},+}(\theta_{\text{app}}) \cos \alpha_{\text{app},-}(\theta_{\text{app}}).$$

Substitution, together with $\rho_{\text{app}} + \rho_{\text{app}}^{-1} = 2v_{\text{app}}$, produces the two identities

$$\mathcal{R}_{\text{app}, m_{\text{app}}}(\cos \theta_{\text{app}}) = 2(v_{\text{app}} - \cos \theta_{\text{app}}) - \frac{2}{\rho_{\text{app}}} (F_{\text{app}}^{\sin}(\theta_{\text{app}}))^2,$$

$$\mathcal{R}_{\text{app}, m_{\text{app}}}(\cos \theta_{\text{app}}) = -2(v_{\text{app}} - \cos \theta_{\text{app}}) + \frac{2}{\rho_{\text{app}}} (F_{\text{app}}^{\cos}(\theta_{\text{app}}))^2.$$

For $z \in [a_{\text{app}}, b_{\text{app}}]$,

$$t_{\text{app}}(z) = \frac{a_{\text{app}} + b_{\text{app}} - 2z}{\Delta_{\text{app}}} \in [-1, 1].$$

Choose $\theta_{\text{app}} \in [0, \pi]$ with $\cos \theta_{\text{app}} = t_{\text{app}}(z)$. The nonnegative squares in the two identities imply

$$-2(v_{\text{app}} - t_{\text{app}}(z)) \leq \mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z)) \leq 2(v_{\text{app}} - t_{\text{app}}(z)).$$

Hence

$$|\mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z))| \leq 2(v_{\text{app}} - t_{\text{app}}(z)) = \frac{4z}{\Delta_{\text{app}}}.$$

5. Every $z \in [a_{\text{app}}, b_{\text{app}}]$ is positive. The residual identity, the preceding bound, and the value of the residual polynomial at v_{app} therefore give

$$\begin{aligned} |z^{-1} - p_{\text{app}}(z)| &= \frac{|\mathcal{R}_{\text{app}, m_{\text{app}}}(t_{\text{app}}(z))|}{z \mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}})} \\ &\leq \frac{4z/\Delta_{\text{app}}}{z \mathcal{R}_{\text{app}, m_{\text{app}}}(v_{\text{app}})} = \frac{2\rho_{\text{app}}^{m_{\text{app}}}}{\Delta_{\text{app}}(v_{\text{app}}^2 - 1)} = E_{\text{app}}. \end{aligned}$$

Taking the supremum yields

$$\sup_{z \in [a_{\text{app}}, b_{\text{app}}]} |z^{-1} - p_{\text{app}}(z)| \leq E_{\text{app}}.$$

6. We next construct alternating equality points. Define the grid

$$\gamma_{\text{app}, \iota} := \frac{\iota\pi}{m_{\text{app}} + 1} \quad (\iota \in \{0, \dots, m_{\text{app}} + 1\}).$$

It is strictly increasing from 0 to π . At its points,

$$\alpha_{\text{app}, +}(\gamma_{\text{app}, \iota}) = \frac{\iota\pi}{2}, \quad \alpha_{\text{app}, -}(\gamma_{\text{app}, \iota}) = \frac{\iota\pi}{2} - \gamma_{\text{app}, \iota}.$$

The sine and cosine subtraction formulas consequently give, for $\eta_{\text{app}} \in \mathbb{N}_0$ whenever the displayed grid indices are in range,

$$\begin{aligned} F_{\text{app}}^{\cos}(\gamma_{\text{app}, 2\eta_{\text{app}}}) &= (-1)^{\eta_{\text{app}}} (1 - \rho_{\text{app}} \cos \gamma_{\text{app}, 2\eta_{\text{app}}}), \\ F_{\text{app}}^{\sin}(\gamma_{\text{app}, 2\eta_{\text{app}}}) &= \rho_{\text{app}} (-1)^{\eta_{\text{app}}} \sin \gamma_{\text{app}, 2\eta_{\text{app}}}, \\ F_{\text{app}}^{\cos}(\gamma_{\text{app}, 2\eta_{\text{app}}+1}) &= -\rho_{\text{app}} (-1)^{\eta_{\text{app}}} \sin \gamma_{\text{app}, 2\eta_{\text{app}}+1}, \\ F_{\text{app}}^{\sin}(\gamma_{\text{app}, 2\eta_{\text{app}}+1}) &= (-1)^{\eta_{\text{app}}} (1 - \rho_{\text{app}} \cos \gamma_{\text{app}, 2\eta_{\text{app}}+1}). \end{aligned}$$

The magnitudes controlling the left and right endpoints satisfy

$$1 - \rho_{\text{app}} \cos \theta_{\text{app}} \geq 1 - \rho_{\text{app}} > 0 \quad (\theta_{\text{app}} \in \mathbb{R}), \quad \rho_{\text{app}} \sin \theta_{\text{app}} \geq 0 \quad (\theta_{\text{app}} \in [0, \pi]).$$

Thus, on a cell with odd right index, the cosine factor has a nonzero left value and a right value of the opposite weak sign. On a cell with positive even right index, the sine factor has this property, since

$$F_{\text{app}}^{\sin}(\gamma_{\text{app}, 2(\eta_{\text{app}}+1)}) = -(-1)^{\eta_{\text{app}}} \rho_{\text{app}} \sin \gamma_{\text{app}, 2(\eta_{\text{app}}+1)}.$$

Continuity and the intermediate value theorem give choices satisfying

$$\theta_{\text{app}, 0} := 0, \quad \gamma_{\text{app}, \iota-1} < \theta_{\text{app}, \iota} \leq \gamma_{\text{app}, \iota} \quad (1 \leq \iota \leq m_{\text{app}} + 1),$$

$$\begin{cases} F_{\text{app}}^{\sin}(\theta_{\text{app},\iota}) = 0, & \iota \text{ even}, \\ F_{\text{app}}^{\cos}(\theta_{\text{app},\iota}) = 0, & \iota \text{ odd}. \end{cases}$$

The left endpoints are excluded because their factor values are nonzero. Also $F_{\text{app}}^{\sin}(0) = 0$. The cell bounds therefore imply

$$0 = \theta_{\text{app},0} < \theta_{\text{app},1} < \cdots < \theta_{\text{app},m_{\text{app}}+1} \leq \pi.$$

This construction applies also when $m_{\text{app}} = 0$.

7. Define the nodes

$$z_{\text{app},\iota} := \frac{a_{\text{app}} + b_{\text{app}} - \Delta_{\text{app}} \cos \theta_{\text{app},\iota}}{2} \quad (0 \leq \iota \leq m_{\text{app}} + 1).$$

Because cosine is strictly decreasing on $[0, \pi]$,

$$a_{\text{app}} \leq z_{\text{app},0} < z_{\text{app},1} < \cdots < z_{\text{app},m_{\text{app}}+1} \leq b_{\text{app}}, \quad t_{\text{app}}(z_{\text{app},\iota}) = \cos \theta_{\text{app},\iota}.$$

For even ι , the sine-square identity has zero square; for odd ι , the cosine-square identity has zero square. Hence

$$\mathcal{R}_{\text{app},m_{\text{app}}}(t_{\text{app}}(z_{\text{app},\iota})) = (-1)^\iota 2(v_{\text{app}} - t_{\text{app}}(z_{\text{app},\iota})) = (-1)^\iota \frac{4z_{\text{app},\iota}}{\Delta_{\text{app}}}.$$

Substituting into the residual identity gives

$$z_{\text{app},\iota}^{-1} - p_{\text{app}}(z_{\text{app},\iota}) = (-1)^\iota \frac{4}{\Delta_{\text{app}} \mathcal{R}_{\text{app},m_{\text{app}}}(v_{\text{app}})} = (-1)^\iota E_{\text{app}}.$$

8. To certify optimality, define the signed Lagrange weights

$$\beta_{\text{app},\iota} := \prod_{\substack{0 \leq j \leq m_{\text{app}}+1 \\ j \neq \iota}} (z_{\text{app},\iota} - z_{\text{app},j})^{-1}, \quad \Lambda_{\text{app}} := \sum_{j=0}^{m_{\text{app}}+1} |\beta_{\text{app},j}|, \quad w_{\text{app},\iota} := \frac{\beta_{\text{app},\iota}}{\Lambda_{\text{app}}} \quad (0 \leq \iota \leq m_{\text{app}}+1).$$

Distinctness of the nodes makes every $\beta_{\text{app},\iota}$ nonzero, so $\Lambda_{\text{app}} > 0$. There are exactly $m_{\text{app}} + 1 - \iota$ negative factors in the product defining $\beta_{\text{app},\iota}$. Therefore

$$\sum_{\iota=0}^{m_{\text{app}}+1} |w_{\text{app},\iota}| = 1, \quad w_{\text{app},\iota} = (-1)^{m_{\text{app}}+1-\iota} |w_{\text{app},\iota}|.$$

For any real polynomial q_{app} of degree at most m_{app} , Lagrange interpolation gives the polynomial identity

$$q_{\text{app}}(z) = \sum_{\iota=0}^{m_{\text{app}}+1} \beta_{\text{app},\iota} q_{\text{app}}(z_{\text{app},\iota}) \prod_{\substack{0 \leq j \leq m_{\text{app}}+1 \\ j \neq \iota}} (z - z_{\text{app},j}).$$

Comparing coefficients of $z^{m_{\text{app}}+1}$, and then dividing by Λ_{app} , yields

$$\sum_{\iota=0}^{m_{\text{app}}+1} \beta_{\text{app},\iota} q_{\text{app}}(z_{\text{app},\iota}) = 0, \quad \sum_{\iota=0}^{m_{\text{app}}+1} w_{\text{app},\iota} q_{\text{app}}(z_{\text{app},\iota}) = 0.$$

This cancellation applies to p_{app} as well. Combining the weight signs with the alternating residuals gives

$$\begin{aligned} \sum_{\iota=0}^{m_{\text{app}}+1} w_{\text{app},\iota} (z_{\text{app},\iota}^{-1} - p_{\text{app}}(z_{\text{app},\iota})) &= (-1)^{m_{\text{app}}+1} E_{\text{app}} \sum_{\iota=0}^{m_{\text{app}}+1} |w_{\text{app},\iota}| \\ &= (-1)^{m_{\text{app}}+1} E_{\text{app}}. \end{aligned}$$

The cancellation identities consequently imply, for every such q_{app} ,

$$\sum_{\iota=0}^{m_{\text{app}}+1} w_{\text{app},\iota} (z_{\text{app},\iota}^{-1} - q_{\text{app}}(z_{\text{app},\iota})) = (-1)^{m_{\text{app}}+1} E_{\text{app}}.$$

Taking absolute values and using the unit absolute mass,

$$\begin{aligned} E_{\text{app}} &\leq \sum_{\iota=0}^{m_{\text{app}}+1} |w_{\text{app},\iota}| |z_{\text{app},\iota}^{-1} - q_{\text{app}}(z_{\text{app},\iota})| \\ &\leq \sup_{z \in [a_{\text{app}}, b_{\text{app}}]} |z^{-1} - q_{\text{app}}(z)|. \end{aligned}$$

The residuals are continuous on this compact positive interval, so these suprema are finite. The lower bound for every admissible polynomial and the upper bound attained by p_{app} establish

$$\inf_{\substack{q_{\text{app}} \text{ real polynomial} \\ \deg(q_{\text{app}}) \leq m_{\text{app}}}} \sup_{z \in [a_{\text{app}}, b_{\text{app}}]} |z^{-1} - q_{\text{app}}(z)| = \frac{2\rho_{\text{app}}^{m_{\text{app}}}}{\Delta_{\text{app}}(v_{\text{app}}^2 - 1)}.$$

9. Specialize to

$$a_{\text{app}} = 1, \quad b_{\text{app}} = H = K^2, \quad m_{\text{app}} = K.$$

These parameters meet the interval and degree conditions because $K \geq 2$. Now

$$v_{\text{app}} = \frac{1 + K^2}{K^2 - 1}, \quad (1 + K^2)^2 - (K^2 - 1)^2 = 4K^2, \quad v_{\text{app}}^2 - 1 = \frac{4K^2}{(K^2 - 1)^2}.$$

Both K and $K^2 - 1$ are positive, so

$$\sqrt{v_{\text{app}}^2 - 1} = \frac{2K}{K^2 - 1}, \quad \rho_{\text{app}} = \frac{1 + K^2 - 2K}{K^2 - 1} = \frac{K - 1}{K + 1}.$$

Finally,

$$\frac{2}{\Delta_{\text{app}}(v_{\text{app}}^2 - 1)} = \frac{K^2 - 1}{2K^2} = \frac{H - 1}{2H}.$$

Substitution into the established interval formula gives

$$\inf_{\substack{p \text{ real polynomial} \\ \deg(p) \leq K}} \sup_{z \in [1, H]} |z^{-1} - p(z)| = \frac{H - 1}{2H} \left(\frac{K - 1}{K + 1} \right)^K,$$

as required. □

Lemma 6 identifies the approximation scale underlying the prior construction. Its mathematical role is to distinguish the reciprocal functional from the polynomial moments used to compare the two experiments. The next input supplies finite probability laws with exactly matching moments and a fixed reciprocal-mean separation.

Lemma 7. `[lem:moment-matched-reciprocal-priors]` (Moment-matched reciprocal priors)
For every integer $K \geq 2$, set $H = K^2$. There exist finitely supported probability laws Π_0 and Π_1 on $[1, H]$ such that

$$\int z^v d\Pi_0(z) = \int z^v d\Pi_1(z) \quad \text{for every integer } v \in \{0, \dots, K\},$$

and

$$\left| \int z^{-1} d\Pi_1(z) - \int z^{-1} d\Pi_0(z) \right| \geq \frac{1}{12}.$$

Proof of Lemma 7. 1. Fix $K \geq 2$, and define

$$H = K^2, \quad \mathcal{I}_\star = [1, H], \quad f_\star : \mathcal{I}_\star \rightarrow \mathbb{R}, \quad f_\star(z) = z^{-1}.$$

Since $H > 1$ and every point of $\mathcal{I}_\star$ is positive, $f_\star$ is continuous. For a continuous function $\varphi : \mathcal{I}_\star \rightarrow \mathbb{R}$, write

$$\|\varphi\|_\infty = \max_{z \in \mathcal{I}_\star} |\varphi(z)|, \quad E_K = \inf_{\substack{\mathbf{p} \in \mathbb{R}[z] \\ \deg \mathbf{p} \leq K}} \|f_\star - \mathbf{p}\|_\infty.$$

In particular, $\|f_\star\|_\infty = 1$.

To obtain a best approximant, consider the finite-dimensional space

$$\mathcal{V}_K = \{\mathbf{p}|_{\mathcal{I}_\star} : \mathbf{p} \in \mathbb{R}[z], \deg \mathbf{p} \leq K\}, \quad \mathcal{B}_\star = \{\varphi \in \mathcal{V}_K : \|\varphi\|_\infty \leq 3\}.$$

The closed ball $\mathcal{B}_\star$ is compact. The continuous error function therefore attains its minimum there at the restriction of a polynomial $p_\star$ of degree at most K . Comparison with the zero polynomial gives

$$\|f_\star - p_\star\|_\infty \leq 1.$$

For every $\varphi \in \mathcal{V}_K \setminus \mathcal{B}_\star$, the reverse triangle inequality gives

$$\|f_\star - \varphi\|_\infty \geq \|\varphi\|_\infty - \|f_\star\|_\infty > 3 - 1 = 2 \geq \|f_\star - p_\star\|_\infty.$$

Thus this minimum is global:

$$\deg p_\star \leq K, \quad \|f_\star - p_\star\|_\infty = E_K \geq 0.$$

2. Define the residual

$$e_\star(z) = f_\star(z) - p_\star(z) \quad (z \in \mathcal{I}_\star).$$

We establish the existence of points and an orientation satisfying

$$1 \leq z_0 < z_1 < \dots < z_{K+1} \leq H, \quad \eta_\star \in \{-1, 1\}, \quad e_\star(z_\iota) = \eta_\star (-1)^\iota E_K \quad (0 \leq \iota \leq K+1).$$

If $E_K = 0$, the residual vanishes throughout the interval, and the choices

$$z_\iota = 1 + \frac{\iota(H-1)}{K+1} \quad (0 \leq \iota \leq K+1), \quad \eta_\star = 1$$

have these properties.

Suppose now that $E_K > 0$. Define the compact extremal sets and the residual sign by

$$\begin{aligned} \mathcal{Z}_\star &= \{z \in \mathcal{I}_\star : |e_\star(z)| = E_K\}, \\ \mathcal{Z}_\star^+ &= \{z \in \mathcal{Z}_\star : e_\star(z) = E_K\}, \\ \mathcal{Z}_\star^- &= \{z \in \mathcal{Z}_\star : e_\star(z) = -E_K\}, \\ \chi_\star(z) &= \begin{cases} 1, & e_\star(z) > 0, \\ 0, & e_\star(z) = 0, \\ -1, & e_\star(z) < 0 \end{cases} \quad (z \in \mathcal{I}_\star). \end{aligned}$$

The set $\mathcal{Z}_\star$ is nonempty because the maximum residual magnitude is attained.

Assume, for a contradiction, that the required $K+2$ alternating extrema do not exist. We first construct a polynomial p_{sign} such that

$$\deg p_{\text{sign}} \leq K, \quad e_\star(z)p_{\text{sign}}(z) > 0 \quad (z \in \mathcal{Z}_\star).$$

If $\mathcal{Z}_\star^+ = \emptyset$, take $p_{\text{sign}}(z) = -1$; if $\mathcal{Z}_\star^- = \emptyset$, take $p_{\text{sign}}(z) = 1$. These choices give

$$e_\star(z)p_{\text{sign}}(z) = E_K > 0 \quad (z \in \mathcal{Z}_\star).$$

When both sign sets are nonempty, their disjointness and compactness give a separation $\delta_{\text{sep}} > 0$ with

$$\delta_{\text{sep}} < |z - z'| \quad (z \in \mathcal{Z}_\star^+, z' \in \mathcal{Z}_\star^-).$$

Set

$$\varepsilon_{\text{sep}} = \frac{\delta_{\text{sep}}}{5}.$$

Choose finitely many distinct centers in $\mathcal{Z}_\star$, ordered as

$$c_0^{\text{ctr}} < \dots < c_{n_{\text{ctr}}-1}^{\text{ctr}}, \quad n_{\text{ctr}} \geq 1,$$

whose ε_{sep} -neighborhoods cover $\mathcal{Z}_\star$. Thus

$$\forall z \in \mathcal{Z}_\star \exists \iota \in \{0, \dots, n_{\text{ctr}} - 1\} : |z - c_\iota^{\text{ctr}}| < \varepsilon_{\text{sep}}.$$

Define the consecutive sign-change indices by

$$\mathcal{D}_{\text{ctr}} = \{\iota \in \{0, \dots, n_{\text{ctr}} - 2\} : \chi_\star(c_\iota^{\text{ctr}}) \neq \chi_\star(c_{\iota+1}^{\text{ctr}})\}.$$

Selecting one center from each successive sign block produces an alternating subsequence of length $|\mathcal{D}_{\text{ctr}}| + 1$. The assumed absence of $K+2$ alternating extrema therefore implies

$$|\mathcal{D}_{\text{ctr}}| \leq K.$$

Define

$$p_{\text{sign}}(z) = \chi_{\star}(c_{n_{\text{ctr}}-1}^{\text{ctr}}) \prod_{\iota \in \mathcal{D}_{\text{ctr}}} \left(z - \frac{c_{\iota}^{\text{ctr}} + c_{\iota+1}^{\text{ctr}}}{2} \right),$$

with an empty product equal to one. Its degree is $|\mathcal{D}_{\text{ctr}}| \leq K$. Each factor changes sign exactly between consecutive blocks, so its sign at every center equals the residual sign there:

$$\chi_{\star}(c_{\iota}^{\text{ctr}}) p_{\text{sign}}(c_{\iota}^{\text{ctr}}) > 0.$$

Oppositely signed centers are more than $5\varepsilon_{\text{sep}}$ apart. Each midpoint appearing as a root is therefore more than $2\varepsilon_{\text{sep}}$ from every center. Consequently,

$$\chi_{\star}(c_{\iota}^{\text{ctr}}) p_{\text{sign}}(z) > 0 \quad \text{whenever } |z - c_{\iota}^{\text{ctr}}| < \varepsilon_{\text{sep}}.$$

An extremal point and a center within ε_{sep} have the same residual sign, by the separation of $\mathcal{Z}_{\star}^{+}$ and $\mathcal{Z}_{\star}^{-}$. Hence

$$e_{\star}(z) p_{\text{sign}}(z) = E_K \chi_{\star}(z) p_{\text{sign}}(z) > 0 \quad (z \in \mathcal{Z}_{\star}).$$

We next turn this sign agreement into a strict improvement. Define

$$\mathcal{B}_{\text{bad}} = \{z \in \mathcal{I}_{\star} : e_{\star}(z) p_{\text{sign}}(z) \leq 0\}, \quad c_{\text{cut}} = \begin{cases} \frac{E_K + \max_{z \in \mathcal{B}_{\text{bad}}} |e_{\star}(z)|}{2}, & \mathcal{B}_{\text{bad}} \neq \emptyset, \\ E_K/2, & \mathcal{B}_{\text{bad}} = \emptyset. \end{cases}$$

When nonempty, $\mathcal{B}_{\text{bad}}$ is compact and contains no extremal point. Its maximum residual magnitude is therefore strictly below E_K . In either case,

$$c_{\text{cut}} < E_K, \quad |e_{\star}(z)| \geq c_{\text{cut}} \implies e_{\star}(z) p_{\text{sign}}(z) > 0.$$

Define

$$\begin{aligned} \mathcal{U}_{\text{high}} &= \{z \in \mathcal{I}_{\star} : |e_{\star}(z)| \geq c_{\text{cut}}\}, \\ \mu_{\text{sign}} &= \min_{z \in \mathcal{U}_{\text{high}}} e_{\star}(z) p_{\text{sign}}(z) > 0, \\ M_{\text{sign}} &= \|p_{\text{sign}}\|_{\infty} > 0, \\ t_{\star} &= \min \left\{ \frac{\mu_{\text{sign}}}{M_{\text{sign}}^2}, \frac{E_K - c_{\text{cut}}}{2M_{\text{sign}}} \right\} > 0. \end{aligned}$$

The minimum defining μ_{sign} exists because $\mathcal{U}_{\text{high}}$ is compact and contains $\mathcal{Z}_{\star}$. The positivity of M_{sign} follows from the strict sign agreement at an extremal point.

For $z \in \mathcal{U}_{\text{high}}$,

$$\begin{aligned} (e_{\star}(z) - t_{\star} p_{\text{sign}}(z))^2 &= e_{\star}(z)^2 - 2t_{\star} e_{\star}(z) p_{\text{sign}}(z) + t_{\star}^2 p_{\text{sign}}(z)^2 \\ &\leq E_K^2 - 2t_{\star} \mu_{\text{sign}} + t_{\star}^2 M_{\text{sign}}^2 \\ &\leq E_K^2 - t_{\star} \mu_{\text{sign}} < E_K^2. \end{aligned}$$

For $z \in \mathcal{I}_{\star} \setminus \mathcal{U}_{\text{high}}$,

$$\begin{aligned} |e_{\star}(z) - t_{\star} p_{\text{sign}}(z)| &\leq |e_{\star}(z)| + t_{\star} M_{\text{sign}} \\ &< c_{\text{cut}} + \frac{E_K - c_{\text{cut}}}{2} = \frac{E_K + c_{\text{cut}}}{2} < E_K. \end{aligned}$$

Continuity and compactness now give

$$\|f_\star - (p_\star + t_\star p_{\text{sign}})\|_\infty < E_K, \quad \deg(p_\star + t_\star p_{\text{sign}}) \leq K,$$

contradicting the definition of E_K . Thus the required alternating extrema exist.

3. Using these nodes, define the index set, nodal weights, normalization, and signed weights by

$$\begin{aligned} \mathcal{N}_K &= \{0, \dots, K+1\}, \\ b_\iota^{\text{lag}} &= \left(\prod_{\substack{\iota' \in \mathcal{N}_K \\ \iota' \neq \iota}} (z_\iota - z_{\iota'}) \right)^{-1}, \\ Z_{\text{lag}} &= \sum_{\iota \in \mathcal{N}_K} |b_\iota^{\text{lag}}|, \\ w_\iota &= \frac{b_\iota^{\text{lag}}}{Z_{\text{lag}}} \quad (\iota \in \mathcal{N}_K). \end{aligned}$$

All nodal weights are nonzero, so $Z_{\text{lag}} > 0$. There are exactly $K+1-\iota$ negative factors in the product defining b_ι^{lag} . It follows that

$$\sum_{\iota \in \mathcal{N}_K} |w_\iota| = 1, \quad w_\iota = (-1)^{K+1-\iota} |w_\iota|.$$

For any polynomial $\mathbf{p}$ of degree at most K , Lagrange interpolation gives

$$\mathbf{p}(z) = \sum_{\iota \in \mathcal{N}_K} \mathbf{p}(z_\iota) b_\iota^{\text{lag}} \prod_{\substack{\iota' \in \mathcal{N}_K \\ \iota' \neq \iota}} (z - z_{\iota'}).$$

Comparing coefficients of z^{K+1} , whose coefficient on the left is zero, and dividing by Z_{lag} , yields

$$\sum_{\iota \in \mathcal{N}_K} w_\iota \mathbf{p}(z_\iota) = 0.$$

In particular,

$$\sum_{\iota \in \mathcal{N}_K} w_\iota z_\iota^v = 0 \quad (0 \leq v \leq K), \quad \sum_{\iota \in \mathcal{N}_K} w_\iota p_\star(z_\iota) = 0.$$

Define the signed reciprocal evaluation by

$$d_\star = \sum_{\iota \in \mathcal{N}_K} w_\iota z_\iota^{-1}.$$

Polynomial cancellation, alternation, and normalization give

$$\begin{aligned} d_\star &= \sum_{\iota \in \mathcal{N}_K} w_\iota (f_\star(z_\iota) - p_\star(z_\iota)) \\ &= \sum_{\iota \in \mathcal{N}_K} (-1)^{K+1-\iota} |w_\iota| \eta_\star (-1)^\iota E_K \\ &= \eta_\star (-1)^{K+1} E_K \sum_{\iota \in \mathcal{N}_K} |w_\iota| = \eta_\star (-1)^{K+1} E_K. \end{aligned}$$

Since $E_K \geq 0$ and $|\eta_\star| = 1$, this proves

$$|d_\star| = E_K.$$

4. Define two nonnegative atomic measures by

$$\Pi^+ = 2 \sum_{\iota \in \mathcal{N}_K} \max(w_\iota, 0) \delta_{z_\iota}, \quad \Pi^- = 2 \sum_{\iota \in \mathcal{N}_K} \max(-w_\iota, 0) \delta_{z_\iota},$$

where the unit point mass is specified, for every Borel set $\mathcal{T} \subseteq \mathbb{R}$, by

$$\delta_z(\mathcal{T}) = \begin{cases} 1, & z \in \mathcal{T}, \\ 0, & z \notin \mathcal{T}. \end{cases}$$

The moment identity at $v = 0$ gives $\sum_\iota w_\iota = 0$. Applying

$$|w_\iota| = 2 \max(w_\iota, 0) - w_\iota = 2 \max(-w_\iota, 0) + w_\iota$$

and summing, we obtain

$$1 = 2 \sum_{\iota \in \mathcal{N}_K} \max(w_\iota, 0) = 2 \sum_{\iota \in \mathcal{N}_K} \max(-w_\iota, 0).$$

Thus Π^+ and Π^- are probability measures.

Orient the pair according to the reciprocal evaluation:

$$(\Pi_0, \Pi_1) = \begin{cases} (\Pi^-, \Pi^+), & d_\star \geq 0, \\ (\Pi^+, \Pi^-), & d_\star < 0. \end{cases}$$

Both oriented measures are therefore probability measures. Define their common finite supporting set by

$$\mathcal{S}_\star = \{z_\iota : \iota \in \mathcal{N}_K\}.$$

Every atom belongs to $\mathcal{S}_\star \subseteq [1, H]$, and hence

$$\Pi^+(\mathcal{S}_\star) = \Pi^-(\mathcal{S}_\star) = 1, \quad \Pi_0(\mathcal{S}_\star) = \Pi_1(\mathcal{S}_\star) = 1.$$

5. For any real-valued function $\varphi_{\text{test}} : \mathbb{R} \rightarrow \mathbb{R}$, the finite atomic formulas and $\max(w_\iota, 0) - \max(-w_\iota, 0) = w_\iota$ give

$$\begin{aligned} \int \varphi_{\text{test}} d\Pi^+ - \int \varphi_{\text{test}} d\Pi^- &= 2 \sum_{\iota \in \mathcal{N}_K} (\max(w_\iota, 0) - \max(-w_\iota, 0)) \varphi_{\text{test}}(z_\iota) \\ &= 2 \sum_{\iota \in \mathcal{N}_K} w_\iota \varphi_{\text{test}}(z_\iota). \end{aligned}$$

Taking $\varphi_{\text{test}}(z) = z^v$, for an integer $0 \leq v \leq K$, yields

$$\int z^v d\Pi^+(z) - \int z^v d\Pi^-(z) = 2 \sum_{\iota \in \mathcal{N}_K} w_\iota z_\iota^v = 0.$$

Either orientation therefore satisfies

$$\int z^v d\Pi_0(z) = \int z^v d\Pi_1(z) \quad (0 \leq v \leq K).$$

For the reciprocal function, the orientation gives

$$\int z^{-1} d\Pi_1(z) - \int z^{-1} d\Pi_0(z) = \begin{cases} 2d_\star, & d_\star \geq 0, \\ -2d_\star, & d_\star < 0 \end{cases} = 2|d_\star| = 2E_K.$$

6. It remains to bound $2E_K$. By [Lemma 6](#),

$$E_K = \frac{K^2 - 1}{2K^2} \left(\frac{K - 1}{K + 1} \right)^K.$$

If $K = 2$, this gives

$$E_2 = \frac{1}{24}, \quad 2E_2 = \frac{1}{12}.$$

Suppose $K \geq 3$. First,

$$\frac{K^2 - 1}{2K^2} - \frac{4}{9} = \frac{K^2 - 9}{18K^2} \geq 0.$$

To bound the power term, define

$$x_{\text{rec}} = \frac{1}{K} \in (0, 1/3].$$

The logarithmic remainder estimate at order zero gives

$$\frac{1}{2} \log \frac{1 + x_{\text{rec}}}{1 - x_{\text{rec}}} \leq \frac{x_{\text{rec}}}{1 - x_{\text{rec}}^2}.$$

Since

$$\frac{1 + x_{\text{rec}}}{1 - x_{\text{rec}}} = \left(\frac{K - 1}{K + 1} \right)^{-1}, \quad 8K^2 \leq 9(K^2 - 1),$$

we obtain

$$-K \log \frac{K - 1}{K + 1} \leq K \frac{2x_{\text{rec}}}{1 - x_{\text{rec}}^2} = \frac{2K^2}{K^2 - 1} \leq \frac{9}{4}.$$

Exponentiating, using the positivity of $(K - 1)/(K + 1)$, gives

$$e^{-9/4} \leq \left(\frac{K - 1}{K + 1} \right)^K.$$

For the numerical calibration, the standard numerical exponential bound and the rational exponential inequality give

$$e < 2.7182818286 < \frac{11}{4}, \quad e^{1/4} \leq \frac{9}{7}.$$

Here the latter estimate is the specialization at $1/4$ of

$$e^y \leq \frac{2 + y}{2 - y} \quad (0 \leq y < 2).$$

Consequently,

$$e^{9/4} = e^2 e^{1/4} \leq \left(\frac{11}{4}\right)^2 \frac{9}{7} = \frac{1089}{112} \leq \frac{32}{3}.$$

Taking reciprocals yields

$$\frac{3}{32} \leq e^{-9/4} \leq \left(\frac{K-1}{K+1}\right)^K.$$

Combining the coefficient and power bounds,

$$2E_K = 2 \frac{K^2 - 1}{2K^2} \left(\frac{K-1}{K+1}\right)^K \geq 2 \cdot \frac{4}{9} \cdot \frac{3}{32} = \frac{1}{12}.$$

Thus, in both cases,

$$\left| \int z^{-1} d\Pi_1(z) - \int z^{-1} d\Pi_0(z) \right| = |2E_K| \geq 2E_K \geq \frac{1}{12}.$$

Together with the finite support and moment identities established above, this proves the result. $\square$

The finite priors Π_0, Π_1 supplied by [Lemma 7](#) satisfy the prior requirements of [Algorithm 2](#). Matching includes the zeroth moment, and reciprocal separation remains bounded below by an absolute constant. Finite support makes the augmented laws finite mixtures of experiments generated conditional on a fixed latent vector.

C.3 A one-category family

For the effective sample-size component, the local family in [Lemma 8](#) places the entire baseline mass on category zero. Both potential surrogates and the control potential outcome are zero; the treated potential outcome is Bernoulli with mean $u \in [1/4, 3/4]$. Treatment is an independent fair coin, and arrival is an independent Bernoulli variable with probability q . Thus P_u , the full-data law indexed by the treated outcome mean and defined through [Lemma 8](#), has average treatment effect u . The following statement records both squared-error and honest-interval comparisons for this family.

Lemma 8. [\[lem:one-cell-testing-family\]](#) (One-cell testing family bounds) *There exist a universal constant $\underline{c} > 0$ and constants $\underline{c}_\alpha > 0$ for every $\alpha \in (0, 1)$, independent of n, d, q , such that the following holds whenever:*

- **(Sample and alphabet sizes.)** n, d are integers with $n \geq 1$ and $d \geq 1$.
- **(Arrival probability.)** $0 < q \leq 1$.
- **(Rare-arrival slice.)** The restriction in [Assumption 8](#) holds.

Write $N = nq$ for the effective sample size. There exists a family of full-data laws $\{P_u : u \in [1/4, 3/4]\}$, chosen independently of α , such that, for every $u \in [1/4, 3/4]$,

$$P_u \in \mathcal{M}_{n,d,q}, \quad P_u(X = 0, S(0) = S(1) = 0, Y(0) = 0) = 1, \quad P_u(R = 1) = q, \quad \tau(P_u) = u.$$

Here $X = 0$ denotes the first baseline category in the alphabet $\{0, \dots, d-1\}$; the model class $\mathcal{M}_{n,d,q}$ and average treatment effect $\tau(P)$ are defined in [Definitions 8](#) and [14](#).

There exist distinct $u_0, u_1 \in [1/4, 3/4]$ such that every possibly randomized estimator $T \in \mathcal{T}_n$, as defined in [Definition 16](#), satisfies

$$\underline{c} \min\{1, N^{-1}\} \leq \max \left\{ \mathbb{E}_{P_{u_0}}[(T - u_0)^2], \mathbb{E}_{P_{u_1}}[(T - u_1)^2] \right\}.$$

Moreover, the minimax squared-error risk in [Definition 16](#) satisfies

$$\mathfrak{R}_{n,d,q} \geq \underline{c} \min\{1, N^{-1}\}.$$

For every $\alpha \in (0, 1)$, there exist an integer $M \geq 1$ and pairwise distinct grid points $u_0, \dots, u_M \in [1/4, 3/4]$ from this same family such that every uniformly honest connected-interval procedure $l \in \mathcal{C}_{n,d,q,\alpha}$, as defined in [Definition 29](#), satisfies

$$\underline{c}_\alpha \min\{1, N^{-1/2}\} \leq \max_{i=0, \dots, M} \mathbb{E}_{P_{u_i}}[u - l],$$

where (l, u) are the random endpoints returned by l . Moreover, the minimax expected interval length in [Definition 29](#) satisfies

$$\mathfrak{L}_{n,d,q,\alpha} \geq \underline{c}_\alpha \min\{1, N^{-1/2}\}.$$

All expectations include the observed sample and the procedure's randomization.

Proof of [Lemma 8](#). 1. **Construction of the family.** Choose

$$\underline{c} := \frac{1}{256}, \quad \underline{c}_\alpha := \frac{3(1-\alpha)^2}{512} \quad (\alpha \in \mathbb{R}).$$

These constants have the required positivity. Fix admissible n, d, q , and write $N = nq > 0$. Define

$$\text{clip}_{[0,1]}(u) := \max\{0, \min\{1, u\}\} \quad (u \in \mathbb{R}).$$

For every $u \in \mathbb{R}$, define P_u by

$$X = 0, \quad S(0) = S(1) = Y(0) = 0,$$

$$A \sim \text{Bernoulli}(1/2), \quad Y(1) \sim \text{Bernoulli}(\text{clip}_{[0,1]}(u)), \quad R \sim \text{Bernoulli}(q),$$

with $A, Y(1), R$ mutually independent. The first baseline category exists because $d \geq 1$. All atom weights are nonnegative, and their sum is

$$\left(\frac{1}{2} + \frac{1}{2}\right) \left(\text{clip}_{[0,1]}(u) + 1 - \text{clip}_{[0,1]}(u)\right) (q + 1 - q) = 1.$$

Thus P_u is a full-data probability law for every real u .

For $u \in [1/4, 3/4]$, clipping leaves u unchanged. Set the realized variables to $S = S(A)$ and $Y = Y(A)$, and use independent observed records with their induced marginal law. This gives the sampling and consistency conditions. Independence and the fair treatment coin give

$$A \perp\!\!\!\perp \{X, S(0), S(1), Y(0), Y(1)\}, \quad P_u(A = 1) = \frac{1}{2}.$$

Moreover, for $a, y, r \in \{0, 1\}$,

$$P_u(R = r, Y = y \mid X = 0, A = a, S = 0) = P_u(R = r) P_u(Y = y \mid A = a),$$

because R is independent of $(A, Y(1))$ and $Y = AY(1)$. This verifies [Assumption 6](#) on both occupied cells; all joint probabilities on the other cells vanish. Direct summation also gives

$$p_{axs}(P_u) = \begin{cases} 1/2, & x = 0, s = 0, \\ 0, & \text{otherwise,} \end{cases} \quad P_u(A = a, X = x, S = s, R = 1) = q p_{axs}(P_u).$$

Consequently, $\rho_{a00}(P_u) = q$, so [Assumption 7](#) holds. The fixed parameters satisfy [Assumption 8](#); hence

$$P_u \in \mathcal{M}_{n,d,q}.$$

Finally, the construction yields

$$P_u(X = 0, S(0) = S(1) = 0, Y(0) = 0) = 1, \quad P_u(R = 1) = q, \quad \tau(P_u) = E_{P_u}[Y(1) - Y(0)] = u.$$

The family is chosen independently of α .

2. **Observed laws and chi-square divergence.** Define the observed-record space, its one-record laws, and its sample laws by

$$\mathcal{O}_d := \{0, \dots, d-1\} \times \{0, 1\}^4, \quad Q_u := \mathcal{L}_{P_u}(X, A, S, R, RY), \quad \mathbb{Q}_u := Q_u^{\otimes n}.$$

For $u \in [1/4, 3/4]$, the potentially nonzero atom probabilities are

$o = (X, A, S, R, RY)$	$Q_u(\{o\})$
$(0, 0, 0, 0, 0)$	$(1 - q)/2$
$(0, 1, 0, 0, 0)$	$(1 - q)/2$
$(0, 0, 0, 1, 0)$	$q/2$
$(0, 1, 0, 1, 0)$	$q(1 - u)/2$
$(0, 1, 0, 1, 1)$	$qu/2$

and all remaining atoms have probability zero. Each full-data atom of positive P_u -mass has positive $P_{1/2}$ -mass: the treatment weights are identical, both central outcome weights are positive, and a zero arrival weight vanishes under both laws. Passing to the observed marginal and then to products gives

$$Q_u \ll Q_{1/2}, \quad \mathbb{Q}_u \ll \mathbb{Q}_{1/2}.$$

For these finite laws, define

$$\chi^2(Q_u \| Q_{1/2}) := \sum_{\substack{o \in \mathcal{O}_d \\ Q_{1/2}(\{o\}) > 0}} \frac{(Q_u(\{o\}) - Q_{1/2}(\{o\}))^2}{Q_{1/2}(\{o\})}.$$

Expanding the square and using the unit total masses gives

$$1 + \chi^2(Q_u \| Q_{1/2}) = \sum_{\substack{o \in \mathcal{O}_d \\ Q_{1/2}(\{o\}) > 0}} \frac{Q_u(\{o\})^2}{Q_{1/2}(\{o\})}.$$

If $0 < q < 1$, summing the five displayed atoms yields

$$1 + \chi^2(Q_u \| Q_{1/2}) = (1 - q) + \frac{q}{2} + q(1 - u)^2 + qu^2 = 1 + 2q \left(u - \frac{1}{2}\right)^2.$$

If $q = 1$, the two nonarrival atoms have zero mass and are omitted; the remaining three terms give the same identity. Thus

$$\chi^2(Q_u \| Q_{1/2}) = 2q \left(u - \frac{1}{2}\right)^2.$$

To record the product calculation, define

$$\ell_u(o) := \begin{cases} Q_u(\{o\})/Q_{1/2}(\{o\}), & Q_{1/2}(\{o\}) > 0, \\ 0, & Q_{1/2}(\{o\}) = 0, \end{cases} \quad L_{u,n}(\mathbf{o}) := \prod_{i=1}^n \ell_u(o_i) \quad (\mathbf{o} \in \mathcal{O}_d^n).$$

These are likelihood ratios almost surely under the corresponding reference laws. Independence gives

$$E_{Q_{1/2}} L_{u,n} = 1, \quad E_{Q_{1/2}} L_{u,n}^2 = \left(E_{Q_{1/2}} \ell_u^2\right)^n.$$

Defining the sample chi-square divergence by

$$\chi^2(Q_u \| Q_{1/2}) := E_{Q_{1/2}} (L_{u,n} - 1)^2,$$

we obtain

$$1 + \chi^2(Q_u \| Q_{1/2}) = \left(1 + \chi^2(Q_u \| Q_{1/2})\right)^n \leq \exp(n \chi^2(Q_u \| Q_{1/2})).$$

The final inequality follows from $1 + x \leq e^x$ for $x \geq 0$. With the total-variation convention of [Lemma 2](#), the finite-space formula and Cauchy–Schwarz give

$$\text{TV}(Q_u, Q_{1/2}) = \frac{1}{2} E_{Q_{1/2}} |L_{u,n} - 1| \leq \frac{1}{2} \sqrt{\chi^2(Q_u \| Q_{1/2})}.$$

3. A two-point testing bound. Define

$$\Delta_{\text{pt}} := \min \left\{ \frac{1}{4}, \frac{1}{4\sqrt{N}} \right\}, \quad u_0^{\text{pt}} := \frac{1}{2}, \quad u_1^{\text{pt}} := \frac{1}{2} + \Delta_{\text{pt}}.$$

Since $N > 0$,

$$0 < \Delta_{\text{pt}} \leq \frac{1}{4}.$$

The two means are therefore distinct and belong to $[1/4, 3/4]$. Also,

$$\Delta_{\text{pt}}^2 \leq \frac{1}{16N}, \quad \chi^2(Q_{u_1^{\text{pt}}} \| Q_{1/2}) = 2q \Delta_{\text{pt}}^2 \leq \frac{1}{8n}.$$

The product calculation above yields

$$\chi^2(Q_{u_1^{\text{pt}}} \| Q_{1/2}) \leq e^{1/8} - 1 < 1.$$

For the numerical bound, use

$$(e^{1/2})^2 = e < 3 < 4, \quad e^{1/8} \leq e^{1/2} < 2.$$

Consequently,

$$\text{TV}(\mathbb{Q}_{u_1^{\text{pt}}}, \mathbb{Q}_{1/2}) \leq \frac{1}{2}.$$

The testing inequality in [Lemma 2](#), applied with the identity kernel, now gives, for every measurable $\mathcal{A} \subseteq \mathcal{O}_d^n$,

$$\mathbb{Q}_{u_1^{\text{pt}}}(\mathcal{A}^c) + \mathbb{Q}_{u_0^{\text{pt}}}(\mathcal{A}) \geq 1 - \text{TV}(\mathbb{Q}_{u_1^{\text{pt}}}, \mathbb{Q}_{u_0^{\text{pt}}}) \geq \frac{1}{2}.$$

4. **From testing to squared risk.** Fix any $\mathsf{T} \in \mathcal{T}_n$, and define its barycenter by

$$b_{\mathsf{T}}(\mathbf{o}) := \int_{[-1,1]} h \mathsf{T}(\mathbf{o}, dh) \quad (\mathbf{o} \in \mathcal{O}_d^n).$$

By [Lemma 1](#), this is a measurable function taking values in $[-1, 1]$, and, for $j \in \{0, 1\}$,

$$E_{\mathbb{Q}_{u_j^{\text{pt}}}} \left[(b_{\mathsf{T}} - u_j^{\text{pt}})^2 \right] \leq E_{P_{u_j^{\text{pt}}}} \left[(\mathsf{T} - u_j^{\text{pt}})^2 \right].$$

Define the sample events

$$\mathcal{E}_j := \left\{ \mathbf{o} : |b_{\mathsf{T}}(\mathbf{o}) - u_j^{\text{pt}}| \geq \frac{\Delta_{\text{pt}}}{2} \right\} \quad (j = 0, 1),$$

$$\mathcal{A}_{\text{pt}} := \left\{ \mathbf{o} : |b_{\mathsf{T}}(\mathbf{o}) - u_1^{\text{pt}}| \leq |b_{\mathsf{T}}(\mathbf{o}) - u_0^{\text{pt}}| \right\}.$$

The triangle inequality gives, at every sample,

$$\Delta_{\text{pt}} \leq |b_{\mathsf{T}} - u_0^{\text{pt}}| + |b_{\mathsf{T}} - u_1^{\text{pt}}|.$$

Thus

$$\mathcal{A}_{\text{pt}} \subseteq \mathcal{E}_0, \quad \mathcal{A}_{\text{pt}}^c \subseteq \mathcal{E}_1, \quad \mathbb{Q}_{u_0^{\text{pt}}}(\mathcal{E}_0) + \mathbb{Q}_{u_1^{\text{pt}}}(\mathcal{E}_1) \geq \frac{1}{2}.$$

Since squared error is at least $\Delta_{\text{pt}}^2/4$ on the corresponding error event,

$$\begin{aligned} \max_{j=0,1} E_{P_{u_j^{\text{pt}}}} \left[(\mathsf{T} - u_j^{\text{pt}})^2 \right] &\geq \frac{\Delta_{\text{pt}}^2}{4} \max_{j=0,1} \mathbb{Q}_{u_j^{\text{pt}}}(\mathcal{E}_j) \\ &\geq \frac{\Delta_{\text{pt}}^2}{16}. \end{aligned}$$

For the scale of this bound, first suppose

$$\frac{1}{4} \leq \frac{1}{4\sqrt{N}}.$$

Then $\Delta_{\text{pt}} = 1/4$, and

$$\Delta_{\text{pt}}^2 = \frac{1}{16} \geq \frac{1}{16} \min\{1, N^{-1}\}.$$

In the complementary case, $\Delta_{\text{pt}} = 1/(4\sqrt{N})$, so

$$\Delta_{\text{pt}}^2 = \frac{1}{16N} \geq \frac{1}{16} \min\{1, N^{-1}\}.$$

Both cases therefore give

$$\frac{1}{256} \min\{1, N^{-1}\} \leq \max_{j=0,1} E_{P_{u_j^{\text{pt}}}} \left[(\mathsf{T} - u_j^{\text{pt}})^2 \right].$$

Both testing laws belong to the model. The estimator class is nonempty, as witnessed by the constant-zero estimator. For every full-data law,

$$\tau(P) = P(Y(1) = 1) - P(Y(0) = 1) \in [-1, 1],$$

and hence every estimator in $\mathcal{T}_n$ has risk between zero and four:

$$0 \leq E_P[(\mathsf{T} - \tau(P))^2] \leq 4.$$

Thus the model supremum is a finite real number and dominates the displayed two-point maximum. Taking the infimum over estimators, as in [Definition 16](#), proves

$$\mathfrak{R}_{n,d,q} \geq \frac{1}{256} \min\{1, N^{-1}\}.$$

5. **Construction and scale of the interval grid.** Fix $\alpha \in (0, 1)$, and define

$$\beta_\alpha := 1 - \alpha, \quad \varepsilon_\alpha := \frac{\beta_\alpha}{8}.$$

Choose an integer

$$M \geq \frac{8}{\beta_\alpha}.$$

Then $M \geq 1$, and $0 < \varepsilon_\alpha \leq 1$. Define

$$w_{\text{grid}} := \min \left\{ \frac{1}{4}, \frac{\varepsilon_\alpha}{4\sqrt{N}} \right\}, \quad \Delta_{\text{grid}} := \frac{w_{\text{grid}}}{M}, \quad u_j^{\text{grid}} := \frac{1}{2} + j\Delta_{\text{grid}} \quad (j \in \{0, \dots, M\}).$$

These quantities satisfy

$$0 < w_{\text{grid}} \leq \frac{1}{4}, \quad \Delta_{\text{grid}} > 0, \quad 2Nw_{\text{grid}}^2 \leq 2N \left(\frac{\varepsilon_\alpha}{4\sqrt{N}} \right)^2 = \frac{\varepsilon_\alpha^2}{8}.$$

For every grid index,

$$0 \leq j\Delta_{\text{grid}} = \frac{j}{M}w_{\text{grid}} \leq w_{\text{grid}}, \quad \frac{1}{2} \leq u_j^{\text{grid}} \leq \frac{3}{4}.$$

Positive spacing makes the grid points pairwise distinct.

We also need

$$\frac{\varepsilon_\alpha}{4} \min\{1, N^{-1/2}\} \leq w_{\text{grid}}.$$

If $1/4 \leq \varepsilon_\alpha/(4\sqrt{N})$, then

$$w_{\text{grid}} = \frac{1}{4}, \quad \frac{\varepsilon_\alpha}{4} \min\{1, N^{-1/2}\} \leq \frac{\varepsilon_\alpha}{4} \leq \frac{1}{4}.$$

In the complementary case,

$$w_{\text{grid}} = \frac{\varepsilon_\alpha}{4\sqrt{N}}, \quad \frac{\varepsilon_\alpha}{4} \min\{1, N^{-1/2}\} \leq \frac{\varepsilon_\alpha}{4\sqrt{N}}.$$

This proves the required scale inequality in both cases.

6. **Coverage and comparison of interval-output laws.** Let $\mathbf{l} \in \mathcal{C}_{n,d,q,\alpha}$. Define its endpoint-output laws by

$$\mu_{\text{ref}} := \mathbb{Q}_{1/2} \mathbf{l}, \quad \mu_j := \mathbb{Q}_{u_j^{\text{grid}}} \mathbf{l} \quad (j = 0, \dots, M),$$

meaning explicitly that, for every measurable endpoint set $\mathcal{A}_{\text{end}}$,

$$\mu_{\text{ref}}(\mathcal{A}_{\text{end}}) = \int \mathbf{l}(\mathbf{o}, \mathcal{A}_{\text{end}}) \mathbb{Q}_{1/2}(d\mathbf{o}), \quad \mu_j(\mathcal{A}_{\text{end}}) = \int \mathbf{l}(\mathbf{o}, \mathcal{A}_{\text{end}}) \mathbb{Q}_{u_j^{\text{grid}}}(d\mathbf{o}).$$

Write the returned endpoint pair as

$$(l, u_{\text{end}}), \quad -1 \leq l \leq u_{\text{end}} \leq 1.$$

Define the coverage events

$$B_j := \{(l, u_{\text{end}}) : -1 \leq l \leq u_{\text{end}} \leq 1, l \leq u_j^{\text{grid}} \leq u_{\text{end}}\}.$$

All grid laws belong to the model and have the corresponding grid means as their average treatment effects. Honesty therefore gives

$$\mu_j(B_j) \geq \beta_\alpha \quad (j = 0, \dots, M).$$

For the comparison with the reference sample law, define

$$\xi_j := 2N(j\Delta_{\text{grid}})^2 \quad (j = 0, \dots, M).$$

The width budget implies

$$0 \leq \xi_j \leq 2Nw_{\text{grid}}^2 \leq \frac{\varepsilon_\alpha^2}{8} \leq \frac{1}{8} < 1.$$

The one-record identity and product calculation give

$$\chi^2(\mathbb{Q}_{u_j^{\text{grid}}} \| \mathbb{Q}_{1/2}) \leq e^{\xi_j} - 1.$$

For $0 \leq \xi_j < 1$, use the exponential bound $e^{\xi_j} \leq (1 - \xi_j)^{-1}$. Furthermore,

$$\xi_j(1 + \varepsilon_\alpha^2) \leq 2\xi_j \leq \frac{\varepsilon_\alpha^2}{4} \leq \varepsilon_\alpha^2.$$

Since $1 - \xi_j > 0$, this implies

$$e^{\xi_j} - 1 \leq \frac{1}{1 - \xi_j} - 1 = \frac{\xi_j}{1 - \xi_j} \leq \varepsilon_\alpha^2.$$

Consequently,

$$\text{TV}(\mathbb{Q}_{1/2}, \mathbb{Q}_{u_j^{\text{grid}}}) \leq \frac{1}{2} \sqrt{\chi^2(\mathbb{Q}_{u_j^{\text{grid}}} \| \mathbb{Q}_{1/2})} \leq \frac{\varepsilon_\alpha}{2} \leq \varepsilon_\alpha.$$

Applying [Lemma 2](#) to the common interval kernel yields

$$\text{TV}(\mu_{\text{ref}}, \mu_j) \leq \varepsilon_\alpha.$$

Thus, by the defining event bound for total variation,

$$\mu_{\text{ref}}(B_j) \geq \mu_j(B_j) - \varepsilon_\alpha \geq \beta_\alpha - \varepsilon_\alpha = \frac{7\beta_\alpha}{8}.$$

7. Counting covered grid points. For every ordered endpoint pair, define

$$C_{\text{grid}}(l, u_{\text{end}}) := \sum_{j=0}^M \mathbf{1}\{l \leq u_j^{\text{grid}} \leq u_{\text{end}}\} \quad (-1 \leq l \leq u_{\text{end}} \leq 1).$$

We claim the pointwise inequality

$$\Delta_{\text{grid}}(C_{\text{grid}}(l, u_{\text{end}}) - 1) \leq u_{\text{end}} - l.$$

If the covered index set is empty, its left-hand side equals $-\Delta_{\text{grid}} \leq 0 \leq u_{\text{end}} - l$. Otherwise define its extreme indices by

$$j_- := \min\{j \in \{0, \dots, M\} : l \leq u_j^{\text{grid}} \leq u_{\text{end}}\}, \quad j_+ := \max\{j \in \{0, \dots, M\} : l \leq u_j^{\text{grid}} \leq u_{\text{end}}\}.$$

Every covered index lies between j_- and j_+ , so

$$C_{\text{grid}}(l, u_{\text{end}}) \leq j_+ - j_- + 1.$$

The two extreme grid points lie in the interval, and hence

$$\Delta_{\text{grid}}(C_{\text{grid}}(l, u_{\text{end}}) - 1) \leq (j_+ - j_-) \Delta_{\text{grid}} = u_{j_+}^{\text{grid}} - u_{j_-}^{\text{grid}} \leq u_{\text{end}} - l.$$

This proves the claim, including the empty and singleton cases.

Summing coverage probabilities under the reference output law gives

$$\int C_{\text{grid}}(l, u_{\text{end}}) \mu_{\text{ref}}(d(l, u_{\text{end}})) = \sum_{j=0}^M \mu_{\text{ref}}(B_j) \geq (M+1)(\beta_\alpha - \varepsilon_\alpha).$$

Integrating the pointwise length bound therefore yields

$$\int (u_{\text{end}} - l) \mu_{\text{ref}}(d(l, u_{\text{end}})) \geq \Delta_{\text{grid}} \left((M+1) \frac{7\beta_\alpha}{8} - 1 \right).$$

The choice of M gives $M\beta_\alpha \geq 8$, and

$$(M+1)\frac{7\beta_\alpha}{8} - 1 - \frac{3\beta_\alpha M}{4} = \frac{M\beta_\alpha}{8} + \frac{7\beta_\alpha}{8} - 1 \geq 0.$$

Since $M\Delta_{\text{grid}} = w_{\text{grid}}$, we conclude

$$\int (u_{\text{end}} - l) \mu_{\text{ref}}(d(l, u_{\text{end}})) \geq \frac{3\beta_\alpha}{4} w_{\text{grid}}.$$

The endpoint length is measurable and bounded by two. Integrating first over the sample and then over the kernel therefore equals integration under its output law, giving

$$E_{P_{1/2}}[u_{\text{end}} - l] = \int (u_{\text{end}} - l) \mu_{\text{ref}}(d(l, u_{\text{end}})) \geq \frac{3\beta_\alpha}{4} w_{\text{grid}}.$$

All randomization is included in this expectation.

8. **Minimax length and constant calibration.** The honest interval class is nonempty. Indeed, for every full-data law,

$$\tau(P) = P(Y(1) = 1) - P(Y(0) = 1) \in [-1, 1],$$

so the constant procedure returning $[-1, 1]$ has coverage one. Every admissible interval procedure also satisfies, for every full-data law,

$$0 \leq E_P[u_{\text{end}} - l] \leq 2,$$

because $0 \leq u_{\text{end}} - l \leq 2$ for every returned endpoint pair. Thus the real-valued model suprema are finite. Since $P_{1/2}$ belongs to the model, the preceding bound implies

$$\sup_{P \in \mathcal{M}_{n,d,q}} E_P[u_{\text{end}} - l] \geq E_{P_{1/2}}[u_{\text{end}} - l] \geq \frac{3\beta_\alpha}{4} w_{\text{grid}}.$$

Taking the infimum over honest procedures in [Definition 29](#) gives

$$\mathfrak{L}_{n,d,q,\alpha} \geq \frac{3\beta_\alpha}{4} w_{\text{grid}}.$$

The width inequality and $\varepsilon_\alpha = \beta_\alpha/8$ give the required calibration:

$$\begin{aligned} c_\alpha \min\{1, N^{-1/2}\} &= \frac{3\beta_\alpha^2}{512} \min\{1, N^{-1/2}\} \\ &\leq \frac{3\beta_\alpha \varepsilon_\alpha}{16} \min\{1, N^{-1/2}\} \\ &\leq \frac{3\beta_\alpha}{4} w_{\text{grid}}. \end{aligned}$$

Finally, $w_0^{\text{grid}} = 1/2$. Hence every honest procedure satisfies

$$c_\alpha \min\{1, N^{-1/2}\} \leq E_{P_{u_0^{\text{grid}}}}[u_{\text{end}} - l] \leq \max_{j=0,\dots,M} E_{P_{u_j^{\text{grid}}}}[u_{\text{end}} - l],$$

and the minimax length satisfies the same lower bound. These are grid points from the family fixed in the first step, completing all assertions. $\square$

The squared-error comparison already applies when all individuals share a baseline category and both potential surrogates are constant. It isolates the information supplied by arrived treated outcomes, whose frequency is governed by balanced assignment and the arrival probability. The same family supports the interval comparison: the family is common across coverage levels, while the finite grid and interval lower constant may depend on the chosen coverage level. Both conclusions retain the randomized decision classes of [Definitions 16](#) and [29](#).

C.4 The many-category comparison in the full experiment

The many-category component uses the augmented laws of [Algorithm 2](#), with the finite priors supplied by [Lemma 7](#). Their membership distribution, treatment allocation, and activation distribution are common to the two hypotheses. The comparison therefore concerns the full fixed-sample experiment, including the information in repeated memberships and in both randomized arms.

Two mathematical inputs have distinct roles in this comparison. Moment matching governs comparison of the integrated observation laws, while reciprocal-mean separation governs separation of the treatment-effect functional across the prior constructions. The common activation mechanism connects these roles while preserving the original record coordinates. Together with the one-category component in [Lemma 8](#), these inputs support the following lower bound for the entire decision class.

Theorem 12. [\[thm:full-experiment-lower\]](#) (Full-experiment minimax lower bound) *There exists a universal lower constant $\underline{c} > 0$ such that, for every $n, d \in \mathbb{N}$ and $q \in \mathbb{R}$ satisfying*

- *(Sample size and dimension.)* $n \geq 1$ and $d \geq 1$.
- *(Arrival parameter.)* $0 < q \leq 1$.
- *(Rare-arrival slice.)* The restriction in [Assumption 8](#) holds.

the minimax squared-error risk $\mathfrak{R}_{n,d,q}$ defined in [Definition 16](#) obeys

$$\mathfrak{R}_{n,d,q} \geq \underline{c} r_{n,d,q},$$

where $r_{n,d,q}$ is the frontier rate defined in [Definition 17](#).

The bound combines effective sample-size uncertainty with uncertainty across many categories. The one-category family accounts for the reciprocal effective sample-size term, and the activated comparison accounts for the squared category term in [Definition 17](#). Full membership observations remain available throughout the experiment. Consequently, the lower bound describes the difficulty of estimating the treatment contrast with all recorded baseline, assignment, surrogate, and arrival information under the stated rare-arrival conditions.

C.5 Large alphabets and the ascertainment deficit

The separate unrestricted comparison is [Theorem 2](#). Its domain is $n \geq 1$, $d \geq 1024n^2$, and $0 < q \leq 1$, with the model and risk defined in [Definitions 7](#) and [12](#). The observed-law comparison concerns all n original records, including their category memberships. Target separation supplies the ascertainment-deficit component, quantified by the lower bound $(1 - q)^2/1024$; the sampling component is bounded below by $1/(256n)$.

These components have complementary interpretations. The sampling term remains present at complete ascertainment. The squared deficit measures the additional difficulty captured by the large-alphabet comparison when the arrival floor is below one. The matching upper bound $4/n + (1 - q)^2$ in [Theorem 2](#) comes from the observed-outcome envelope of [Theorem 1](#). Thus the large-alphabet result supplies a comparison with universal constants for the joint sampling-and-deficit scale.

C.6 Squared-error comparisons and their consequences

The lower bound in [Theorem 12](#) and the constructive upper comparison in [Theorem 6](#) provide the two sides of [Theorem 4](#). Under [Assumption 8](#), the minimax squared-error risk is therefore comparable, with universal constants, to the rate in [Definition 17](#). Conditional averaging preserves the upper guarantee by [Lemma 1](#), while the lower comparison covers the full class of possibly randomized estimators.

For a fixed alphabet and growing effective sample size, [Proposition 4](#) records risk of order N^{-1} , including within the constant-surrogate submodel. Along growing-alphabet sequences satisfying the rare-arrival restriction and $N \rightarrow \infty$, [Proposition 5](#) gives consistency exactly when $d = o\{N \log(e + N)\}$, and risk of order N^{-1} exactly when $d = O\{\sqrt{N} \log(e + N)\}$. These consequences distinguish the category-growth range permitting consistency from the range preserving effective sample-size precision.

For the unrestricted model, [Proposition 1](#) combines the large-alphabet lower comparison with the dimension-free upper envelope. An eventual risk bound of order n^{-1} , simultaneous over every alphabet size, is equivalent to an ascertainment deficit $1 - q_n = O(n^{-1/2})$. The numerical comparison along arbitrary alphabet-size sequences is given by [Proposition 2](#). At complete arrival, [Theorem 3](#) and [Proposition 3](#) establish the same sampling order uniformly over category growth.

Finally, the fixed-floor comparison in [Theorem 5](#) concerns each fixed $q \in (0, 1/2)$, with constants depending on that floor. Its lower comparison invokes the discrete observational result of [Zeng et al. \[2026, Theorem 2, Section 3.2, p. 9, and its proof in Section C.5, pp. 39–44\]](#), including the submodel with zero control regression. The resulting category-growth consequences are stated in [Proposition 6](#): consistency corresponds to $d = o\{n \log(e + n)\}$, and parametric squared-error order corresponds to $d = O\{\sqrt{n} \log(e + n)\}$. These comparisons preserve the distinct parameter domains and constant dependencies of the rare-arrival, fixed-floor, and near-complete ascertainment results.

D Interval and observational proofs

The interval and observational comparisons use two features of the constructive analysis: a deterministic bound on squared-error risk and a parameter-independent transformation preserving the target. The decision-theoretic results in [Lemmas 1](#) and [2](#) supply the corresponding comparisons for procedures using auxiliary randomization.

D.1 Coverage and expected interval length

Fix $\alpha \in (0, 1)$, and suppose that $n, d \geq 1$, $0 < q \leq 1$, and [Assumption 8](#) holds. The deterministic envelope $\mathfrak{U}_{n,d,q}$ in [Theorem 6](#) controls squared-error risk uniformly over the model in [Definition 14](#). Its role in [Definition 30](#) is to determine the radius before observing the sample.

Markov's inequality converts this risk control into coverage at least $1 - \alpha$. Because the target belongs to $[-1, 1]$, intersecting the interval with that range preserves coverage.

The mixed-count estimator $\hat{\tau}_{n,d,q}^{\text{mix}}$ is defined in [Algorithm 1](#). The resulting interval I_α^{mix} has length bounded by

$$\text{length}(I_\alpha^{\text{mix}}) \leq \min \left\{ 2, 2\sqrt{\frac{\mathfrak{L}_{n,d,q}}{\alpha}} \right\}.$$

Thus the envelope comparison with the rate $r_{n,d,q}$ in [Definition 17](#) supplies the upper expected-length comparison in [Theorem 7](#). The upper constant $\bar{C}_\alpha$ may depend on the fixed miscoverage level and is uniform over the sample sizes, dimensions, and arrival floors satisfying these conditions.

The lower comparison concerns every endpoint procedure admitted by [Definition 29](#). Its sampling component is supplied directly by [Lemma 8](#). For each fixed α , that result supplies a finite grid of distinct target values within the same one-category family and bounds maximal expected length below at order

$$\min\{1, (nq)^{-1/2}\}.$$

The underlying family is common across coverage levels; the grid and its comparison constant may depend on α . This finite-family formulation supports every coverage level $1 - \alpha \in (0, 1)$.

The many-category component of [Theorem 7](#) uses the separation and observation-law comparison associated with [Algorithm 2](#) and [Lemma 7](#). Moment matching controls statistical distinguishability, while reciprocal separation determines the target scale. Connectedness supplies the geometric link to expected length: an interval containing separated target values spans their separation. Finite testing families at the specified coverage level retain this link throughout the coverage range in the theorem.

Auxiliary endpoint randomization enters through the joint observation–action laws defined in [Lemma 2](#). Apply that result with the action space equal to the ordered endpoint pairs and the common decision rule equal to the interval procedure. The joint-law total variation equals the total variation of the observation laws. Consequently, coverage events can be compared while retaining both the observed sample and the reported endpoints. Expected length is evaluated under this same joint law, exactly as required by [Definition 29](#).

Together, these components give the lower constant $\underline{c}_\alpha$ in [Theorem 7](#). The minimax expected length $\mathfrak{L}_{n,d,q,\alpha}$ is therefore comparable to $\sqrt{r_{n,d,q}}$, with constants depending on α . The square-root scale reflects both sources of uncertainty: effective sample size and the number of covariate categories. Along sequences satisfying the rare-arrival restriction, at fixed α , optimal expected length tends to zero exactly when $r_{n,d,q}$ tends to zero.

D.2 Synthetic assignment and observational risk

Fix a strict positivity margin $0 < \epsilon < 1/2$. The observational class $\mathcal{D}_{d,\epsilon}$ is defined in [Definition 31](#), and its target is the contrast in [Definition 32](#). The transformation in [Lemma 5](#), with arrival floor $q = \epsilon$, assigns an independent fair arm to each observational record and retains the outcome when that arm agrees with the observational treatment. Positivity supplies the arrival floor in both arms.

The induced full-data law P_Z^* , defined by [Lemma 5](#), belongs to the unrestricted randomized class and preserves the target:

$$\tau(P_Z^*) = \theta_Z(P_Z).$$

The same statement identifies the synthetic sample law with the product observed-record law of this completion. These two properties make the unrestricted risk certificate in [Theorem 6](#) applicable to the mixed-count estimator evaluated on the synthetic records.

The observational estimator $\hat{\theta}_{n,d,\epsilon}^{\text{poly}}$ in [Theorem 8](#) averages over the independent synthetic assignments conditional on the observational sample. The barycenter comparison in [Lemma 1](#) weakly reduces squared-error risk under this averaging. Accordingly, the constructive bound has a universal upper constant and retains the positivity dependence explicitly through

$$r_{n,d,\epsilon} = \min \left\{ 1, \frac{1}{n\epsilon} + \left[\frac{d}{n\epsilon \log(e + n\epsilon)} \right]^2 \right\}.$$

Synthetic assignment converts observational treatment imbalance into selective outcome arrival; conditional averaging then removes the variation introduced by that assignment. Target preservation connects both operations to the same observational contrast.

The matching lower comparison invokes [Zeng et al. \[2026, Theorem 2, Section 3.2, p. 9, together with its proof in Section C.5, pp. 39–44\]](#). The relevant published conclusion concerns fixed strict positivity, sufficiently large n , and $1 \leq d \leq c_0 n \log n$, where $c_0 > 0$ is a constant in that conclusion. Its proof uses the submodel with zero control regression on occupied categories and supplies the lower order

$$\frac{1}{n} + \left(\frac{d}{n \log n} \right)^2.$$

This conclusion supplies the observational lower-bound input. The matched comparison in [Theorem 8](#) covers every $n, d \geq 1$, incorporates bounded-risk truncation, and uses $\log(e + n)$.

For each fixed ϵ , the observational minimax risk $\mathfrak{R}_{n,d,\epsilon}^Z$ consequently has the two-sided comparison stated in [Theorem 8](#), with positive finite constants c_ϵ and C_ϵ depending on that margin. This dependence is distinct from the universal constant in the constructive bound expressed through $r_{n,d,\epsilon}$.

D.3 Fixed arrival and category-growth consequences

The same transformation determines the direction of the fixed-arrival experiment comparison. Fix $q \in (0, 1/2)$. For every observational law in $\mathcal{D}_{d,q}$, [Lemma 5](#) supplies a target-preserving completion in the unrestricted class $\mathcal{M}_{n,d,q}^+$ of [Definition 7](#). Composing any randomized estimation procedure with the synthetic assignment gives an observational procedure with the same risk at the corresponding completion. Hence the unrestricted randomized minimax risk $\mathfrak{R}_{n,d,q}^+$ dominates the observational minimax risk at positivity margin q . The published observational lower comparison therefore transfers to this randomized class.

Combined with the constructive upper comparison, this transfer supports [Theorem 5](#). Its constants $\underline{c}(q)$ and $\overline{C}(q)$ depend only on the fixed arrival floor and apply uniformly over $n, d \geq 1$. The resulting category-growth conclusions are recorded in [Proposition 6](#): consistency is equivalent to

$$d_n = o\{n \log(e + n)\},$$

and parametric squared-error order is equivalent to

$$d_n = O\{\sqrt{n} \log(e + n)\}.$$

For an eventual bound $d_n \leq M\sqrt{n} \log(e + n)$, the lower risk constant may depend on q , while the upper constant may depend on both q and M . The eventual sample-size threshold may depend

on the category sequence. These dependencies specify the fixed-arrival scope of the matched comparison and its sequence-level consequences.

E Verification note

The theorem and lemma statements and proofs attached to the frozen results are machine-checked in Lean 4 under their stated hypotheses and cited dependencies. The checked conclusions cover the identification identities in [Lemmas 3](#) and [4](#), the decision-rule comparisons in [Lemmas 1](#) and [2](#), and the synthetic-assignment construction in [Lemma 5](#). They also cover the estimator risk certificates in [Theorems 9](#) to [11](#), the approximation and testing conclusions in [Lemmas 6](#) to [8](#), and the lower bound in [Theorem 12](#). The point-risk comparisons and their stated phase-boundary consequences, the connected-interval comparison in [Theorem 7](#), and the observational comparison in [Theorem 8](#) are checked with the conditions and constant dependencies recorded in their statements.

The model restrictions enter as assumptions. These include independent sampling, randomized assignment and balance, surrogate and outcome consistency, conditional independence of outcome arrival, and the occupied-cell arrival floor in [Assumptions 1](#) to [7](#). Results for rare arrival additionally use [Assumption 8](#); the observational comparison uses the positivity class specified in [Definition 31](#). Presentation definitions specify the laws, targets, decision classes, and procedures to which the mathematical assertions apply.

Published mathematical conclusions used through theorem-local cited dependencies remain part of the assumptions of the corresponding checks. The generated local verification disclosures identify those conclusions, preserve their source locators, and distinguish them from proofs formalized in Lean. Together, the stated hypotheses and those disclosures specify the scope of the machine-checked conditional conclusions.

F Proofs of the main results

Proof of [Theorem 1](#). 1. Choose the universal constant $\bar{C} > 0$ supplied by [Theorem 6](#). For every admissible law P and the specified sampling law, that result gives

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \bar{C} r_{n,d,q},$$

and

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq E_P \left[E_\omega \left[(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P))^2 \mid \mathbf{O}_n \right] \right].$$

Its envelope is exactly the branchwise envelope in the statement, and its auxiliary output includes the zero output on the Poisson tail. We retain this same constant throughout.

2. We first record how a deterministic risk bound yields a minimax bound. Write

$$\mathcal{O}_d := \{0, \dots, d-1\} \times \{0, 1\}^4$$

for the observed-record space. For a bounded estimator kernel T , define its risk by

$$\mathcal{R}_P(\mathsf{T}) := \int_{\mathcal{O}_d^n} \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}(\mathbf{o}, dh) \mathcal{L}_P(\mathbf{O}_n)(d\mathbf{o}).$$

These risks are nonnegative.

Let $f_{\det} : \mathcal{O}_d^n \rightarrow [-1, 1]$ be measurable, and let $b_{\text{up}} \geq 0$. Suppose

$$E_P \left[(f_{\det}(\mathbf{O}_n) - \tau(P))^2 \right] \leq b_{\text{up}} \quad \text{for every } P \in \mathcal{M}_{n,d,q}^+.$$

The parameter-independent kernel

$$\mathbb{T}_{\det}(\mathbf{o}, \cdot) := \text{Dirac}(f_{\det}(\mathbf{o}))$$

belongs to the decision class in [Definition 12](#), and satisfies

$$\mathcal{R}_P(\mathbb{T}_{\det}) = E_P \left[(f_{\det}(\mathbf{O}_n) - \tau(P))^2 \right].$$

Consequently,

$$\mathfrak{R}_{n,d,q}^+ \leq \sup_{P \in \mathcal{M}_{n,d,q}^+} E_P \left[(f_{\det}(\mathbf{O}_n) - \tau(P))^2 \right] \leq b_{\text{up}}.$$

For a nonempty model class, the last inequality follows by taking the supremum of the pointwise bounds. For an empty class, the real supremum is defined to be zero, and the inequality follows from $b_{\text{up}} \geq 0$.

3. Fix $n, d \geq 1$ and $0 < q \leq 1$. Binary potential outcomes and [Definition 8](#) give

$$\tau(P) = P(Y(1) = 1) - P(Y(0) = 1) \in [-1, 1].$$

Indeed, each probability belongs to $[0, 1]$. The zero map is measurable and takes values in $[-1, 1]$, and its risk under the canonical product probability law is

$$E_P \left[(0 - \tau(P))^2 \right] = \tau(P)^2 \leq 1.$$

Applying the deterministic comparison just established yields

$$\mathfrak{R}_{n,d,q}^+ \leq 1.$$

4. The rate in [Definition 17](#) satisfies

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{nq} + \left[\frac{d}{nq \log(e + nq)} \right]^2 \right\} \geq 0, \quad \bar{C} r_{n,d,q} \geq 0.$$

The estimator in [Algorithm 1](#) is measurable. Its auxiliary output lies in $[-1, 1]$ for every sample and every randomization, so averaging gives

$$-1 \leq E_{\omega}[\tilde{\tau}(\mathbf{o}; \omega) \mid \mathbf{o}] = \hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{o}) \leq 1.$$

Under the canonical product law, the coordinate observations are measurable and independent, with

$$\mathcal{L}_P(O_i) = \mathcal{L}_P(O_1), \quad i = 1, \dots, n,$$

so they satisfy [Assumption 1](#). The first step therefore applies to every law in the model class. The deterministic comparison, with $b_{\text{up}} = \bar{C} r_{n,d,q}$, gives

$$\mathfrak{R}_{n,d,q}^+ \leq \bar{C} r_{n,d,q}.$$

5. We next establish the observed-contrast bound. Fix $P \in \mathcal{M}_{n,d,q}^+$. For an observed record, write its coordinates as

$$o = (X(o), A(o), S(o), R(o), (RY)(o)),$$

and define

$$\xi_{\text{obs}}(o) := 2(2A(o) - 1)\mathbf{1}\{R(o) = 1, (RY)(o) = 1\}.$$

For a sample $\mathbf{o} = (o_1, \dots, o_n) \in \mathcal{O}_d^n$, define

$$\bar{\xi}_n(\mathbf{o}) := \frac{1}{n} \sum_{i=1}^n \xi_{\text{obs}}(o_i).$$

The canonical product law gives

$$E_P[\bar{\xi}_n(\mathbf{O}_n)] = E_P[\xi_{\text{obs}}(O_1)], \quad \text{Var}_P(\bar{\xi}_n(\mathbf{O}_n)) = \frac{1}{n} \text{Var}_P(\xi_{\text{obs}}(O_1)).$$

Since every observed coordinate is binary,

$$-2 \leq \xi_{\text{obs}}(o) \leq 2, \quad \text{Var}_P(\xi_{\text{obs}}(O_1)) \leq E_P[\xi_{\text{obs}}(O_1)^2] \leq 4.$$

Expanding the squared error around the mean now gives

$$E_P[(\bar{\xi}_n(\mathbf{O}_n) - \tau(P))^2] = \frac{\text{Var}_P(\xi_{\text{obs}}(O_1))}{n} + (E_P[\xi_{\text{obs}}(O_1)] - \tau(P))^2.$$

By [Definition 18](#),

$$\hat{\tau}_n^{\text{HT}}(\mathbf{o}) = \text{clip}_{[-1,1]}(\bar{\xi}_n(\mathbf{o})).$$

Because $\tau(P) \in [-1, 1]$, considering whether $\bar{\xi}_n(\mathbf{o}) < -1$, belongs to $[-1, 1]$, or exceeds 1, gives

$$(\hat{\tau}_n^{\text{HT}}(\mathbf{o}) - \tau(P))^2 \leq (\bar{\xi}_n(\mathbf{o}) - \tau(P))^2.$$

6. To bound the bias, define the missing-outcome masses for both arms by

$$\Delta_a(P) := 2P(A = a, R = 0, Y = 1), \quad a \in \{0, 1\}.$$

For every cell indexed by a, x, s , the occupied-cell arrival condition implies

$$q p_{axs}(P) \leq P(A = a, X = x, S = s, R = 1).$$

For an occupied cell this is the arrival-floor inequality in [Assumption 7](#); for a null cell its left side is zero and its right side is nonnegative. Partitioning a cell according to R therefore gives

$$\begin{aligned} P(A = a, X = x, S = s, R = 0) &= p_{axs}(P) - P(A = a, X = x, S = s, R = 1) \\ &\leq (1 - q)p_{axs}(P). \end{aligned}$$

Balanced assignment gives $P(A = 1) = 1/2$ and $P(A = 0) = 1 - P(A = 1) = 1/2$. Thus, for each $a \in \{0, 1\}$,

$$\sum_{x=0}^{d-1} \sum_{s=0}^1 p_{axs}(P) = \frac{1}{2},$$

and the finite cell partition yields

$$\begin{aligned} P(A = a, R = 0) &= \sum_{x=0}^{d-1} \sum_{s=0}^1 P(A = a, X = x, S = s, R = 0) \\ &\leq (1 - q) \sum_{x=0}^{d-1} \sum_{s=0}^1 p_{axs}(P) = \frac{1 - q}{2}. \end{aligned}$$

The event $\{A = a, R = 0, Y = 1\}$ is contained in $\{A = a, R = 0\}$, so

$$0 \leq \Delta_a(P) \leq 2P(A = a, R = 0) \leq 1 - q.$$

Outcome consistency, randomized independence, and balanced assignment give

$$\begin{aligned} P(A = a, Y = 1) &= P(A = a, Y(a) = 1) \\ &= P(A = a)P(Y(a) = 1) = \frac{1}{2}P(Y(a) = 1). \end{aligned}$$

Using the observed-record map and partitioning the outcome-one event according to arrival, we obtain

$$\begin{aligned} E_P[\xi_{\text{obs}}(O_1)] &= 2P(A = 1, R = 1, Y = 1) - 2P(A = 0, R = 1, Y = 1) \\ &= 2P(A = 1, Y = 1) - 2P(A = 0, Y = 1) - \Delta_1(P) + \Delta_0(P) \\ &= \tau(P) - \Delta_1(P) + \Delta_0(P). \end{aligned}$$

The bounds on the two deficits imply

$$-(1 - q) \leq E_P[\xi_{\text{obs}}(O_1)] - \tau(P) \leq 1 - q,$$

and hence, since $1 - q \geq 0$,

$$(E_P[\xi_{\text{obs}}(O_1)] - \tau(P))^2 \leq (1 - q)^2.$$

Together with the preceding variance and clipping bounds, this gives

$$\begin{aligned} E_P\left[(\hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P))^2\right] &\leq E_P\left[(\bar{\xi}_n(\mathbf{O}_n) - \tau(P))^2\right] \\ &\leq \frac{4}{n} + (1 - q)^2. \end{aligned}$$

The clipped contrast is measurable on the finite sample space and takes values in $[-1, 1]$. Since $4/n + (1 - q)^2 \geq 0$, the deterministic comparison gives both

$$\mathfrak{R}_{n,d,q}^+ \leq \frac{4}{n} + (1 - q)^2$$

and

$$\sup_{P \in \mathcal{M}_{n,d,q}^+} E_P\left[(\hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P))^2\right] \leq \frac{4}{n} + (1 - q)^2.$$

7. For the lower bound, first construct a complete-arrival pair. Define

$$\delta_{\text{gap}} := \frac{1}{4\sqrt{n}}, \quad 0 < \delta_{\text{gap}} \leq \frac{1}{4}, \quad \delta_{\text{gap}}^2 = \frac{1}{16n}.$$

For $\iota \in \{0, 1\}$, let $P_{\text{test}, \iota}$ be the full-data law specified by

$$\begin{aligned} X = 0, \quad S(0) = S(1) = 0, \quad Y(0) = 0, \quad R = 1, \\ A \sim \text{Bernoulli}(1/2), \quad Y(1) \sim \text{Bernoulli}(1/2 + \iota\delta_{\text{gap}}), \quad A \perp\!\!\!\perp Y(1), \\ S = S(A) = 0, \quad Y = Y(A) = AY(1). \end{aligned}$$

The baseline value 0 is available for every $d \geq 1$, and the outcome parameters lie in $[1/2, 3/4]$. Independence and the fair treatment coin verify randomized independence and balanced assignment. The displayed definitions verify consistency. Constant arrival verifies arrival MAR and gives arrival probability one in every occupied cell. Therefore

$$P_{\text{test}, 0}, P_{\text{test}, 1} \in \mathcal{M}_{n, d, 1}^+, \quad \tau(P_{\text{test}, \iota}) = \frac{1}{2} + \iota\delta_{\text{gap}}.$$

Define the observed laws and their sample laws by

$$Q_{\text{test}, \iota} := \mathcal{L}_{P_{\text{test}, \iota}}(O_1), \quad Q_{\text{test}, \iota}^{(n)} := Q_{\text{test}, \iota}^{\otimes n}.$$

Their common support consists of the three records

$$\begin{aligned} o_{\text{test}, \text{control}} &:= (0, 0, 0, 1, 0), \\ o_{\text{test}, \text{zero}} &:= (0, 1, 0, 1, 0), \quad \mathcal{Z}_{\text{test}} := \{o_{\text{test}, \text{control}}, o_{\text{test}, \text{zero}}, o_{\text{test}, \text{one}}\}, \\ o_{\text{test}, \text{one}} &:= (0, 1, 0, 1, 1), \end{aligned}$$

The atom probabilities are

$$\begin{aligned} Q_{\text{test}, \iota}(\{o_{\text{test}, \text{control}}\}) &= \frac{1}{2}, \\ Q_{\text{test}, \iota}(\{o_{\text{test}, \text{zero}}\}) &= \frac{1}{4} - \frac{\iota\delta_{\text{gap}}}{2}, \\ Q_{\text{test}, \iota}(\{o_{\text{test}, \text{one}}\}) &= \frac{1}{4} + \frac{\iota\delta_{\text{gap}}}{2}, \end{aligned}$$

with probability zero elsewhere. In particular, $Q_{\text{test}, 1} \ll Q_{\text{test}, 0}$.

For finite laws with a positive reference law on their common support, define

$$\chi^2(Q_{\text{test}, 1} \| Q_{\text{test}, 0}) := \sum_{o \in \mathcal{Z}_{\text{test}}} \frac{(Q_{\text{test}, 1}(\{o\}) - Q_{\text{test}, 0}(\{o\}))^2}{Q_{\text{test}, 0}(\{o\})}.$$

Summing the squared probability ratios gives

$$\begin{aligned} 1 + \chi^2(Q_{\text{test}, 1} \| Q_{\text{test}, 0}) &= \sum_{o \in \mathcal{Z}_{\text{test}}} \frac{Q_{\text{test}, 1}(\{o\})^2}{Q_{\text{test}, 0}(\{o\})} \\ &= \frac{(1/2)^2}{1/2} + \frac{(1/4 - \delta_{\text{gap}}/2)^2}{1/4} + \frac{(1/4 + \delta_{\text{gap}}/2)^2}{1/4} \\ &= 1 + 2\delta_{\text{gap}}^2. \end{aligned}$$

Thus

$$\chi^2(Q_{\text{test}, 1} \| Q_{\text{test}, 0}) = 2\delta_{\text{gap}}^2 = \frac{1}{8n}.$$

8. Define the sample likelihood ratio on $\mathcal{O}_d^n$ by

$$L_{\text{test},n}(\mathbf{o}) := \begin{cases} \prod_{i=1}^n \frac{Q_{\text{test},1}(\{o_i\})}{Q_{\text{test},0}(\{o_i\})}, & \mathbf{o} \in \mathcal{Z}_{\text{test}}^n, \\ 0, & \mathbf{o} \notin \mathcal{Z}_{\text{test}}^n. \end{cases}$$

The product laws satisfy

$$E_{Q_{\text{test},0}^{(n)}}[L_{\text{test},n}] = 1,$$

and independence gives

$$\begin{aligned} 1 + \chi^2(Q_{\text{test},1}^{(n)} \| Q_{\text{test},0}^{(n)}) &= E_{Q_{\text{test},0}^{(n)}}[L_{\text{test},n}^2] \\ &= (1 + \chi^2(Q_{\text{test},1} \| Q_{\text{test},0}))^n. \end{aligned}$$

Here the product divergence is defined by the same finite-sum formula on $\mathcal{Z}_{\text{test}}^n$. Since $1 + t \leq e^t$ for $t \geq 0$,

$$\chi^2(Q_{\text{test},1}^{(n)} \| Q_{\text{test},0}^{(n)}) \leq e^{1/8} - 1 < 1.$$

For the numerical inequality, monotonicity gives $e^{1/8} \leq e^{1/2}$, while

$$(e^{1/2})^2 = e < 3 < 4,$$

so $e^{1/8} < 2$.

Define total variation for these sample laws by

$$\text{TV}(Q_{\text{test},1}^{(n)}, Q_{\text{test},0}^{(n)}) := \frac{1}{2} \sum_{\mathbf{o} \in \mathcal{Z}_{\text{test}}^n} \left| Q_{\text{test},1}^{(n)}(\{\mathbf{o}\}) - Q_{\text{test},0}^{(n)}(\{\mathbf{o}\}) \right|.$$

Cauchy–Schwarz applied to the likelihood ratio yields

$$\begin{aligned} \text{TV}(Q_{\text{test},1}^{(n)}, Q_{\text{test},0}^{(n)}) &= \frac{1}{2} E_{Q_{\text{test},0}^{(n)}} |L_{\text{test},n} - 1| \\ &\leq \frac{1}{2} \sqrt{E_{Q_{\text{test},0}^{(n)}} [(L_{\text{test},n} - 1)^2]} \\ &= \frac{1}{2} \sqrt{\chi^2(Q_{\text{test},1}^{(n)} \| Q_{\text{test},0}^{(n)})} \leq \frac{1}{2}. \end{aligned}$$

Consequently, for every measurable $\mathcal{E} \subseteq \mathcal{O}_d^n$,

$$\begin{aligned} Q_{\text{test},1}^{(n)}(\mathcal{E}^c) + Q_{\text{test},0}^{(n)}(\mathcal{E}) &= 1 - (Q_{\text{test},1}^{(n)}(\mathcal{E}) - Q_{\text{test},0}^{(n)}(\mathcal{E})) \\ &\geq 1 - \text{TV}(Q_{\text{test},1}^{(n)}, Q_{\text{test},0}^{(n)}) \geq \frac{1}{2}. \end{aligned}$$

9. Take any parameter-independent bounded kernel $\mathbb{T}$ in the decision class, and define its barycenter by

$$b_{\mathbb{T}}(\mathbf{o}) := \int_{[-1,1]} h \mathbb{T}(\mathbf{o}, dh).$$

By Lemma 1, this is a measurable map into $[-1, 1]$, and

$$E_{P_{\text{test},\iota}} \left[(b_{\mathbf{T}}(\mathbf{O}_n) - \tau(P_{\text{test},\iota}))^2 \right] \leq \mathcal{R}_{P_{\text{test},\iota}}(\mathbf{T}), \quad \iota \in \{0, 1\}.$$

Define the two large-error events and the nearest-target test by

$$\mathcal{E}_{\text{test},\iota} := \left\{ \mathbf{o} : \frac{\delta_{\text{gap}}}{2} \leq |b_{\mathbf{T}}(\mathbf{o}) - \tau(P_{\text{test},\iota})| \right\}, \quad \iota \in \{0, 1\},$$

and

$$\mathcal{A}_{\text{test}} := \{ \mathbf{o} : |b_{\mathbf{T}}(\mathbf{o}) - \tau(P_{\text{test},1})| \leq |b_{\mathbf{T}}(\mathbf{o}) - \tau(P_{\text{test},0})| \}.$$

The target separation and the triangle inequality give, for every sample,

$$\delta_{\text{gap}} \leq |b_{\mathbf{T}}(\mathbf{o}) - \tau(P_{\text{test},0})| + |b_{\mathbf{T}}(\mathbf{o}) - \tau(P_{\text{test},1})|.$$

On $\mathcal{A}_{\text{test}}$, the first error is at least $\delta_{\text{gap}}/2$; on its complement, the second error is at least $\delta_{\text{gap}}/2$. Hence

$$\mathcal{A}_{\text{test}} \subseteq \mathcal{E}_{\text{test},0}, \quad \mathcal{A}_{\text{test}}^c \subseteq \mathcal{E}_{\text{test},1}.$$

Applying the testing bound and then event monotonicity gives

$$\frac{1}{2} \leq Q_{\text{test},1}^{(n)}(\mathcal{A}_{\text{test}}^c) + Q_{\text{test},0}^{(n)}(\mathcal{A}_{\text{test}}) \leq Q_{\text{test},1}^{(n)}(\mathcal{E}_{\text{test},1}) + Q_{\text{test},0}^{(n)}(\mathcal{E}_{\text{test},0}),$$

so

$$\max_{\iota \in \{0,1\}} Q_{\text{test},\iota}^{(n)}(\mathcal{E}_{\text{test},\iota}) \geq \frac{1}{4}.$$

Integrating squared error over each large-error event yields

$$\left(\frac{\delta_{\text{gap}}}{2} \right)^2 Q_{\text{test},\iota}^{(n)}(\mathcal{E}_{\text{test},\iota}) \leq E_{P_{\text{test},\iota}} \left[(b_{\mathbf{T}}(\mathbf{O}_n) - \tau(P_{\text{test},\iota}))^2 \right].$$

Whether the probability for $\iota = 0$ is smaller than that for $\iota = 1$, or conversely, multiplication by the nonnegative factor $(\delta_{\text{gap}}/2)^2$ preserves their ordering. Therefore

$$\begin{aligned} \frac{\delta_{\text{gap}}^2}{16} &\leq \left(\frac{\delta_{\text{gap}}}{2} \right)^2 \max_{\iota \in \{0,1\}} Q_{\text{test},\iota}^{(n)}(\mathcal{E}_{\text{test},\iota}) \\ &= \max_{\iota \in \{0,1\}} \left\{ \left(\frac{\delta_{\text{gap}}}{2} \right)^2 Q_{\text{test},\iota}^{(n)}(\mathcal{E}_{\text{test},\iota}) \right\} \\ &\leq \max_{\iota \in \{0,1\}} E_{P_{\text{test},\iota}} \left[(b_{\mathbf{T}}(\mathbf{O}_n) - \tau(P_{\text{test},\iota}))^2 \right] \\ &\leq \max_{\iota \in \{0,1\}} \mathcal{R}_{P_{\text{test},\iota}}(\mathbf{T}). \end{aligned}$$

For every bounded kernel and every full-data law, bounded actions and $\tau(P) \in [-1, 1]$ also give

$$0 \leq \mathcal{R}_P(\mathbf{T}) \leq 4.$$

Thus the worst-case risks are finite. The decision class is nonempty because it contains the zero estimator, and the complete-arrival class contains the constructed pair. Taking the supremum over that class and then the infimum over kernels gives

$$\mathfrak{R}_{n,d,1}^+ \geq \frac{\delta_{\text{gap}}^2}{16} = \frac{1}{256n}.$$

10. Finally, every complete-arrival law belongs to the model with arrival floor q . Indeed, all other conditions are shared, and for each occupied cell of such a law,

$$q p_{axs}(P) \leq p_{axs}(P) \leq P(A = a, X = x, S = s, R = 1).$$

Together with $0 < q \leq 1$, this proves

$$\mathcal{M}_{n,d,1}^+ \subseteq \mathcal{M}_{n,d,q}^+.$$

For every kernel $\mathbb{T}$, class inclusion gives

$$\sup_{P \in \mathcal{M}_{n,d,1}^+} \mathcal{R}_P(\mathbb{T}) \leq \sup_{P \in \mathcal{M}_{n,d,q}^+} \mathcal{R}_P(\mathbb{T}).$$

The risks lie in $[0, 4]$, and both classes contain the complete-arrival pair. Taking infima over the same nonempty decision class therefore yields

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,1}^+ \leq \mathfrak{R}_{n,d,q}^+.$$

Combining this with the three upper bounds proves

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+ \leq \min \left\{ 1, \bar{C} r_{n,d,q}, \frac{4}{n} + (1 - q)^2 \right\}.$$

The first step supplies the mixed-count envelope and its auxiliary-risk comparison, and the observed-contrast calculation supplies the asserted uniform witness bound. $\square$

Proof of Theorem 2. 1. Put

$$\delta := 1 - q \in [0, 1).$$

Since $d \geq 1024n^2 \geq 1$, the hypotheses of Theorem 1 hold. Its lower bound and the last term of its upper envelope give

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+, \quad \mathfrak{R}_{n,d,q}^+ \leq \frac{4}{n} + \delta^2.$$

If $\delta^2 \leq n^{-1}$, then

$$\frac{\delta^2}{1024} \leq \frac{1}{1024n} \leq \frac{1}{256n} \leq \mathfrak{R}_{n,d,q}^+.$$

This case includes $q = 1$. For the remaining case, assume

$$\delta^2 > n^{-1}, \quad \text{so in particular} \quad n^{-1} \leq \delta^2.$$

We establish the deficit lower bound in this regime.

2. Write an ordered observed sample as

$$\mathbf{o} = (o_1, \dots, o_n) \in (\{0, \dots, d-1\} \times \{0, 1\}^4)^n,$$

with each record ordered as (X, A, S, R, RY) . The decision class contains the constant zero kernel

$$\mathbb{T}_0(\mathbf{o}, \cdot) := \text{Dirac}(0).$$

Fix an arbitrary $\mathsf{T} \in \mathcal{T}_n$, the bounded, parameter-independent kernel class of [Definition 12](#), and define its risk under a full-data law P by

$$\mathcal{R}_{\mathsf{T}}(P) := \int \int_{[-1,1]} (w - \tau(P))^2 \mathsf{T}(\mathbf{o}, dw) \mathcal{L}_P(\mathbf{O}_n)(d\mathbf{o}).$$

Define the separation between the two target centers by

$$h_{\text{sep}} := \frac{1}{2} - \frac{q}{1+q} = \frac{\delta}{2(1+q)}.$$

Because $1 < 1+q \leq 2$ and $\delta \geq 0$, this identity gives

$$\frac{\delta}{4} \leq h_{\text{sep}} \leq \frac{\delta}{2}, \quad h_{\text{sep}} \geq 0.$$

Construct a comparison law $P_{\star}$ by taking a uniform baseline, zero surrogates and control outcome, and a Bernoulli treated potential outcome:

$$X \sim \text{Unif}\{0, \dots, d-1\}, \quad S(0) = S(1) = Y(0) = 0, \quad Y(1) \mid X \sim \text{Bernoulli}\left(\frac{q}{1+q}\right).$$

Draw $A \sim \text{Bernoulli}(1/2)$ independently of the baseline and potential variables, set

$$S = S(A), \quad Y = Y(A),$$

and draw the arrival indicator conditionally independently of $Y(1)$, with

$$P_{\star}(R = 1 \mid X, A, Y(1)) = \begin{cases} 1, & A = 0, \\ (1+q)/2, & A = 1. \end{cases}$$

These distributions are well defined because

$$0 < \frac{q}{1+q} \leq \frac{1}{2}, \quad q \leq \frac{1+q}{2} \leq 1.$$

Evaluating the average treatment effect of [Definition 8](#) yields

$$\tau(P_{\star}) = E_{P_{\star}}[Y(1) - Y(0)] = \frac{q}{1+q} = \frac{1}{2} - h_{\text{sep}}.$$

3. For each

$$\xi = (\xi_x)_{x=0}^{d-1} \in \{0, 1\}^{\{0, \dots, d-1\}},$$

construct a law P_{ξ} with the same uniform baseline and independent fair treatment assignment, and with

$$S(0) = S(1) = Y(0) = 0, \quad Y(1) = \xi_X, \quad S = S(A), \quad Y = Y(A).$$

Its arrival draw satisfies, for $x \in \{0, \dots, d-1\}$ and $a \in \{0, 1\}$,

$$P_{\xi}(R = 1 \mid X = x, A = a) = 1 - a\delta\xi_x.$$

Thus every arrival probability is either q or 1, and

$$\tau(P_\xi) = \frac{1}{d} \sum_{x=0}^{d-1} \xi_x.$$

Put the product fair-coin prior on these vectors:

$$\nu_{\text{cell}} := \bigotimes_{x=0}^{d-1} \text{Bernoulli}(1/2), \quad \nu_{\text{cell}}(\{\xi\}) = 2^{-d}.$$

Define the sample laws and their mixture by

$$Q_\xi := \mathcal{L}_{P_\xi}(\mathbf{O}_n), \quad Q_\star := \mathcal{L}_{P_\star}(\mathbf{O}_n), \quad Q_{\text{mix}} := \int Q_\xi \nu_{\text{cell}}(d\xi) = \sum_{\xi} 2^{-d} Q_\xi.$$

For each fixed full-data law, the sample consists of n independent observed records as in [Assumption 1](#). The mixture draws ξ once for the entire sample.

Define the test event, the exceptional prior vectors, and the two risks by

$$E_{\text{test}} := \left\{ (\mathbf{o}, w) : w \leq \frac{1}{2} - \frac{h_{\text{sep}}}{2} \right\}, \quad w \in \mathbb{R},$$

$$F_{\text{bad}} := \left\{ \xi : \left| \tau(P_\xi) - \frac{1}{2} \right| > \frac{h_{\text{sep}}}{4} \right\},$$

$$\mathcal{R}_{\text{prior}}(\mathbb{T}) := \int \mathcal{R}_{\mathbb{T}}(P_\xi) \nu_{\text{cell}}(d\xi), \quad \mathcal{R}_\star(\mathbb{T}) := \mathcal{R}_{\mathbb{T}}(P_\star).$$

The notation $Q \ltimes \mathbb{T}$ denotes the joint sample-action law used in [Lemma 2](#).

If $\xi \notin F_{\text{bad}}$ and $(\mathbf{o}, w) \in E_{\text{test}}$, then

$$\tau(P_\xi) - w \geq \left(\frac{1}{2} - \frac{h_{\text{sep}}}{4} \right) - \left(\frac{1}{2} - \frac{h_{\text{sep}}}{2} \right) = \frac{h_{\text{sep}}}{4}.$$

Consequently, for every ξ ,

$$\frac{h_{\text{sep}}^2}{16} (Q_\xi \ltimes \mathbb{T})(E_{\text{test}}) \leq \mathcal{R}_{\mathbb{T}}(P_\xi) + \frac{h_{\text{sep}}^2}{16} \mathbf{1}_{F_{\text{bad}}}(\xi).$$

For exceptional vectors this follows from the probability bound by one and nonnegativity of risk; for the other vectors it follows by integrating the displayed pointwise loss bound. Since the same kernel is used for every ξ , finite averaging gives

$$(Q_{\text{mix}} \ltimes \mathbb{T})(E_{\text{test}}) = \int (Q_\xi \ltimes \mathbb{T})(E_{\text{test}}) \nu_{\text{cell}}(d\xi).$$

Averaging the member-wise risk inequality therefore yields

$$\frac{h_{\text{sep}}^2}{16} (Q_{\text{mix}} \ltimes \mathbb{T})(E_{\text{test}}) \leq \mathcal{R}_{\text{prior}}(\mathbb{T}) + \frac{h_{\text{sep}}^2}{16} \nu_{\text{cell}}(F_{\text{bad}}).$$

Under $P_\star$, an action in E_{test}^c satisfies

$$w - \tau(P_\star) > \left(\frac{1}{2} - \frac{h_{\text{sep}}}{2}\right) - \left(\frac{1}{2} - h_{\text{sep}}\right) = \frac{h_{\text{sep}}}{2}.$$

In particular, its squared loss is at least $h_{\text{sep}}^2/16$, so

$$\frac{h_{\text{sep}}^2}{16}(Q_\star \ltimes \mathbb{T})(E_{\text{test}}^c) \leq \mathcal{R}_\star(\mathbb{T}).$$

4. We next compare the two sample laws. For a single treated visit to any baseline cell x , averaging the fair coordinate ξ_x gives the following observed-mark probabilities:

(R, RY)	$\int P_\xi((R, RY) \mid X = x, A = 1) \nu_{\text{cell}}(d\xi)$	$P_\star((R, RY) \mid X = x, A = 1)$
$(1, 1)$	$q/2$	$\frac{1+q}{2} \frac{q}{1+q} = q/2$
$(1, 0)$	$1/2$	$\frac{1+q}{2} \left(1 - \frac{q}{1+q}\right) = 1/2$
$(0, 0)$	$\delta/2$	$1 - \frac{1+q}{2} = \delta/2$
$(0, 1)$	0	0

In the prior column, the first probability comes from $\xi_x = 1$ and arrival, the second from $\xi_x = 0$, and the third from $\xi_x = 1$ and nonarrival. A control visit has mark $(R, RY) = (1, 0)$ with probability one under every law. All surrogates equal zero.

Define the event of distinct treated baseline cells by

$$E_{\text{distinct}} := \bigcap_{1 \leq i < j \leq n} \{A_i \neq 1 \text{ or } A_j \neq 1 \text{ or } X_i \neq X_j\}.$$

For any sample atom $\mathbf{o} = (o_1, \dots, o_n)$ in this event, each treated-record probability depends on a different prior coordinate, while each control-record probability is constant in ξ . Independence of the prior coordinates and the one-visit calculation therefore give

$$\begin{aligned} Q_{\text{mix}}(\{\mathbf{o}\}) &= \int \prod_{i=1}^n \mathcal{L}_{P_\xi}(O_1)(\{o_i\}) \nu_{\text{cell}}(d\xi) \\ &= \prod_{i=1}^n \int \mathcal{L}_{P_\xi}(O_1)(\{o_i\}) \nu_{\text{cell}}(d\xi) \\ &= \prod_{i=1}^n \mathcal{L}_{P_\star}(O_1)(\{o_i\}) = Q_\star(\{\mathbf{o}\}). \end{aligned}$$

The equality also covers atoms with impossible surrogate or arrived-outcome marks, whose probabilities are zero under both constructions.

5. Under each P_ξ and under $P_\star$, the probability of any treated baseline cell is

$$P_\xi(A = 1, X = x) = P_\star(A = 1, X = x) = \frac{1}{2d}, \quad x \in \{0, \dots, d-1\}.$$

Thus independence of distinct observations gives, for $1 \leq i < j \leq n$,

$$Q_\xi\{A_i = A_j = 1, X_i = X_j\} \leq \sum_{x=0}^{d-1} \left(\frac{1}{2d}\right)^2 = \frac{1}{4d},$$

$$Q_\star\{A_i = A_j = 1, X_i = X_j\} \leq \frac{1}{4d}.$$

Taking a union over the $\binom{n}{2}$ pairs yields

$$Q_\xi(E_{\text{distinct}}^c), Q_\star(E_{\text{distinct}}^c) \leq \binom{n}{2} \frac{1}{4d} \leq \frac{n^2}{8d} \leq \frac{1}{8192}.$$

When $n = 1$, the pair set is empty and these collision probabilities are zero. Averaging the first bound over the prior gives

$$Q_{\text{mix}}(E_{\text{distinct}}^c) = \int Q_\xi(E_{\text{distinct}}^c) \nu_{\text{cell}}(d\xi) \leq \frac{1}{8192}.$$

On the finite sample space, agreement of the atom masses on E_{distinct} implies

$$\begin{aligned} \text{TV}(Q_{\text{mix}}, Q_\star) &\leq \frac{1}{2} \sum_{\mathbf{o}} |Q_{\text{mix}}(\{\mathbf{o}\}) - Q_\star(\{\mathbf{o}\})| \\ &\leq \frac{1}{2} \{Q_{\text{mix}}(E_{\text{distinct}}^c) + Q_\star(E_{\text{distinct}}^c)\} \leq \frac{1}{8192}. \end{aligned}$$

Indeed, the summands vanish on E_{distinct} , and elsewhere each absolute difference is bounded by the sum of the two atom masses. Applying [Lemma 2](#) to the common kernel and the measurable event E_{test} now gives

$$1 - \frac{1}{8192} \leq 1 - \text{TV}(Q_{\text{mix}}, Q_\star) \leq (Q_{\text{mix}} \times \mathbb{T})(E_{\text{test}}) + (Q_\star \times \mathbb{T})(E_{\text{test}}^c).$$

6. Under ν_{cell} , the independent fair coordinates have

$$E_{\nu_{\text{cell}}}[\xi_x] = \frac{1}{2}, \quad \text{Var}_{\nu_{\text{cell}}}(\xi_x) = \frac{1}{4}.$$

Consequently,

$$E_{\nu_{\text{cell}}} \left[\sum_{x=0}^{d-1} \xi_x \right] = \frac{d}{2}, \quad \text{Var}_{\nu_{\text{cell}}} \left(\sum_{x=0}^{d-1} \xi_x \right) = \frac{d}{4}.$$

The regime $n^{-1} \leq \delta^2$, together with $\delta \geq 0$, implies

$$\delta > 0, \quad h_{\text{sep}} \geq \frac{\delta}{4} > 0, \quad 1 \leq n\delta^2, \quad \delta^2 \leq 16h_{\text{sep}}^2.$$

These inequalities and the alphabet-size condition give the concentration budget

$$dh_{\text{sep}}^2 \geq 1024n^2h_{\text{sep}}^2 \geq 64n^2\delta^2 \geq 64n \geq 64.$$

Since $\tau(P_\xi) = d^{-1} \sum_x \xi_x$, Chebyshev's inequality applied to the sum of the prior coordinates yields

$$\begin{aligned} \nu_{\text{cell}}(F_{\text{bad}}) &= \nu_{\text{cell}} \left\{ \left| \sum_{x=0}^{d-1} \xi_x - \frac{d}{2} \right| > \frac{dh_{\text{sep}}}{4} \right\} \\ &\leq \frac{d/4}{(dh_{\text{sep}}/4)^2} = \frac{4}{dh_{\text{sep}}^2} \leq \frac{1}{16}. \end{aligned}$$

7. Multiplying the testing and concentration bounds by the nonnegative factor $h_{\text{sep}}^2/16$ gives

$$\begin{aligned} \frac{h_{\text{sep}}^2}{16} \left(1 - \frac{1}{8192}\right) &\leq \frac{h_{\text{sep}}^2}{16} \{(Q_{\text{mix}} \times \mathbb{T})(E_{\text{test}}) + (Q_{\star} \times \mathbb{T})(E_{\text{test}}^c)\}, \\ \frac{h_{\text{sep}}^2}{16} \nu_{\text{cell}}(F_{\text{bad}}) &\leq \frac{h_{\text{sep}}^2}{256}. \end{aligned}$$

Combining these with the two test-risk inequalities established above, we obtain

$$\begin{aligned} \mathcal{R}_{\text{prior}}(\mathbb{T}) + \mathcal{R}_{\star}(\mathbb{T}) &\geq \frac{h_{\text{sep}}^2}{16} \left\{1 - \frac{1}{8192} - \frac{1}{16}\right\} \\ &\geq \frac{h_{\text{sep}}^2}{32}, \end{aligned}$$

where the last inequality uses $1 - 1/8192 - 1/16 \geq 1/2$.

8. To pass to worst-case risk, define

$$\mathcal{R}_{\text{sup}}(\mathbb{T}) := \sup_{P \in \mathcal{M}_{n,d,q}^+} \mathcal{R}_{\mathbb{T}}(P).$$

Binary potential outcomes give $\tau(P) \in [-1, 1]$. Since the kernel takes values in $[-1, 1]$, its loss and risk satisfy

$$0 \leq (w - \tau(P))^2 \leq 4 \quad (w \in [-1, 1]), \quad 0 \leq \mathcal{R}_{\mathbb{T}}(P) \leq 4.$$

Thus the member-law risks are bounded above.

Every constructed law belongs to the model class in [Definition 7](#). The independent fair treatment draw verifies [Assumptions 2](#) and [3](#), and the defining identities for the realized variables verify [Assumptions 4](#) and [5](#). Within an occupied cell, the outcome under P_{ξ} is deterministic, while under $P_{\star}$ its arrival draw is independent of the outcome; these properties verify [Assumption 6](#). For every $a \in \{0, 1\}$ and $x \in \{0, \dots, d-1\}$, the cell probabilities of [Definition 6](#) are

$$p_{ax0}(P_{\xi}) = p_{ax0}(P_{\star}) = \frac{1}{2d}, \quad p_{ax1}(P_{\xi}) = p_{ax1}(P_{\star}) = 0.$$

On the occupied cells, their arrival probabilities are

$$\rho_{ax0}(P_{\xi}) = 1 - a\delta\xi_x \geq q, \quad \rho_{ax0}(P_{\star}) = \begin{cases} 1, & a = 0, \\ (1+q)/2, & a = 1, \end{cases} \geq q.$$

This verifies [Assumption 7](#), and hence

$$P_{\xi}, P_{\star} \in \mathcal{M}_{n,d,q}^+.$$

Each member-law risk is therefore at most $\mathcal{R}_{\text{sup}}(\mathbb{T})$. Since ν_{cell} is a probability law,

$$\mathcal{R}_{\text{prior}}(\mathbb{T}) \leq \int \mathcal{R}_{\text{sup}}(\mathbb{T}) \nu_{\text{cell}}(d\xi) = \mathcal{R}_{\text{sup}}(\mathbb{T}), \quad \mathcal{R}_{\star}(\mathbb{T}) \leq \mathcal{R}_{\text{sup}}(\mathbb{T}).$$

It follows that

$$2\mathcal{R}_{\text{sup}}(\mathbb{T}) \geq \mathcal{R}_{\text{prior}}(\mathbb{T}) + \mathcal{R}_{\star}(\mathbb{T}) \geq \frac{h_{\text{sep}}^2}{32}.$$

Using $h_{\text{sep}} \geq \delta/4 \geq 0$, we conclude

$$\mathcal{R}_{\text{sup}}(\mathbb{T}) \geq \frac{h_{\text{sep}}^2}{64} \geq \frac{\delta^2}{1024}.$$

This holds for every kernel in the stated decision class. Taking its infimum, as in [Definition 12](#), proves

$$\frac{\delta^2}{1024} \leq \mathfrak{R}_{n,d,q}^+$$

in the remaining regime.

9. The two cases establish the deficit lower bound throughout the stated parameter range. Combining it with the initial bounds gives

$$\max \left\{ \frac{1}{256n}, \frac{\delta^2}{1024} \right\} \leq \mathfrak{R}_{n,d,q}^+ \leq \frac{4}{n} + \delta^2.$$

Adding the two lower inequalities and dividing by two yields

$$\mathfrak{R}_{n,d,q}^+ \geq \frac{1}{512n} + \frac{\delta^2}{2048} \geq \frac{1}{2048} \left\{ \frac{1}{n} + \delta^2 \right\}.$$

Finally, $\delta^2 \geq 0$ gives

$$\mathfrak{R}_{n,d,q}^+ \leq \frac{4}{n} + \delta^2 \leq 4 \left\{ \frac{1}{n} + \delta^2 \right\}.$$

Substituting $\delta = 1 - q$ proves the asserted envelope. □

Proof of Proposition 1. 1. We first record two bounds. By [Theorem 1](#), for every $n, d \geq 1$,

$$\mathfrak{R}_{n,d,q_n}^+ \leq \frac{4}{n} + (1 - q_n)^2.$$

For $n \geq 1$, the alphabet size $1024n^2$ satisfies

$$1 \leq 1024n^2.$$

Thus [Theorem 2](#), applied at this alphabet size, supplies

$$\frac{1}{2048} \left\{ \frac{1}{n} + (1 - q_n)^2 \right\} \leq \mathfrak{R}_{n,1024n^2,q_n}^+.$$

Multiplication by $n > 0$, using

$$n \cdot \frac{1}{2048} \left\{ \frac{1}{n} + (1 - q_n)^2 \right\} = \frac{1}{2048} \{1 + n(1 - q_n)^2\},$$

gives

$$\frac{1}{2048} \{1 + n(1 - q_n)^2\} \leq n\mathfrak{R}_{n,1024n^2,q_n}^+.$$

2. Multiplying the upper bound from the preceding step by $n > 0$ yields, for every $d \geq 1$,

$$n\mathfrak{R}_{n,d,q_n}^+ \leq n \left\{ \frac{4}{n} + (1 - q_n)^2 \right\} = 4 + n(1 - q_n)^2.$$

For any real b satisfying the stated strict inequality, the same step gives

$$b < \frac{1}{2048} \{1 + n(1 - q_n)^2\} \leq n\mathfrak{R}_{n,1024n^2,q_n}^+.$$

Hence $d = 1024n^2$ is the required integer witness.

3. Suppose first that the arrival-rate condition holds with $M \geq 0$. For all sufficiently large $n \geq 1$,

$$1 - q_n \leq \frac{M}{\sqrt{n}}.$$

Since $q_n \leq 1$ and $\sqrt{n} > 0$, this implies

$$0 \leq (1 - q_n)\sqrt{n} \leq M.$$

Squaring and using $(\sqrt{n})^2 = n$, we obtain

$$n(1 - q_n)^2 = ((1 - q_n)\sqrt{n})^2 \leq M^2, \quad (1 - q_n)^2 \leq \frac{M^2}{n}.$$

Consequently, for every $d \geq 1$ and all such n ,

$$\mathfrak{R}_{n,d,q_n}^+ \leq \frac{4}{n} + (1 - q_n)^2 \leq \frac{4}{n} + \frac{M^2}{n} = \frac{4 + M^2}{n}.$$

This proves the uniform risk bound with $D = 4 + M^2$.

Conversely, suppose the uniform risk bound holds with a real constant D . Define the nonnegative real constant

$$M_{\text{risk}} := \sqrt{\max\{2048D, 0\}}.$$

For all sufficiently large $n \geq 1$, applying that bound at $d = 1024n^2$ and multiplying by n gives

$$n\mathfrak{R}_{n,1024n^2,q_n}^+ \leq D.$$

Combining this with the lower bound from the first step yields

$$\frac{1}{2048} \{1 + n(1 - q_n)^2\} \leq D.$$

In particular,

$$(\sqrt{n}(1 - q_n))^2 = n(1 - q_n)^2 \leq 2048D \leq \max\{2048D, 0\} = M_{\text{risk}}^2.$$

Both $\sqrt{n}(1 - q_n)$ and M_{risk} are nonnegative, so

$$\sqrt{n}(1 - q_n) \leq M_{\text{risk}}, \quad 1 - q_n \leq \frac{M_{\text{risk}}}{\sqrt{n}}.$$

This is the arrival-rate condition.

4. The lower bound in [Theorem 1](#) directly supplies, for every $n, d \geq 1$,

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,q_n}^+.$$

5. Finally, assume that $(\sqrt{n}(1 - q_n))_{n \geq 0}$ is unbounded above, and fix $C \in \mathbb{R}$. Suppose, for contradiction, that the claimed strict risk inequality holds for only finitely many integers n . Choose an integer $n_{\text{tail}} \geq 1$ such that

$$n\mathfrak{R}_{n,1024n^2,q_n}^+ \leq C \quad \text{for every } n \geq n_{\text{tail}}.$$

Define the nonnegative real constant

$$M_{\text{tail}} := \sqrt{\max\{2048C, 0\}}.$$

For every $n \geq n_{\text{tail}}$, the first step implies

$$\frac{1}{2048} \{1 + n(1 - q_n)^2\} \leq C,$$

and hence

$$(\sqrt{n}(1 - q_n))^2 = n(1 - q_n)^2 \leq 2048C \leq \max\{2048C, 0\} = M_{\text{tail}}^2.$$

Nonnegativity therefore gives

$$\sqrt{n}(1 - q_n) \leq M_{\text{tail}} \quad (n \geq n_{\text{tail}}).$$

The remaining terms form a finite set. Explicitly, the finite maximum below exists and bounds every term:

$$\sqrt{n}(1 - q_n) \leq \max \left(\{M_{\text{tail}}\} \cup \left\{ \sqrt{n'}(1 - q_{n'}) : n' \in \mathbb{N}, n' < n_{\text{tail}} \right\} \right) \quad (n \in \mathbb{N}).$$

This contradicts unboundedness above. Thus there are infinitely many integers n for which

$$C < n\mathfrak{R}_{n,1024n^2,q_n}^+.$$

□

Proof of [Proposition 2](#). 1. For all sufficiently large n , we have

$$n \geq 1, \quad 1 - q_n \leq \frac{M}{\sqrt{n}}.$$

Fix such an n . Since $d_n \geq 1$ and $0 < q_n \leq 1$, [Theorem 1](#) applies. Retaining the last term in its upper-bound minimum gives

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d_n,q_n}^+ \leq \frac{4}{n} + (1 - q_n)^2.$$

2. Since $n > 0$, we have $\sqrt{n} > 0$, so multiplying the deficit bound by $\sqrt{n}$ yields

$$(1 - q_n)\sqrt{n} \leq M.$$

Moreover, $q_n \leq 1$ implies

$$0 \leq (1 - q_n)\sqrt{n}.$$

Thus, using $M \geq 0$, we may square the inequality to obtain

$$((1 - q_n)\sqrt{n})^2 \leq M^2.$$

The identity $(\sqrt{n})^2 = n$ then gives

$$n(1 - q_n)^2 = ((1 - q_n)\sqrt{n})^2 \leq M^2.$$

Dividing by $n > 0$, we conclude that

$$(1 - q_n)^2 \leq \frac{M^2}{n}.$$

3. Substituting this bound into the upper estimate gives

$$\mathfrak{R}_{n,d,q_n}^+ \leq \frac{4}{n} + (1 - q_n)^2 \leq \frac{4}{n} + \frac{M^2}{n} = \frac{4 + M^2}{n}.$$

Together with the lower estimate, this proves both asserted inequalities for all sufficiently large n . □

Proof of Theorem 3. Fix integers $n, d \geq 1$, with sampling as in Assumption 1.

1. Since $0 < 1 \leq 1$, specialize Theorem 1 to $q = 1$. Its lower bound and observed-outcome contrast bound give

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,1}^+, \quad \sup_{P \in \mathcal{M}_{n,d,1}^+} E_P \left[\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \}^2 \right] \leq \frac{4}{n} + (1 - 1)^2 = \frac{4}{n}.$$

2. The estimator in Definition 18 is measurable, takes values in $[-1, 1]$, and depends on the observed sample and the fixed parameters n, d . Define its deterministic probability kernel by

$$\mathsf{T}_{\text{HT}}(\mathbf{o}, E_{\text{out}}) := \mathbf{1} \{ \hat{\tau}_n^{\text{HT}}(\mathbf{o}) \in E_{\text{out}} \}, \quad \mathbf{o} \in (\{0, \dots, d-1\} \times \{0, 1\}^4)^n, \quad E_{\text{out}} \subseteq [-1, 1] \text{ Borel}.$$

This parameter-independent kernel belongs to the decision class $\mathcal{T}_n$ of Definition 12. For every $\mathsf{T} \in \mathcal{T}_n$ and $P \in \mathcal{M}_{n,d,1}^+$, squared loss gives

$$0 \leq \int \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}(\mathbf{o}, dh) \mathcal{L}_P(\mathbf{O}_n)(d\mathbf{o}).$$

Evaluating the defining minimax infimum at T_{HT} therefore yields

$$\mathfrak{R}_{n,d,1}^+ \leq \sup_{P \in \mathcal{M}_{n,d,1}^+} \int \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}_{\text{HT}}(\mathbf{o}, dh) \mathcal{L}_P(\mathbf{O}_n)(d\mathbf{o}).$$

By the definition of this kernel, for every such P ,

$$\int \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}_{\text{HT}}(\mathbf{o}, dh) \mathcal{L}_P(\mathbf{O}_n)(d\mathbf{o}) = E_P \left[\left\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right].$$

Consequently,

$$\mathfrak{R}_{n,d,1}^+ \leq \sup_{P \in \mathcal{M}_{n,d,1}^+} E_P \left[\left\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right].$$

3. Combining these inequalities gives

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d,1}^+ \leq \sup_{P \in \mathcal{M}_{n,d,1}^+} E_P \left[\left\{ \hat{\tau}_n^{\text{HT}}(\mathbf{O}_n) - \tau(P) \right\}^2 \right] \leq \frac{4}{n}.$$

The constants $1/256$ and 4 are independent of n and d , establishing order n^{-1} uniformly over $d \geq 1$.

□

Proof of Proposition 3. 1. Choose the real comparison constants

$$c := \frac{1}{256}, \quad C := 4.$$

Then $0 < c \leq C$. Fix any sequence of natural alphabet sizes $(d_n)_{n \geq 0}$ satisfying $d_n \geq 1$ for every n . Define the arrival-floor sequence and deficit constant by

$$q_n := 1 \quad (n \geq 0), \quad M := 0.$$

Thus $M \geq 0$ and $q_n \in (0, 1]$ for every n . For every $n \geq 1$,

$$1 - q_n = 0 = \frac{M}{\sqrt{n}},$$

so the arrival-deficit condition holds for all sufficiently large n .

2. Applying Proposition 2 with these choices gives, for all sufficiently large n ,

$$\frac{1}{256n} \leq \mathfrak{R}_{n,d_n,1}^+ \leq \frac{4}{n},$$

since $q_n = 1$ and $4 + M^2 = 4$. These are precisely the required bounds with the chosen c and C . Both constants were fixed before choosing the alphabet-size sequence, so they are independent of that sequence.

□

Proof of Theorem 4. 1. Choose universal constants $\underline{c}$ and C_0 supplied by Theorems 11 and 12, respectively, and set

$$0 < \underline{c}, \quad 0 < C_0, \quad \overline{C} := \max\{\underline{c}, C_0\}.$$

Thus

$$0 < \underline{c} \leq \overline{C} < \infty, \quad C_0 \leq \overline{C}.$$

Fix n, d, q satisfying the hypotheses of the theorem. By Theorem 12,

$$\underline{c} r_{n,d,q} \leq \mathfrak{R}_{n,d,q}.$$

2. We next construct an admissible decision kernel from the deterministic estimator in [Algorithm 1](#). For each observed sample o , its defining finite average is

$$\hat{\tau}_{n,d,q}^{\text{mix}}(o) = \sum_{\nu=0}^n \Pr_{\omega}(K_0 = \nu) E_{\omega}[\tilde{\tau}(o; \omega) \mid K_0 = \nu].$$

The omitted Poisson tail contributes zero by the estimator's definition. Each conditional expectation is the uniform average over the 3^{ν} stream assignments. Clipping, together with the zero-output branches, gives

$$-1 \leq \tilde{\tau}(o; \omega) \leq 1, \quad -1 \leq E_{\omega}[\tilde{\tau}(o; \omega) \mid K_0 = \nu] \leq 1 \quad (0 \leq \nu \leq n).$$

The prefix weights satisfy

$$\Pr_{\omega}(K_0 = \nu) \geq 0, \quad \sum_{\nu=0}^n \Pr_{\omega}(K_0 = \nu) = \Pr_{\omega}(K_0 \leq n) \leq 1.$$

Consequently, at every observed sample,

$$-1 \leq -\Pr_{\omega}(K_0 \leq n) \leq \hat{\tau}_{n,d,q}^{\text{mix}}(o) \leq \Pr_{\omega}(K_0 \leq n) \leq 1.$$

The observed-sample space is finite with measurable singletons, so this deterministic map is measurable. Its Dirac kernel,

$$\mathsf{T}_{\text{mix}}(o, dh) := \delta_{\hat{\tau}_{n,d,q}^{\text{mix}}(o)}(dh),$$

therefore belongs to the decision class $\mathcal{T}_n$ of [Definition 16](#).

3. For every $\mathsf{T} \in \mathcal{T}_n$ and $P \in \mathcal{M}_{n,d,q}$, the joint squared-error risk is nonnegative:

$$E_P[(\mathsf{T} - \tau(P))^2] = \int \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}(o, dh) \mathcal{L}_P(\mathbf{O}_n)(do) \geq 0,$$

because both integrals have nonnegative integrands. For the Dirac kernel just constructed,

$$E_P[(\mathsf{T}_{\text{mix}} - \tau(P))^2] = E_P[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2].$$

Moreover,

$$nq > 0, \quad r_{n,d,q} = \min \left\{ 1, \frac{1}{nq} + \left[\frac{d}{nq \log(e + nq)} \right]^2 \right\} \geq 0.$$

For each $P \in \mathcal{M}_{n,d,q}$, [Theorem 11](#) now yields the uniform bound

$$E_P[(\mathsf{T}_{\text{mix}} - \tau(P))^2] \leq \mathfrak{U}_{n,d,q} \leq C_0 r_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

The last inequality uses $C_0 \leq \overline{C}$ and $r_{n,d,q} \geq 0$.

4. Nonnegativity of all the risks permits comparison of the minimax value with the worst-case risk of any admissible estimator. In particular,

$$\mathfrak{R}_{n,d,q} \leq \sup_{P \in \mathcal{M}_{n,d,q}} E_P[(\mathsf{T}_{\text{mix}} - \tau(P))^2].$$

If $\mathcal{M}_{n,d,q}$ is nonempty, the uniform bound from the preceding step gives

$$\sup_{P \in \mathcal{M}_{n,d,q}} E_P[(T_{\text{mix}} - \tau(P))^2] \leq \bar{C} r_{n,d,q}.$$

If $\mathcal{M}_{n,d,q}$ is empty, the real-supremum convention assigns its worst-case risk the value zero. Since $\bar{C} \geq C_0 > 0$ and $r_{n,d,q} \geq 0$, in this case

$$\sup_{P \in \mathcal{M}_{n,d,q}} E_P[(T_{\text{mix}} - \tau(P))^2] = 0 \leq \bar{C} r_{n,d,q}.$$

Thus, in either case,

$$\underline{c} r_{n,d,q} \leq \mathfrak{R}_{n,d,q} \leq \bar{C} r_{n,d,q}.$$

The constants were chosen independently of n, d, q , as required. □

Proof of Proposition 4. Write

$$N(t) = n(t)q(t) > 0, \quad \ell(t) = \log(e + N(t)).$$

Define the restricted model class and its minimax risk by

$$\mathcal{M}_{n,d,q}^{(0)} := \{P \in \mathcal{M}_{n,d,q} : P(S(0) = 0, S(1) = 0) = 1\}, \quad \mathfrak{R}_{n,d,q}^{(0)} := \inf_{T \in \mathcal{T}_n} \sup_{P \in \mathcal{M}_{n,d,q}^{(0)}} E_P[(T - \tau(P))^2].$$

Here the estimator class and joint sample-and-kernel expectation are those of Definition 16.

1. Since $N(t) \rightarrow \infty$, eventually

$$N(t) \geq \max\{2, d^2\}.$$

Also $\ell(t) \geq 1$, because $e + N(t) \geq e$. Consequently, on this eventual range,

$$\left[\frac{d}{N(t)\ell(t)} \right]^2 = \frac{d^2}{N(t)^2 \ell(t)^2} \leq \frac{1}{N(t)\ell(t)^2} \leq N(t)^{-1}.$$

The expression in Definition 17 is nonnegative for every t . Eventually $N(t)^{-1} \leq 1$, and both entries of its minimum are at least $N(t)^{-1}$. Thus

$$N(t)^{-1} \leq r_{n(t),d,q(t)} = \min \left\{ 1, N(t)^{-1} + \left[\frac{d}{N(t)\ell(t)} \right]^2 \right\} \leq 2N(t)^{-1}.$$

These inequalities prove the first order comparison.

2. Let

$$0 < \underline{c}_{\text{match}} \leq \bar{C}_{\text{match}} < \infty$$

be the constants supplied by Theorem 4. Its hypotheses hold at every t : the sample and alphabet sizes are positive, $0 < q(t) \leq 1$, and the rare-arrival restriction is assumed. Hence

$$\underline{c}_{\text{match}} r_{n(t),d,q(t)} \leq \mathfrak{R}_{n(t),d,q(t)} \leq \bar{C}_{\text{match}} r_{n(t),d,q(t)} \quad (t \in \mathbb{N}).$$

In particular, both quantities are nonnegative, and the lower inequality also gives

$$r_{n(t),d,q(t)} \leq \underline{c}_{\text{match}}^{-1} \mathfrak{R}_{n(t),d,q(t)}.$$

Combining these bounds with the eventual frontier bounds from the preceding step yields

$$\underline{c}_{\text{match}} N(t)^{-1} \leq \mathfrak{R}_{n(t),d,q(t)} \leq 2\bar{C}_{\text{match}} N(t)^{-1}.$$

3. We next obtain a pointwise upper bound for the restricted risk. For each t , use the estimator of [Algorithm 1](#) with parameters $n(t), d, q(t)$. Its auxiliary output is zero or a value clipped to $[-1, 1]$, so its conditional average satisfies

$$-1 = E_\omega[-1 \mid \mathbf{O}_{n(t)}] \leq \hat{\tau}_{n(t), d, q(t)}^{\text{mix}}(\mathbf{O}_{n(t)}) \leq E_\omega[1 \mid \mathbf{O}_{n(t)}] = 1.$$

The average is measurable because it is a finite sum over the admissible prefix lengths and assignments. Therefore the deterministic kernel

$$\mathbb{T}_t^{\text{mix}}(o, dh) := \text{Dirac}_{\hat{\tau}_{n(t), d, q(t)}^{\text{mix}}(o)}(dh)$$

belongs to $\mathcal{T}_{n(t)}$. Its joint risk equals its deterministic risk:

$$E_P[(\mathbb{T}_t^{\text{mix}} - \tau(P))^2] = E_P\left[\left(\hat{\tau}_{n(t), d, q(t)}^{\text{mix}}(\mathbf{O}_{n(t)}) - \tau(P)\right)^2\right].$$

Let

$$0 < \overline{C}_{\text{mix}} < \infty$$

be supplied by [Theorem 11](#). For every $P \in \mathcal{M}_{n(t), d, q(t)}^{(0)}$, membership in the full model class allows us to apply its risk-envelope bound:

$$E_P[(\mathbb{T}_t^{\text{mix}} - \tau(P))^2] \leq \mathfrak{U}_{n(t), d, q(t)} \leq \overline{C}_{\text{mix}} r_{n(t), d, q(t)}.$$

All squared risks are nonnegative, so the minimax value is bounded above by the worst-case risk of this admissible kernel. If the restricted class is nonempty, taking its supremum gives

$$\mathfrak{R}_{n(t), d, q(t)}^{(0)} \leq \sup_{P \in \mathcal{M}_{n(t), d, q(t)}^{(0)}} E_P[(\mathbb{T}_t^{\text{mix}} - \tau(P))^2] \leq \overline{C}_{\text{mix}} r_{n(t), d, q(t)}. \quad (2)$$

If the restricted class is empty, its real-valued worst-case risk is zero under the empty-supremum convention. The same inequality follows from

$$0 \leq \overline{C}_{\text{mix}} r_{n(t), d, q(t)}.$$

Thus the upper bound holds for every t .

4. To derive the restricted lower bound, fix t . For $\mu_{\text{test}} \in [1/4, 3/4]$, define $P_{\mu_{\text{test}}}^{(t)}$ by

$$X = 0, \quad S(0) = S(1) = Y(0) = 0, \quad A \sim \text{Bernoulli}(1/2), \quad Y(1) \sim \text{Bernoulli}(\mu_{\text{test}}), \quad R \sim \text{Bernoulli}(q)$$

where $A, Y(1), R$ are mutually independent, and set $S = S(A), Y = Y(A)$. The category 0 exists because $d \geq 1$. Take the observed sample independently from the induced observed-record law.

This construction satisfies randomization and consistency, and independence of R from (X, A, S, Y) implies the required conditional arrival independence. Its occupied cells satisfy

$$p_{a, 0, 0}(P_{\mu_{\text{test}}}^{(t)}) = \frac{1}{2}, \quad \rho_{a, 0, 0}(P_{\mu_{\text{test}}}^{(t)}) = q(t) \quad (a \in \{0, 1\}),$$

while every other cell has zero membership probability. Together with the assumed parameter restrictions and rare-arrival slice, these facts establish

$$P_{\mu_{\text{test}}}^{(t)} \in \mathcal{M}_{n(t), d, q(t)}^{(0)}, \quad \tau(P_{\mu_{\text{test}}}^{(t)}) = E_{P_{\mu_{\text{test}}}^{(t)}} [Y(1) - Y(0)] = \mu_{\text{test}}.$$

Define the separation and the two outcome means by

$$\delta(t) := \min \left\{ \frac{1}{4}, \frac{1}{4\sqrt{N(t)}} \right\}, \quad u_0(t) := \frac{1}{2}, \quad u_1(t) := \frac{1}{2} + \delta(t).$$

Since $N(t) > 0$,

$$0 < \delta(t) \leq \frac{1}{4}, \quad u_0(t), u_1(t) \in [1/4, 3/4].$$

For $i \in \{0, 1\}$, define the single-record and sample laws by

$$Q_i^{\text{obs}}(t) := \mathcal{L}_{P_{u_i(t)}^{(t)}}(O_1), \quad Q_i^{\text{sam}}(t) := \mathcal{L}_{P_{u_i(t)}^{(t)}}(\mathbf{O}_{n(t)}) = (Q_i^{\text{obs}}(t))^{\otimes n(t)}.$$

For finite probability laws $Q_1 \ll Q_0$, use the divergence and total-variation conventions

$$\chi^2(Q_1 \| Q_0) := \sum_{o: Q_0(\{o\}) > 0} \frac{(Q_1(\{o\}) - Q_0(\{o\}))^2}{Q_0(\{o\})}, \quad \text{TV}(Q_1, Q_0) := \sup_{\mathcal{A}} |Q_1(\mathcal{A}) - Q_0(\mathcal{A})| = \frac{1}{2} \sum_o |Q_1(\{o\}) - Q_0(\{o\})|.$$

Under $P_{\mu_{\text{test}}}^{(t)}$, every observed record has $X = S = 0$. The remaining coordinates, with arrived outcome $RY = RY$, have probabilities

(A, R, RY)	$(0, 0, 0)$	$(1, 0, 0)$	$(0, 1, 0)$	$(1, 1, 0)$	$(1, 1, 1)$
Probability	$\frac{1 - q(t)}{2}$	$\frac{1 - q(t)}{2}$	$\frac{q(t)}{2}$	$\frac{q(t)(1 - \mu_{\text{test}})}{2}$	$\frac{q(t)\mu_{\text{test}}}{2}$

and all other records have probability zero. At $q(t) = 1$, the first two atoms have zero probability under both laws and are omitted from the divergence sum. Thus $Q_1^{\text{obs}}(t) \ll Q_0^{\text{obs}}(t)$, and summing the atom probabilities gives

$$\begin{aligned} 1 + \chi^2(Q_1^{\text{obs}}(t) \| Q_0^{\text{obs}}(t)) &= 1 - q(t) + \frac{q(t)}{2} + q(t)((1 - u_1(t))^2 + u_1(t)^2) \\ &= 1 + 2q(t)\delta(t)^2. \end{aligned}$$

The definition of $\delta(t)$ gives

$$16N(t)\delta(t)^2 \leq 1, \quad \chi^2(Q_1^{\text{obs}}(t) \| Q_0^{\text{obs}}(t)) = 2q(t)\delta(t)^2 \leq \frac{1}{8n(t)}.$$

Factoring the finite product sums and using $1 + x \leq e^x$ now yields

$$\begin{aligned} 1 + \chi^2(Q_1^{\text{sam}}(t) \| Q_0^{\text{sam}}(t)) &= \left(1 + \chi^2(Q_1^{\text{obs}}(t) \| Q_0^{\text{obs}}(t))\right)^{n(t)} \\ &\leq \exp\left(n(t)\chi^2(Q_1^{\text{obs}}(t) \| Q_0^{\text{obs}}(t))\right) \leq e^{1/8}. \end{aligned}$$

Since

$$e^{1/2}e^{1/2} = e < 3, \quad e^{1/8} \leq e^{1/2} < \sqrt{3} < 2,$$

the sample divergence is less than 1. Cauchy–Schwarz applied to the finite total-variation sum therefore gives

$$\text{TV}(Q_1^{\text{sam}}(t), Q_0^{\text{sam}}(t)) \leq \frac{1}{2} \sqrt{\chi^2(Q_1^{\text{sam}}(t) \| Q_0^{\text{sam}}(t))} \leq \frac{1}{2}.$$

Consequently, every measurable sample event $\mathcal{A}$ satisfies

$$Q_1^{\text{sam}}(t)(\mathcal{A}^c) + Q_0^{\text{sam}}(t)(\mathcal{A}) \geq 1 - \text{TV}(Q_1^{\text{sam}}(t), Q_0^{\text{sam}}(t)) \geq \frac{1}{2}.$$

5. Let $\mathsf{T} \in \mathcal{T}_{n(t)}$ be arbitrary, and define its barycenter and the relevant sample events by

$$b_{\mathsf{T}}(o) := \int_{[-1,1]} h \mathsf{T}(o, dh),$$

$$\mathcal{E}_{\mathsf{T},i} := \left\{ o : \frac{\delta(t)}{2} \leq |b_{\mathsf{T}}(o) - u_i(t)| \right\} \quad (i \in \{0,1\}), \quad \mathcal{A}_{\mathsf{T}} := \{o : |b_{\mathsf{T}}(o) - u_1(t)| \leq |b_{\mathsf{T}}(o) - u_0(t)|\}.$$

These events are measurable. The triangle inequality gives, at every sample,

$$\delta(t) = |u_1(t) - u_0(t)| \leq |b_{\mathsf{T}}(o) - u_0(t)| + |b_{\mathsf{T}}(o) - u_1(t)|.$$

It follows that

$$\mathcal{A}_{\mathsf{T}} \subseteq \mathcal{E}_{\mathsf{T},0}, \quad \mathcal{A}_{\mathsf{T}}^c \subseteq \mathcal{E}_{\mathsf{T},1}.$$

Applying the testing bound from the preceding step and then these inclusions,

$$\frac{1}{2} \leq Q_1^{\text{sam}}(t)(\mathcal{A}_{\mathsf{T}}^c) + Q_0^{\text{sam}}(t)(\mathcal{A}_{\mathsf{T}}) \leq Q_1^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},1}) + Q_0^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},0}).$$

Thus

$$\frac{1}{4} \leq \max_{i \in \{0,1\}} Q_i^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},i}).$$

On $\mathcal{E}_{\mathsf{T},i}$, the barycenter's squared error is at least $(\delta(t)/2)^2$. The barycenter inequality of [Lemma 1](#) therefore supplies

$$\left(\frac{\delta(t)}{2}\right)^2 Q_i^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},i}) \leq \int (b_{\mathsf{T}}(o) - u_i(t))^2 Q_i^{\text{sam}}(t)(do) \leq E_{P_{u_i(t)}^{(t)}} [(\mathsf{T} - u_i(t))^2].$$

Whether the probability for $i = 0$ is at most the probability for $i = 1$, or conversely, multiplication by the nonnegative factor $(\delta(t)/2)^2$ preserves their order. Hence

$$\begin{aligned} \frac{\delta(t)^2}{16} &\leq \left(\frac{\delta(t)}{2}\right)^2 \max_{i \in \{0,1\}} Q_i^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},i}) \\ &= \max_{i \in \{0,1\}} \left\{ \left(\frac{\delta(t)}{2}\right)^2 Q_i^{\text{sam}}(t)(\mathcal{E}_{\mathsf{T},i}) \right\} \\ &\leq \max_{i \in \{0,1\}} E_{P_{u_i(t)}^{(t)}} [(\mathsf{T} - u_i(t))^2]. \end{aligned}$$

6. The separation also has the required lower calibration. If $1/4 \leq 1/(4\sqrt{N(t)})$, then

$$\delta(t)^2 = \frac{1}{16} \geq \frac{1}{16} \min\{1, N(t)^{-1}\}.$$

In the other case,

$$\delta(t)^2 = \frac{1}{16N(t)} \geq \frac{1}{16} \min\{1, N(t)^{-1}\}.$$

Combining either inequality with the preceding risk bound gives, for every $\mathbb{T} \in \mathcal{T}_{n(t)}$,

$$\frac{1}{256} \min\{1, N(t)^{-1}\} \leq \frac{\delta(t)^2}{16} \leq \max_{i \in \{0,1\}} E_{P_{u_i(t)}^{(t)}} [(\mathbb{T} - u_i(t))^2].$$

The estimator class is nonempty: the constant kernel

$$\mathbb{T}^0(o, dh) := \text{Dirac}_0(dh)$$

is admissible. Moreover, binary potential outcomes imply $\tau(P) \in [-1, 1]$, and the kernel outputs lie in $[-1, 1]$, so

$$0 \leq E_P[(\mathbb{T} - \tau(P))^2] \leq 4$$

for every full-data law P and admissible $\mathbb{T}$. Both testing laws belong to the full model and to its restricted subclass. Their maximum risk therefore bounds each corresponding worst-case risk from below; the uniform bound by 4 ensures these suprema are finite. Taking the infimum over estimators gives

$$\frac{1}{256} \min\{1, N(t)^{-1}\} \leq \mathfrak{R}_{n(t),d,q(t)}, \quad \frac{1}{256} \min\{1, N(t)^{-1}\} \leq \mathfrak{R}_{n(t),d,q(t)}^{(0)}.$$

7. The last lower bound establishes nonnegativity of the restricted risk for every t . Its upper bound in [Equation \(2\)](#), combined with the eventual frontier bound, gives

$$0 \leq \mathfrak{R}_{n(t),d,q(t)}^{(0)} \leq 2\overline{C}_{\text{mix}} N(t)^{-1} \quad \text{eventually.}$$

Also $N(t) \rightarrow \infty$ implies $N(t) \geq 1$ eventually, so

$$\min\{1, N(t)^{-1}\} = N(t)^{-1}.$$

The lower bound consequently becomes

$$\frac{1}{256} N(t)^{-1} \leq \mathfrak{R}_{n(t),d,q(t)}^{(0)}, \quad \text{equivalently} \quad N(t)^{-1} \leq 256 \mathfrak{R}_{n(t),d,q(t)}^{(0)}.$$

Together these inequalities prove the restricted-risk order comparison, and hence all three assertions.

□

Proof of Proposition 5. 1. For every $t \in \mathbb{N}$, write

$$\ell_t := \log(e + N_t), \quad r_t := r_{n_t, d_t, q_t} = \min \left\{ 1, N_t^{-1} + \left(\frac{d_t}{N_t \ell_t} \right)^2 \right\}, \quad \mathfrak{R}_t := \mathfrak{R}_{n_t, d_t, q_t}.$$

The expression for r_t is supplied by Definition 17. The size and arrival assumptions give

$$N_t = n_t q_t > 0, \quad N_t^{-1} + \left(\frac{d_t}{N_t \ell_t} \right)^2 \geq 0, \quad r_t \geq 0.$$

All hypotheses of Theorem 4 hold at every t . Consequently, there are constants $0 < \underline{c} \leq \overline{C} < \infty$, independent of t , such that

$$\underline{c} r_t \leq \mathfrak{R}_t \leq \overline{C} r_t.$$

In particular,

$$0 \leq r_t \leq \underline{c}^{-1} \mathfrak{R}_t, \quad 0 \leq \mathfrak{R}_t \leq \overline{C} r_t.$$

The first inequality and the squeeze theorem show that $\mathfrak{R}_t \rightarrow 0$ implies $r_t \rightarrow 0$; the second gives the converse. Thus

$$\mathfrak{R}_t \longrightarrow 0 \iff r_t \longrightarrow 0.$$

2. Define, for every $t \in \mathbb{N}$,

$$\nu_t := N_t^{-1}, \quad \xi_t := \frac{d_t}{N_t \ell_t}.$$

Since $e > 1$ and $N_t > 0$,

$$e + N_t > 1, \quad \ell_t > 0, \quad N_t \ell_t > 0, \quad \nu_t \geq 0, \quad \xi_t \geq 0.$$

Effective sample growth gives

$$\nu_t \longrightarrow 0.$$

Continuity of the square root, together with

$$\sqrt{\xi_t^2} = |\xi_t| = \xi_t,$$

shows that $\xi_t^2 \rightarrow 0$ implies $\xi_t \rightarrow 0$. Continuity of squaring gives the reverse implication. Hence

$$\xi_t^2 \longrightarrow 0 \iff \xi_t \longrightarrow 0.$$

Because $N_t \ell_t > 0$ for every t , the quotient characterization of little o gives

$$d_t = o(N_t \ell_t) \iff \xi_t \longrightarrow 0.$$

We next establish the truncation equivalence

$$r_t = \min\{1, \nu_t + \xi_t^2\} \longrightarrow 0 \iff \xi_t^2 \longrightarrow 0.$$

For the forward direction, $r_t \rightarrow 0$ implies $r_t < 1$ eventually. At each such index, $\nu_t + \xi_t^2 \geq 1$ would give $r_t = 1$. Therefore, eventually,

$$\nu_t + \xi_t^2 < 1, \quad r_t = \nu_t + \xi_t^2.$$

The sum consequently tends to zero, and

$$0 \leq \xi_t^2 \leq \nu_t + \xi_t^2$$

implies $\xi_t^2 \rightarrow 0$. Conversely, if $\xi_t^2 \rightarrow 0$, then

$$\nu_t + \xi_t^2 \rightarrow 0, \quad \min\{1, \nu_t + \xi_t^2\} \rightarrow \min\{1, 0\} = 0,$$

by continuity of addition and minimum. Combining these equivalences with [Item 1](#) yields

$$\mathfrak{R}_t \rightarrow 0 \iff d_t = o(N_t \ell_t) = o(N_t \log(e + N_t)).$$

3. The comparison in [Item 1](#) also gives a two-sided asymptotic comparison. Indeed, nonnegativity allows us to write, for every t ,

$$|\mathfrak{R}_t| \leq \overline{C} |r_t|, \quad |r_t| \leq \underline{c}^{-1} |\mathfrak{R}_t|.$$

These are the defining bounds for

$$\mathfrak{R}_t = O(r_t), \quad r_t = O(\mathfrak{R}_t), \quad \text{and hence} \quad \mathfrak{R}_t = \Theta(r_t).$$

4. Since $N_t \rightarrow \infty$, eventually $N_t \geq 1$. At those indices,

$$0 \leq \nu_t \leq 1, \quad \nu_t \leq \nu_t + \xi_t^2.$$

Thus both entries in the minimum defining r_t are at least ν_t , and

$$0 \leq \nu_t \leq \min\{1, \nu_t + \xi_t^2\} = r_t.$$

This proves the baseline bound

$$\nu_t = O(r_t).$$

5. Define the dimension scale, for every $t \in \mathbb{N}$, by

$$B_{\text{phase},t} := \sqrt{N_t} \ell_t.$$

The positivity established in [Item 2](#) gives

$$B_{\text{phase},t} > 0, \quad N_t \ell_t > 0.$$

Moreover, using $(\sqrt{N_t})^2 = N_t$,

$$\nu_t (N_t \ell_t)^2 = N_t^{-1} N_t^2 \ell_t^2 = N_t \ell_t^2 = B_{\text{phase},t}^2.$$

Suppose first that $r_t = O(\nu_t)$. Choose a real constant β_{rate} such that, eventually,

$$0 \leq r_t \leq \beta_{\text{rate}} \nu_t.$$

Since $\beta_{\text{rate}} \nu_t \rightarrow 0$, the squeeze theorem gives $r_t \rightarrow 0$, and hence $r_t < 1$ eventually. As in [Item 2](#), this forces

$$\nu_t + \xi_t^2 < 1, \quad r_t = \nu_t + \xi_t^2$$

eventually. Intersecting these eventual conditions with the assumed bound gives

$$\nu_t + \xi_t^2 \leq \beta_{\text{rate}} \nu_t, \quad \xi_t^2 \leq \beta_{\text{rate}} \nu_t,$$

where the latter inequality uses $\nu_t \geq 0$. Multiplication by the positive quantity $(N_t \ell_t)^2$ then yields

$$d_t^2 = \xi_t^2 (N_t \ell_t)^2 \leq \beta_{\text{rate}} \nu_t (N_t \ell_t)^2 = \beta_{\text{rate}} B_{\text{phase},t}^2.$$

Set

$$\beta_{\text{bound}} := \max\{1, \beta_{\text{rate}}\} \in \mathbb{R}.$$

Then

$$\beta_{\text{bound}} \geq 1, \quad \beta_{\text{rate}} \leq \beta_{\text{bound}} \leq \beta_{\text{bound}}^2.$$

Consequently, eventually,

$$d_t^2 \leq \beta_{\text{bound}}^2 B_{\text{phase},t}^2.$$

Both d_t and $\beta_{\text{bound}} B_{\text{phase},t}$ are nonnegative, so this squared inequality gives

$$d_t \leq \beta_{\text{bound}} B_{\text{phase},t}.$$

Therefore

$$r_t = O(\nu_t) \implies d_t = O(B_{\text{phase},t}).$$

6. Conversely, suppose that $d_t = O(B_{\text{phase},t})$. Choose a real constant β_{dim} such that, eventually,

$$0 \leq d_t \leq \beta_{\text{dim}} B_{\text{phase},t}.$$

The right-hand side is consequently nonnegative, so squaring gives

$$d_t^2 \leq \beta_{\text{dim}}^2 B_{\text{phase},t}^2.$$

Dividing by $(N_t \ell_t)^2 > 0$ and using the identity from [Item 5](#) yields

$$\xi_t^2 = \frac{d_t^2}{(N_t \ell_t)^2} \leq \beta_{\text{dim}}^2 \frac{B_{\text{phase},t}^2}{(N_t \ell_t)^2} = \beta_{\text{dim}}^2 \nu_t.$$

It follows that, eventually,

$$0 \leq r_t = \min\{1, \nu_t + \xi_t^2\} \leq \nu_t + \xi_t^2 \leq (1 + \beta_{\text{dim}}^2) \nu_t.$$

Thus $r_t = O(\nu_t)$. Together with [Item 5](#), this establishes

$$r_t = O(\nu_t) \iff d_t = O(B_{\text{phase},t}) = O\left(\sqrt{N_t} \log(e + N_t)\right).$$

7. If $\mathfrak{R}_t = \Theta(\nu_t)$, then

$$r_t = O(\mathfrak{R}_t), \quad \mathfrak{R}_t = O(\nu_t) \implies r_t = O(\nu_t),$$

by transitivity of big O . The equivalence in [Item 6](#) therefore gives $d_t = O(B_{\text{phase},t})$.

Conversely, if $d_t = O(B_{\text{phase},t})$, then [Item 6](#) gives $r_t = O(\nu_t)$. Combining this with [Items 3](#) and [4](#) gives

$$\mathfrak{R}_t = O(r_t), \quad r_t = O(\nu_t) \implies \mathfrak{R}_t = O(\nu_t),$$

and

$$\nu_t = O(r_t), \quad r_t = O(\mathfrak{R}_t) \quad \implies \quad \nu_t = O(\mathfrak{R}_t).$$

These two bounds prove

$$\mathfrak{R}_t = \Theta(N_t^{-1}) \quad \iff \quad d_t = O\left(\sqrt{N_t} \log(e + N_t)\right),$$

which is the second claimed equivalence. □

Proof of Theorem 5. Fix $0 < q < 1/2$. For $n, d \geq 1$, put

$$\mathfrak{h}_{n,d}^\circ := \frac{1}{n} + \left(\frac{d}{n \log(e + n)} \right)^2, \quad \mathfrak{h}_{n,d} := \min\{1, \mathfrak{h}_{n,d}^\circ\}.$$

For a bounded estimator kernel T and a full-data law P , write

$$\mathcal{R}_P(\mathsf{T}) := \int \int_{[-1,1]} (h - \tau(P))^2 \mathsf{T}(o, dh) \mathcal{L}_P(\mathbf{O}_n)(do).$$

Thus $\mathcal{R}_P(\mathsf{T})$ integrates both the sample and the estimator draw.

1. The deterministic estimator in Algorithm 1 defines an admissible decision kernel

$$\mathsf{T}_{\text{mix}}(o, dh) := \delta_{\hat{\tau}_{n,d,q}^{\text{mix}}(o)}(dh).$$

By Theorem 6, there is a universal constant $\bar{C}_{\text{mix}} > 0$ such that

$$0 \leq \mathcal{R}_P(\mathsf{T}_{\text{mix}}) \leq \bar{C}_{\text{mix}} r_{n,d,q} \quad \text{for every } P \in \mathcal{M}_{n,d,q}^+.$$

Taking the supremum over the model and then using this kernel in the defining infimum gives

$$\mathfrak{R}_{n,d,q}^+ \leq \sup_{P \in \mathcal{M}_{n,d,q}^+} \mathcal{R}_P(\mathsf{T}_{\text{mix}}) \leq \bar{C}_{\text{mix}} r_{n,d,q}.$$

If the model class is empty, its real-valued worst-case risk is taken to be zero, and the same inequality follows from

$$r_{n,d,q} = \min \left\{ 1, \frac{1}{nq} + \left(\frac{d}{nq \log(e + nq)} \right)^2 \right\} \geq 0.$$

2. We next convert this upper bound to the fixed- q rate. Define

$$\beta_q := \frac{1 - \log q}{q}, \quad N := nq, \quad \ell := \log(e + N), \quad \ell_{n,1} := \log(e + n).$$

Throughout the present domain,

$$N > 0, \quad \ell \geq 1, \quad \ell_{n,1} > 0, \quad \log q \leq 0, \quad \beta_q \geq 1.$$

Since

$$e + n \leq \frac{e + N}{q},$$

monotonicity of the logarithm yields

$$\ell_{n,1} \leq \ell - \log q \leq (1 - \log q)\ell.$$

The second inequality follows from

$$(-\log q)(\ell - 1) \geq 0.$$

Consequently,

$$n\ell_{n,1} \leq n(1 - \log q)\ell = \beta_q N\ell.$$

Moreover,

$$q\beta_q = 1 - \log q \geq 1, \quad q\beta_q^2 = (1 - \log q)\beta_q \geq 1,$$

so

$$\frac{1}{N} \leq \frac{\beta_q^2}{n}.$$

The positive denominator comparison also gives

$$0 \leq \frac{d}{N\ell} \leq \beta_q \frac{d}{n\ell_{n,1}}, \quad \left(\frac{d}{N\ell}\right)^2 \leq \beta_q^2 \left(\frac{d}{n\ell_{n,1}}\right)^2.$$

Adding these bounds,

$$\frac{1}{N} + \left(\frac{d}{N\ell}\right)^2 \leq \beta_q^2 \mathfrak{h}_{n,d}^\circ.$$

Truncation therefore gives

$$r_{n,d,q} \leq \min\{1, \beta_q^2 \mathfrak{h}_{n,d}^\circ\} \leq \beta_q^2 \mathfrak{h}_{n,d}.$$

Indeed, if $\mathfrak{h}_{n,d}^\circ \leq 1$, the last inequality is

$$\min\{1, \beta_q^2 \mathfrak{h}_{n,d}^\circ\} \leq \beta_q^2 \mathfrak{h}_{n,d}^\circ = \beta_q^2 \mathfrak{h}_{n,d};$$

if $\mathfrak{h}_{n,d}^\circ \geq 1$, it is

$$\min\{1, \beta_q^2 \mathfrak{h}_{n,d}^\circ\} \leq 1 \leq \beta_q^2 = \beta_q^2 \mathfrak{h}_{n,d}.$$

Define the positive fixed- q constant

$$\overline{C}_{\text{poly}}(q) := \overline{C}_{\text{mix}}\beta_q^2.$$

The upper bound from the preceding step now implies

$$\mathfrak{R}_{n,d,q}^+ \leq \overline{C}_{\text{mix}} r_{n,d,q} \leq \overline{C}_{\text{poly}}(q) \mathfrak{h}_{n,d}.$$

3. For the lower bound, invoke the published fixed-positivity lower bound on the control-zero observational subclass [Zeng et al., 2026, Theorem 2, Section 3.2, p. 9, with its proof in Section C.5, pp. 39–44]. Its truncated, all- n, d comparison is supplied by Theorem 8: at $\epsilon = q$, there is a constant $\underline{c}_Z(q) > 0$ such that

$$\underline{c}_Z(q) \mathfrak{h}_{n,d} \leq \mathfrak{R}_{n,d,q}^Z \quad (n, d \geq 1).$$

We transfer this lower bound to the unrestricted experiment. For an observational estimator kernel $\widehat{\theta}$, define

$$\mathcal{R}_{P_Z}^Z(\widehat{\theta}) := \int \int_{[-1,1]} (h - \theta_Z(P_Z))^2 \widehat{\theta}(\mathbf{z}, dh) \mathcal{L}_{P_Z}(\mathbf{Z}_n)(d\mathbf{z}).$$

Given any admissible unrestricted kernel T , define its observational pullback by

$$\widehat{\theta}_{\mathsf{T}}(\mathbf{z}, \mathcal{V}) := 2^{-n} \sum_{\mathbf{u} \in \{0,1\}^n} \mathsf{T}((o(x_i, b_i, y_i; u_i))_{i=1}^n, \mathcal{V}),$$

where

$$\mathbf{z} = ((x_i, b_i, y_i))_{i=1}^n, \quad \mathbf{u} = (u_i)_{i=1}^n, \quad \mathcal{V} \in \mathcal{B}([-1, 1]),$$

and $o(x, b, y; u)$ is the synthetic observation map of [Lemma 5](#). This finite mixture composes the parameter-independent fair-coin transformation with T , and hence is an admissible observational kernel.

For every $P_Z \in \mathcal{D}_{d,q}$, [Lemma 5](#) supplies a completion $P_Z^* \in \mathcal{M}_{n,d,q}^+$ whose observed product law is exactly the transformed sample law and whose target satisfies

$$\tau(P_Z^*) = \theta_Z(P_Z).$$

Thus

$$\mathcal{R}_{P_Z}^Z(\widehat{\theta}_{\mathsf{T}}) = \mathcal{R}_{P_Z^*}(\mathsf{T}).$$

Both targets and all kernel outputs lie in $[-1, 1]$, giving

$$0 \leq \mathcal{R}_{P_Z}^Z(\widehat{\theta}_{\mathsf{T}}) \leq 4, \quad 0 \leq \mathcal{R}_P(\mathsf{T}) \leq 4.$$

The supremum comparison and the observational minimax definition therefore give, for every T ,

$$\mathfrak{R}_{n,d,q}^Z \leq \sup_{P_Z \in \mathcal{D}_{d,q}} \mathcal{R}_{P_Z}^Z(\widehat{\theta}_{\mathsf{T}}) \leq \sup_{P \in \mathcal{M}_{n,d,q}^+} \mathcal{R}_P(\mathsf{T}).$$

Taking the infimum over T yields

$$\mathfrak{R}_{n,d,q}^Z \leq \mathfrak{R}_{n,d,q}^+.$$

Now define

$$\underline{c}(q) := \frac{\underline{c}_Z(q)}{\beta_q^2} > 0.$$

Using the rate comparison established in the preceding step,

$$\underline{c}(q)r_{n,d,q} \leq \frac{\underline{c}_Z(q)}{\beta_q^2} \beta_q^2 \mathfrak{h}_{n,d} = \underline{c}_Z(q) \mathfrak{h}_{n,d} \leq \mathfrak{R}_{n,d,q}^Z \leq \mathfrak{R}_{n,d,q}^+.$$

4. Choose the common upper constant

$$\overline{C}(q) := \max\{\overline{C}_{\text{mix}}, \overline{C}_{\text{poly}}(q)\} + \underline{c}(q).$$

These are finite real constants depending only on q , and

$$0 < \underline{c}(q) \leq \overline{C}(q).$$

The rates are nonnegative:

$$\mathfrak{h}_{n,d} \geq 0, \quad r_{n,d,q} \geq 0,$$

and the chosen constant satisfies

$$\overline{C}_{\text{mix}} \leq \overline{C}(q), \quad \overline{C}_{\text{poly}}(q) \leq \overline{C}(q).$$

To obtain the fixed-rate lower bound with the same constant, observe that

$$0 < N \leq n, \quad 0 < \ell \leq \ell_{n,1}, \quad 0 < N\ell \leq n\ell_{n,1}.$$

It follows that

$$\frac{1}{n} \leq \frac{1}{N}, \quad 0 \leq \frac{d}{n\ell_{n,1}} \leq \frac{d}{N\ell},$$

and hence

$$\left(\frac{d}{n\ell_{n,1}} \right)^2 \leq \left(\frac{d}{N\ell} \right)^2.$$

Adding and truncating at one proves

$$\mathfrak{h}_{n,d} = \min \left\{ 1, \frac{1}{n} + \left(\frac{d}{n\ell_{n,1}} \right)^2 \right\} \leq \min \left\{ 1, \frac{1}{N} + \left(\frac{d}{N\ell} \right)^2 \right\} = r_{n,d,q}.$$

Combining this comparison with the preceding bounds gives

$$\begin{aligned} \underline{c}(q)\mathfrak{h}_{n,d} &\leq \underline{c}(q)r_{n,d,q} \leq \mathfrak{R}_{n,d,q}^+, \\ \mathfrak{R}_{n,d,q}^+ &\leq \overline{C}_{\text{poly}}(q)\mathfrak{h}_{n,d} \leq \overline{C}(q)\mathfrak{h}_{n,d}, \\ \underline{c}(q)r_{n,d,q} &\leq \mathfrak{R}_{n,d,q}^+, \\ \mathfrak{R}_{n,d,q}^+ &\leq \overline{C}_{\text{mix}}r_{n,d,q} \leq \overline{C}(q)r_{n,d,q}. \end{aligned}$$

These are the two asserted comparisons. □

Proof of Proposition 6. Fix q , the arrival floor, with $0 < q < 1/2$. Throughout the proof, n ranges over positive integers, and all limits are taken as $n \rightarrow \infty$.

1. For a sequence of integers $d_n \geq 1$, define

$$\ell_n^{\text{fix}} := \log(e + n), \quad r_n^{\text{fix}} := \min \left\{ 1, \frac{1}{n} + \left(\frac{d_n}{n\ell_n^{\text{fix}}} \right)^2 \right\}.$$

Apply Theorem 5, whose lower comparison uses the published zero-control-regression lower bound [Zeng et al., 2026, Theorem 2, Section 3.2, p. 9, and its proof in Section C.5, pp. 39–44]. It supplies constants $0 < \underline{c}(q) \leq \overline{C}(q)$, depending only on q , such that

$$\underline{c}(q)r_n^{\text{fix}} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \overline{C}(q)r_n^{\text{fix}} \quad (n \geq 1).$$

These constants are common to every dimension sequence. Moreover,

$$0 \leq r_n^{\text{fix}}, \quad 0 \leq \overline{C}(q).$$

2. We first establish the consistency equivalence. The comparison above gives

$$0 \leq r_n^{\text{fix}} \leq \underline{c}(q)^{-1} \mathfrak{R}_{n,d_n,q}^+, \quad 0 \leq \mathfrak{R}_{n,d_n,q}^+ \leq \overline{C}(q) r_n^{\text{fix}}.$$

Applying the squeeze theorem in each direction yields

$$\mathfrak{R}_{n,d_n,q}^+ \longrightarrow 0 \iff r_n^{\text{fix}} \longrightarrow 0.$$

To characterize the latter limit, define

$$\nu_n := \frac{1}{n}, \quad \xi_n := \frac{d_n}{n\ell_n^{\text{fix}}}.$$

Since $e + n > 1$, we have

$$\ell_n^{\text{fix}} > 0, \quad n\ell_n^{\text{fix}} > 0, \quad \xi_n \geq 0, \quad \nu_n \geq 0, \quad \nu_n \longrightarrow 0.$$

The last limit follows from $n \rightarrow \infty$ and continuity of inversion at infinity. Continuity of the square root and nonnegativity of ξ_n give

$$\xi_n^2 \longrightarrow 0 \implies \xi_n = \sqrt{\xi_n^2} \longrightarrow 0.$$

Conversely, continuity of squaring gives

$$\xi_n \longrightarrow 0 \implies \xi_n^2 \longrightarrow 0.$$

The positive denominator also gives the usual quotient characterization of little- o :

$$d_n = o\left(n\ell_n^{\text{fix}}\right) \iff \xi_n \longrightarrow 0.$$

It remains to handle the truncation. If $r_n^{\text{fix}} \rightarrow 0$, then eventually

$$r_n^{\text{fix}} = \min\{1, \nu_n + \xi_n^2\} < 1.$$

At every such index,

$$\nu_n + \xi_n^2 < 1, \quad r_n^{\text{fix}} = \nu_n + \xi_n^2 :$$

indeed, $\nu_n + \xi_n^2 \geq 1$ would make the minimum equal to 1. Thus $\nu_n + \xi_n^2 \rightarrow 0$, and

$$0 \leq \xi_n^2 \leq \nu_n + \xi_n^2$$

implies $\xi_n^2 \rightarrow 0$. Conversely, if $\xi_n^2 \rightarrow 0$, then

$$\nu_n + \xi_n^2 \longrightarrow 0, \quad \min\{1, \nu_n + \xi_n^2\} \longrightarrow \min\{1, 0\} = 0,$$

by continuity of addition and the minimum.

Consequently,

$$r_n^{\text{fix}} \longrightarrow 0 \iff \xi_n^2 \longrightarrow 0 \iff \xi_n \longrightarrow 0 \iff d_n = o(n \log(e + n)).$$

Together with the risk comparison, this proves the first assertion.

3. For the sufficient condition for the parametric rate, choose

$$c := \underline{c}(q) > 0.$$

Given $M > 0$, define

$$C := \overline{C}(q)(1 + M^2).$$

These constants satisfy

$$c = \underline{c}(q) \leq \overline{C}(q) \leq \overline{C}(q)(1 + M^2) = C.$$

Suppose that eventually

$$d_n \leq M\sqrt{n} \ell_n^{\text{fix}}.$$

Both sides are nonnegative, so squaring this inequality and using $(\sqrt{n})^2 = n$ gives

$$d_n^2 \leq \left(M\sqrt{n} \ell_n^{\text{fix}}\right)^2 = M^2 n \left(\ell_n^{\text{fix}}\right)^2.$$

Dividing by the positive squared denominator yields

$$\xi_n^2 = \frac{d_n^2}{(n\ell_n^{\text{fix}})^2} \leq \frac{M^2}{n}.$$

Here the cancellation uses

$$\frac{1}{n}(n\ell_n^{\text{fix}})^2 = n \left(\ell_n^{\text{fix}}\right)^2.$$

Since $1/n \leq 1$ for $n \geq 1$, both entries of the minimum defining the rate are at least $1/n$. Hence

$$\frac{1}{n} \leq r_n^{\text{fix}} = \min \left\{ 1, \frac{1}{n} + \xi_n^2 \right\} \leq \frac{1}{n} + \xi_n^2 \leq \frac{1 + M^2}{n}.$$

Multiplying the lower and upper bounds by the corresponding frontier constants gives, eventually,

$$\frac{c}{n} \leq \underline{c}(q) r_n^{\text{fix}} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \overline{C}(q) r_n^{\text{fix}} \leq \frac{C}{n}.$$

The choice of c depends only on q , and the choice of C depends only on q and M . The eventual dimension bound determines the threshold along each sequence.

4. For the converse, suppose there exist constants $0 < c_* \leq C_*$ such that eventually

$$\frac{c_*}{n} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \frac{C_*}{n}.$$

In particular, $C_* \geq 0$. Combining the upper risk bound with the lower frontier comparison gives

$$\underline{c}(q) r_n^{\text{fix}} \leq \mathfrak{R}_{n,d_n,q}^+ \leq \frac{C_*}{n},$$

and therefore

$$0 \leq r_n^{\text{fix}} \leq \frac{C_*}{\underline{c}(q)} \frac{1}{n} \longrightarrow 0.$$

Thus eventually $r_n^{\text{fix}} < 1$.

Define the dimension-bound constant

$$K_{\text{dim}} := \max \left\{ 1, \frac{C_*}{\underline{c}(q)} \right\}.$$

Its calibration gives

$$0 \leq K_{\text{dim}}, \quad 1 \leq K_{\text{dim}}, \quad \frac{C_*}{\underline{c}(q)} \leq K_{\text{dim}} \leq K_{\text{dim}}^2.$$

At all sufficiently large indices, $r_n^{\text{fix}} < 1$ implies

$$\frac{1}{n} + \xi_n^2 < 1, \quad r_n^{\text{fix}} = \frac{1}{n} + \xi_n^2,$$

because a sum at least 1 would make the truncated rate equal to 1. The preceding rate bound and $1/n \geq 0$ now yield

$$\xi_n^2 \leq \frac{1}{n} + \xi_n^2 = r_n^{\text{fix}} \leq \frac{C_*}{\underline{c}(q)} \frac{1}{n}.$$

Multiplying by $(n\ell_n^{\text{fix}})^2 > 0$ gives

$$d_n^2 \leq \frac{C_*}{\underline{c}(q)} n \left(\ell_n^{\text{fix}} \right)^2 \leq K_{\text{dim}}^2 n \left(\ell_n^{\text{fix}} \right)^2.$$

The second inequality uses $n(\ell_n^{\text{fix}})^2 \geq 0$. Also,

$$K_{\text{dim}} \sqrt{n} \ell_n^{\text{fix}} \geq 0, \quad \left(K_{\text{dim}} \sqrt{n} \ell_n^{\text{fix}} \right)^2 = K_{\text{dim}}^2 n \left(\ell_n^{\text{fix}} \right)^2.$$

Since $d_n \geq 0$, comparison of these squares implies

$$d_n \leq K_{\text{dim}} \sqrt{n} \ell_n^{\text{fix}} \quad \text{eventually.}$$

Both the dimension and the comparison scale are nonnegative, so this is precisely the required bound

$$d_n = O(\sqrt{n} \log(e + n)).$$

□

Proof of Theorem 6. 1. By Theorem 9, choose a universal constant $0 < \overline{C} < \infty$. For every n, d, q, P satisfying the stated domain, model, and sampling conditions, this constant gives

$$E_P \left[\left(\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q},$$

where $\Phi(P)$ is the signed cell functional of Definition 10. The envelope here has precisely the branch formulas in the statement.

2. The model hypothesis permits application of Lemma 3, which supplies

$$\tau(P) = \Phi(P).$$

Substituting this identity into the preceding risk bound yields

$$E_P \left[\left(\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

3. To establish the auxiliary-risk comparison, fix an observed sample

$$\mathbf{o} = (o_i)_{i=1}^n \in (\{0, \dots, d-1\} \times \{0, 1\}^4)^n, \quad o_i = (X_i, A_i, S_i, R_i, R_i Y_i).$$

Define the prefix probabilities and remaining probability by

$$w_{k_0} := P_\omega(K_0 = k_0) = e^{-n/2} \frac{(n/2)^{k_0}}{k_0!} \quad (k_0 \in \mathbb{Z}_{\geq 0}), \quad \delta_{\text{tail}} := 1 - \sum_{k_0=0}^n w_{k_0} = P_\omega(K_0 > n).$$

The disjoint prefix events imply

$$w_{k_0} \geq 0, \quad \sum_{k_0=0}^n w_{k_0} \leq 1, \quad \delta_{\text{tail}} \geq 0.$$

For each $0 \leq k_0 \leq n$, define the assignment set and its output average by

$$\Sigma_{k_0} := \{\sigma : \{1, \dots, k_0\} \rightarrow \{1, 2, 3\}\}, \quad \bar{\tau}_{k_0}(\mathbf{o}) := 3^{-k_0} \sum_{\sigma \in \Sigma_{k_0}} \tilde{\tau}(\mathbf{o}; (k_0, \sigma)),$$

where the three assignment values denote the membership, pilot, and estimation streams. The notation (k_0, σ) specifies the auxiliary prefix and assignment in [Algorithm 1](#).

The weights $w_0, \dots, w_n, \delta_{\text{tail}}$ sum to one, and

$$\sum_{k_0=0}^n w_{k_0} \bar{\tau}_{k_0}(\mathbf{o}) - \tau(P) = \delta_{\text{tail}}(-\tau(P)) + \sum_{k_0=0}^n w_{k_0} (\bar{\tau}_{k_0}(\mathbf{o}) - \tau(P)).$$

Convexity of the square therefore gives

$$\left(\sum_{k_0=0}^n w_{k_0} \bar{\tau}_{k_0}(\mathbf{o}) - \tau(P) \right)^2 \leq \delta_{\text{tail}} \tau(P)^2 + \sum_{k_0=0}^n w_{k_0} (\bar{\tau}_{k_0}(\mathbf{o}) - \tau(P))^2.$$

4. Since $|\Sigma_{k_0}| = 3^{k_0}$, centering the assignment average gives

$$\bar{\tau}_{k_0}(\mathbf{o}) - \tau(P) = 3^{-k_0} \sum_{\sigma \in \Sigma_{k_0}} (\tilde{\tau}(\mathbf{o}; (k_0, \sigma)) - \tau(P)).$$

The finite mean-square inequality, obtained by Cauchy–Schwarz, consequently yields

$$(\bar{\tau}_{k_0}(\mathbf{o}) - \tau(P))^2 \leq 3^{-k_0} \sum_{\sigma \in \Sigma_{k_0}} (\tilde{\tau}(\mathbf{o}; (k_0, \sigma)) - \tau(P))^2.$$

This also covers $k_0 = 0$: the empty prefix has one assignment, so the averaging denominator equals one.

5. Multiplying these inequalities by the nonnegative prefix probabilities and summing gives

$$\begin{aligned} & \sum_{k_0=0}^n w_{k_0} (\bar{\tau}_{k_0}(\mathbf{o}) - \tau(P))^2 \\ & \leq \sum_{k_0=0}^n w_{k_0} 3^{-k_0} \sum_{\sigma \in \Sigma_{k_0}} (\tilde{\tau}(\mathbf{o}; (k_0, \sigma)) - \tau(P))^2. \end{aligned}$$

The estimator's finite averaging formula in [Algorithm 1](#) is

$$\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{o}) = \sum_{k_0=0}^n w_{k_0} \bar{\tau}_{k_0}(\mathbf{o}).$$

Combining this identity with the preceding two bounds establishes, for every $\mathbf{o}$,

$$\begin{aligned} & \left(\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{o}) - \tau(P) \right)^2 \\ & \leq \sum_{k_0=0}^n w_{k_0} 3^{-k_0} \sum_{\sigma \in \Sigma_{k_0}} \left(\tilde{\tau}(\mathbf{o}; (k_0, \sigma)) - \tau(P) \right)^2 + \delta_{\text{tail}} \tau(P)^2 \\ & = E_{\omega} \left[\left(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P) \right)^2 \middle| \mathbf{O}_n = \mathbf{o} \right]. \end{aligned}$$

The last term in the sum is the squared loss of the zero output on $K_0 > n$. All functions in this inequality are measurable and integrable on the finite observed-sample space. Integrating under the canonical product law yields

$$E_P \left[\left(\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq E_P \left[E_{\omega} \left[\left(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P) \right)^2 \middle| \mathbf{O}_n \right] \right],$$

as required. □

Proof of [Theorem 7](#). Fix $\alpha \in (0, 1)$. Throughout the proof, write

$$\beta := 1 - \alpha > 0, \quad N := nq > 0, \quad \ell := \log(e + N) \geq 1,$$

where $n, d \geq 1$, $0 < q \leq 1$, and $q\ell^2 \leq 1/64$. Define the two length scales by

$$s_{\text{par}} := \min\{1, N^{-1/2}\}, \quad s_{\text{many}} := \min\left\{1, \frac{d}{N\ell}\right\}.$$

1. **Upper bound for the risk-envelope interval.** By [Theorem 11](#), there is a universal $\bar{C} > 0$ such that, for every $P \in \mathcal{M}_{n,d,q}$,

$$E_P \left[\left(\widehat{\tau}_{n,d,q}^{\text{mix}} - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \bar{C} r_{n,d,q}.$$

For every observed sample, define the envelope radius by

$$\rho_{\text{env}} := \sqrt{\frac{\mathfrak{U}_{n,d,q}}{\alpha}}.$$

The endpoints in [Definition 30](#) satisfy

$$\text{length}(I_{\alpha}^{\text{mix}}) = \min\{1, \widehat{\tau}_{n,d,q}^{\text{mix}} + \rho_{\text{env}}\} - \max\{-1, \widehat{\tau}_{n,d,q}^{\text{mix}} - \rho_{\text{env}}\} \leq 2\rho_{\text{env}}.$$

Consequently, with

$$C_{\alpha} := 2\sqrt{\frac{\bar{C}}{\alpha}} > 0,$$

we have

$$E_P[\text{length}(I_\alpha^{\text{mix}})] \leq 2\sqrt{\frac{\mathfrak{U}_{n,d,q}}{\alpha}} \leq C_\alpha \sqrt{r_{n,d,q}}.$$

Taking the supremum gives

$$\sup_{P \in \mathcal{M}_{n,d,q}} E_P[\text{length}(I_\alpha^{\text{mix}})] \leq C_\alpha \sqrt{r_{n,d,q}}.$$

For an empty model class, the supremum is zero by the convention in [Definition 29](#), and the same inequality follows from $r_{n,d,q} \geq 0$.

2. Calibration of the activated regime. Define constants depending only on α by

$$\begin{aligned} \eta &:= e^{-32}, \quad M_{\text{act},\alpha} := \left\lceil \frac{8}{\beta} \right\rceil + 1, \quad D_{\text{act},\alpha} := \left\lceil \frac{18432M_{\text{act},\alpha}^2}{\beta} \right\rceil + 1, \\ T_{\text{act},\alpha} &:= \max \left\{ 1, e^2, 4\eta, 2\eta D_{\text{act},\alpha}, \exp \left(\frac{32 - \log(\beta/16)}{28} \right) \right\}, \quad c_{\text{act},\alpha} := \frac{21\beta}{32} \frac{\eta}{24} > 0. \end{aligned}$$

The ceiling inequalities give

$$M_{\text{act},\alpha} \geq 1, \quad \frac{1}{M_{\text{act},\alpha}} \leq \frac{\beta}{8}, \quad D_{\text{act},\alpha} \geq 1, \quad \frac{18432M_{\text{act},\alpha}^2}{\beta} \leq D_{\text{act},\alpha}.$$

Use the quantities from [Algorithm 2](#):

$$K := \lceil \ell \rceil, \quad H := K^2, \quad b_0 := \frac{\eta}{N\ell}, \quad J := \min \left\{ d-1, \left\lfloor \frac{1}{2b_0} \right\rfloor \right\}.$$

They are defined for all the parameters under consideration, including $d = 1$, when $J = 0$.

Suppose first that $N \geq T_{\text{act},\alpha}$. Since $\ell \geq 1$,

$$D_{\text{act},\alpha} \leq \frac{N}{2\eta} \leq \frac{N\ell}{2\eta} = \frac{1}{2b_0}.$$

Also,

$$b_0 \leq \frac{\eta}{N} \leq \frac{1}{4}, \quad \ell \geq 2, \quad K \geq 2.$$

The ceiling bound $K \leq \ell + 1 \leq 2\ell$, together with the slice restriction, gives

$$qH = qK^2 \leq 4q\ell^2 \leq \frac{1}{16}.$$

By [Lemma 7](#), choose finitely supported probability laws Π_0, Π_1 on $[1, H]$ with matching moments through degree K . Interchange their names if necessary and define

$$\mu_{\text{rec},i} := \int z^{-1} d\Pi_i(z) \quad (i = 0, 1).$$

Then

$$\int z^v d\Pi_0(z) = \int z^v d\Pi_1(z) \quad (0 \leq v \leq K), \quad \mu_{\text{rec},1} - \mu_{\text{rec},0} \geq \frac{1}{12}.$$

We will show that whenever $J \geq D_{\text{act},\alpha}$,

$$c_{\text{act},\alpha} s_{\text{many}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

3. **The activated grid and its likelihood comparison.** Continue under $N \geq T_{\text{act},\alpha}$ and $J \geq D_{\text{act},\alpha}$. For each integer $0 \leq \iota \leq M_{\text{act},\alpha}$, define

$$\Pi^{(\iota)} := \left(1 - \frac{\iota}{M_{\text{act},\alpha}}\right) \Pi_0 + \frac{\iota}{M_{\text{act},\alpha}} \Pi_1.$$

These are finitely supported probability laws on $[1, H]$, with the same moments through degree K .

For completeness, the full-data laws underlying the activated experiment can be specified as follows. For a fixed $\mathbf{z} = (z_0, \dots, z_{d-1}) \in [1, H]^d$, let $P_{\mathbf{z}}^{\text{act}}$ have baseline probabilities

$$P_{\mathbf{z}}^{\text{act}}(X = x) = \begin{cases} b_0, & 0 \leq x < J, \\ 1 - Jb_0, & x = J, \\ 0, & J < x < d. \end{cases}$$

Let treatment be an independent fair coin, and set

$$S(0) = S(1) = Y(0) = 0, \quad S = 0, \quad Y = AY(1).$$

Conditional on $X = x$, generate the treated potential outcome and arrival independently, with

$$P_{\mathbf{z}}^{\text{act}}(Y(1) = 1 \mid X = x) = \begin{cases} z_x^{-1}, & x < J, \\ 0, & x \geq J, \end{cases} \quad P_{\mathbf{z}}^{\text{act}}(R = 1 \mid X = x, A) = \begin{cases} qz_x, & x < J, \\ qH, & x \geq J. \end{cases}$$

The inequalities $Jb_0 \leq 1/2$ and $qH \leq 1/16$ ensure valid probabilities. Independence of treatment, consistency, conditional independence of arrival and outcome, and the arrival floor follow from this construction. Thus

$$P_{\mathbf{z}}^{\text{act}} \in \mathcal{M}_{n,d,q}, \quad \tau(P_{\mathbf{z}}^{\text{act}}) = b_0 \sum_{x=0}^{J-1} z_x^{-1}.$$

Let $\tilde{\mathbb{Q}}_{\text{act},\iota}$ be the augmented sample law obtained from the rule in [Algorithm 2](#), using $(\Pi^{(\iota)})^{\otimes d}$ for the latent vector. Let $\mathbb{Q}_{\text{act},\iota}$ be its projection onto the observed sample. The conditional probabilities in that construction give

$$\mathbb{Q}_{\text{act},\iota} = \int \mathcal{L}_{P_{\mathbf{z}}^{\text{act}}}(\mathbf{O}_n) d(\Pi^{(\iota)})^{\otimes d}(\mathbf{z}).$$

Define the activation count in each rare category and the common cutoff event by

$$V_x := \sum_{i=1}^n \mathbf{1}\{X_i = x, B_{\text{act},i} = 1\} \quad (0 \leq x < J), \quad G_{\text{act}} := \{V_x \leq K \text{ for every } x < J\}.$$

Conditional on the membership, treatment, and activation pattern, every activated observation in category x contributes an affine factor in z_x , and every inactive observation contributes a constant factor. The conditional likelihood therefore has degree at most V_x

in each z_x . On G_{act} , integration against the product priors and moment matching imply equality of all augmented sample atoms:

$$\tilde{\mathbb{Q}}_{\text{act},\iota}(\{\mathbf{o}_{\text{aug}}\}) = \tilde{\mathbb{Q}}_{\text{act},0}(\{\mathbf{o}_{\text{aug}}\}) \quad (\mathbf{o}_{\text{aug}} \in G_{\text{act}}).$$

Hence the two laws also assign the same mass to G_{act} and its complement. Splitting any event into its intersections with these two sets, and then projecting, yields

$$\text{TV}(\mathbb{Q}_{\text{act},0}, \mathbb{Q}_{\text{act},\iota}) \leq \tilde{\mathbb{Q}}_{\text{act},0}(G_{\text{act}}^c).$$

To bound this last probability, define

$$K_{\text{tail}} := K + 1, \quad \lambda_{\text{act}} := nb_0qH = \frac{\eta K^2}{\ell}.$$

The membership and activation pattern has a common law under all the priors, and each V_x has distribution $\text{Binomial}(n, b_0qH)$. Counting subsets of K_{tail} successful observations gives

$$E\binom{V_x}{K_{\text{tail}}} = \binom{n}{K_{\text{tail}}}(b_0qH)^{K_{\text{tail}}},$$

with both sides zero when $K_{\text{tail}} > n$. Since $\mathbf{1}\{V_x > K\} \leq \binom{V_x}{K_{\text{tail}}}$,

$$\Pr(V_x > K) \leq \frac{\lambda_{\text{act}}^{K_{\text{tail}}}}{K_{\text{tail}}!}.$$

The ceiling inequalities imply

$$\lambda_{\text{act}} = \frac{\eta K^2}{\ell} \leq 2\eta K_{\text{tail}} \leq e^{-31} K_{\text{tail}}, \quad K_{\text{tail}} \geq \ell.$$

Using the K_{tail} -th term of the exponential series,

$$\frac{\lambda_{\text{act}}^{K_{\text{tail}}}}{K_{\text{tail}}!} \leq e^{-31K_{\text{tail}}} \frac{K_{\text{tail}}^{K_{\text{tail}}}}{K_{\text{tail}}!} \leq e^{-30K_{\text{tail}}} \leq e^{-30\ell}.$$

A union bound, followed by the mass cap and $N \leq e^\ell$, $\ell \leq e^\ell$, now gives

$$\tilde{\mathbb{Q}}_{\text{act},0}(G_{\text{act}}^c) \leq J e^{-30\ell} \leq \frac{N\ell}{2e^{-32}} e^{-30\ell} \leq e^{32-28\ell}.$$

Finally, the definition of $T_{\text{act},\alpha}$ implies

$$\ell \geq \frac{32 - \log(\beta/16)}{28}, \quad e^{32-28\ell} \leq \frac{\beta}{16}.$$

Thus, for every grid index,

$$\text{TV}(\mathbb{Q}_{\text{act},0}, \mathbb{Q}_{\text{act},\iota}) \leq \frac{\beta}{16}.$$

4. **Coverage of the prior neighborhoods and their length cost.** Define the target separation, centers, and neighborhood radius by

$$\Delta_{\text{act}} := Jb_0(\mu_{\text{rec},1} - \mu_{\text{rec},0}) \geq \frac{Jb_0}{12} > 0,$$

$$\theta_{\text{act},\iota} := Jb_0 \left[\left(1 - \frac{\iota}{M_{\text{act},\alpha}}\right) \mu_{\text{rec},0} + \frac{\iota}{M_{\text{act},\alpha}} \mu_{\text{rec},1} \right] = \theta_{\text{act},0} + \frac{\iota \Delta_{\text{act}}}{M_{\text{act},\alpha}}, \quad w_{\text{act}} := \frac{\Delta_{\text{act}}}{8M_{\text{act},\alpha}}.$$

Under the ι -th product prior, the actual target has mean $\theta_{\text{act},\iota}$. Independence of the latent coordinates and $z_x^{-1} \in [0, 1]$ give

$$\text{Var} \left(b_0 \sum_{x=0}^{J-1} z_x^{-1} \right) \leq \frac{Jb_0^2}{4}.$$

Indeed, a variable in $[0, 1]$ has variance at most $1/4$, and the variances of these independent summands add. Chebyshev's inequality therefore gives

$$\Pr_{\Pi^{(\iota)} \otimes d} (|\tau(P_{\mathbf{z}}^{\text{act}}) - \theta_{\text{act},\iota}| > w_{\text{act}}) \leq \frac{Jb_0^2/4}{w_{\text{act}}^2} \leq \frac{2304M_{\text{act},\alpha}^2}{J} \leq \frac{\beta}{8}.$$

Fix any $\mathbf{l} \in \mathcal{C}_{n,d,q,\alpha}$, and let $\nu_{\text{act},\iota}$ denote its returned-interval law when the sample has law $\mathbb{Q}_{\text{act},\iota}$:

$$\nu_{\text{act},\iota} := \mathbb{Q}_{\text{act},\iota} \mathbf{l}.$$

For each grid index, define the interval event

$$F_{\text{act},\iota} := \{[l, u] : [l, u] \cap [\theta_{\text{act},\iota} - w_{\text{act}}, \theta_{\text{act},\iota} + w_{\text{act}}] \neq \emptyset\}.$$

Honesty holds for every fixed completion. Averaging that coverage over the finite prior and subtracting the target-concentration failure probability yields

$$\nu_{\text{act},\iota}(F_{\text{act},\iota}) \geq \beta - \frac{\beta}{8} = \frac{7\beta}{8}.$$

By [Lemma 2](#) and the preceding likelihood comparison,

$$\text{TV}(\nu_{\text{act},0}, \nu_{\text{act},\iota}) \leq \frac{\beta}{16}, \quad \nu_{\text{act},0}(F_{\text{act},\iota}) \geq \frac{13\beta}{16}.$$

Define the number of neighborhoods met by a returned interval by

$$\mathcal{N}_{\text{hit}}([l, u]) := \sum_{\iota=0}^{M_{\text{act},\alpha}} \mathbf{1}_{F_{\text{act},\iota}}([l, u]).$$

Meeting the ι -th neighborhood is equivalent to $\theta_{\text{act},\iota} \in [l - w_{\text{act}}, u + w_{\text{act}}]$. The centers are spaced by $\Delta_{\text{act}}/M_{\text{act},\alpha}$. Thus the distance between the extreme centers contained in this expanded interval gives

$$(u - l) + 2w_{\text{act}} \geq \frac{\Delta_{\text{act}}}{M_{\text{act},\alpha}} (\mathcal{N}_{\text{hit}}([l, u]) - 1).$$

For zero or one contained center, the right side is nonpositive or zero, so the inequality covers those cases as well. Taking expectations,

$$\begin{aligned} E_{\nu_{\text{act},0}}[u - l] &\geq \frac{\Delta_{\text{act}}}{M_{\text{act},\alpha}} \left[(M_{\text{act},\alpha} + 1) \frac{13\beta}{16} - 1 \right] - 2w_{\text{act}} \\ &= \Delta_{\text{act}} \left[\frac{13\beta}{16} + \frac{13\beta}{16M_{\text{act},\alpha}} - \frac{5}{4M_{\text{act},\alpha}} \right] \\ &\geq \frac{21\beta}{32} \Delta_{\text{act}}, \end{aligned}$$

where the last step uses $1/M_{\text{act},\alpha} \leq \beta/8$. The left side is a prior average of the expected lengths of this same procedure over laws in $\mathcal{M}_{n,d,q}$. Therefore

$$\frac{21\beta}{32} \Delta_{\text{act}} \leq \sup_{P \in \mathcal{M}_{n,d,q}} E_P[\text{length}(l)].$$

The constant procedure returning $[-1, 1]$ is honest, so the procedure class is nonempty. Taking its defining infimum gives

$$\frac{21\beta}{32} \Delta_{\text{act}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

To express the separation in terms of d , split according to the minimum defining J . If $d - 1 \leq \lfloor (2b_0)^{-1} \rfloor$, then $J = d - 1$, and $J \geq D_{\text{act},\alpha} \geq 1$ implies $d \geq 2$ and $J \geq d/2$. Otherwise, $J = \lfloor (2b_0)^{-1} \rfloor$, and

$$Jb_0 \geq \frac{1}{2} - b_0 \geq \frac{1}{4}.$$

In both cases,

$$Jb_0 \geq \min \left\{ \frac{db_0}{2}, \frac{1}{4} \right\}.$$

Since $\eta \leq 1/2$, the two inequalities

$$\frac{\eta}{24} s_{\text{many}} \leq \frac{db_0}{24}, \quad \frac{\eta}{24} s_{\text{many}} \leq \frac{1}{48}$$

give

$$\frac{\eta}{24} s_{\text{many}} \leq \frac{1}{12} \min \left\{ \frac{db_0}{2}, \frac{1}{4} \right\} \leq \frac{Jb_0}{12} \leq \Delta_{\text{act}}.$$

Consequently,

$$c_{\text{act},\alpha} s_{\text{many}} \leq \mathfrak{L}_{n,d,q,\alpha} \quad (N \geq T_{\text{act},\alpha}, \quad J \geq D_{\text{act},\alpha}).$$

5. **An explicit one-cell floor.** Use the family P_u supplied by [Lemma 8](#). We record the calibration needed here:

$$c_{\text{par},\alpha} := \frac{3\beta^2}{512} > 0, \quad c_{\text{par},\alpha} s_{\text{par}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

To obtain this explicit constant, define

$$\eta_{\text{cell}} := \frac{\beta}{8}, \quad w_{\text{cell}} := \min \left\{ \frac{1}{4}, \frac{\eta_{\text{cell}}}{4\sqrt{N}} \right\}.$$

Choose an integer $M_{\text{cell},\alpha} \geq 8/\beta$, and set

$$\delta_{\text{cell}} := \frac{w_{\text{cell}}}{M_{\text{cell},\alpha}} > 0, \quad u_{\text{cell},\iota} := \frac{1}{2} + \iota\delta_{\text{cell}} \quad (0 \leq \iota \leq M_{\text{cell},\alpha}).$$

These grid points are distinct and belong to $[1/4, 3/4]$, and

$$2Nw_{\text{cell}}^2 \leq \frac{\eta_{\text{cell}}^2}{8}, \quad w_{\text{cell}} \geq \frac{\eta_{\text{cell}}}{4}s_{\text{par}}.$$

Indeed, $N > 0$ and $0 < \eta_{\text{cell}} = \beta/8 < 1$, so $w_{\text{cell}} > 0$. The defining minimum gives

$$w_{\text{cell}} \leq \frac{\eta_{\text{cell}}}{4\sqrt{N}}, \quad 2Nw_{\text{cell}}^2 \leq 2N \left(\frac{\eta_{\text{cell}}}{4\sqrt{N}} \right)^2 = \frac{\eta_{\text{cell}}^2}{8},$$

where the equality uses $(\sqrt{N})^2 = N$. For the lower bound, if $1/4 \leq \eta_{\text{cell}}/(4\sqrt{N})$, then $w_{\text{cell}} = 1/4$, and $s_{\text{par}} \leq 1$ gives

$$\frac{\eta_{\text{cell}}}{4}s_{\text{par}} \leq \frac{\eta_{\text{cell}}}{4} \leq \frac{1}{4} = w_{\text{cell}}.$$

Otherwise, $w_{\text{cell}} = \eta_{\text{cell}}/(4\sqrt{N})$, and $s_{\text{par}} \leq N^{-1/2}$ gives

$$\frac{\eta_{\text{cell}}}{4}s_{\text{par}} \leq \frac{\eta_{\text{cell}}}{4\sqrt{N}} = w_{\text{cell}}.$$

Write the one-record and sample laws of this family as

$$\mathbb{Q}_{\text{cell},u}^{(1)} := \mathcal{L}_{P_u}(O_1), \quad \mathbb{Q}_{\text{cell},u}^{(n)} := (\mathbb{Q}_{\text{cell},u}^{(1)})^{\otimes n}.$$

In this family, the arrival floor and $P_u(R = 1) = q$, together with balanced assignment, force arrival probability q in each treatment arm. Conditional independence of arrival and outcome then gives the two treated arrived-outcome atom probabilities $qu/2$ and $q(1-u)/2$; all other observed atom probabilities are constant in u . Consequently, defining the one-record chi-squared divergence by its finite sum,

$$\mathcal{X}_{\text{cell},\iota}^{(1)} := \sum_{\substack{o: \\ \mathbb{Q}_{\text{cell},1/2}^{(1)}(\{o\}) > 0}} \frac{\left[\mathbb{Q}_{\text{cell},u_{\text{cell},\iota}}^{(1)}(\{o\}) - \mathbb{Q}_{\text{cell},1/2}^{(1)}(\{o\}) \right]^2}{\mathbb{Q}_{\text{cell},1/2}^{(1)}(\{o\})} = 2q(\iota\delta_{\text{cell}})^2.$$

This computation also covers $q = 1$, since the vanished nonarrival atoms are omitted from the sum. Define $\mathcal{X}_{\text{cell},\iota}^{(n)}$ by the analogous sum for the sample laws, and put

$$t_{\text{cell},\iota} := 2N(\iota\delta_{\text{cell}})^2 \leq \frac{\eta_{\text{cell}}^2}{8}.$$

Independence of the observations gives

$$1 + \mathcal{X}_{\text{cell},\iota}^{(n)} = (1 + \mathcal{X}_{\text{cell},\iota}^{(1)})^n \leq e^{t_{\text{cell},\iota}}.$$

For $0 \leq t_{\text{cell},\iota} < 1$, comparison of the exponential and geometric series gives

$$e^{t_{\text{cell},\iota}} \leq \frac{1}{1 - t_{\text{cell},\iota}}, \quad \frac{t_{\text{cell},\iota}}{1 - t_{\text{cell},\iota}} \leq \eta_{\text{cell}}^2,$$

using $t_{\text{cell},\iota} \leq \eta_{\text{cell}}^2/8$ and $\eta_{\text{cell}} \leq 1$. Thus Cauchy–Schwarz applied to the finite likelihood-ratio sum yields

$$\text{TV}(\mathbb{Q}_{\text{cell},1/2}^{(n)}, \mathbb{Q}_{\text{cell},u_{\text{cell},\iota}}^{(n)}) \leq \frac{1}{2} \sqrt{\mathcal{X}_{\text{cell},\iota}^{(n)}} \leq \eta_{\text{cell}}.$$

For any honest procedure $\mathbf{l}$, coverage of each grid point is at least β under its corresponding sample law. By [Lemma 2](#), under the reference law $P_{1/2}$ each point is contained in the returned interval with probability at least $\beta - \eta_{\text{cell}} = 7\beta/8$. A connected interval containing a given number of grid points has length at least their spacing times that number minus one. Summing the inclusion probabilities therefore gives

$$\begin{aligned} E_{P_{1/2}}[\text{length}(\mathbf{l})] &\geq \delta_{\text{cell}} \left[(M_{\text{cell},\alpha} + 1) \frac{7\beta}{8} - 1 \right] \\ &\geq \frac{3\beta}{4} w_{\text{cell}}, \end{aligned}$$

because $1/M_{\text{cell},\alpha} \leq \beta/8$. All interval risks lie in $[0, 2]$, and the constant interval $[-1, 1]$ is honest. Hence taking the supremum over models and the infimum over honest procedures preserves this lower bound. Finally,

$$c_{\text{par},\alpha} s_{\text{par}} \leq \frac{3\beta}{4} \frac{\eta_{\text{cell}}}{4} s_{\text{par}} \leq \frac{3\beta}{4} w_{\text{cell}} \leq \mathfrak{L}_{n,d,q,\alpha},$$

which proves the asserted explicit floor.

6. Extension of the many-cell scale to all regimes. Define

$$c_{\text{many},\alpha} := \min \left\{ c_{\text{act},\alpha}, c_{\text{par},\alpha} \min\{1, T_{\text{act},\alpha}^{-1/2}\}, \frac{c_{\text{par},\alpha}}{D_{\text{act},\alpha}} \right\} > 0.$$

We claim

$$c_{\text{many},\alpha} s_{\text{many}} \leq \mathfrak{L}_{n,d,q,\alpha}$$

throughout the parameter range.

First suppose $N \geq T_{\text{act},\alpha}$. If $J \geq D_{\text{act},\alpha}$, the activated bound gives

$$c_{\text{many},\alpha} s_{\text{many}} \leq c_{\text{act},\alpha} s_{\text{many}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

If $J < D_{\text{act},\alpha}$, the capacity inequality from [Item 2](#) implies

$$\left\lfloor \frac{1}{2b_0} \right\rfloor \geq D_{\text{act},\alpha}.$$

Thus the minimum defining J forces $d - 1 < D_{\text{act},\alpha}$, and hence $d \leq D_{\text{act},\alpha}$. Since $N \geq 1$ and $\ell \geq 1$,

$$s_{\text{many}} \leq \frac{d}{N\ell} \leq \frac{D_{\text{act},\alpha}}{\sqrt{N}} = D_{\text{act},\alpha} s_{\text{par}}.$$

It follows that

$$c_{\text{many},\alpha} s_{\text{many}} \leq c_{\text{many},\alpha} D_{\text{act},\alpha} s_{\text{par}} \leq c_{\text{par},\alpha} s_{\text{par}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

This case includes $d = 1$ and $J = 0$.

Finally, if $N < T_{\text{act},\alpha}$, monotonicity of the square root and reciprocal gives

$$\min\{1, T_{\text{act},\alpha}^{-1/2}\} \leq s_{\text{par}}.$$

Using $s_{\text{many}} \leq 1$, we obtain

$$c_{\text{many},\alpha} s_{\text{many}} \leq c_{\text{many},\alpha} \leq c_{\text{par},\alpha} \min\{1, T_{\text{act},\alpha}^{-1/2}\} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

7. Combination of the lower scales. The identities

$$\sqrt{\min\{1, N^{-1}\}} = s_{\text{par}}, \quad \sqrt{\min\left\{1, \left(\frac{d}{N\ell}\right)^2\right\}} = s_{\text{many}}$$

follow by splitting at $N^{-1} = 1$ and $d/(N\ell) = 1$, respectively. Define

$$c_{\text{both},\alpha} := \min\{c_{\text{par},\alpha}, c_{\text{many},\alpha}\} > 0.$$

The preceding bounds give

$$c_{\text{both},\alpha} \max\{s_{\text{par}}, s_{\text{many}}\} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

Moreover,

$$r_{n,d,q} \leq 2 \max\{s_{\text{par}}^2, s_{\text{many}}^2\}.$$

Indeed, when both untruncated terms N^{-1} and $(d/(N\ell))^2$ are at most one, their sum is at most twice their maximum. If either exceeds one, the maximum of the truncated terms equals one, while $r_{n,d,q} \leq 1$. Since the maximum is nonnegative,

$$\sqrt{r_{n,d,q}} \leq 2 \max\{s_{\text{par}}, s_{\text{many}}\}.$$

Therefore, defining

$$c_{\text{low},\alpha} := \frac{c_{\text{both},\alpha}}{2} > 0,$$

we obtain

$$c_{\text{low},\alpha} \sqrt{r_{n,d,q}} \leq \mathfrak{L}_{n,d,q,\alpha}.$$

8. Honesty and the minimax comparison. Set

$$\underline{c}_\alpha := \min\{c_{\text{low},\alpha}, C_\alpha\}, \quad \overline{C}_\alpha := C_\alpha.$$

Then $0 < \underline{c}_\alpha \leq \overline{C}_\alpha < \infty$, and [Item 7](#) supplies the required lower inequality.

Fix $P \in \mathcal{M}_{n,d,q}$. Define the squared error by

$$V_{\text{env}} := \left(\widehat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P)\right)^2.$$

The risk bound implies

$$0 \leq E_P[V_{\text{env}}] \leq \mathfrak{U}_{n,d,q}.$$

If $\mathfrak{U}_{n,d,q} = 0$, nonnegativity gives $V_{\text{env}} = 0$ almost surely, so the interval contains $\tau(P)$ almost surely. If $\mathfrak{U}_{n,d,q} > 0$, define

$$t_{\text{env}} := \frac{\mathfrak{U}_{n,d,q}}{\alpha} > 0.$$

Integrating the pointwise inequality $V_{\text{env}} \geq t_{\text{env}} \mathbf{1}\{V_{\text{env}} \geq t_{\text{env}}\}$ gives

$$t_{\text{env}} \Pr_P(V_{\text{env}} \geq t_{\text{env}}) \leq E_P[V_{\text{env}}] \leq \mathfrak{U}_{n,d,q}, \quad \Pr_P(V_{\text{env}} \geq t_{\text{env}}) \leq \alpha.$$

Because $\tau(P) \in [-1, 1]$, squared error at most t_{env} implies inclusion in the clipped interval. Consequently,

$$\{\tau(P) \notin I_{\alpha}^{\text{mix}}\} \subseteq \{V_{\text{env}} \geq t_{\text{env}}\}, \quad \Pr_P\{\tau(P) \in I_{\alpha}^{\text{mix}}\} \geq 1 - \alpha.$$

The mixed-count estimator takes values in $[-1, 1]$ by [Algorithm 1](#). Its envelope interval therefore has ordered endpoints in $[-1, 1]$. The finite observed-sample space makes its endpoint map measurable, and its Dirac kernel is an honest member of $\mathcal{C}_{n,d,q,\alpha}$. For this kernel, joint sample-and-kernel expected length equals $E_P[\text{length}(I_{\alpha}^{\text{mix}})]$. The definition in [Definition 29](#) hence gives

$$\mathfrak{L}_{n,d,q,\alpha} \leq \sup_{P \in \mathcal{M}_{n,d,q}} E_P[\text{length}(I_{\alpha}^{\text{mix}})].$$

Combining this comparison with [Items 1 and 7](#) proves

$$\mathfrak{C}_{\alpha} \sqrt{r_{n,d,q}} \leq \mathfrak{L}_{n,d,q,\alpha} \leq \sup_{P \in \mathcal{M}_{n,d,q}} E_P[\text{length}(I_{\alpha}^{\text{mix}})] \leq \overline{C}_{\alpha} \sqrt{r_{n,d,q}},$$

together with the asserted coverage.

□

Proof of Theorem 8. 1. *Uniform risk bound for the deterministic estimator.* Let $\overline{C} > 0$ be the universal constant supplied by [Theorem 6](#). Fix $n, d \geq 1$, $0 < \epsilon < 1/2$, and $P_Z \in \mathcal{D}_{d,\epsilon}$. Define the auxiliary assignment vector and synthetic sample by

$$\mathbf{u} \sim \text{Unif}(\{0, 1\}^n), \quad \mathbf{O}_n^{\text{syn}} := (o_i(u_i))_{i=1}^n,$$

with $\mathbf{u}$ independent of $\mathbf{Z}_n$. By [Lemma 5](#), there exists $P_Z^* \in \mathcal{M}_{n,d,\epsilon}^+$ such that

$$\mathcal{L}_{P_Z, \mathbf{u}}(\mathbf{O}_n^{\text{syn}}) = (\mathcal{L}_{P_Z^*}(X, A, S, R, RY))^{\otimes n}, \quad \tau(P_Z^*) = \theta_Z(P_Z).$$

Thus the synthetic sample satisfies the sampling and model hypotheses of [Theorem 6](#) at $q = \epsilon$.

For each fixed sample realization $\mathbf{z} = ((x_i, b_i, y_i))_{i=1}^n$, Cauchy–Schwarz on the 2^n assignment vectors gives

$$\begin{aligned} & \left[\sum_{\mathbf{u} \in \{0,1\}^n} \{ \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)) - \theta_Z(P_Z) \} \right]^2 \\ & \leq 2^n \sum_{\mathbf{u} \in \{0,1\}^n} \{ \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)) - \theta_Z(P_Z) \}^2. \end{aligned}$$

Dividing by 2^{2n} and using the defining finite average yields

$$\begin{aligned} & \left(\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{z}) - \theta_Z(P_Z) \right)^2 \\ & \leq \frac{1}{2^n} \sum_{\mathbf{u} \in \{0,1\}^n} \{ \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)) - \theta_Z(P_Z) \}^2. \end{aligned}$$

Integrating this inequality, identifying the uniform coin average with the synthetic sample law, and then applying [Theorem 6](#), we obtain

$$\begin{aligned} & E_{P_Z} \left[\left(\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{Z}_n) - \theta_Z(P_Z) \right)^2 \right] \\ & \leq E_{P_Z, \mathbf{u}} \left[\left(\hat{\tau}_{n,d,\epsilon}^{\text{mix}}(\mathbf{O}_n^{\text{syn}}) - \theta_Z(P_Z) \right)^2 \right] \\ & = E_{P_Z^*} \left[\left(\hat{\tau}_{n,d,\epsilon}^{\text{mix}}(\mathbf{O}_n) - \tau(P_Z^*) \right)^2 \right] \leq \mathfrak{U}_{n,d,\epsilon} \leq \bar{C} r_{n,d,\epsilon}. \end{aligned}$$

Taking the supremum over a nonempty observational class proves the first assertion. If that class is empty, its real-valued supremum is zero, and the same bound follows from

$$0 \leq r_{n,d,\epsilon} = \min \left\{ 1, \frac{1}{n\epsilon} + \left[\frac{d}{n\epsilon \log(e + n\epsilon)} \right]^2 \right\}, \quad 0 < \bar{C}.$$

2. *Upper bound at fixed positivity.* Fix $0 < \epsilon < 1/2$. For $n, d \geq 1$, define

$$r_{n,d}^{\text{raw}} := \frac{1}{n} + \left[\frac{d}{n \log(e + n)} \right]^2, \quad r_{n,d}^{\text{fix}} := \min\{1, r_{n,d}^{\text{raw}}\},$$

and set

$$\ell_{\epsilon,n} := \log(e + n\epsilon), \quad \ell_{1,n} := \log(e + n), \quad \gamma_{\epsilon,\text{cmp}} := \frac{1 - \log \epsilon}{\epsilon}.$$

All denominators below are positive. In particular,

$$\ell_{\epsilon,n} \geq 1, \quad \ell_{1,n} > 0, \quad \log \epsilon \leq 0, \quad \gamma_{\epsilon,\text{cmp}} \geq 1.$$

Since $\epsilon \leq 1$,

$$e + n \leq \frac{e + n\epsilon}{\epsilon}.$$

Monotonicity of the logarithm therefore gives

$$\ell_{1,n} \leq \ell_{\epsilon,n} - \log \epsilon \leq (1 - \log \epsilon) \ell_{\epsilon,n},$$

where the second inequality follows from

$$(-\log \epsilon)(\ell_{\epsilon,n} - 1) \geq 0.$$

Consequently,

$$n\ell_{1,n} \leq \gamma_{\epsilon,\text{cmp}} n\ell_{\epsilon,n}.$$

Also,

$$\epsilon\gamma_{\epsilon,\text{cmp}}^2 \geq \epsilon\gamma_{\epsilon,\text{cmp}} = 1 - \log \epsilon \geq 1.$$

These two comparisons imply, respectively,

$$\left[\frac{d}{n\ell_{\epsilon,n}} \right]^2 \leq \gamma_{\epsilon,\text{cmp}}^2 \left[\frac{d}{n\ell_{1,n}} \right]^2, \quad \frac{1}{n\epsilon} \leq \gamma_{\epsilon,\text{cmp}}^2 \frac{1}{n}.$$

Adding gives

$$\frac{1}{n\epsilon} + \left[\frac{d}{n\ell_{\epsilon,n}} \right]^2 \leq \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{raw}}.$$

Truncation preserves the needed comparison. Indeed, if $r_{n,d}^{\text{raw}} \leq 1$, then

$$\min\{1, \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{raw}}\} \leq \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{raw}} = \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{fix}};$$

if $r_{n,d}^{\text{raw}} \geq 1$, then

$$\min\{1, \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{raw}}\} \leq 1 \leq \gamma_{\epsilon,\text{cmp}}^2 = \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{fix}}.$$

Thus

$$r_{n,d,\epsilon} \leq \gamma_{\epsilon,\text{cmp}}^2 r_{n,d}^{\text{fix}}.$$

The mixed-count estimator takes values in $[-1, 1]$, by its clipped output and averaging construction in [Algorithm 1](#). Hence, for every sample,

$$-2^n \leq \sum_{\mathbf{u} \in \{0,1\}^n} \hat{\tau}_{n,d,\epsilon}^{\text{mix}}(o_1(u_1), \dots, o_n(u_n)) \leq 2^n,$$

so

$$-1 \leq \hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{z}) \leq 1.$$

This finite-sample map is measurable and is an admissible deterministic procedure in [Definition 33](#). The defining infimum and the preceding bounds therefore give

$$\begin{aligned} \mathfrak{R}_{n,d,\epsilon}^Z &\leq \sup_{P_Z \in \mathcal{D}_{d,\epsilon}} E_{P_Z} \left[\left(\hat{\theta}_{n,d,\epsilon}^{\text{poly}}(\mathbf{Z}_n) - \theta_Z(P_Z) \right)^2 \right] \\ &\leq \overline{C} r_{n,d,\epsilon} \leq C_{\epsilon,\text{up}} r_{n,d}^{\text{fix}}, \quad C_{\epsilon,\text{up}} := \overline{C} \gamma_{\epsilon,\text{cmp}}^2 > 0. \end{aligned}$$

3. *Published lower bound on the zero-control subclass.* Define

$$\mathcal{D}_{d,\epsilon}^{(0)} := \{P_Z \in \mathcal{D}_{d,\epsilon} : \mu_{0x} = 0 \text{ for every } x \text{ with } p_x > 0\},$$

and its minimax risk by

$$\mathfrak{R}_{n,d,\epsilon}^{Z,0} := \inf_{\hat{\theta}} \sup_{P_Z \in \mathcal{D}_{d,\epsilon}^{(0)}} E_{P_Z} \left[\left(\hat{\theta} - \theta_Z(P_Z) \right)^2 \right],$$

using the same procedure class as in [Definition 33](#). The subclass inclusion gives

$$\mathfrak{R}_{n,d,\epsilon}^{Z,0} \leq \mathfrak{R}_{n,d,\epsilon}^Z.$$

The published lower bound [[Zeng et al., 2026](#), Theorem 2, Section 3.2, p. 9, and its zero-control construction in Section C.5, pp. 39–44] supplies constants

$$c_{\epsilon,\text{lit}} > 0, \quad c_{\epsilon,\text{dim}} > 0, \quad n_{\epsilon,\text{lit}} \in \mathbb{N}$$

such that

$$\mathfrak{R}_{n,d,\epsilon}^{Z,0} \geq c_{\epsilon,\text{lit}} \left\{ \frac{1}{n} + \left[\frac{d}{n \log n} \right]^2 \right\}$$

whenever

$$n \geq n_{\epsilon,\text{lit}}, \quad 1 \leq d \leq c_{\epsilon,\text{dim}} n \log n.$$

The target belongs to $[-1, 1]$, since it is a probability-weighted average of differences of two means in $[0, 1]$. By [Lemma 1](#), clipping and then taking conditional barycenters preserve or reduce squared-error risk at every law. Thus the deterministic and randomized minimax values agree, and the published bound applies to the procedure class above.

4. *A uniform parametric lower bound.* We next establish a universal constant $c_{\text{par}} > 0$ such that

$$\frac{c_{\text{par}}}{n} \leq \mathfrak{R}_{n,d,\epsilon}^{Z,0} \leq \mathfrak{R}_{n,d,\epsilon}^Z \quad (n, d \geq 1, \ 0 < \epsilon < 1/2).$$

For $n \geq 1$, define

$$\delta_n := \frac{1}{4\sqrt{n}}, \quad \vartheta_0^{\text{test}} := \frac{1}{2}, \quad \vartheta_1^{\text{test}} := \frac{1}{2} + \delta_n.$$

Then $0 < \delta_n \leq 1/4$, so both target values lie in $[0, 1]$. For $\iota \in \{0, 1\}$, define the one-record law $P_{Z,\iota}^{\text{test}}$ on $\{1, \dots, d\} \times \{0, 1\}^2$ by

$$P_{Z,\iota}^{\text{test}}\{(x, b, y)\} = \begin{cases} 1/2, & (x, b, y) = (1, 0, 0), \\ (1 - \vartheta_\iota^{\text{test}})/2, & (x, b, y) = (1, 1, 0), \\ \vartheta_\iota^{\text{test}}/2, & (x, b, y) = (1, 1, 1), \\ 0, & \text{otherwise.} \end{cases}$$

Its sole occupied baseline category has treatment probability $1/2$, control mean zero, and treated mean $\vartheta_\iota^{\text{test}}$. Thus

$$P_{Z,\iota}^{\text{test}} \in \mathcal{D}_{d,\epsilon}^{(0)}, \quad \theta_Z(P_{Z,\iota}^{\text{test}}) = \vartheta_\iota^{\text{test}}.$$

The two laws have common support. Define their chi-square divergence by

$$\chi^2(P_{Z,1}^{\text{test}} \| P_{Z,0}^{\text{test}}) := \sum_{\substack{z \in \{1, \dots, d\} \times \{0, 1\}^2: \\ P_{Z,0}^{\text{test}}\{z\} > 0}} \frac{(P_{Z,1}^{\text{test}}\{z\} - P_{Z,0}^{\text{test}}\{z\})^2}{P_{Z,0}^{\text{test}}\{z\}}.$$

Only the two treated-outcome probabilities change. Each changes in absolute value by $\delta_n/2$, and each has reference probability $1/4$. Hence

$$\chi^2(P_{Z,1}^{\text{test}} \| P_{Z,0}^{\text{test}}) = 2\delta_n^2 = \frac{1}{8n}.$$

Write the sample laws and their likelihood ratio as

$$Q_\iota^{\text{test}} := (P_{Z,\iota}^{\text{test}})^{\otimes n}, \quad \iota \in \{0, 1\},$$

$$L_{\text{test}}(\mathbf{z}) := \begin{cases} Q_1^{\text{test}}\{\mathbf{z}\}/Q_0^{\text{test}}\{\mathbf{z}\}, & Q_0^{\text{test}}\{\mathbf{z}\} > 0, \\ 0, & Q_0^{\text{test}}\{\mathbf{z}\} = 0. \end{cases}$$

Independence gives the second-moment identity and bound

$$E_{Q_0^{\text{test}}}[L_{\text{test}}^2] = [1 + \chi^2(P_{Z,1}^{\text{test}} \| P_{Z,0}^{\text{test}})]^n = \left(1 + \frac{1}{8n}\right)^n \leq e^{1/8}.$$

Define the universal testing constant by

$$c_{\text{test}} := \frac{1}{4e^{1/8}} > 0.$$

For any event $\mathcal{A}$ in the finite sample space, set

$$\eta_{\mathcal{A}} := Q_1^{\text{test}}(\mathcal{A}^c) + Q_0^{\text{test}}(\mathcal{A}).$$

Cauchy–Schwarz yields

$$1 - \eta_{\mathcal{A}} \leq Q_1^{\text{test}}(\mathcal{A}) = E_{Q_0^{\text{test}}}[L_{\text{test}}\mathbf{1}_{\mathcal{A}}] \leq \sqrt{e^{1/8}Q_0^{\text{test}}(\mathcal{A})}.$$

If $\eta_{\mathcal{A}} < c_{\text{test}}$, then $Q_0^{\text{test}}(\mathcal{A}) \leq \eta_{\mathcal{A}}$ implies

$$\sqrt{e^{1/8}Q_0^{\text{test}}(\mathcal{A})} < \frac{1}{2}, \quad 1 - \eta_{\mathcal{A}} > \frac{3}{4},$$

because $c_{\text{test}} \leq 1/4$. This contradicts the preceding inequality. Therefore every event satisfies

$$Q_1^{\text{test}}(\mathcal{A}^c) + Q_0^{\text{test}}(\mathcal{A}) \geq c_{\text{test}}.$$

Consider an arbitrary estimator. By [Lemma 1](#), it suffices to consider its clipped conditional action kernel κ , supported on $[-1, 1]$. Define its barycenter by

$$b_{\kappa}(\mathbf{z}) := \int_{[-1,1]} h \kappa(\mathbf{z}, dh).$$

The same cited result gives

$$b_{\kappa}(\mathbf{z}) \in [-1, 1], \quad E_{Q_\iota^{\text{test}}}[(b_{\kappa} - \vartheta_\iota^{\text{test}})^2] \leq E_{P_{Z,\iota}^{\text{test}}}[(\hat{\theta} - \vartheta_\iota^{\text{test}})^2], \quad \iota \in \{0, 1\},$$

where the right-hand expectation includes the estimator's randomization.

Define the nearest-target test and its large-error events by

$$\mathcal{A}_{\text{test}} := \left\{ \mathbf{z} : |b_\kappa(\mathbf{z}) - \vartheta_1^{\text{test}}| \leq |b_\kappa(\mathbf{z}) - \vartheta_0^{\text{test}}| \right\},$$

$$\mathcal{A}_{\text{bad},\iota} := \left\{ \mathbf{z} : |b_\kappa(\mathbf{z}) - \vartheta_\iota^{\text{test}}| \geq \frac{\delta_n}{2} \right\}, \quad \iota \in \{0, 1\}.$$

The triangle inequality gives

$$\delta_n = |\vartheta_1^{\text{test}} - \vartheta_0^{\text{test}}| \leq |b_\kappa - \vartheta_0^{\text{test}}| + |b_\kappa - \vartheta_1^{\text{test}}|.$$

Consequently,

$$\mathcal{A}_{\text{test}} \subseteq \mathcal{A}_{\text{bad},0}, \quad \mathcal{A}_{\text{test}}^c \subseteq \mathcal{A}_{\text{bad},1}.$$

Applying the testing bound and then these inclusions yields

$$c_{\text{test}} \leq Q_1^{\text{test}}(\mathcal{A}_{\text{bad},1}) + Q_0^{\text{test}}(\mathcal{A}_{\text{bad},0}),$$

and hence

$$\frac{c_{\text{test}}}{2} \leq \max_{\iota \in \{0,1\}} Q_\iota^{\text{test}}(\mathcal{A}_{\text{bad},\iota}).$$

On each large-error event, squared loss is at least $(\delta_n/2)^2$. Therefore

$$\begin{aligned} \frac{c_{\text{test}} \delta_n^2}{8} &\leq \left(\frac{\delta_n}{2} \right)^2 \max_{\iota \in \{0,1\}} Q_\iota^{\text{test}}(\mathcal{A}_{\text{bad},\iota}) \\ &\leq \max_{\iota \in \{0,1\}} E_{Q_\iota^{\text{test}}} [(b_\kappa - \vartheta_\iota^{\text{test}})^2] \\ &\leq \max_{\iota \in \{0,1\}} E_{P_{Z,\iota}^{\text{test}}} [(\hat{\theta} - \vartheta_\iota^{\text{test}})^2]. \end{aligned}$$

Both testing laws belong to the zero-control subclass, so taking the minimax infimum gives

$$\frac{c_{\text{par}}}{n} = \frac{c_{\text{test}} \delta_n^2}{8} \leq \mathfrak{R}_{n,d,\epsilon}^{Z,0} \leq \mathfrak{R}_{n,d,\epsilon}^Z, \quad c_{\text{par}} := \frac{c_{\text{test}}}{128} > 0.$$

5. *Calibration and the large-sample bound within the published range.* Choose an integer $n_{\epsilon,1}$ satisfying

$$n_{\epsilon,1} \geq \frac{2}{c_{\epsilon,\text{dim}}},$$

and define

$$n_{\epsilon,*} := \max\{n_{\epsilon,\text{lit}}, 3, n_{\epsilon,1}\}, \quad c_{\epsilon,\text{sat}} := c_{\epsilon,\text{lit}} \left(\frac{c_{\epsilon,\text{dim}}}{2} \right)^2,$$

$$c_\epsilon := \min \left\{ \frac{c_{\text{par}}}{n_{\epsilon,*}}, \min\{c_{\epsilon,\text{lit}}, c_{\epsilon,\text{sat}}\} \right\} > 0.$$

For all $n, d \geq 1$,

$$0 \leq r_{n,d}^{\text{fix}} \leq 1.$$

First suppose $n \geq n_{\epsilon,*}$. Then

$$n \geq n_{\epsilon,\text{lit}}, \quad n \geq 3, \quad \log n > 1, \quad n \log n > 0.$$

Furthermore, the choice of $n_{\epsilon,1}$ gives

$$2 \leq c_{\epsilon,\dim} n \leq c_{\epsilon,\dim} n \log n.$$

If

$$d \leq c_{\epsilon,\dim} n \log n,$$

the published bound and subclass inclusion imply

$$c_{\epsilon,\text{lit}} \left\{ \frac{1}{n} + \left[\frac{d}{n \log n} \right]^2 \right\} \leq \mathfrak{R}_{n,d,\epsilon}^{Z,0} \leq \mathfrak{R}_{n,d,\epsilon}^Z.$$

Since

$$0 < \log n \leq \log(e+n), \quad 0 \leq \frac{d}{n \log(e+n)} \leq \frac{d}{n \log n},$$

we have

$$r_{n,d}^{\text{fix}} \leq \frac{1}{n} + \left[\frac{d}{n \log(e+n)} \right]^2 \leq \frac{1}{n} + \left[\frac{d}{n \log n} \right]^2.$$

Using $c_\epsilon \leq c_{\epsilon,\text{lit}}$ gives

$$\begin{aligned} c_\epsilon r_{n,d}^{\text{fix}} &\leq c_\epsilon \left\{ \frac{1}{n} + \left[\frac{d}{n \log n} \right]^2 \right\} \\ &\leq c_{\epsilon,\text{lit}} \left\{ \frac{1}{n} + \left[\frac{d}{n \log n} \right]^2 \right\} \leq \mathfrak{R}_{n,d,\epsilon}^Z. \end{aligned}$$

6. *The large-alphabet case.* Continue to assume $n \geq n_{\epsilon,*}$, and now suppose

$$d > c_{\epsilon,\dim} n \log n.$$

Define the smaller alphabet size by

$$d_{\epsilon,n}^{\text{pad}} := \lfloor c_{\epsilon,\dim} n \log n \rfloor.$$

The preceding lower bound of two on the quantity inside the floor gives

$$1 \leq d_{\epsilon,n}^{\text{pad}} \leq d, \quad d_{\epsilon,n}^{\text{pad}} \leq c_{\epsilon,\dim} n \log n.$$

Also, the elementary floor bound gives

$$\begin{aligned} \frac{c_{\epsilon,\dim}}{2} n \log n &\leq c_{\epsilon,\dim} n \log n - 1 \\ &< d_{\epsilon,n}^{\text{pad}}. \end{aligned}$$

The published lower bound therefore applies at this smaller alphabet:

$$c_{\epsilon,\text{lit}} \left\{ \frac{1}{n} + \left[\frac{d_{\epsilon,n}^{\text{pad}}}{n \log n} \right]^2 \right\} \leq \mathfrak{R}_{n,d_{\epsilon,n}^{\text{pad}},\epsilon}^{Z,0} \leq \mathfrak{R}_{n,d_{\epsilon,n}^{\text{pad}},\epsilon}^Z.$$

To compare the two alphabet sizes, extend each law $P_Z \in \mathcal{D}_{d_{\epsilon,n}^{\text{pad}}, \epsilon}$ by zero masses. Explicitly, define the d -category law $\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z$ by

$$(\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z)\{(x, b, y)\} := \begin{cases} P_Z\{(x, b, y)\}, & 1 \leq x \leq d_{\epsilon,n}^{\text{pad}}, \\ 0, & d_{\epsilon,n}^{\text{pad}} < x \leq d, \end{cases} \quad b, y \in \{0, 1\}.$$

On the original categories, all baseline, treatment, and outcome probabilities are preserved; the added categories have zero mass. Thus

$$\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z \in \mathcal{D}_{d, \epsilon}, \quad \theta_Z(\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z) = \theta_Z(P_Z).$$

For a d -category estimator $\hat{\theta}$, define $\hat{\theta}^{\text{res}}$ on the smaller sample space by

$$\hat{\theta}^{\text{res}}((x_i, b_i, y_i)_{i=1}^n) := \hat{\theta}((x_i, b_i, y_i)_{i=1}^n), \quad x_i \in \{1, \dots, d_{\epsilon,n}^{\text{pad}}\},$$

preserving its conditional action distribution if it is randomized. The product sample law is preserved under this inclusion, so

$$\begin{aligned} E_{P_Z} \left[\left(\hat{\theta}^{\text{res}} - \theta_Z(P_Z) \right)^2 \right] \\ = E_{\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z} \left[\left(\hat{\theta} - \theta_Z(\text{pad}_{d_{\epsilon,n}^{\text{pad}}, d} P_Z) \right)^2 \right]. \end{aligned}$$

Taking suprema and then infima proves

$$\mathfrak{R}_{n, d_{\epsilon,n}^{\text{pad}}, \epsilon}^Z \leq \mathfrak{R}_{n, d, \epsilon}^Z.$$

Since $n \log n > 0$, the floor bound also gives

$$\frac{c_{\epsilon, \text{dim}}}{2} \leq \frac{d_{\epsilon,n}^{\text{pad}}}{n \log n}.$$

Squaring and using $1/n \geq 0$, we obtain

$$c_{\epsilon, \text{sat}} \leq c_{\epsilon, \text{lit}} \left\{ \frac{1}{n} + \left[\frac{d_{\epsilon,n}^{\text{pad}}}{n \log n} \right]^2 \right\}.$$

Finally, $r_{n,d}^{\text{fix}} \leq 1$ and $c_{\epsilon} \leq c_{\epsilon, \text{sat}}$ give

$$\begin{aligned} c_{\epsilon} r_{n,d}^{\text{fix}} &\leq c_{\epsilon} \leq c_{\epsilon, \text{sat}} \\ &\leq c_{\epsilon, \text{lit}} \left\{ \frac{1}{n} + \left[\frac{d_{\epsilon,n}^{\text{pad}}}{n \log n} \right]^2 \right\} \\ &\leq \mathfrak{R}_{n, d_{\epsilon,n}^{\text{pad}}, \epsilon}^{Z, 0} \leq \mathfrak{R}_{n, d_{\epsilon,n}^{\text{pad}}, \epsilon}^Z \leq \mathfrak{R}_{n, d, \epsilon}^Z. \end{aligned}$$

7. *Small samples and the final constants.* If $1 \leq n < n_{\epsilon,*}$, the parametric bound and the calibration of c_ϵ imply

$$c_\epsilon r_{n,d}^{\text{fix}} \leq c_\epsilon \leq \frac{c_{\text{par}}}{n_{\epsilon,*}} \leq \frac{c_{\text{par}}}{n} \leq \mathfrak{R}_{n,d,\epsilon}^Z.$$

Together with the two large-sample cases, this establishes the lower bound for every $n, d \geq 1$. Define

$$C_\epsilon := \max\{C_{\epsilon,\text{up}}, c_\epsilon\}.$$

Then $0 < c_\epsilon \leq C_\epsilon < \infty$. Since $r_{n,d}^{\text{fix}} \geq 0$, the upper bound already proved gives

$$\mathfrak{R}_{n,d,\epsilon}^Z \leq C_{\epsilon,\text{up}} r_{n,d}^{\text{fix}} \leq C_\epsilon r_{n,d}^{\text{fix}}.$$

Thus, for every $n, d \geq 1$,

$$c_\epsilon \min \left\{ 1, \frac{1}{n} + \left[\frac{d}{n \log(e+n)} \right]^2 \right\} \leq \mathfrak{R}_{n,d,\epsilon}^Z \leq C_\epsilon \min \left\{ 1, \frac{1}{n} + \left[\frac{d}{n \log(e+n)} \right]^2 \right\},$$

as claimed. □

Proof of Theorem 9. We take $\bar{C} = 10^{30}$. We first compare the deterministic envelope with the rate, and then establish the risk bound.

1. Define the untruncated rate expression by

$$r_{n,d,q}^\circ := N^{-1} + \left(\frac{d}{N\ell} \right)^2, \quad N = nq, \quad m = \frac{n}{6}, \quad N_0 = \frac{N}{6}, \quad \ell = \log(e + N).$$

The domain assumptions give

$$0 < N \leq n, \quad m > 0, \quad N_0 > 0, \quad r_{n,d,q} = \min\{1, r_{n,d,q}^\circ\}.$$

If $N \leq 1$, then $N^{-1} \geq 1$, so

$$\mathfrak{U}_{n,d,q} = r_{n,d,q} = 1 \leq 10^{30} r_{n,d,q}.$$

For the envelope comparison, henceforth suppose $N > 1$.

2. The logarithmic scale and stream sizes satisfy

$$1 < \ell, \quad e^\ell = e + N, \quad e^{-\ell} \leq N^{-1}, \quad \frac{1}{N_0} = 6N^{-1}, \quad \frac{1}{m} = \frac{6}{n} \leq 6N^{-1}.$$

Indeed, $e + N > e$, and inversion of $e^\ell = e + N > N$ gives the exponential bound.

The elementary numerical bound for $\log 2$ gives

$$\gamma_0 = \log 2 - \frac{1}{2} \geq \frac{1}{8}.$$

For every real $t_{\text{dec}} > 0$, the inequality $t_{\text{dec}} \leq e^{t_{\text{dec}}}$ implies $e^{-t_{\text{dec}}} \leq t_{\text{dec}}^{-1}$. Applying this with $t_{\text{dec}} = n\gamma_0$ yields

$$e^{-n\gamma_0} \leq \frac{1}{n\gamma_0} \leq \frac{8}{n} \leq 8N^{-1}.$$

3. Suppose first that $\ell \geq 128$ and $d \geq \sqrt{N}$. On this branch,

$$k = \lfloor \ell/64 \rfloor, \quad B = 256\ell, \quad \frac{\ell}{128} \leq k \leq \frac{\ell}{64}.$$

For completeness, for every real $t_{\text{floor}} \geq 1$,

$$\lfloor t_{\text{floor}} \rfloor \geq \frac{t_{\text{floor}}}{2}.$$

If $t_{\text{floor}} < 2$, its floor is at least one; if $t_{\text{floor}} \geq 2$, use $\lfloor t_{\text{floor}} \rfloor > t_{\text{floor}} - 1 \geq t_{\text{floor}}/2$. This proves the lower bound on k .

The inequality $2t_{\text{exp}} \leq e^{t_{\text{exp}}}$, for real t_{exp} , gives

$$\ell \leq 16e^{\ell/32}, \quad \ell^4 \leq 65536e^{\ell/8}.$$

Since $e < 3$ and $N > 1$,

$$e^\ell = e + N \leq 4N \leq 4d^2, \quad e^{\ell/2} \leq 2d.$$

Consequently,

$$\begin{aligned} (e^{\ell/8} + 1)\ell^4 &\leq 2e^{\ell/8} 65536e^{\ell/8} \\ &= 131072e^{\ell/4} \leq 131072e^{\ell/2} \leq 262144d, \end{aligned} \quad \ell \leq e^{\ell/2} \leq 2d.$$

4. On the same branch, the preceding bounds imply

$$e^{-16\ell} \leq N^{-1}, \quad e^{-32\ell} \leq N^{-1}, \quad N^{-1} \leq \frac{2d}{N\ell}.$$

Moreover,

$$\begin{aligned} \frac{4dB}{N_0k^2} &= \frac{6144d\ell}{Nk^2} \\ &\leq \frac{6144d\ell}{N(\ell/128)^2} = 100663296 \frac{d}{N\ell}. \end{aligned}$$

Thus

$$\frac{4dB}{N_0k^2} + 3e^{-16\ell} \leq 100663302 \frac{d}{N\ell},$$

and, because both sides are nonnegative,

$$b_{n,d,q}^2 \leq 40532401478172816 \left(\frac{d}{N\ell} \right)^2.$$

5. Since $N_0 \leq m$ and $B \geq 1$,

$$\frac{B}{mN_0} \leq \frac{B}{N_0^2} \leq \frac{B^2}{N_0^2}.$$

Therefore

$$\frac{B^2}{N_0^2} + \frac{B}{mN_0} \leq \frac{2B^2}{N_0^2} = 4718592 \frac{\ell^2}{N^2}.$$

The growth estimate above also gives

$$\frac{d(e^{\ell/8} + 1)\ell^2}{N^2} = \frac{d(e^{\ell/8} + 1)\ell^4}{N^2\ell^2} \leq 262144 \left(\frac{d}{N\ell} \right)^2.$$

Combining these inequalities,

$$64d(e^{\ell/8} + 1) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right) \leq 79164837199872 \left(\frac{d}{N\ell} \right)^2.$$

The remaining variance terms satisfy

$$8 \left(\frac{4}{N_0} + \frac{1}{m} \right) \leq 240N^{-1},$$

and, using $N^{-1} \leq 1$ and $1/m \leq 6$,

$$32e^{-32\ell} \left(1 + \frac{1}{m} \right) \leq 224N^{-1}.$$

Together with the prefix-tail bound, this proves

$$b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0} \leq 40611566315372688 \left(\frac{d}{N\ell} \right)^2 + 496N^{-1} \leq 10^{30}r_{n,d,q}^\circ.$$

If $r_{n,d,q}^\circ \leq 1$, take the minimum with four on the left. If $r_{n,d,q}^\circ > 1$, use $\mathfrak{U}_{n,d,q} \leq 4 \leq 10^{30}$. In either case,

$$\mathfrak{U}_{n,d,q} \leq 10^{30}r_{n,d,q}.$$

6. On the remaining branch, $\ell < 128$ or $d < \sqrt{N}$. Since $e \geq 1$,

$$0 \leq \frac{8d}{eN_0} \leq \frac{48d}{N}, \quad \left(\frac{8d}{eN_0} \right)^2 \leq 2304 \left(\frac{d}{N} \right)^2.$$

If $\ell < 128$, then

$$\left(\frac{d}{N} \right)^2 = \ell^2 \left(\frac{d}{N\ell} \right)^2 \leq 16384 \left(\frac{d}{N\ell} \right)^2.$$

If $d < \sqrt{N}$, then $d^2 \leq N$, so

$$\left(\frac{d}{N} \right)^2 \leq N^{-1}.$$

Hence throughout this branch,

$$\left(\frac{d}{N} \right)^2 \leq N^{-1} + 16384 \left(\frac{d}{N\ell} \right)^2.$$

Also,

$$4 \left(\frac{4}{N_0} + \frac{1}{m} \right) \leq 120N^{-1}.$$

It follows that

$$\begin{aligned} \left(\frac{8d}{eN_0}\right)^2 + 4\left(\frac{4}{N_0} + \frac{1}{m}\right) + 4e^{-n\gamma_0} &\leq 2456N^{-1} + 37748736\left(\frac{d}{N\ell}\right)^2 \\ &\leq 10^{30}r_{n,d,q}^\circ. \end{aligned}$$

The same split at $r_{n,d,q}^\circ = 1$ proves

$$\mathfrak{U}_{n,d,q} \leq 10^{30}r_{n,d,q}.$$

This completes the deterministic comparison over the stated domain.

7. Fix now $P \in \mathcal{M}_{n,d,q}^+$. By [Lemma 3](#) and [Definition 8](#),

$$\Phi(P) = \tau(P) = P(Y(1) = 1) - P(Y(0) = 1) \in [-1, 1].$$

The auxiliary output in [Algorithm 1](#) lies in $[-1, 1]$. Its conditional average lies in that interval as well: for each prefix length $\nu \leq n$, there are 3^ν equally weighted stream assignments, and

$$\Pr(K_0 = \nu) \geq 0, \quad \sum_{\nu=0}^n \Pr(K_0 = \nu) \leq 1,$$

with the remaining probability assigned output zero. Thus

$$\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) \in [-1, 1], \quad E_P[(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] \leq 4.$$

If $N \leq 1$, the auxiliary output and its conditional average are zero, so

$$E_P[(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] = \Phi(P)^2 \leq 1 = \mathfrak{U}_{n,d,q}.$$

If $N > 1$ and $\mathfrak{U}_{n,d,q} = 4$, the displayed bound by four proves the required inequality. We henceforth consider $N > 1$ and $\mathfrak{U}_{n,d,q} \neq 4$.

8. We first establish the transfer to independent Poisson streams. Write

$$\mathbf{Q}_{\text{str}} := \bigotimes_{h_{\text{str}} \in \{0,1,2\}} \left[\sum_{\nu=0}^{\infty} e^{-m} \frac{m^\nu}{\nu!} (\mathcal{L}_P(O_1))^{\otimes \nu} \right],$$

where each sum is a probability law on the disjoint union of finite observed-record sequences. The stream labels 0, 1, 2 denote membership, pilot, and estimation, respectively.

An uncapped prefix of length $K_0 \sim \text{Poisson}(n/2)$, marked independently with these three labels, has law $\mathbf{Q}_{\text{str}}$ after separation into streams. Indeed, for any prescribed nonnegative integer stream sizes ν_0, ν_1, ν_2 , put $\nu_{\text{tot}} := \nu_0 + \nu_1 + \nu_2$. Their joint probability is

$$e^{-n/2} \frac{(n/2)^{\nu_{\text{tot}}}}{\nu_{\text{tot}}!} \frac{\nu_{\text{tot}}!}{\nu_0! \nu_1! \nu_2!} 3^{-\nu_{\text{tot}}} = \prod_{h_{\text{str}}=0}^2 e^{-m} \frac{m^{\nu_{h_{\text{str}}}}}{\nu_{h_{\text{str}}}!}.$$

Conditional on these sizes, the records in the three streams have their respective product observed-data laws. Equality on all count fibres proves the asserted stream law.

Let

$$\tilde{\tau}_{\text{Pois}} := \text{clip}_{[-1,1]} \left(2 \sum_{j=(a,x,s) \in \mathcal{J}_d} g(a) W_j G_j \right)$$

be the uncapped output, using the statistics of [Definitions 22 to 28](#) on these finite streams. On the ratio branch, $G_j = D_j$. Here and below E_{Pois} and $\Pr_{\text{Pois}}$ denote expectation and probability under $\mathbf{Q}_{\text{str}}$.

For the capped auxiliary output, conditional averaging gives

$$\begin{aligned} E_{\omega} [(\tilde{\tau}(\mathbf{O}_n; \omega) - \Phi(P))^2 \mid \mathbf{O}_n] \\ = (\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P))^2 + E_{\omega} [(\tilde{\tau}(\mathbf{O}_n; \omega) - \hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n))^2 \mid \mathbf{O}_n]. \end{aligned}$$

On each prefix-count fibre $\nu \leq n$, its stream law agrees with the corresponding uncapped law. Summing these fibres, discarding the nonnegative uncapped loss on the remaining fibres, and bounding the overflow loss by four yields

$$E_P [(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] \leq E_{\text{Pois}} [(\tilde{\tau}_{\text{Pois}} - \Phi(P))^2] + 4 \Pr(K_0 > n).$$

The Poisson probability series evaluates the exponential moment as

$$E[2^{K_0}] = e^{-n/2} \sum_{\nu=0}^{\infty} \frac{n^{\nu}}{\nu!} = e^{n/2}.$$

Since $\{K_0 > n\} \subseteq \{2^{K_0} \geq 2^n\}$, exponential Markov gives

$$\Pr(K_0 > n) \leq \Pr(2^{K_0} \geq 2^n) \leq 2^{-n} E[2^{K_0}] = e^{-n \log 2 + n/2}.$$

Multiplying by $e^{-n/2}$ cancels the intensity factor, so

$$\Pr(K_0 > n) e^{-n/2} \leq e^{-n \log 2}, \quad \Pr(K_0 > n) \leq e^{-n \gamma_0}.$$

Consequently,

$$E_P [(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] \leq E_{\text{Pois}} [(\tilde{\tau}_{\text{Pois}} - \Phi(P))^2] + 4e^{-n \gamma_0}.$$

All these expectations are well defined; the capped sample space is finite and the clipped outputs are bounded.

9. For every $j = (a, x, s) \in \mathcal{J}_d$, define

$$p_j := p_{axs}(P), \quad \mu_j := \mu_{axs}(P), \quad t_j := mP(A = a, X = x, S = s, R = 1).$$

These definitions include null cells, with the mean convention of [Definition 10](#). Probability monotonicity and the occupied-cell arrival condition give

$$0 \leq \mu_j \leq 1, \quad t_j \geq N_0 p_j, \quad \sum_j p_j = 1, \quad |\mathcal{J}_d| = 4d.$$

For $p_j = 0$, the membership, arrival, and arrived-one probabilities are zero, and all the associated counts vanish almost surely.

Applying the same Poisson partition calculation to stream and cell categories gives independent counts across distinct cells and streams, with

$$M_j \sim \text{Poisson}(mp_j), \quad C'_j, C_j \sim \text{Poisson}(t_j).$$

The estimation arrived-one and arrived-zero intensities are, respectively, $t_j\mu_j$ and $t_j(1-\mu_j)$. Poisson moments therefore yield

$$E_{\text{Pois}}W_j = p_j, \quad E_{\text{Pois}}W_j^2 = p_j^2 + \frac{p_j}{m}.$$

Also, the membership statistic is independent of every statistic formed from the pilot and estimation streams.

10. Consider the polynomial branch $\ell \geq 128$, $d \geq \sqrt{N}$. Define the pilot-selection event by

$$\mathbf{L}_j := \{C'_j \leq B/4\}.$$

The coefficient and cell-statistic definitions in [Definitions 21, 24](#) and [25](#) give

$$1 - Q_k(t) = \sum_{v=1}^{k-1} a_v t^v, \quad G_j = \mathbf{1}_{\mathbf{L}_j} H_j + \mathbf{1}_{\mathbf{L}_j^c} D_j \quad (t \geq 0).$$

To justify the expansion at zero and its degree, the numerator $1 - T_k(1 - 2t/B)$ vanishes at zero and has degree at most k . The recurrence for T_k , differentiated at one, gives

$$T'_0(1) = 0, \quad T'_1(1) = 1, \quad T'_{h+1}(1) = 2 + 2T'_h(1) - T'_{h-1}(1) \quad (h \geq 1).$$

Induction yields $T'_h(1) = h^2$. Division of the numerator by t therefore gives

$$Q_k(0) = 1, \quad \deg(1 - Q_k) \leq k - 1,$$

which proves the displayed finite expansion.

For integers $v \geq 1$, write

$$(C_j)_v := \prod_{h=0}^{v-1} \max\{C_j - h, 0\}, \quad \mathbf{F}_{j,v} := U_j \prod_{h=1}^{v-1} \max\{C_j - h, 0\}.$$

Empty products equal one. Counting ordered distinct tuples with the first record an arrived one and the remaining records arrived gives

$$E_{\text{Pois}}\mathbf{F}_{j,v} = t_j^v \mu_j.$$

More explicitly, condition on the marked prefix size, count its ordered distinct v -tuples, and then use the Poisson factorial moment $E(K_0)_v = (n/2)^v$. The one-record probabilities are $P(A = a, X = x, S = s, R = 1, Y = 1)/3$ for the distinguished record and $P(A = a, X = x, S = s, R = 1)/3$ for the others. Their product gives the stated identity. When the arrived-cell probability is zero, both sides vanish. Thus

$$E_{\text{Pois}}H_j = \mu_j\{1 - Q_k(t_j)\}.$$

For the ratio statistic, conditioning on the pattern of estimation arrivals gives mean μ_j when $C_j > 0$; its zero-count value is zero. Consequently,

$$E_{\text{Pois}} D_j = \mu_j(1 - e^{-t_j}), \quad 0 \leq D_j \leq 1.$$

Pilot independence now gives the exact selected-statistic bias:

$$E_{\text{Pois}} G_j - \mu_j = -\mu_j \left\{ \Pr_{\text{Pois}}(\mathbf{L}_j) Q_k(t_j) + \Pr_{\text{Pois}}(\mathbf{L}_j^c) e^{-t_j} \right\}.$$

11. We next derive the coefficient bound needed for the polynomial moments. For a real polynomial F_{pol} , define

$$\|F_{\text{pol}}\|_{\text{coef}} := \sum_{h \geq 0} |[t_{\text{pol}}^h] F_{\text{pol}}|,$$

where t_{pol} is the polynomial indeterminate and the sum is finite. For a second real polynomial and coefficient indices, take

$$F_{\text{in}} \in \mathbb{R}[t_{\text{pol}}], \quad h_{\text{coef}}, i_{\text{coef}}, j_{\text{coef}} \in \mathbb{N}_0.$$

All coefficient sums below are finite, with coefficients beyond the polynomial degree equal to zero. The coefficientwise triangle inequality gives

$$\begin{aligned} \|F_{\text{pol}} + F_{\text{in}}\|_{\text{coef}} &= \sum_{h_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{h_{\text{coef}}}] F_{\text{pol}} + [t_{\text{pol}}^{h_{\text{coef}}}] F_{\text{in}} \right| \\ &\leq \sum_{h_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{h_{\text{coef}}}] F_{\text{pol}} \right| + \sum_{h_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{h_{\text{coef}}}] F_{\text{in}} \right| \\ &= \|F_{\text{pol}}\|_{\text{coef}} + \|F_{\text{in}}\|_{\text{coef}}. \end{aligned}$$

Applying this inequality successively to the finite monomial expansion of the product yields

$$\begin{aligned} \|F_{\text{pol}} F_{\text{in}}\|_{\text{coef}} &= \left\| \sum_{i_{\text{coef}} \geq 0} \sum_{j_{\text{coef}} \geq 0} ([t_{\text{pol}}^{i_{\text{coef}}}] F_{\text{pol}}) ([t_{\text{pol}}^{j_{\text{coef}}}] F_{\text{in}}) t_{\text{pol}}^{i_{\text{coef}} + j_{\text{coef}}} \right\|_{\text{coef}} \\ &\leq \sum_{i_{\text{coef}} \geq 0} \sum_{j_{\text{coef}} \geq 0} \left\| ([t_{\text{pol}}^{i_{\text{coef}}}] F_{\text{pol}}) ([t_{\text{pol}}^{j_{\text{coef}}}] F_{\text{in}}) t_{\text{pol}}^{i_{\text{coef}} + j_{\text{coef}}} \right\|_{\text{coef}} \\ &= \sum_{i_{\text{coef}} \geq 0} \sum_{j_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{i_{\text{coef}}}] F_{\text{pol}} \right| \left| [t_{\text{pol}}^{j_{\text{coef}}}] F_{\text{in}} \right| \\ &= \left(\sum_{i_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{i_{\text{coef}}}] F_{\text{pol}} \right| \right) \left(\sum_{j_{\text{coef}} \geq 0} \left| [t_{\text{pol}}^{j_{\text{coef}}}] F_{\text{in}} \right| \right) \\ &= \|F_{\text{pol}}\|_{\text{coef}} \|F_{\text{in}}\|_{\text{coef}}. \end{aligned}$$

Here the norm of each monomial is the absolute value of its coefficient, by the defining coefficient sum. For every real polynomial F_{in} , submultiplicativity gives by induction

$$\|F_{\text{in}}^h\|_{\text{coef}} \leq \|F_{\text{in}}\|_{\text{coef}}^h \quad (h \in \mathbb{N}_0).$$

For $F_{\text{pol}} \neq 0$, expand the composition as the finite sum

$$F_{\text{pol}} \circ F_{\text{in}} = \sum_{h=0}^{\deg F_{\text{pol}}} ([t_{\text{pol}}^h] F_{\text{pol}}) F_{\text{in}}^h.$$

For the composition bound, assume $\|F_{\text{in}}\|_{\text{coef}} \geq 1$. Subadditivity and the power bound then give

$$\begin{aligned} \|F_{\text{pol}} \circ F_{\text{in}}\|_{\text{coef}} &\leq \sum_{h=0}^{\deg F_{\text{pol}}} \|[t_{\text{pol}}^h] F_{\text{pol}}\| \|F_{\text{in}}^h\|_{\text{coef}} \\ &\leq \sum_{h=0}^{\deg F_{\text{pol}}} \|[t_{\text{pol}}^h] F_{\text{pol}}\| \|F_{\text{in}}\|_{\text{coef}}^h \\ &\leq \sum_{h=0}^{\deg F_{\text{pol}}} \|[t_{\text{pol}}^h] F_{\text{pol}}\| \|F_{\text{in}}\|_{\text{coef}}^{\deg F_{\text{pol}}} \\ &= \|F_{\text{pol}}\|_{\text{coef}} \|F_{\text{in}}\|_{\text{coef}}^{\deg F_{\text{pol}}}. \end{aligned}$$

When $F_{\text{pol}} = 0$, the composition has coefficient norm zero. The Chebyshev recurrence implies

$$\|T_{h+2}\|_{\text{coef}} \leq 2\|T_{h+1}\|_{\text{coef}} + \|T_h\|_{\text{coef}}, \quad \|T_0\|_{\text{coef}} = \|T_1\|_{\text{coef}} = 1.$$

Since $(1 + \sqrt{2})^2 = 2(1 + \sqrt{2}) + 1$, induction gives

$$\|T_h\|_{\text{coef}} \leq (1 + \sqrt{2})^h \quad (h \in \mathbb{N}).$$

Define

$$\mathcal{P}_k(t_{\text{pol}}) := 1 - T_k(1 - 2t_{\text{pol}}).$$

Then

$$\|\mathcal{P}_k\|_{\text{coef}} \leq 1 + (1 + \sqrt{2})^k 3^k \leq 1 + 8^k,$$

using $\|1 - 2t_{\text{pol}}\|_{\text{coef}} = 3$ and $\sqrt{2} \leq 3/2$. The scaled quotient satisfies

$$Q_k(Bt_{\text{pol}}) = \frac{1}{2k^2} \frac{\mathcal{P}_k(t_{\text{pol}})}{t_{\text{pol}}},$$

where division is exact because $\mathcal{P}_k(0) = 0$. Removing the zero constant coefficient cannot increase the coefficient norm. Hence, using $k \geq 2$ and $8^k \geq 64$,

$$\begin{aligned} \|1 - Q_k(Bt_{\text{pol}})\|_{\text{coef}} &\leq 1 + \frac{1 + 8^k}{2k^2} \\ &\leq 1 + \frac{1 + 8^k}{8} \leq 8^k. \end{aligned}$$

In particular,

$$\sum_{v=1}^{k-1} |a_v B^v| \leq 8^k.$$

12. For $t \geq 0$ and integers $v, w \geq 1$, define the factorial overlap

$$\mathcal{A}_{v,w}(t) := \sum_{h=0}^{\min\{v,w\}} \binom{v}{h} \binom{w}{h} h! t^{v+w-h}.$$

Counting the h shared entries of two ordered distinct tuples gives

$$(C_j)_v (C_j)_w = \sum_{h=0}^{\min\{v,w\}} \binom{v}{h} \binom{w}{h} h! (C_j)_{v+w-h}.$$

Taking Poisson factorial moments yields

$$E_{\text{Pois}}[(C_j)_v (C_j)_w] = \mathcal{A}_{v,w}(t_j).$$

Since $0 \leq F_{j,v} \leq (C_j)_v$, expansion of H_j^2 gives

$$E_{\text{Pois}} H_j^2 \leq \sum_{v=1}^{k-1} \sum_{w=1}^{k-1} |a_v a_w| \mathcal{A}_{v,w}(t_j).$$

For $0 \leq t \leq B$ and $v, w \leq k$, the bound

$$\binom{v}{h} \binom{w}{h} h! \leq \frac{(vw)^h}{h!}$$

implies

$$\begin{aligned} \mathcal{A}_{v,w}(t) &\leq B^{v+w} \sum_{h=0}^{\min\{v,w\}} \frac{(k^2/B)^h}{h!} \\ &\leq B^{v+w} e^{k^2/B}. \end{aligned}$$

Therefore, when $t_j \leq B$,

$$E_{\text{Pois}} H_j^2 \leq e^{k^2/B} \left(\sum_{v=1}^{k-1} |a_v B^v| \right)^2 \leq e^{k^2/B} 64^k.$$

The tuning bounds give

$$\frac{k^2}{B} \leq \frac{\ell}{64}, \quad 64^k = 2^{6k} \leq e^{6k},$$

so

$$E_{\text{Pois}} H_j^2 \leq e^{k^2/B+6k} \leq e^{\ell/8} \quad (t_j \leq B).$$

13. A second overlap estimate controls arbitrarily large intensities. Using $\binom{w}{h} h! \leq w^h$ and the binomial theorem,

$$\begin{aligned} \mathcal{A}_{v,w}(t) &\leq \sum_{h=0}^v \binom{v}{h} w^h t^{v-h} t^w \\ &= (t+w)^v t^w \leq (t+v+w)^{v+w}. \end{aligned}$$

For $v, w \leq k$, split according as $(t + 2k)/B \leq 1$ or exceeds one. In both cases,

$$\left(\frac{t + 2k}{B}\right)^{v+w} \leq \max \left\{ 1, \left(\frac{t + 2k}{B}\right)^{2k} \right\}.$$

The coefficient bound thus gives

$$E_{\text{Pois}} H_j^2 \leq 64^k \max \left\{ 1, \left(\frac{t_j + 2k}{B}\right)^{2k} \right\}.$$

14. Suppose $t_j > B$. Define the integer pilot cutoff by

$$\nu_{\text{cut}} := \lfloor B/4 \rfloor.$$

Then $\nu_{\text{cut}} < t_j/4$. The Poisson lower-tail estimate used here follows from the centered exponential moment. To see its relevant form, put

$$a_{\text{tail}} := \frac{1}{\sqrt{2}}.$$

The elementary exponential inequality

$$e^{-a_{\text{tail}}} \leq 1 - a_{\text{tail}} + \frac{a_{\text{tail}}^2}{2}$$

follows by multiplying $e^{a_{\text{tail}}} \geq 1 + a_{\text{tail}} + a_{\text{tail}}^2/2$ by the nonnegative quadratic on the right and using

$$\left(1 + a_{\text{tail}} + \frac{a_{\text{tail}}^2}{2}\right) \left(1 - a_{\text{tail}} + \frac{a_{\text{tail}}^2}{2}\right) = 1 + \frac{a_{\text{tail}}^4}{4} \geq 1.$$

For $C'_j \sim \text{Poisson}(t_j)$, exponential Markov therefore gives

$$\begin{aligned} \Pr_{\text{Pois}} \left(t_j - C'_j > \frac{t_j}{\sqrt{2}} \right) &\leq \exp \left\{ -\frac{t_j}{2} + t_j(e^{-a_{\text{tail}}} - 1 + a_{\text{tail}}) \right\} \\ &\leq e^{-t_j/4}. \end{aligned}$$

On $\{C'_j \leq \nu_{\text{cut}}\}$, $t_j - C'_j > 3t_j/4 > t_j/\sqrt{2}$. Consequently,

$$\Pr_{\text{Pois}}(\mathbf{L}_j) \leq e^{-t_j/4}.$$

Since $t_j > B$, the maximum in the preceding moment bound selects its second argument. The inequalities $64^k \leq e^{6k}$ and $t_{\text{grow}} \leq e^{t_{\text{grow}}}$ for $t_{\text{grow}} := (t_j + 2k)/B$ give

$$\Pr_{\text{Pois}}(\mathbf{L}_j) E_{\text{Pois}} H_j^2 \leq \exp \left\{ -\frac{t_j}{4} + 6k + \frac{2k(t_j + 2k)}{B} \right\}.$$

Here

$$k \leq \frac{t_j}{16384}, \quad \frac{2k}{B} \leq \frac{1}{8192},$$

and hence

$$6k + \frac{2k(t_j + 2k)}{B} \leq t_j \left(\frac{6}{16384} + \frac{1}{8192} + \frac{1}{67108864} \right) \leq \frac{t_j}{8}.$$

Thus

$$\Pr_{\text{Pois}}(\mathbf{L}_j) E_{\text{Pois}} H_j^2 \leq e^{-t_j/8} \leq e^{-32\ell}.$$

Also $\Pr_{\text{Pois}}(\mathbf{L}_j) \leq e^{-32\ell}$.

15. We now bound the two terms in the selected-statistic bias. For $0 \leq t \leq B$, the polynomial quotient in [Definition 20](#) satisfies

$$tQ_k(t) = \frac{B}{2k^2} \{1 - T_k(1 - 2t/B)\}.$$

Because $|1 - 2t/B| \leq 1$, the cosine characterization of T_k gives $|T_k(1 - 2t/B)| \leq 1$, and therefore

$$t|Q_k(t)| \leq \frac{B}{k^2}.$$

For $t_j \leq B$, use $N_0 p_j \mu_j \leq t_j$ to obtain

$$p_j \mu_j \Pr_{\text{Pois}}(\mathbf{L}_j) |Q_k(t_j)| \leq \frac{B}{N_0 k^2}.$$

This includes $t_j = 0$, because then $p_j = 0$.

For $t_j > B$, Cauchy–Schwarz and the heavy moment estimate give

$$\begin{aligned} \left\{ \Pr_{\text{Pois}}(\mathbf{L}_j) |E_{\text{Pois}} H_j| \right\}^2 &\leq \Pr_{\text{Pois}}(\mathbf{L}_j) E_{\text{Pois}} H_j^2 \\ &\leq e^{-t_j/8}. \end{aligned}$$

Using $\mu_j Q_k(t_j) = \mu_j - E_{\text{Pois}} H_j$, it follows that

$$\begin{aligned} \mu_j \Pr_{\text{Pois}}(\mathbf{L}_j) |Q_k(t_j)| &\leq \Pr_{\text{Pois}}(\mathbf{L}_j) + \Pr_{\text{Pois}}(\mathbf{L}_j) |E_{\text{Pois}} H_j| \\ &\leq 2e^{-t_j/16} \leq 2e^{-16\ell}. \end{aligned}$$

Splitting the cell sum at $t_j = B$, using $|g(a)| = 1$ from [Definition 9](#), and then using $4d$ cells and $\sum_j p_j = 1$, yields

$$\left| 2 \sum_j g(a) p_j \mu_j \Pr_{\text{Pois}}(\mathbf{L}_j) Q_k(t_j) \right| \leq \frac{8dB}{N_0 k^2} + 4e^{-16\ell}.$$

16. The complementary-pilot term uses an upper-tail bound with the same integer cutoff. The Poisson probability series gives

$$E_{\text{Pois}}[2^{C'_j}] = e^{-t_j} \sum_{\nu=0}^{\infty} \frac{(2t_j)^\nu}{\nu!} = e^{t_j}.$$

The threshold inclusion and exponential Markov inequality imply

$$\begin{aligned}\Pr_{\text{Pois}}(C'_j > \nu_{\text{cut}}) &\leq \Pr_{\text{Pois}}(2^{C'_j} \geq 2^{\nu_{\text{cut}}}) \\ &\leq 2^{-\nu_{\text{cut}}} E_{\text{Pois}}[2^{C'_j}] = e^{-\nu_{\text{cut}} \log 2 + t_j}.\end{aligned}$$

Multiplying by e^{-t_j} cancels the intensity factor and yields

$$\Pr_{\text{Pois}}(C'_j > \nu_{\text{cut}}) e^{-t_j} \leq e^{-\nu_{\text{cut}} \log 2}.$$

Since

$$64\ell < \nu_{\text{cut}} + 1, \quad \ell \geq 128, \quad \log 2 > \frac{5}{8},$$

we have $\nu_{\text{cut}} \log 2 \geq 16\ell$. Thus

$$\Pr_{\text{Pois}}(\mathbf{L}_j^c) e^{-t_j} \leq e^{-16\ell},$$

including $t_j = 0$, when the upper-tail probability is zero. Consequently,

$$\left| 2 \sum_j g(a) p_j \mu_j \Pr_{\text{Pois}}(\mathbf{L}_j^c) e^{-t_j} \right| \leq 2e^{-16\ell}.$$

Define the ideal raw contrast by

$$\tau_{\text{raw}} := 2 \sum_j g(a) W_j G_j.$$

Membership independence gives $E_{\text{Pois}}(W_j G_j) = p_j E_{\text{Pois}} G_j$. Combining the exact bias identity with the last two aggregate bounds,

$$|E_{\text{Pois}} \tau_{\text{raw}} - \Phi(P)| \leq \frac{8dB}{N_0 k^2} + 6e^{-16\ell} = b_{n,d,q}.$$

17. We next establish the ratio variance contribution. For a fixed marked prefix length ν , define the estimation-arrival index set by

$$\mathbf{l}_j^{(\nu)} := \{i \in \{1, \dots, \nu\} : i \text{ has estimation label and is an arrival in cell } j\}.$$

On a prescribed pattern $\mathbf{l}_j^{(\nu)} = \mathbf{l}_{\text{arr}}$ with $\nu_{\text{arr}} := |\mathbf{l}_{\text{arr}}| > 0$, the restricted product law makes the arrived-outcome indicators independent with mean μ_j . Their centered first moments vanish, and their centered squares are at most one. Expanding the square therefore gives, for patterns of positive probability,

$$E\left[(U_j - \nu_{\text{arr}} \mu_j)^2 \mid \mathbf{l}_j^{(\nu)} = \mathbf{l}_{\text{arr}}\right] \leq \nu_{\text{arr}},$$

and hence

$$E\left[(D_j - \mu_j)^2 \mid \mathbf{l}_j^{(\nu)} = \mathbf{l}_{\text{arr}}\right] \leq \nu_{\text{arr}}^{-1}.$$

Zero-probability patterns contribute zero. On the zero-count pattern, $(D_j - \mu_j)^2 = \mu_j^2 \leq 1$. Summing first over arrival patterns and then over the Poisson prefix size gives

$$E_{\text{Pois}}(D_j - \mu_j)^2 \leq E_{\text{Pois}} \left[\mathbf{1}_{\{C_j=0\}} + \frac{\mathbf{1}_{\{C_j>0\}}}{C_j} \right],$$

where the quotient is zero at $C_j = 0$.

If $t_j \leq 1$, the integrand on the right is at most one, so its expectation is at most $1 \leq 4/(1+t_j)$. If $t_j > 1$, its value is at most $2/(C_j+1)$. For every nonnegative integer h_{count} , the Poisson probabilities satisfy

$$\Pr_{\text{Pois}}(C_j = h_{\text{count}}) = e^{-t_j} \frac{t_j^{h_{\text{count}}}}{h_{\text{count}}!}, \quad \frac{t_j}{h_{\text{count}}+1} \Pr_{\text{Pois}}(C_j = h_{\text{count}}) = \Pr_{\text{Pois}}(C_j = h_{\text{count}} + 1).$$

The probability series sums to one, and the reciprocal series is bounded termwise by it. Summing the recurrence and shifting the index therefore gives

$$\begin{aligned} t_j E_{\text{Pois}} \frac{1}{C_j + 1} &= \sum_{h_{\text{count}}=0}^{\infty} \frac{t_j}{h_{\text{count}}+1} \Pr_{\text{Pois}}(C_j = h_{\text{count}}) \\ &= \sum_{h_{\text{count}}=0}^{\infty} \Pr_{\text{Pois}}(C_j = h_{\text{count}} + 1) \\ &= \Pr_{\text{Pois}}(C_j > 0) = 1 - e^{-t_j}. \end{aligned}$$

Dividing by $t_j > 1$ yields

$$E_{\text{Pois}} \frac{1}{C_j + 1} = \frac{1 - e^{-t_j}}{t_j}.$$

Therefore

$$E_{\text{Pois}}(D_j - \mu_j)^2 \leq \frac{2}{t_j} \leq \frac{4}{1+t_j} \quad (t_j > 1).$$

When the arrived-cell probability is zero, $C_j = U_j = \mu_j = 0$ almost surely, so the centered second moment is zero. In all cases,

$$\text{Var}_{\text{Pois}}(D_j) = \text{Var}_{\text{Pois}}(D_j - \mu_j) \leq E_{\text{Pois}}(D_j - \mu_j)^2 \leq \frac{4}{1+t_j}.$$

Expanding the variance of the independent product $W_j D_j$ gives

$$\text{Var}_{\text{Pois}}(W_j D_j) = p_j^2 \text{Var}_{\text{Pois}}(D_j) + \frac{p_j}{m} E_{\text{Pois}} D_j^2.$$

Using $D_j^2 \leq 1$ and $N_0 p_j \leq t_j$,

$$\frac{p_j^2}{1+t_j} \leq \frac{p_j}{N_0},$$

including $p_j = 0$. Summing over cells proves

$$\sum_j \text{Var}_{\text{Pois}}(W_j D_j) \leq \frac{4}{N_0} + \frac{1}{m}.$$

18. On the polynomial branch, define the weighted correction by

$$\Delta_j := W_j(G_j - D_j) = W_j \mathbf{1}_{L_j}(H_j - D_j).$$

Pilot and membership independence give

$$E_{\text{Pois}} \Delta_j^2 = \left(p_j^2 + \frac{p_j}{m} \right) \Pr_{\text{Pois}}(L_j) E_{\text{Pois}}(H_j - D_j)^2.$$

Since $(H_j - D_j)^2 \leq 2(H_j^2 + D_j^2)$, for $t_j \leq B$,

$$E_{\text{Pois}} \Delta_j^2 \leq 2 \left(p_j^2 + \frac{p_j}{m} \right) (e^{\ell/8} + 1).$$

Here $p_j \leq B/N_0$, so

$$p_j^2 + \frac{p_j}{m} \leq \frac{B^2}{N_0^2} + \frac{B}{mN_0}.$$

Summing over at most $4d$ cells,

$$8 \sum_{j: t_j \leq B} E_{\text{Pois}} \Delta_j^2 \leq 64d(e^{\ell/8} + 1) \left(\frac{B^2}{N_0^2} + \frac{B}{mN_0} \right).$$

For $t_j > B$, the two heavy-pilot estimates instead yield

$$\Pr_{\text{Pois}}(L_j) E_{\text{Pois}}(H_j - D_j)^2 \leq 4e^{-32\ell},$$

and therefore

$$8 \sum_{j: t_j > B} E_{\text{Pois}} \Delta_j^2 \leq 32e^{-32\ell} \left(1 + \frac{1}{m} \right).$$

In the last step we used

$$\sum_j \left(p_j^2 + \frac{p_j}{m} \right) \leq 1 + \frac{1}{m},$$

because $0 \leq p_j \leq 1$ and $\sum_j p_j = 1$.

19. Distinct cells are independent in the uncapped marked experiment. Since $g(a)^2 = 1$,

$$\text{Var}_{\text{Pois}}(\tau_{\text{raw}}) = 4 \sum_j \text{Var}_{\text{Pois}}(W_j G_j).$$

For each cell, $W_j G_j = W_j D_j + \Delta_j$. Nonnegativity of $\text{Var}_{\text{Pois}}(W_j D_j - \Delta_j)$ implies

$$\text{Var}_{\text{Pois}}(W_j G_j) \leq 2 \text{Var}_{\text{Pois}}(W_j D_j) + 2 \text{Var}_{\text{Pois}}(\Delta_j) \leq 2 \text{Var}_{\text{Pois}}(W_j D_j) + 2E_{\text{Pois}} \Delta_j^2.$$

The preceding aggregate estimates thus give

$$\text{Var}_{\text{Pois}}(\tau_{\text{raw}}) \leq v_{n,d,q}.$$

The raw contrast has finite second moment by the finite polynomial moment bounds already established, and its squared error decomposes as

$$E_{\text{Pois}}(\tau_{\text{raw}} - \Phi(P))^2 = \text{Var}_{\text{Pois}}(\tau_{\text{raw}}) + (E_{\text{Pois}} \tau_{\text{raw}} - \Phi(P))^2 \leq v_{n,d,q} + b_{n,d,q}^2.$$

Clipping decreases squared distance to every point of $[-1, 1]$: below -1 it replaces the value by -1 , above 1 by 1 , and inside the interval it preserves the value. Since $\Phi(P) \in [-1, 1]$,

$$E_{\text{Pois}}(\tilde{\tau}_{\text{Pois}} - \Phi(P))^2 \leq b_{n,d,q}^2 + v_{n,d,q}.$$

Combining this with the prefix transfer and the bound by four proves

$$E_P[(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] \leq \min\{4, b_{n,d,q}^2 + v_{n,d,q} + 4e^{-n\gamma_0}\} = \mathfrak{U}_{n,d,q}.$$

20. Finally, consider the ratio branch. Its ideal raw contrast is

$$\tau_{\text{raw}}^{\text{ratio}} := 2 \sum_j g(a) W_j D_j.$$

The ratio mean and membership independence give

$$E_{\text{Pois}} \tau_{\text{raw}}^{\text{ratio}} - \Phi(P) = -2 \sum_j g(a) p_j \mu_j e^{-t_j}.$$

Because $t_j \geq N_0 p_j$, $0 \leq \mu_j \leq 1$, and $t_{\text{bias}} e^{-t_{\text{bias}}} \leq e^{-1}$ for every real $t_{\text{bias}} \geq 0$,

$$p_j \mu_j e^{-t_j} \leq p_j e^{-N_0 p_j} \leq \frac{1}{e N_0}.$$

The exponential inequality follows from $e^{t_{\text{bias}}-1} \geq t_{\text{bias}}$. Thus

$$|E_{\text{Pois}} \tau_{\text{raw}}^{\text{ratio}} - \Phi(P)| \leq \frac{8d}{e N_0}.$$

Independence across cells and the ratio variance aggregate give

$$\text{Var}_{\text{Pois}}(\tau_{\text{raw}}^{\text{ratio}}) = 4 \sum_j \text{Var}_{\text{Pois}}(W_j D_j) \leq 4 \left(\frac{4}{N_0} + \frac{1}{m} \right).$$

The squared-error decomposition, clipping, prefix transfer, and bound by four therefore imply

$$\begin{aligned} E_P[(\hat{\tau}_{n,d,q}^{\text{mix}} - \Phi(P))^2] \\ \leq \min \left\{ 4, \left(\frac{8d}{e N_0} \right)^2 + 4 \left(\frac{4}{N_0} + \frac{1}{m} \right) + 4e^{-n\gamma_0} \right\} = \mathfrak{U}_{n,d,q}. \end{aligned}$$

Together with the zero and capped-envelope cases, this establishes the risk bound for every admissible P . Combining it with the deterministic comparison proved above gives

$$E_P[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P))^2] \leq \mathfrak{U}_{n,d,q} \leq 10^{30} r_{n,d,q}.$$

The expectation is under the canonical product observed-sample law specified in the statement, and 10^{30} is positive, finite, and independent of P, n, d, q .

□

Proof of Theorem 10. 1. Choose the universal constant supplied by Theorem 9, so that

$$0 < \overline{C} < \infty.$$

Fix n, d, q and $P \in \mathcal{M}_{n,d,q}$ satisfying the hypotheses.

2. The parameter, randomization, consistency, and arrival conditions in Definition 14 give

$$P \in \mathcal{M}_{n,d,q}^+,$$

by Definition 7. Moreover, Definition 14 supplies Assumption 1 for the observed sample. Applying Theorem 9 therefore yields

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q}, \quad \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q},$$

where $\Phi(P)$ is the signed cell functional of Definition 10. The estimator, rate, and deterministic envelope in that certificate are precisely those in the present statement, with the same formulas on each branch.

3. Since $P \in \mathcal{M}_{n,d,q}^+$, Lemma 3 gives

$$\tau(P) = \Phi(P).$$

Substituting this identity into the risk bound proves

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] = E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \Phi(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

The chosen constant is universal, so these inequalities hold throughout the stated model class. □

Proof of Theorem 11. 1. Choose the universal constant $\overline{C} > 0$ supplied by Theorem 10. Fix n, d, q and a full-data law P satisfying the stated conditions. Membership in $\mathcal{M}_{n,d,q}$ supplies all hypotheses of that certificate. By Lemma 4, the causal target equals the total observed-cell functional on this model class, which is the identification step used by the certificate. The certificate therefore gives

$$E_P \left[\left(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) \right)^2 \right] \leq \mathfrak{U}_{n,d,q} \leq \overline{C} r_{n,d,q}.$$

The envelope and branch rules in that certificate are exactly those in the present statement. It remains to establish the comparison with the auxiliary randomized risk.

2. Write the prefix probabilities and their remaining mass as

$$w_{k_0} := \Pr(K_0 = k_0) = e^{-n/2} \frac{(n/2)^{k_0}}{k_0!}, \quad k_0 \in \mathbb{N}, \quad w_{\text{tail}} := 1 - \sum_{k_0=0}^n w_{k_0}.$$

The singleton prefix events are disjoint, so

$$w_{k_0} \geq 0, \quad \sum_{k_0=0}^n w_{k_0} \leq 1, \quad w_{\text{tail}} = \Pr(K_0 > n) \geq 0.$$

For each integer $0 \leq k_0 \leq n$, define the assignment set by

$$\mathcal{A}_{\text{str},k_0} := \{\sigma : \{1, \dots, k_0\} \longrightarrow \{\text{membership, pilot, estimation}\}\}, \quad |\mathcal{A}_{\text{str},k_0}| = 3^{k_0}.$$

Let the output at a prescribed prefix and assignment, and its assignment average, be

$$\tilde{\tau}_{k_0,\sigma}(\mathbf{O}_n) := \tilde{\tau}(\mathbf{O}_n; \omega) \quad \text{when } \omega \text{ prescribes } K_0 = k_0 \text{ and assignment } \sigma,$$

$$\bar{\tau}_{\text{str},k_0}(\mathbf{O}_n) := \frac{1}{3^{k_0}} \sum_{\sigma \in \mathcal{A}_{\text{str},k_0}} \tilde{\tau}_{k_0,\sigma}(\mathbf{O}_n).$$

These outputs use the branch rules of [Algorithm 1](#). For $k_0 = 0$, the assignment set contains the unique empty assignment.

The zero output on the prefix tail and the conditional-average definition give

$$\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) = \sum_{k_0=0}^n w_{k_0} \bar{\tau}_{\text{str},k_0}(\mathbf{O}_n).$$

Pointwise in the observed sample, its centered value therefore satisfies

$$\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P) = w_{\text{tail}}(-\tau(P)) + \sum_{k_0=0}^n w_{k_0} (\bar{\tau}_{\text{str},k_0}(\mathbf{O}_n) - \tau(P)).$$

The weights on the right are nonnegative and sum to one. Convexity of the square yields

$$(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2 \leq w_{\text{tail}}\tau(P)^2 + \sum_{k_0=0}^n w_{k_0} (\bar{\tau}_{\text{str},k_0}(\mathbf{O}_n) - \tau(P))^2.$$

3. For every $0 \leq k_0 \leq n$, centering the assignment average gives

$$\bar{\tau}_{\text{str},k_0}(\mathbf{O}_n) - \tau(P) = \frac{1}{3^{k_0}} \sum_{\sigma \in \mathcal{A}_{\text{str},k_0}} (\tilde{\tau}_{k_0,\sigma}(\mathbf{O}_n) - \tau(P)).$$

The finite Cauchy–Schwarz inequality consequently gives

$$(\bar{\tau}_{\text{str},k_0}(\mathbf{O}_n) - \tau(P))^2 \leq \frac{1}{3^{k_0}} \sum_{\sigma \in \mathcal{A}_{\text{str},k_0}} (\tilde{\tau}_{k_0,\sigma}(\mathbf{O}_n) - \tau(P))^2.$$

Multiplying these inequalities by the nonnegative w_{k_0} , summing, and inserting the result into the preceding bound yields

$$\begin{aligned} & (\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2 \\ & \leq w_{\text{tail}}\tau(P)^2 + \sum_{k_0=0}^n \frac{w_{k_0}}{3^{k_0}} \sum_{\sigma \in \mathcal{A}_{\text{str},k_0}} (\tilde{\tau}_{k_0,\sigma}(\mathbf{O}_n) - \tau(P))^2. \end{aligned}$$

4. Conditional on the observed sample, the prefix has probabilities w_{k_0} , each assignment has probability 3^{-k_0} , and the tail output is zero. Thus

$$\begin{aligned} E_\omega \left[(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P))^2 \mid \mathbf{O}_n \right] \\ = w_{\text{tail}} \tau(P)^2 + \sum_{k_0=0}^n \frac{w_{k_0}}{3^{k_0}} \sum_{\sigma \in \mathcal{A}_{\text{str}, k_0}} (\tilde{\tau}_{k_0, \sigma}(\mathbf{O}_n) - \tau(P))^2. \end{aligned}$$

The observed-sample space is finite, so the displayed functions are measurable and integrable. Integrating the pointwise inequality establishes

$$E_P \left[(\hat{\tau}_{n,d,q}^{\text{mix}}(\mathbf{O}_n) - \tau(P))^2 \right] \leq E_{P,\omega} \left[(\tilde{\tau}(\mathbf{O}_n; \omega) - \tau(P))^2 \right].$$

Together with the certificate bound, this proves all the asserted inequalities with the same universal constant $\bar{C}$. □

Proof of Theorem 12. Let $\underline{c}_{\text{one}} > 0$ be the universal constant supplied by Lemma 8. Define the universal constants

$$\begin{aligned} \eta &= e^{-32}, \quad N_{\text{act}} = \max \left\{ 1, e^2, 4\eta, 4096\eta, e^{(32+\log 8)/28} \right\}, \\ \underline{c}_{\text{act}} &= \min \left\{ \frac{1}{256N_{\text{act}}}, \frac{1}{256 \cdot 2048^2}, \frac{\eta^2}{16384} \right\}, \quad \underline{c}_* = \min \{ \underline{c}_{\text{one}}, \underline{c}_{\text{act}} \}, \quad \underline{c} = \frac{\underline{c}_*}{4}. \end{aligned}$$

These constants are positive and independent of n, d, q .

Fix parameters satisfying the hypotheses, and define

$$N = nq, \quad \ell = \log(e + N), \quad u_{\text{act}} = \frac{d}{N\ell}, \quad a_{\text{eff}} = N^{-1}, \quad b_{\text{dim}} = u_{\text{act}}^2.$$

Then

$$N > 0, \quad \ell \geq \log e = 1, \quad a_{\text{eff}} \geq 0, \quad b_{\text{dim}} \geq 0.$$

By Definition 17,

$$r_{n,d,q} = \min\{1, a_{\text{eff}} + b_{\text{dim}}\}.$$

1. *The one-cell contribution.* By Lemma 8,

$$\underline{c}_{\text{one}} \min\{1, a_{\text{eff}}\} \leq \mathfrak{R}_{n,d,q}.$$

We next establish the complementary bound

$$\sqrt{N} \leq d \implies \underline{c}_{\text{act}} \min\{1, b_{\text{dim}}\} \leq \mathfrak{R}_{n,d,q}.$$

2. *A numerical one-cell estimate for constant calibration.* We record explicitly the numerical estimate used below:

$$\frac{1}{256} \min\{1, N^{-1}\} \leq \mathfrak{R}_{n,d,q}.$$

For $\mu_{\text{one}} \in [1/4, 3/4]$, define $P_{\text{one}, \mu_{\text{one}}}$ by

$$X = 0, \quad S(0) = S(1) = Y(0) = 0, \quad (A, Y(1), R) \sim \text{Bernoulli}(1/2) \otimes \text{Bernoulli}(\mu_{\text{one}}) \otimes \text{Bernoulli}(q),$$

$$S = 0, \quad Y = AY(1).$$

Treatment is independent of the potential variables, and arrival is independent of the outcome conditional on the observed cell. Every occupied cell has arrival probability q . Consequently,

$$P_{\text{one}, \mu_{\text{one}}} \in \mathcal{M}_{n, d, q}, \quad \tau(P_{\text{one}, \mu_{\text{one}}}) = \mu_{\text{one}}.$$

Write

$$Q_{\text{one}, \mu_{\text{one}}} = \mathcal{L}_{P_{\text{one}, \mu_{\text{one}}}}(O_1), \quad \delta_{\text{one}} = \min \left\{ \frac{1}{4}, \frac{1}{4\sqrt{N}} \right\}.$$

Thus $0 < \delta_{\text{one}} \leq 1/4$, and the two target values $1/2$ and $1/2 + \delta_{\text{one}}$ belong to $[1/4, 3/4]$. Considering the two branches of this minimum gives

$$\delta_{\text{one}}^2 \geq \frac{1}{16} \min\{1, N^{-1}\}, \quad 16N\delta_{\text{one}}^2 \leq 1.$$

For probability laws $Q' \ll Q$, use the convention

$$\chi^2(Q' \| Q) = \int \left(\frac{dQ'}{dQ} - 1 \right)^2 dQ.$$

The two one-record laws differ precisely at the treated, arrived records with outcome zero or one. Each of these atoms has probability $q/4$ under $Q_{\text{one}, 1/2}$, and its probability changes in absolute value by $q\delta_{\text{one}}/2$. Summing their contributions gives

$$\chi^2(Q_{\text{one}, 1/2 + \delta_{\text{one}}} \| Q_{\text{one}, 1/2}) = 2q\delta_{\text{one}}^2 \leq \frac{1}{8n}.$$

Independence of the records factors the second moment of the likelihood ratio, so

$$1 + \chi^2(Q_{\text{one}, 1/2 + \delta_{\text{one}}}^{\otimes n} \| Q_{\text{one}, 1/2}^{\otimes n}) = [1 + \chi^2(Q_{\text{one}, 1/2 + \delta_{\text{one}}} \| Q_{\text{one}, 1/2})]^n \leq e^{1/8}.$$

Since $e^{1/8} \leq e^{1/2} < \sqrt{3} < 2$, Cauchy–Schwarz applied to the likelihood ratio yields

$$\text{TV}(Q_{\text{one}, 1/2 + \delta_{\text{one}}}^{\otimes n}, Q_{\text{one}, 1/2}^{\otimes n}) \leq \frac{1}{2} \sqrt{\chi^2(Q_{\text{one}, 1/2 + \delta_{\text{one}}}^{\otimes n} \| Q_{\text{one}, 1/2}^{\otimes n})} \leq \frac{1}{2}.$$

For every $\mathsf{T} \in \mathcal{T}_n$, define its barycenter on the observed-sample space by

$$b_{\mathsf{T}}(o) = \int_{[-1, 1]} h \mathsf{T}(o, dh).$$

By [Lemma 1](#),

$$b_{\mathsf{T}}(o) \in [-1, 1], \quad E_P[(b_{\mathsf{T}}(\mathbf{O}_n) - \tau(P))^2] \leq E_P[(\mathsf{T} - \tau(P))^2].$$

Define the nearer-target test

$$\psi_{\text{one}}(o) = \mathbf{1} \left\{ \left| b_{\mathsf{T}}(o) - \left(\frac{1}{2} + \delta_{\text{one}} \right) \right| \leq \left| b_{\mathsf{T}}(o) - \frac{1}{2} \right| \right\}.$$

An error under either target entails absolute estimation error at least $\delta_{\text{one}}/2$. The testing inequality in [Lemma 2](#), followed by the barycenter risk comparison, therefore gives

$$\begin{aligned} & \max \left\{ E_{P_{\text{one}, 1/2}} \left[(\mathbb{T} - \tfrac{1}{2})^2 \right], E_{P_{\text{one}, 1/2 + \delta_{\text{one}}}} \left[(\mathbb{T} - \tfrac{1}{2} - \delta_{\text{one}})^2 \right] \right\} \\ & \geq \frac{\delta_{\text{one}}^2}{8} \left[1 - \text{TV} \left(Q_{\text{one}, 1/2 + \delta_{\text{one}}}^{\otimes n}, Q_{\text{one}, 1/2}^{\otimes n} \right) \right] \geq \frac{\delta_{\text{one}}^2}{16} \geq \frac{1}{256} \min\{1, N^{-1}\}. \end{aligned}$$

Both laws belong to the model class. Taking the supremum over that class and then the infimum over $\mathbb{T}$ proves the displayed numerical estimate.

3. *Large effective size and many rare cells.* To prove the complementary bound, suppose $\sqrt{N} \leq d$. Use the construction parameters of [Algorithm 2](#):

$$K = \lceil \ell \rceil, \quad H = K^2, \quad b_0 = \frac{\eta}{N\ell}, \quad J = \min \left\{ d - 1, \left\lfloor \frac{1}{2b_0} \right\rfloor \right\}.$$

Here $b_0 > 0$, $0 \leq J \leq d - 1$, and

$$Jb_0 \leq \frac{1}{2}.$$

The ceiling bound and $\ell \geq 1$ give

$$\ell \leq K < \ell + 1 \leq 2\ell, \quad H \geq 1.$$

The rare-arrival restriction then implies

$$0 < qH \leq 4q\ell^2 \leq \frac{1}{16}.$$

First suppose $N_{\text{act}} \leq N$. In particular,

$$N \geq 1, \quad N \geq e^2, \quad N \geq 4\eta, \quad N \geq 4096\eta, \quad N \geq e^{(32 + \log 8)/28}.$$

The capacity bound is

$$2048(2b_0) = \frac{4096\eta}{N\ell} \leq 1, \quad 2048 \leq \frac{1}{2b_0}.$$

Within this branch, first suppose $J \geq 2048$. Since $N \geq e^2$,

$$\ell = \log(e + N) \geq 2, \quad K \geq 2.$$

By [Lemma 7](#), there are finitely supported probability laws Π_0, Π_1 on $[1, H]$ satisfying

$$\int z^v d\Pi_0(z) = \int z^v d\Pi_1(z) \quad (0 \leq v \leq K, v \in \mathbb{N}).$$

If the reciprocal mean under Π_0 exceeds that under Π_1 , interchange the priors; otherwise retain their order. Their separation then reads

$$\int z^{-1} d\Pi_1(z) - \int z^{-1} d\Pi_0(z) \geq \frac{1}{12}.$$

For each $\mathbf{z} = (z_x)_{x=0}^{d-1} \in [1, H]^d$, define a full-data law $P_{\text{act}, \mathbf{z}}$ as follows. Its baseline probabilities are

$$p_{\text{act}, x} = \begin{cases} b_0, & 0 \leq x < J, \\ 1 - Jb_0, & x = J, \\ 0, & J < x < d. \end{cases}$$

Set $S(0) = S(1) = Y(0) = 0$, draw $A \sim \text{Bernoulli}(1/2)$ independently of the baseline and potential variables, and impose $S = 0$ and $Y = AY(1)$. Conditional on $X = x$, draw $Y(1)$ and R independently with

$$Y(1) \mid X = x \sim \begin{cases} \text{Bernoulli}(z_x^{-1}), & x < J, \\ \text{Bernoulli}(0), & x \geq J, \end{cases} \quad R \mid X = x \sim \begin{cases} \text{Bernoulli}(qz_x), & x < J, \\ \text{Bernoulli}(qH), & x \geq J. \end{cases}$$

The baseline masses sum to one and are nonnegative. Since $1 \leq z_x \leq H$, the occupied-cell arrival probabilities belong to $[q, 1]$. The specified independences give randomized assignment and conditional independence of arrival and outcome, and the definitions of S, Y give consistency. With independent sampling from this fixed law, all model conditions hold:

$$P_{\text{act}, \mathbf{z}} \in \mathcal{M}_{n, d, q}, \quad \tau_{\text{act}}(\mathbf{z}) := \tau(P_{\text{act}, \mathbf{z}}) = b_0 \sum_{x=0}^{J-1} z_x^{-1}.$$

For $\iota \in \{0, 1\}$, draw

$$\mathbf{Z} = (Z_x)_{x=0}^{d-1} \sim \Pi_\iota^{\otimes d}$$

once for the entire sample. Define the observed mixture law

$$Q_{\text{act}, \iota} = \int_{[1, H]^d} \mathcal{L}_{P_{\text{act}, \mathbf{z}}}(\mathbf{O}_n) d\Pi_\iota^{\otimes d}(\mathbf{z}),$$

and let $\tilde{Q}_{\text{act}, \iota}$ be the corresponding augmented law constructed in [Algorithm 2](#). Thus

$$\tilde{Q}_{\text{act}, \iota} = \int_{[1, H]^d} \mathcal{L}((\mathbf{B}_{\text{act}, t}, O_t)_{t=1}^n \mid \mathbf{Z} = \mathbf{z}) d\Pi_\iota^{\otimes d}(\mathbf{z}).$$

The activation flags have probability qH , independently of the latent vector and the baseline and treatment pattern.

At the record level, forgetting the flag gives the following probabilities for the three marks $(R, RY) = (1, 1), (1, 0), (0, 0)$:

	(1, 1)	(1, 0)	(0, 0)
$A = 1, X = x < J$	q	$q(z_x - 1)$	$1 - qz_x$
$A = 0, X = x < J$	0	qz_x	$1 - qz_x$
$X \geq J$	0	qH	$1 - qH$

These are exactly the probabilities induced by $P_{\text{act}, \mathbf{z}}$. Define the forgetting map by

$$\text{pr}_{\text{obs}}((\mathbf{B}_{\text{act}, t}, O_t)_{t=1}^n) = (O_t)_{t=1}^n.$$

Conditional independence of the records and integration over the common latent vector therefore give

$$Q_{\text{act}, \iota} = \tilde{Q}_{\text{act}, \iota} \circ \text{pr}_{\text{obs}}^{-1}.$$

4. *Equality on the moment-matching event.* Continue in the branch $N \geq N_{\text{act}}, J \geq 2048$. For an augmented sample, define the activated count in cell x and the cutoff event by

$$V_x = \sum_{t=1}^n \mathbf{1}\{X_t = x, \mathbf{B}_{\text{act},t} = 1\}, \quad \mathcal{G} = \{V_x \leq K \text{ for every } 0 \leq x < J\}.$$

Fix a possible augmented sample $\mathbf{o}_{\text{aug}}$, and denote its coordinates by the corresponding record symbols. Its baseline, treatment, surrogate, and flag pattern has the common probability

$$\varpi_{\mathbf{o}_{\text{aug}}} = \mathbf{1}\{S_t = 0 \text{ for every } t\} \prod_{t=1}^n \frac{p_{\text{act},X_t}}{2} (qH)^{\mathbf{B}_{\text{act},t}} (1 - qH)^{1-\mathbf{B}_{\text{act},t}}.$$

For each cell x , define $\mathcal{P}_{\mathbf{o}_{\text{aug}},x}(z)$ as the product, over records with $X_t = x$, of their conditional mark probabilities from [Algorithm 2](#):

$$\mathcal{P}_{\mathbf{o}_{\text{aug}},x}(z) = \prod_{t: X_t=x} \Pr((R, RY) = (R_t, R_t Y_t) \mid X = x, A = A_t, \mathbf{B}_{\text{act}} = \mathbf{B}_{\text{act},t}, Z_x = z).$$

Empty products equal one. The row entries extend these expressions to polynomials on $\mathbb{R}$; their probability interpretation applies on $[1, H]$. Impossible marks have weight zero. For a rare cell, an inactive factor is constant and an activated factor is affine in z . For every other cell the factors are constant. Hence

$$\deg \mathcal{P}_{\mathbf{o}_{\text{aug}},x} \leq V_x \quad (x < J), \quad \deg \mathcal{P}_{\mathbf{o}_{\text{aug}},x} \leq 0 \quad (x \geq J),$$

where a zero polynomial satisfies each stated degree bound.

On $\mathcal{G}$, define the coefficients $\beta_{\mathbf{o}_{\text{aug}},x,v}$ by the polynomial expansion

$$\mathcal{P}_{\mathbf{o}_{\text{aug}},x}(z) = \sum_{v=0}^K \beta_{\mathbf{o}_{\text{aug}},x,v} z^v \quad (z \in \mathbb{R}),$$

padding with zero coefficients when necessary. Product-prior independence then gives

$$\tilde{Q}_{\text{act},\iota}\{\mathbf{o}_{\text{aug}}\} = \varpi_{\mathbf{o}_{\text{aug}}} \prod_{x=0}^{d-1} \left[\sum_{v=0}^K \beta_{\mathbf{o}_{\text{aug}},x,v} \int z^v d\Pi_{\iota}(z) \right] \quad (\mathbf{o}_{\text{aug}} \in \mathcal{G}).$$

The matched moments make the right-hand side identical for $\iota = 0, 1$. Thus

$$\tilde{Q}_{\text{act},0}\{\mathbf{o}_{\text{aug}}\} = \tilde{Q}_{\text{act},1}\{\mathbf{o}_{\text{aug}}\} \quad (\mathbf{o}_{\text{aug}} \in \mathcal{G}).$$

5. *The cutoff probability and total variation.* The retained baseline, treatment, and flag pattern has the same distribution under both priors. In particular, for $x < J$,

$$V_x \sim \text{Binomial}(n, b_0 q H).$$

Define

$$k_{\text{tail}} = K + 1, \quad \nu_{\text{act}} = n b_0 q H = \frac{\eta K^2}{\ell}.$$

For an integer $v \geq 0$, let its falling factorial be

$$(v)_{k_{\text{tail}}} = \begin{cases} \prod_{t=0}^{k_{\text{tail}}-1} (v-t), & v \geq k_{\text{tail}}, \\ 0, & v < k_{\text{tail}}. \end{cases}$$

Expanding over ordered distinct record indices gives

$$E[(V_x)_{k_{\text{tail}}}] = (n)_{k_{\text{tail}}} (b_0 q H)^{k_{\text{tail}}} \leq \nu_{\text{act}}^{k_{\text{tail}}}.$$

When $k_{\text{tail}} > n$, both falling-factorial moments are zero. On $V_x > K$, the falling factorial is at least $k_{\text{tail}}!$. Therefore

$$\tilde{Q}_{\text{act},\iota}(V_x > K) \leq \frac{\nu_{\text{act}}^{k_{\text{tail}}}}{k_{\text{tail}}!}.$$

The bounds $K \leq 2\ell$ and $K \leq k_{\text{tail}}$ imply

$$K^2 \leq 2\ell k_{\text{tail}}, \quad \nu_{\text{act}} \leq 2\eta k_{\text{tail}} \leq e^{-31} k_{\text{tail}}, \quad k_{\text{tail}} \geq \ell,$$

where $2 \leq e$ gives the second inequality involving the exponential. The exponential series gives $k_{\text{tail}}^{k_{\text{tail}}}/k_{\text{tail}}! \leq e^{k_{\text{tail}}}$, whence

$$\frac{\nu_{\text{act}}^{k_{\text{tail}}}}{k_{\text{tail}}!} \leq e^{-31k_{\text{tail}}} \frac{k_{\text{tail}}^{k_{\text{tail}}}}{k_{\text{tail}}!} \leq e^{-30k_{\text{tail}}} \leq e^{-30\ell}.$$

A union bound over the rare cells yields

$$\tilde{Q}_{\text{act},\iota}(\mathcal{G}^c) \leq J e^{-30\ell}.$$

The mass cap and elementary exponential bounds give

$$J \leq \frac{N\ell}{2\eta}, \quad N \leq e^\ell, \quad \ell \leq e^\ell, \quad N\ell \leq e^{2\ell}.$$

Consequently,

$$\tilde{Q}_{\text{act},\iota}(\mathcal{G}^c) \leq \frac{N\ell}{2\eta} e^{-30\ell} \leq \frac{e^{2\ell}}{\eta} e^{-30\ell} = e^{32-28\ell}.$$

The threshold $N \geq e^{(32+\log 8)/28}$ implies

$$\ell \geq \frac{32 + \log 8}{28}, \quad \tilde{Q}_{\text{act},\iota}(\mathcal{G}^c) \leq \frac{1}{8}.$$

Equality of the atoms on $\mathcal{G}$ implies equality of the probabilities of $\mathcal{G}$, and hence of its complement. For every augmented-sample event $\mathcal{B}$,

$$\begin{aligned} & \left| \tilde{Q}_{\text{act},0}(\mathcal{B}) - \tilde{Q}_{\text{act},1}(\mathcal{B}) \right| \\ &= \left| \tilde{Q}_{\text{act},0}(\mathcal{B} \cap \mathcal{G}^c) - \tilde{Q}_{\text{act},1}(\mathcal{B} \cap \mathcal{G}^c) \right| \leq \tilde{Q}_{\text{act},0}(\mathcal{G}^c). \end{aligned}$$

Applying this inequality to preimages under the forgetting map gives

$$\text{TV}(Q_{\text{act},0}, Q_{\text{act},1}) \leq \text{TV}(\tilde{Q}_{\text{act},0}, \tilde{Q}_{\text{act},1}) \leq \frac{1}{8}.$$

6. *From center separation to squared-error risk.* For $\iota \in \{0, 1\}$, define the prior center and its separation by

$$\theta_{\text{act}, \iota} = E_{\Pi_\iota^{\otimes d}}[\tau_{\text{act}}(\mathbf{Z})] = Jb_0 \int z^{-1} d\Pi_\iota(z), \quad \Delta_{\text{act}} = |\theta_{\text{act}, 1} - \theta_{\text{act}, 0}|.$$

The reciprocal-mean separation implies

$$\Delta_{\text{act}} \geq \frac{Jb_0}{12}.$$

Since $Z_x^{-1} \in [0, 1]$, its variance satisfies

$$\text{Var}_{\Pi_\iota}(Z_x^{-1}) = E_{\Pi_\iota}[(Z_x^{-1} - \frac{1}{2})^2] - (E_{\Pi_\iota}[Z_x^{-1}] - \frac{1}{2})^2 \leq \frac{1}{4}.$$

Independence of the latent coordinates therefore yields

$$E_{\Pi_\iota^{\otimes d}}[(\tau_{\text{act}}(\mathbf{Z}) - \theta_{\text{act}, \iota})^2] = b_0^2 \sum_{x=0}^{J-1} \text{Var}_{\Pi_\iota}(Z_x^{-1}) \leq \frac{Jb_0^2}{4}.$$

Fix $\mathsf{T} \in \mathcal{T}_n$, and define its maximal model risk and its centered mixture risks by

$$\mathcal{W}(\mathsf{T}) = \sup_{P \in \mathcal{M}_{n,d,q}} E_P[(\mathsf{T} - \tau(P))^2], \quad C_{\text{act}, \iota}(\mathsf{T}) = \int (b_{\mathsf{T}}(o) - \theta_{\text{act}, \iota})^2 dQ_{\text{act}, \iota}(o).$$

The bounded action and target make these risks finite. Class membership of every prior support law gives

$$\int E_{P_{\text{act}, \mathbf{z}}}[(\mathsf{T} - \tau_{\text{act}}(\mathbf{z}))^2] d\Pi_\iota^{\otimes d}(\mathbf{z}) \leq \mathcal{W}(\mathsf{T}).$$

Using the barycenter comparison from [Lemma 1](#) and

$$(b_{\mathsf{T}}(o) - \theta_{\text{act}, \iota})^2 \leq 2(b_{\mathsf{T}}(o) - \tau_{\text{act}}(\mathbf{z}))^2 + 2(\tau_{\text{act}}(\mathbf{z}) - \theta_{\text{act}, \iota})^2$$

inside the mixture integral gives

$$C_{\text{act}, \iota}(\mathsf{T}) \leq 2\mathcal{W}(\mathsf{T}) + \frac{Jb_0^2}{2}.$$

Define the nearer-center test

$$\psi_{\text{act}}(o) = \mathbf{1} \{|b_{\mathsf{T}}(o) - \theta_{\text{act}, 1}| \leq |b_{\mathsf{T}}(o) - \theta_{\text{act}, 0}|\}.$$

The triangle inequality shows that a testing error under center ι entails

$$|b_{\mathsf{T}}(o) - \theta_{\text{act}, \iota}| \geq \frac{\Delta_{\text{act}}}{2}.$$

By [Lemma 2](#),

$$Q_{\text{act}, 0}(\psi_{\text{act}} = 1) + Q_{\text{act}, 1}(\psi_{\text{act}} = 0) \geq 1 - \text{TV}(Q_{\text{act}, 0}, Q_{\text{act}, 1}).$$

Bounding each error probability by its centered squared risk consequently gives

$$\frac{\Delta_{\text{act}}^2}{8} [1 - \text{TV}(Q_{\text{act},0}, Q_{\text{act},1})] \leq \max_{\iota \in \{0,1\}} C_{\text{act},\iota}(\mathbb{T}).$$

Together with the total variation and centered-risk bounds, this implies

$$\frac{7\Delta_{\text{act}}^2}{128} - \frac{Jb_0^2}{4} \leq \mathcal{W}(\mathbb{T}).$$

Now $J \geq 2048$ and $\Delta_{\text{act}} \geq Jb_0/12$, so

$$\mathcal{W}(\mathbb{T}) \geq (Jb_0)^2 \left(\frac{7}{18432} - \frac{1}{4J} \right) \geq \frac{(Jb_0)^2}{4096}.$$

Taking the infimum over $\mathbb{T}$ gives the calibrated fuzzy bound

$$\frac{(Jb_0)^2}{4096} \leq \mathfrak{R}_{n,d,q}.$$

7. *Converting rare-cell mass to the dimensional contribution.* Still suppose $N \geq N_{\text{act}}$ and $J \geq 2048$. We have

$$b_0 = \frac{\eta}{N\ell} \leq \frac{1}{4}, \quad 0 < \eta = e^{-32} \leq \frac{1}{2}, \quad d \geq 2.$$

The first inequality uses $N \geq 4\eta$ and $\ell \geq 1$; the second follows from $e^{32} \geq 1 + 32$.

If $d - 1 \leq \lfloor (2b_0)^{-1} \rfloor$, then

$$J = d - 1 \geq \frac{d}{2}, \quad Jb_0 \geq \frac{db_0}{2}.$$

Otherwise the floor determines J , and

$$Jb_0 = \left\lfloor \frac{1}{2b_0} \right\rfloor b_0 \geq \frac{1}{2} - b_0 \geq \frac{1}{4}.$$

Thus both branches give

$$\min \left\{ \frac{db_0}{2}, \frac{1}{4} \right\} \leq Jb_0.$$

Moreover,

$$\frac{\eta}{24} \min\{1, u_{\text{act}}\} \leq \frac{\eta u_{\text{act}}}{24} = \frac{db_0}{24}, \quad \frac{\eta}{24} \min\{1, u_{\text{act}}\} \leq \frac{\eta}{24} \leq \frac{1}{48}.$$

It follows that

$$\frac{\eta}{24} \min\{1, u_{\text{act}}\} \leq \frac{1}{12} \min \left\{ \frac{db_0}{2}, \frac{1}{4} \right\} \leq \frac{Jb_0}{12},$$

and hence

$$\frac{\eta}{2} \min\{1, u_{\text{act}}\} \leq Jb_0.$$

For $u_{\text{act}} \geq 0$, considering $u_{\text{act}} \leq 1$ and $u_{\text{act}} \geq 1$ gives

$$(\min\{1, u_{\text{act}}\})^2 = \min\{1, u_{\text{act}}^2\}.$$

Squaring the mass bound therefore yields

$$\frac{\eta^2}{4} \min\{1, u_{\text{act}}^2\} \leq (Jb_0)^2.$$

The choice of $\underline{c}_{\text{act}}$ and the calibrated fuzzy bound now imply

$$\underline{c}_{\text{act}} \min\{1, b_{\text{dim}}\} \leq \frac{\eta^2}{16384} \min\{1, u_{\text{act}}^2\} \leq \frac{(Jb_0)^2}{4096} \leq \mathfrak{R}_{n,d,q}.$$

8. *Large effective size with fewer rare cells.* Suppose instead that $N \geq N_{\text{act}}$ and $J < 2048$. The capacity bound established above gives

$$\left\lfloor \frac{1}{2b_0} \right\rfloor \geq 2048.$$

The definition of J therefore forces

$$d - 1 < 2048, \quad d \leq 2048.$$

This includes $J = 0$, for which the same argument gives $d = 1$.

Since $N \geq 1$ and $\ell \geq 1$,

$$\sqrt{N} \leq N \leq N\ell.$$

Consequently,

$$\min\{1, u_{\text{act}}\} \leq \frac{d}{N\ell} \leq \frac{2048}{\sqrt{N}}, \quad \min\{1, u_{\text{act}}^2\} \leq \frac{2048^2}{N}.$$

Also,

$$\min\{1, N^{-1}\} = N^{-1}, \quad \underline{c}_{\text{act}} 2048^2 \leq \frac{1}{256}.$$

The numerical one-cell estimate thus gives

$$\underline{c}_{\text{act}} \min\{1, b_{\text{dim}}\} \leq \frac{\underline{c}_{\text{act}} 2048^2}{N} \leq \frac{1}{256N} \leq \mathfrak{R}_{n,d,q}.$$

9. *Small effective size.* In the remaining effective-size branch, $N < N_{\text{act}}$. Because $N > 0$ and $N_{\text{act}} \geq 1$,

$$N_{\text{act}}^{-1} \leq 1, \quad N_{\text{act}}^{-1} \leq N^{-1}, \quad N_{\text{act}}^{-1} \leq \min\{1, N^{-1}\}.$$

The definition of $\underline{c}_{\text{act}}$ and the numerical one-cell estimate give

$$\underline{c}_{\text{act}} \min\{1, b_{\text{dim}}\} \leq \underline{c}_{\text{act}} \leq \frac{1}{256N_{\text{act}}} \leq \frac{1}{256} \min\{1, N^{-1}\} \leq \mathfrak{R}_{n,d,q}.$$

The three effective-size and rare-count branches establish the complementary bound whenever $\sqrt{N} \leq d$.

10. *Combining the bounds when $d \geq \sqrt{N}$.* First suppose $\sqrt{N} \leq d$. Both component bounds apply, and their nonnegative capped factors give

$$\underline{c}_* \min\{1, a_{\text{eff}}\} \leq \underline{c}_{\text{one}} \min\{1, a_{\text{eff}}\} \leq \mathfrak{R}_{n,d,q},$$

$$\underline{c}_* \min\{1, b_{\text{dim}}\} \leq \underline{c}_{\text{act}} \min\{1, b_{\text{dim}}\} \leq \mathfrak{R}_{n,d,q}.$$

For the nonnegative quantities $a_{\text{eff}}, b_{\text{dim}}$,

$$\min\{1, a_{\text{eff}} + b_{\text{dim}}\} \leq \min\{1, a_{\text{eff}}\} + \min\{1, b_{\text{dim}}\}.$$

Indeed, if both quantities are at most one, the right-hand side is their sum; if either is at least one, the right-hand side is at least one. Each capped quantity is at most their maximum, so

$$\min\{1, a_{\text{eff}} + b_{\text{dim}}\} \leq 2 \max\{\min\{1, a_{\text{eff}}\}, \min\{1, b_{\text{dim}}\}\},$$

while the component risk bounds imply

$$\underline{c}_* \max\{\min\{1, a_{\text{eff}}\}, \min\{1, b_{\text{dim}}\}\} \leq \mathfrak{R}_{n,d,q}.$$

Since $\mathfrak{R}_{n,d,q} \geq 0$, we conclude that

$$\underline{c} r_{n,d,q} \leq \frac{\underline{c}_*}{2} \max\{\min\{1, a_{\text{eff}}\}, \min\{1, b_{\text{dim}}\}\} \leq \frac{1}{2} \mathfrak{R}_{n,d,q} \leq \mathfrak{R}_{n,d,q}.$$

11. *Combining the bounds when $d < \sqrt{N}$.* Suppose $\sqrt{N} > d$. Since $d \geq 0$,

$$d^2 \leq (\sqrt{N})^2 = N.$$

Also $\ell \geq 1$, so

$$\ell^2 \geq 1, \quad (N\ell)^2 > 0, \quad N^{-1}(N\ell)^2 = N\ell^2.$$

Therefore

$$b_{\text{dim}} = \frac{d^2}{(N\ell)^2} \leq \frac{N\ell^2}{(N\ell)^2} = N^{-1} = a_{\text{eff}},$$

and monotonicity of the minimum gives

$$\min\{1, b_{\text{dim}}\} \leq \min\{1, a_{\text{eff}}\}.$$

Applying the capped-sum inequality established above,

$$r_{n,d,q} \leq \min\{1, a_{\text{eff}}\} + \min\{1, b_{\text{dim}}\} \leq 2 \min\{1, a_{\text{eff}}\}.$$

Finally, the one-cell contribution and $\underline{c}_* \leq \underline{c}_{\text{one}}$ yield

$$\underline{c} r_{n,d,q} \leq \frac{\underline{c}_*}{2} \min\{1, a_{\text{eff}}\} \leq \underline{c}_{\text{one}} \min\{1, a_{\text{eff}}\} \leq \mathfrak{R}_{n,d,q}.$$

The two dimension branches cover all admissible parameters.

□

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